OPTIMUS ORIEN

Part 1

Product Foundations & Core Requirements

Objective: Define the product, its purpose, users, core functions, boundaries, and essential design principles before designing its technical architecture.

Section 1. What exactly is Optimus Orien?

Let's start with the most important question.

Imagine you own a group of schools.

You have 10 schools, 500 teachers, and 8,000 students.

Every day, these schools generate information:

- Students receive marks in examinations.
- Teachers record attendance.
- Students complete or miss homework.
- Teachers observe student participation.
- Students develop different academic skills.
- Some students improve, while others struggle.
- Some classes perform better than others.
- Some schools perform better than others.

The problem is that collecting this information is not the same as understanding it.

You might know that a student scored 42 out of 100 in Mathematics. But that number alone doesn't tell you everything you need to know.

For example:

- Is the student improving or declining?
- Has the student consistently struggled with Mathematics?
- Is the student attending classes regularly?

- Does the student perform well in other subjects?
- Is the problem limited to one topic or spread across several subjects?
- Has the student recently started missing homework?
- Does the teacher need to intervene?
- Should the parents be informed?
- Is the student's performance improving after receiving support?

Now imagine trying to answer these questions for 8,000 students across 10 schools.

You would need to examine thousands of records, compare results, identify patterns, and decide which problems require attention.

This is the problem Optimus Orien is intended to address.

Our proposed definition

Optimus Orien is a **School Performance Intelligence and Early Intervention Platform**.

In simple English, it is a system that helps schools:

1. Collect and organise important school performance information.
2. Understand student, class, teacher, and school performance.
3. Identify students and areas that may need attention.
4. Track changes over time.
5. Present information through understandable charts, reports, and indicators.
6. Help school leaders decide where support or improvement may be needed.

The platform should make it easier for schools to understand what is happening, why it might be happening, and where they should investigate further.

Notice the distinction: the platform helps identify possible problems. It should not automatically assume that every low score means a student is lazy, that a teacher is ineffective, or that a school is poorly managed.

The purpose is to support better decisions, not to blame people.

What does "Performance Intelligence" mean?

Let's break down the phrase.

Performance means how well something is working or progressing.

Examples:

- A student's academic performance.
- A class's average performance.
- A school's attendance performance.
- A student's progress in a particular subject.

Intelligence, in this context, means turning raw information into useful understanding.

For example:

Raw information	Useful interpretation
A student scored 85% in the previous examination and 52% in the latest one.	The student's results have declined, so the school may need to investigate.
A student has missed several classes recently.	Attendance may be affecting the student's learning.
A class has a Mathematics average of 48%, while its other subject averages are substantially higher.	Mathematics may need additional attention at the class level.
A student's results have improved over three consecutive assessments.	The student is showing a positive performance trend.

The interpretation is not necessarily a definitive explanation. It is a starting point for investigation.

Important distinction: Optimus Orien should initially use transparent calculations and documented rules. It does not need an artificial intelligence system that makes mysterious predictions to provide useful performance intelligence.

We can consider advanced artificial intelligence in the future, but it should not be a prerequisite for the core product.

Section 2. What problem are we solving?

Before deciding which features to build, we need to identify the problems they are supposed to solve.

Otherwise, we risk creating a website full of features without a clear purpose.

I propose that Optimus Orien address six core problems.

Problem 1: School information is difficult to understand collectively

A school might keep examination results in one spreadsheet, attendance records in another, and homework information somewhere else.

Even when the information exists, school leaders may struggle to see the complete picture.

Our solution: Bring relevant information into one organised platform and present it through understandable dashboards, reports, and visualisations.

For example, a principal should be able to examine a student's marks, attendance, and academic progress without manually collecting information from several different files.

This does not necessarily mean replacing every existing school system immediately. We can eventually support importing information from spreadsheets and connecting with other software.

Problem 2: Struggling students can go unnoticed

A student may experience several small problems that become serious when combined.

For example:

- Attendance begins to decline.
- Homework completion becomes inconsistent.
- Examination marks fall.
- The student performs poorly in a particular subject.

Each individual issue might appear manageable. Together, they may indicate that the student needs additional support.

Our solution: Provide clear indicators that help authorised staff identify students who may need attention.

For the first version, the platform could use school-configured rules.

For example, a school might configure an indicator when a student's attendance falls below a specified percentage.

That percentage would be a configurable threshold, not a universal rule imposed on every school.

We must also ensure that an indicator is treated as a reason to investigate rather than a diagnosis or an automatic judgement about the student.

Problem 3: School leaders lack a clear picture of performance

A principal may know that a school performed poorly in its latest examinations without knowing which subjects, classes, or groups contributed to the result.

Our solution: Allow school leaders to move from an overall school summary to more detailed information.

For example:

School overview → Grade 8 → Mathematics → Individual student results

This is called *drill-down*: starting with a broad summary and opening progressively more detailed information.

The user should not need to understand the technical term. The interface can simply let them click on a chart or category to see the underlying information.

Problem 4: Improvement over time is difficult to measure

A single examination tells us how a student performed at one point in time.

It does not tell us whether the student has improved over six months or several academic years.

Our solution: Maintain appropriate historical records and show performance trends over time.

For example:

Academic year	Mathematics result
Year 1	45%
Year 2	54%
Year 3	63%
Year 4	71%

This student has improved across the recorded assessments.

However, we must be careful when comparing results from different examinations, curricula, or grading systems. A change in percentage does not always mean that two assessments were equally difficult or measured exactly the same knowledge.

The platform should display comparisons responsibly and identify limitations where relevant.

Problem 5: Schools need better visibility into their own operations

Schools also need to understand operational information, such as:

- Student attendance.
- Teacher attendance.
- Homework completion.
- Assessment completion.
- Class and subject performance.
- Selected financial information, where appropriate.

Our solution: Give authorised users a consolidated view of the information relevant to their responsibilities.

Financial reporting, payroll, and full accounting are separate and potentially substantial products. We should not assume they belong in the first version simply because schools use financial information.

We can define a limited financial overview if it is part of the intended product, while postponing full accounting functionality.

Problem 6: Different schools operate differently

A school in Pakistan may use a different educational structure from a school in Singapore or the United States.

Schools may also differ in their assessment methods, grading systems, academic calendars, and reporting requirements.

Our solution: Design Optimus Orien so schools can configure their educational arrangements rather than forcing every school into one rigid structure.

This is one of our most important long-term requirements.

We will study it in detail in later parts. For now, we need to agree that international flexibility is a fundamental product requirement, not an optional feature to add after launch.

Section 3. Who will use Optimus Orien?

Different people in a school have different responsibilities.

A parent does not need the same information as a principal. A teacher should not automatically have access to the private information of every student in the school.

Therefore, Optimus Orien needs different user experiences for different roles.

A *role* simply means the responsibility a person has within the platform.

I propose six primary user categories.

3.1 Platform Owner

This is the person or team responsible for the Optimus Orien software itself.

For example, your team would need a way to manage the platform, maintain its configuration, and oversee its operation.

The platform owner is not necessarily the owner of every school using the system.

Your team should not automatically have unrestricted access to every school's student records merely because it operates the software. Any necessary support access must be appropriately authorised, restricted, and audited.

Potential responsibilities:

- Manage the platform's overall configuration.
- Manage customer organisations.
- Monitor platform health.
- Manage approved platform-wide templates and settings.
- Investigate technical problems through controlled support mechanisms.

3.2 Organisation Owner or Administrator

Imagine a company owns five schools.

Its leadership may want to examine performance across all five institutions.

An organisation-level administrator needs access to the information permitted by their responsibilities.

They might need to see:

- Overall organisation performance.
- Individual school performance.
- Comparisons between campuses.
- Attendance summaries.
- Academic trends.
- Areas requiring intervention.
- Relevant administrative reports.

However, a person who administers one independent school should not automatically see information belonging to another school.

We will design these boundaries carefully.

3.3 Principal or School Administrator

The principal generally needs a school-wide overview.

For example:

- How are students performing?
- Which classes require attention?
- Which subjects have lower average results?
- Is attendance improving or declining?
- Which students may need additional support?
- How is performance changing over time?

The principal should be able to examine relevant details without manually requesting reports from every teacher.

Access must remain limited to the institution or institutions assigned to that person.

3.4 Teacher

Teachers need tools that help them manage their assigned academic responsibilities.

Depending on the features implemented, they might:

- Record assessment results.
- Record or review attendance.
- Manage assigned homework.
- View the students they teach.
- Examine class and subject performance.
- Review student progress.
- Record permitted observations about student skills or participation.

A teacher should not automatically be able to view confidential staff information or modify another teacher's records.

The precise permissions will depend on the school's configuration and the teacher's assignments.

3.5 Student

Students need an understandable view of their own academic information.

They might see:

- Their results.
- Their academic progress.
- Their attendance.
- Homework status.
- Subject-specific performance.
- Relevant feedback or progress indicators.

A student should not automatically be able to access another student's private records.

3.6 Parent or Guardian

Parents and guardians need an appropriate view of the student or students they are authorised to access.

They might see:

- Examination results.
- Attendance.
- Homework completion.

- Academic progress.
- Important school notifications.
- Indicators that their child may need additional support.

A guardian may have more than one child enrolled at the same school or at different schools.

The platform should accommodate this without creating separate parent accounts for every child.

Access should also respect the school's policies and applicable legal requirements.

A crucial requirement

These roles must not be hard-coded as if every school has exactly the same organisational structure.

A person might be a teacher at one institution and a principal at another.

A school owner might also be its administrator.

A person might manage several campuses but have no responsibility for other campuses in the same organisation.

The platform must support these legitimate differences while keeping permissions understandable and secure.

We will translate this into precise technical requirements later.

Section 4. What should Optimus Orien actually measure?

This is where we need to distinguish between the information schools collect and the indicators Optimus Orien calculates.

They are related, but they are not the same thing.

First, the school records information

Examples include:

- Examination marks.
- Attendance records.
- Homework submissions.
- Assessment completion.
- Teacher observations.
- Subject-specific skill ratings, if the school uses them.

These are recorded facts or observations.

Then, Optimus Orien calculates indicators

For example, it might calculate:

- Average marks for a subject.
- Average performance for a class.
- Attendance percentage.
- Change in performance over time.
- Homework completion percentage.
- A school-defined performance indicator.
- Whether a configured intervention threshold has been reached.

The system should explain how important calculations are produced.

For example, if a class has 25 students, its Mathematics average should be calculated using the defined results for those students, according to a documented rule.

We must decide how the system handles missing marks, different maximum marks, weighted assessments, withdrawn students, and other situations. We should not silently treat missing data as zero or combine incompatible results.

Those decisions will be made when we define the assessment and calculation requirements.

Proposed initial measurement categories

Category	Information collected	Possible indicators
Academic performance	Marks, grades, and assessment results	Subject averages, performance trends
Attendance	Present, absent, late, and other configured attendance statuses	Attendance percentage, changes in attendance
Homework	Assigned work and recorded completion status	Completion percentage, incomplete work
Assessment	Tests, examinations, assignments, and other assessment activities	Assessment results and completion
Skills	Configured subject-related skills and teacher observations	Skill ratings and changes over time
Class performance	Student results within a defined class or group	Class averages and subject comparisons
School performance	Aggregated information across the institution	School-level trends and configured indicators
Intervention	Recorded concerns, follow-up actions, and outcomes	Open interventions and progress following support

This is a proposed initial scope, not a claim that every school uses all these measures.

For instance, schools may define skills differently. Some may use numerical ratings, while others may have no formal skills-rating system at all.

Optimus Orien should allow supported features to be configured appropriately.

What about rankings and leaderboards?

Your earlier product concept included student rankings, school rankings, and a school performance or health score.

These features require special care.

A ranking can be useful in some contexts, but it can also misrepresent performance when schools use different curricula, examinations, student populations, or assessment methods.

We should therefore distinguish between:

1. **Internal comparisons:** Comparing classes or students using appropriately comparable measures within a school.
2. **Organisation-level comparisons:** Comparing campuses using clearly defined and compatible indicators.
3. **Public rankings:** Publishing comparisons across schools or countries.

The third category requires the most careful design.

My recommendation is that we build the platform's underlying reporting and comparison capabilities first. Public rankings should only be introduced when we have a defensible methodology, appropriate permissions, and a clear explanation of what the ranking does and does not mean.

A single universal score should not be presented as an objective measure of educational quality unless the methodology genuinely supports that claim.

Section 5. What should the user experience feel like?

We now know what the platform is supposed to accomplish. Next, we need to establish how people should interact with it.

Your goal is not merely to build a powerful system.

It is to build a powerful system that school staff can use without needing extensive technical training.

I propose the following principles.

Principle 1: Simple English

Use familiar terms wherever possible.

For example:

Avoid displaying	Prefer displaying
Assessment aggregation	How results are combined
Learner progression configuration	How students move to the next level
Curriculum mapping	Connect subjects to learning requirements
Intervention threshold	When to flag a concern
Performance variance	Change in performance
Institutional analytics	School performance
Configure academic periods	Set up the school calendar

Technical terminology can still appear in developer documentation. It should not unnecessarily appear in the school's everyday interface.

Principle 2: Guided setup

A school should not need to understand database design or educational architecture to configure its account.

The system should ask questions in a sensible order, explain unfamiliar options, and show only the questions relevant to the school's choices.

For example, a school choosing a particular qualification pathway should not be asked to configure unrelated pathways unless it offers them too.

Principle 3: Show important information clearly

A principal should not have to open 15 pages to discover that a subject's average performance has declined.

The dashboard should highlight important information and provide a straightforward route to the underlying records.

Principle 4: Explain indicators

If the platform flags a student for possible intervention, the user should be able to see why.

For example:

> **Attendance requires attention**
>
> Recorded attendance: 72%

School-configured threshold: 80%

Suggested next step: Review the attendance record and investigate the reasons for absence.

This is more useful than displaying a red warning without explaining what caused it.

The suggested next step should not be mistaken for a diagnosis. The school needs to investigate the circumstances.

Principle 5: Avoid unnecessary complexity

Not every user needs every feature.

A teacher should not be confronted with organisation-wide financial reports. A parent should not need to navigate curriculum configuration pages.

The interface should reflect the user's responsibilities.

Principle 6: Make the data trustworthy

Users need confidence that the information they see is accurate.

That means the platform should:

- Validate information when it is entered.
- Explain important calculations.
- Handle missing information responsibly.
- Preserve appropriate historical records.
- Restrict access to authorised users.
- Record important changes.
- Clearly identify when information is incomplete or not directly comparable.

A beautiful dashboard is of little value if its calculations are misleading.

Section 6. What should we build first?

This is one of the most important decisions in the entire project.

Optimus Orien could eventually support many types of schools, many educational systems, multiple countries, complex reporting, and integrations with other software.

But trying to build everything at once would create unnecessary risk.

We need a distinction between the **foundation** of the platform and the **features released in its first version**.

The foundation should anticipate international expansion. The first version should remain manageable.

Proposed first-version scope

Feature	Initial objective
User accounts	Let authorised people sign in securely
Organisations and schools	Support individual schools and organisations with multiple campuses
Roles and permissions	Control what each user can see and change
School setup	Allow schools to configure their basic educational arrangements
Students and staff	Maintain appropriate records for students, teachers, and other staff
Classes and subjects	Organise students and teaching activities
Academic calendar	Record school years and relevant academic periods
Assessments and results	Record assessments, marks, and grades using supported configurations
Attendance	Record and report attendance
Homework	Record assigned work and completion status
Performance dashboards	Show understandable academic and operational indicators
Progress tracking	Show appropriate historical trends
Early intervention	Flag configured concerns and record follow-up actions
Reporting	Provide appropriate summaries for authorised users
Data import and export	Support practical exchange of information, beginning with common formats such as spreadsheets

The detailed scope must be refined as we work through the education concepts.

For example, we should not promise support for every possible international grading structure in the first release before we understand what those structures require.

Instead, we will define a flexible foundation and specify which supported configurations are ready for launch.

Features I would postpone

Unless you have a compelling business reason to prioritise them earlier, I recommend postponing:

- Complex artificial intelligence predictions.
- Full payroll and accounting.
- A complete mobile application.
- An international public school-ranking system.
- A large third-party integration marketplace.

- Complicated government-wide analytics.
- A large library claiming to contain every country's complete curriculum.

These are not necessarily bad ideas. They are simply not the best starting point for a team that is still establishing its product foundations.

Section 7. What must never be compromised?

I recommend that the following become core requirements for Optimus Orien.

1. International extensibility

The platform must not assume that every school follows the same educational structure, academic calendar, curriculum, or grading method.

2. Clear user experience

Complexity should be handled by the system wherever possible, rather than pushed onto teachers and administrators.

3. Accurate calculations

Every important calculation must have a defined, testable method.

4. Secure access

Users should only access the information they are authorised to see.

5. Historical integrity

Changes to future school configurations must not silently rewrite previously finalised results.

6. Transparent indicators

The platform should explain why it flags a concern and distinguish recorded facts from interpretations.

7. Configurable rules

Where legitimate differences exist between schools, the platform should support appropriate configuration rather than relying on hard-coded assumptions.

8. Responsible comparisons

Results should only be compared in ways that are meaningful, with limitations made clear.

9. Maintainability

The system should be designed so your developer, or a future developer, can understand and extend it without having to rebuild everything whenever you support a new school model.

10. Testability

Important behaviours must have clear conditions and expected outcomes so the developer's AI coding tools can generate and run meaningful tests.

These principles will guide our subsequent technical decisions.

Section 8. Developer specification — Part 1

Everything above explains the product in plain English. The following is the initial requirement set your developer should eventually use as the foundation for implementation.

This is not the complete technical specification. It is the starting baseline that later parts will refine.

Product objective

Build Optimus Orien as a school performance intelligence and early intervention platform that enables authorised users to record, organise, analyse, and review academic and operational information.

The platform must support different types of educational institutions and provide configurable educational structures, assessment methods, grading approaches, and reporting rules.

Functional requirements

PR-001 — Organisation management

The system must support independent school organisations and organisations containing multiple institutions or campuses.

PR-002 — User roles

The system must support role-based access for platform operators, organisation administrators, school administrators, teachers, students, and parents or guardians.

The role model must allow appropriate variation in responsibilities and scope.

PR-003 — Student and staff records

The system must support the creation and management of student and staff records, including relevant relationships with schools, academic years, classes, and teaching assignments.

PR-004 — Educational configuration

The system must support configurable educational arrangements without assuming that all schools use the same model.

The exact educational configuration model will be defined in subsequent parts.

PR-005 — Assessment and grading

The system must support assessment records and results using explicitly configured assessment and grading rules.

The system must not assume that every assessment is marked out of 100 or that every school uses the same grading method.

PR-006 — Attendance

The system must support attendance recording and reporting using configurable attendance statuses and documented calculation rules.

PR-007 — Performance intelligence

The system must calculate and display defined performance indicators using documented formulas and appropriate underlying data.

PR-008 — Early intervention

The system must allow authorised users to identify configured concerns, review supporting information, and record follow-up actions.

PR-009 — Historical records

The system must preserve appropriate historical academic records and retain sufficient information to explain how finalised results were calculated.

PR-010 — Data security

The system must enforce permissions on the server side. Hiding a button in the user interface must not be treated as sufficient access control.

PR-011 — Usability

The interface must prioritise simple language, guided workflows, understandable validation messages, and clear explanations of important indicators.

PR-012 — Extensibility

The system must allow additional educational configurations to be introduced without requiring the entire application to be redesigned.

PR-013 — Data exchange

The system should support practical data import and export, with validation and understandable error reporting.

PR-014 — Auditability

Important changes to records, permissions, and educational configurations must be traceable through appropriately protected audit records.

PR-015 — Responsible reporting

The system must distinguish missing information from zero values and must not silently compare results that use incompatible definitions or grading methods.

Initial acceptance criteria

The initial product foundation will be considered adequately addressed when:

1. An independent school can be represented without creating a multi-campus organisation.
2. An organisation can manage multiple campuses with appropriate access restrictions.
3. A teacher can access their authorised teaching information without automatically gaining access to all school records.
4. A student or guardian can only access records they are authorised to see.
5. The system can represent more than one supported assessment and grading configuration.
6. Performance indicators have documented formulas.
7. Historical finalised results are not silently overwritten when future configurations change.
8. Users can understand why a configured concern has been flagged.
9. Important permissions are enforced by the backend.
10. The system can introduce additional educational configurations without requiring a fundamental redesign.

These criteria will need to be expanded into detailed test cases as we work through the later parts.

Section 9. What Part 1 has established — and what it has not

At this point, we have established the proposed identity of Optimus Orien, its main problems, its users, its initial capabilities, and the principles that should guide its development.

We have **not** yet established:

- Exactly how Pakistan's different examination boards structure their qualifications.
- How O Levels, A Levels, and IGCSE relate to other educational pathways.
- How curricula, syllabuses, subjects, year groups, and qualifications relate to one another.
- How different assessment and grading systems work.
- How final results should be calculated under different educational models.
- How schools should configure these differences in the software.
- The precise database structure needed to support them.

Those questions require dedicated treatment. Guessing at them now would undermine the goal of building a robust international product.

The next step is therefore to study the education domain before finalising the technical design.

Part 2

Understanding Education Systems from Zero

Document Type: Education Domain Fundamentals and Software Requirements
Audience: Product Owner, Developer, AI Coding Assistant
Status: Foundational Specification
Prerequisite: Part 1 — Product Foundations
Purpose: Establish a shared understanding of how different education systems work and define how Optimus Orien must represent them.

1. Why This Document Exists

Optimus Orien is intended to serve schools in Pakistan and eventually schools internationally. The difficulty is that schools do not all organise education in the same way.

Consider four schools. School A follows a Pakistani provincial examination board. School B follows the Federal Board of Intermediate and Secondary Education (FBISE). School C follows Cambridge International qualifications. School D follows an American-style educational programme.

All four schools might teach Mathematics and English. All four might have teachers, students, classes, examinations, and report cards. However, they might differ in the names and structure of their school levels, the subjects students study, the educational content they teach, the examinations students take, the way marks and grades are awarded, the qualifications students receive, the rules for moving students to the next level, and the academic calendar.

If Optimus Orien assumes that all four schools work identically, it will eventually fail to meet the needs of some customers.

The solution is not to build four separate applications. Instead, we need a platform with a shared foundation that can accommodate different educational arrangements.

Before designing that foundation, we must understand what those arrangements actually mean.

1.1 The Objective of Part 2

By the end of this document, the product owner and developer should understand the difference between a country, an education system, a curriculum, an examination board, and a qualification. They should understand what terms such as IGCSE, O Level, A Level, Matriculation, and IB mean at a basic level. They should also understand how educational levels, subjects, syllabuses, assessments, and grades relate to one another.

This document will explain why schools within the same country may follow different educational arrangements, why one school may use more than one educational programme, and how Optimus Orien should represent these differences. It will also establish the information the system must collect during school setup and the requirements the developer must implement without assuming that every school follows the same structure.

Important boundary: This document establishes the fundamental concepts and initial requirements. It does not yet define every examination board's grading rules, every country's curriculum, or the final database schema. Those require separate specifications and authoritative source material.

2. Start With the Simplest Possible Example

Imagine a school named Greenfield School in Karachi.

The school has students, teachers, subjects, examinations, and academic records. A student named Ali studies Mathematics. His teacher gives him an examination, and he receives a result.

At first glance, the process seems simple. Ali is enrolled in the school, studies Mathematics, takes an examination, receives a result, and has that result recorded in the school's system.

However, several important questions arise.

What exactly is Ali studying? Perhaps he is studying Mathematics under a Pakistani curriculum. Perhaps he is studying Mathematics under Cambridge International IGCSE. Perhaps his school follows an American-style programme. These arrangements may have different content and learning expectations.

Who determines what Ali should learn? The answer might involve a government education authority, an examination board, an awarding organisation, a curriculum framework, or the school's own educational decisions.

Who determines how Ali is assessed? His school might conduct internal tests, while an external awarding organisation might establish or administer the assessment arrangements for a qualification. The school might use both.

How is Ali's result expressed? He might receive marks out of 100, a percentage, a letter grade, a grade point, or another form of result.

Finally, what happens after Ali finishes this level? He might progress to another school year, move into a different programme, or prepare for a qualification examination.

These questions concern different parts of education. Our first task is to understand each part separately and then understand how they connect.

3. The Nine Fundamental Concepts

The following concepts form the foundation for understanding the educational structures Optimus Orien must support.

3.1 Country

A country is the geographical and political setting in which a school operates. Examples include Pakistan, the United Arab Emirates, Singapore, the United Kingdom, the United States, and Malaysia.

The country matters because different countries have different education authorities, policies, legal requirements, languages, and recognised educational arrangements.

However, country and education system are not the same thing.

A country can contain multiple educational arrangements. Pakistan is a good example. Some schools follow provincial or federal examination-board pathways, while others offer Cambridge International qualifications, the International Baccalaureate, or other programmes.

Therefore, selecting Pakistan as the country must not force a school into one specific educational structure.

Software Requirement

The system must record a school's country independently of its educational programmes. The country selection may help the interface suggest relevant options, but it must not automatically determine every other educational setting.

For example, a school could be located in Pakistan, offer Cambridge International qualifications, and also use school-defined internal assessments. This must be a valid configuration.

3.2 Education System

An education system is the broader arrangement through which education is organised. It can include educational stages, national or regional policies, school structures, curriculum expectations, assessment arrangements, qualifications, rules governing progression, and relevant education authorities.

There is no single universal definition of the phrase "education system" that perfectly describes every international arrangement. For Optimus Orien, we will use it as a broad product concept.

In Pakistan, schools can follow different educational pathways, including provincial or federal examination-board arrangements and Cambridge International qualifications. The relevant structures and requirements depend on the particular pathway.

The United Kingdom also does not have one completely uniform school curriculum. Education policy differs among England, Scotland, Wales, and Northern Ireland. These differences affect curricula and qualifications. Further information is available in the UK Parliament's briefing on comparing school curricula across the United Kingdom: https://commonslibrary.parliament.uk/research-briefings/cbp-9834/.

In the United States, schools commonly use grade levels such as Grade 1, Grade 2, and Grade 12. However, school arrangements, curricula, assessments, and graduation requirements can vary by state, school district, and school type.

The existence of a broadly recognisable American school structure does not mean that every school follows identical rules.

Software Requirement

Optimus Orien must not create one rigid structure called an education system and assume that it contains every detail needed to operate a school.

Instead, the platform must distinguish the different elements that make up a school's educational arrangement. A school may need to configure several of these elements independently.

3.3 Educational Programme or Pathway

A programme or pathway is a more specific arrangement that defines what a student studies and, often, how the student progresses toward particular educational outcomes or qualifications.

The terminology differs between organisations, so we must not assume that every organisation uses the word programme in exactly the same way.

Examples include Cambridge International IGCSE studies, Cambridge International AS & A Level studies, the International Baccalaureate Diploma Programme, a school's internally defined academic programme, and a national or regional secondary-school pathway.

A school might offer more than one programme. For example, one school could offer a national curriculum pathway for some students and Cambridge International qualifications for others. It might also offer different pathways at different educational stages.

If Optimus Orien allows only one programme for an entire institution, the school will not be represented accurately.

Software Requirement

The platform must allow an institution to offer multiple educational programmes or pathways. The system must associate students with the relevant programme or programmes according to the school's configuration. It must not assume that every student in an institution follows the same programme.

3.4 Examination Board or Awarding Organisation

An examination board or awarding organisation is an organisation responsible for specified examinations, qualifications, assessments, or awards. The precise responsibilities depend on the organisation.

An organisation might define or publish subject specifications, establish assessment requirements, set or administer examinations, mark examinations or arrange marking, determine grades, issue certificates or qualifications, or publish official examination results. Not every organisation performs all these functions in exactly the same way.

Pakistan has provincial and federal examination-board arrangements. Examples include the Punjab Boards of Intermediate and Secondary Education and the Federal Board of Intermediate and Secondary Education.

These organisations are not simply names for school curricula. They are examination and certification authorities operating under their respective arrangements.

Cambridge International is an international education and assessment organisation that offers qualifications such as Cambridge IGCSE, Cambridge O Level, and Cambridge International AS & A Level. Its qualifications have defined subject syllabuses and assessment arrangements. Official information is available at https://www.cambridgeinternational.org/.

What Is the Difference Between a School and an Examination Board?

A school teaches students and manages their education. An examination board or awarding organisation establishes or administers specified assessment and qualification arrangements.

A school may teach a course and prepare students for an external examination. The organisation awarding the qualification may be different from the school.

Software Requirement

The system must distinguish between the institution delivering education, the programme a student follows, the organisation responsible for a particular examination or qualification, and the school that records or imports the student's result.

These may be related, but they must not be treated as the same entity.

A school should be able to record the relevant examination organisation when necessary. The platform must not assume that every assessment conducted by a teacher is an external board examination.

3.5 Qualification

A qualification is a recognised award or credential that represents a specified level or type of achievement.

Examples include Cambridge IGCSE qualifications, Cambridge O Level qualifications, Cambridge International AS & A Level qualifications, and qualifications awarded through relevant Pakistani examination-board arrangements.

A qualification may be based on one subject or a combination of subjects and other requirements, depending on the qualification.

Imagine a student taking Cambridge IGCSE Mathematics. The student studies the relevant subject, completes the required assessment arrangements, and receives a

result. That result contributes to the student's record of achievement in the qualification.

The student may also take IGCSE English, Biology, Chemistry, and other subjects. Each subject may have its own assessment requirements and result.

Why Qualifications Matter to Optimus Orien

A school may need to record which qualification a student is studying toward, which subjects are associated with that qualification, which awarding organisation is responsible, which examination session applies, what results the student receives, and whether the student has completed the school's relevant requirements.

However, the school management platform should not assume it is itself the authority that awards the external qualification.

Software Requirement

The platform must be able to represent a qualification separately from the school year, subject, assessment, and individual result. The precise relationships will be defined in the later education-model and database specifications.

3.6 Curriculum

Curriculum is one of the most important terms in this document.

A curriculum describes the intended learning students should experience. Depending on the educational arrangement, it can describe what students should learn, which subjects or areas of learning they study, what knowledge they should develop, which skills they should acquire, how learning is organised or sequenced, and what learning outcomes are expected.

A curriculum can be broad and flexible, or it can specify detailed requirements.

Imagine a Mathematics curriculum for students at a particular level. It might include fractions, decimals, percentages, ratios, basic algebra, geometry, and problem-solving.

The curriculum helps define what students are expected to learn. It does not necessarily determine every examination question or every mark the student can receive.

Curriculum Versus Education System

The education system is the broader arrangement. The curriculum is the intended learning content and structure within that arrangement.

Curriculum Versus Qualification

A qualification defines an award and its associated requirements. The curriculum may describe what students study while preparing for that qualification, but the qualification and curriculum are not identical concepts.

Curriculum Versus Syllabus

These words are sometimes used differently by different education authorities and organisations. For Optimus Orien, we need a practical distinction.

A curriculum is the broader learning plan or framework. A syllabus is a more specific description of the content and requirements associated with a particular subject or course.

For example, a school's broader curriculum might include English, Mathematics, and Science. The syllabus for its Mathematics course might specify which topics students must study and which learning objectives or assessment requirements apply.

Cambridge International publishes subject-specific syllabuses for its qualifications. These define subject content and assessment arrangements. Official information is available at https://www.cambridgeinternational.org/.

A Critical Distinction

Optimus Orien does not need to become the organisation that creates every curriculum in the world.

Its first responsibility is to represent the educational arrangements its customers actually use. That could mean allowing schools to configure their own subjects and levels, providing supported templates for selected educational programmes, recording relevant curriculum information, connecting subjects to the appropriate learning requirements when needed, and supporting assessment and performance reporting based on those arrangements.

If we later provide a catalogue of official curricula or syllabuses, we must ensure that the information is accurate, appropriately sourced, current, and legally usable. We should not invent curriculum content merely to populate the platform.

Software Requirement

The platform must distinguish the curriculum or learning framework from the subject and the assessments used to measure learning. It must also support schools that use a locally defined curriculum without forcing them to select an official international curriculum.

3.7 Subject

A subject is an area of study. Examples include Mathematics, English, Biology, Chemistry, Physics, History, Geography, Computer Science, Urdu, and Economics.

This appears simple, but there are details to consider.

A school might teach English Language and English Literature as separate subjects. Another school might use a broader English course. A qualification provider might distinguish between different Mathematics courses or levels.

Therefore, the platform should not assume that two subjects with similar names are necessarily identical.

Software Requirement

The system must support school-defined subject names, subjects linked to relevant educational programmes, subjects associated with appropriate levels or courses, different subject specifications where necessary, and the ability to distinguish subjects that share a name but have different requirements.

For example, two schools might both have a subject called Mathematics. That does not automatically mean their courses cover identical content or use comparable assessments.

3.8 School Level, Grade, Year Group, or Stage

These terms describe where a student is within an educational structure. However, they are not universally interchangeable.

Examples include Grade 5, Year 7, Primary 4, Secondary 2, Key Stage 3, Cambridge IGCSE study, and the International Baccalaureate Middle Years Programme.

Some terms describe a school year or grade. Others describe a broader stage, programme, or qualification pathway.

In an American-style structure, a school might organise students into Grade 1 through Grade 12.

A school in England might use Year 1, Year 2, and later year groups, with Key Stages grouping certain year groups together.

Cambridge IGCSE is a qualification pathway typically intended for learners aged 14–16. It should not be treated as merely another name for Year 10 or Year 11.

A student may prepare for IGCSE examinations while the school also uses its own year-group labels.

Imagine a school with Year 10 students preparing for IGCSE examinations, Year 11 students taking another set of IGCSE subjects, and a different group of students following another pathway. The school year, the qualification pathway, and the subject-level course are related but distinct.

If Optimus Orien combines them into one field called grade, it may become difficult to represent real situations correctly.

Software Requirement

The system must distinguish between the school's own year-group or grade label, the broader educational stage where applicable, the programme or qualification pathway, and the course or subject level where applicable.

The exact model will be designed in a later part.

3.9 Assessment and Grading

We will examine these in considerable detail in Part 3, but the basic distinction is essential here.

Assessment is the process of gathering evidence about what a student knows, understands, or can do. Examples include written examinations, quizzes, homework assignments, projects, oral examinations, practical work, and presentations.

Grading is the process of representing or classifying performance according to defined rules. Examples include marks out of 100, percentages, letter grades, defined grade bands, and grade points.

A school may use several forms of assessment and grading. An examination board may also have specific assessment and grading rules for its qualifications.

Software Requirement

Optimus Orien must not assume that all assessments use the same maximum mark, have the same weighting, use the same grading scale, are conducted by the same organisation, or contribute to final results in the same way.

These differences must be represented through explicit, documented configuration.

4. How These Concepts Connect

We have defined the terms individually. Now we need to understand their relationships.

Imagine a school in Pakistan that offers Cambridge International qualifications. A simplified relationship might look like this:

The country is Pakistan. The institution is Greenfield International School. The educational programme is a Cambridge International qualification pathway. One qualification is Cambridge IGCSE. One subject is Mathematics. The subject syllabus is the applicable Cambridge IGCSE Mathematics syllabus and version. The assessment is an examination or another assessment component specified by the

applicable syllabus. The result is the student's recorded mark or grade according to the relevant rules.

This is an explanatory sequence, not a claim that every school must use this exact structure. In practice, the school may also maintain internal year groups, conduct additional classroom assessments, or offer other programmes.

Each element answers a different question.

The country answers where the school operates. The school identifies the institution delivering the education. The programme describes the educational pathway being followed. The qualification identifies the award or credential the student is pursuing or receiving. The subject identifies the area of study. The syllabus identifies the specific content and requirements applicable to the subject or course. The assessment describes how evidence of learning is collected. The grading rules describe how performance is represented or classified. The result records the outcome achieved by the student.

What the Developer Must Understand

These concepts are not simply different labels for the same record.

The developer must not design one database field that attempts to represent the country, curriculum, examination board, qualification, subject, school year, and grading method simultaneously.

The software must represent the important distinctions explicitly.

This does not mean creating a separate database table for every word in this document. The eventual design must be proportionate to the actual requirements.

It means the underlying model must not erase meaningful educational differences.

5. Pakistan: Understanding the Main Educational Arrangements

Pakistan is our starting point because it makes the problem tangible.

We will not attempt to document every Pakistani board, qualification, or current regulation in this section. The purpose is to understand the different categories of arrangements.

5.1 Matriculation and Intermediate Pathways

In Pakistan, the terms Matriculation and Intermediate are commonly used in connection with secondary and higher-secondary education and the relevant examination-board pathways.

Broadly speaking, students may study toward secondary-school qualifications and then proceed to higher-secondary qualifications. However, the exact structure, subject requirements, examination arrangements, and grading rules depend on the applicable board and current regulations.

For example, the Federal Board of Intermediate and Secondary Education operates within the federal examination-board framework. Provincial examination boards operate within their respective jurisdictions and arrangements.

What Must Optimus Orien Do?

The platform must be able to represent a school that follows an applicable Pakistani board pathway.

At a minimum, the configuration must accommodate the school's own year groups and classes, the applicable educational pathway, relevant subjects, the assessment arrangements the school uses, marks or grades recorded under the applicable rules, relevant academic years and examination periods, and historical student results.

We must not assume that every board has identical subjects, examination structures, or grading rules. We must also avoid treating the names of provincial boards as interchangeable with curricula or qualifications.

A Practical Example

Suppose Greenfield School prepares students for a relevant provincial board examination.

Its principal wants to see the marks students received, average results for each subject, the distribution of results across classes, student progress across academic years, and students who may need additional support.

Optimus Orien should support those reporting needs without pretending to replace the board's own examination or certification system.

The exact assessment configuration must reflect the school's actual arrangement.

5.2 What Is FBISE?

FBISE stands for Federal Board of Intermediate and Secondary Education.

It is an examination board, not the name of a school curriculum that automatically describes every educational decision a school makes.

A school may be associated with the relevant federal-board examination arrangements. For Optimus Orien, FBISE must be represented as an organisation with a particular role in the educational arrangement, rather than as a generic school level.

The platform may also need to record which applicable qualification or examination arrangement is associated with the school or student. The exact relationship should be verified against current official information when implementing support for FBISE.

5.3 What Are Punjab, Sindh, and Khyber Pakhtunkhwa Boards?

These names refer to provincial examination-board arrangements and the relevant boards operating within them.

For example, Punjab has multiple Boards of Intermediate and Secondary Education. Sindh and Khyber Pakhtunkhwa also have their respective examination-board arrangements.

Selecting a province does not necessarily identify the exact board or educational pathway.

The system must allow a school to identify the relevant organisation and configure its actual educational arrangements.

A school in Punjab might select Pakistan as its country, Punjab as its province, the relevant Pakistani board pathway as its educational arrangement, the applicable board as its examination organisation, and the school's own defined year groups.

The interface should show only relevant choices where the system has reliable reference data.

If a particular board has not yet been added to Optimus Orien's supported catalogue, the platform should not invent its official rules. A controlled custom configuration may be possible, depending on the available features, but the system must distinguish school-defined settings from verified official rules.

5.4 What Are O Levels?

O Level means Ordinary Level.

Cambridge O Level is an internationally recognised qualification offered by Cambridge International. It is generally intended for learners around the 14–16 age range and is offered in individual subjects. Cambridge O Level assessment can include written, oral, and practical examinations, depending on the subject and requirements.

Official information is available at https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-upper-secondary/cambridge-o-level/.

What Optimus Orien Must Understand

O Level is a qualification category, not simply a synonym for Grade 9, Grade 10, or a generic secondary-school level.

A school may use internal year groups to organise its students while preparing them for O Level examinations.

The platform must be able to represent both arrangements.

5.5 What Is IGCSE?

IGCSE stands for International General Certificate of Secondary Education.

Cambridge International IGCSE is an international qualification for learners typically aged 14–16. Students can take individual subjects, and the available assessment methods depend on the applicable subject syllabus. Cambridge IGCSE subjects may include written examinations, oral or practical assessments, and coursework options where specified.

Official information is available at https://help.cambridgeinternational.org/hc/en-gb/articles/115004310029-What-is-Cambridge-IGCSE.

Is IGCSE the Same as O Level?

They are related but should not be treated as identical labels.

Cambridge International states that its Cambridge IGCSE and Cambridge O Level qualifications are equivalent grade for grade. However, the qualifications can differ in features such as available grades, coursework options, and subject-specific arrangements.

Official information is available at https://help.cambridgeinternational.org/hc/en-gb/articles/115004449425-What-is-the-difference-between-Cambridge-IGCSE-and-Cambridge-O-Levels.

This illustrates an important principle for Optimus Orien: two qualifications may be recognised as equivalent for a particular purpose without having identical structures or assessment arrangements.

Example

Imagine a school offers Cambridge IGCSE Mathematics.

The system may need to record the school year in which the student is enrolled, the Cambridge IGCSE qualification, the relevant Mathematics subject, the applicable syllabus or syllabus version, the student's internal assessments, the relevant external examination session, and the results available to the school.

The student's school year and qualification pathway must remain separate concepts.

5.6 What Are AS Levels and A Levels?

AS Level and A Level are advanced qualifications offered by Cambridge International and other awarding organisations, with the precise arrangements depending on the provider.

For Cambridge International, AS stands for Advanced Subsidiary and A Level stands for Advanced Level.

The AS Level is a qualification at a different standard from the full A Level. Cambridge International A Level includes more demanding content and assessment associated with the second half of the course, commonly referred to as A2.

The applicable structure and assessment arrangements must be verified against the relevant qualification and syllabus information. Official information is available at https://www.cambridgeinternational.org/.

Students often take a selection of subjects rather than studying every possible subject.

For example, one student might study Mathematics, Physics, and Chemistry. Another student might study Economics, Business, and Mathematics.

The student's subject selection and resulting qualifications depend on the relevant programme and requirements.

What Optimus Orien Must Understand

The platform must not assume that every student at the same school level studies the same subjects. It must also not assume that every subject uses identical assessment components or grading rules.

6. International Education Models

We have examined several familiar Pakistani arrangements. Now we need to understand why the international market makes the problem more complicated.

The examples below are representative starting points, not a complete catalogue of global education.

6.1 Cambridge International

Cambridge International offers programmes and qualifications across several stages of school education. Its offerings include Cambridge Primary, Cambridge Lower Secondary, Cambridge IGCSE, Cambridge O Level, and Cambridge International AS & A Level.

The precise structure and available subjects depend on the programme and qualification.

Cambridge International publishes programme and qualification information through its official website: https://www.cambridgeinternational.org/programmes-and-qualifications/.

Implication for Optimus Orien

The platform should not treat Cambridge International as one single qualification.

A school might offer several Cambridge programmes or qualifications. The software must allow the school to configure the relevant offerings and connect them to the appropriate subjects, levels, assessments, and results.

6.2 The International Baccalaureate

The International Baccalaureate, commonly called the IB, offers four educational programmes: the Primary Years Programme (PYP), Middle Years Programme (MYP), Diploma Programme (DP), and Career-related Programme (CP).

The IB describes these programmes as serving students across the approximate ages of 3–19, with different programmes designed for different educational stages. Schools must be authorised to offer IB programmes.

Official information is available at https://ibo.org/programmes/.

Why the IB Is Important for Our Design

The IB is an example of why Optimus Orien cannot assume that every educational model is simply a collection of conventional school grades followed by one final examination.

Different programmes can have different learning frameworks, subject arrangements, assessment practices, and completion requirements.

For example, the Diploma Programme is not merely another name for Grade 12.

A school offering the IB Diploma Programme may also maintain internal year groups and student records.

Implication for Optimus Orien

The system must distinguish between the school's administrative structure and the educational programme being delivered.

It must also allow supported programmes to have their own relevant subject, assessment, and reporting configurations.

We will not attempt to implement every IB-specific rule in this document. That requires a dedicated model specification and authoritative programme information.

6.3 The American-Style Education Model

American schools commonly use grade levels such as Grade 1, Grade 2, Grade 3, and later grades, often extending to Grade 12.

However, this does not mean that every school in the United States follows identical curriculum, assessment, or graduation rules. The United States has state and local variation, and school arrangements differ.

Implication for Optimus Orien

The system should support grade-based structures without assuming that all American schools use the same subjects, grading scales, or promotion requirements.

For example, one school might use percentage marks and letter grades. Another might use a different grading policy.

The platform must support the actual configuration selected by the school.

6.4 The British Education Model: An Important Warning

People often use the phrase British curriculum as if it describes one uniform system. That is an oversimplification.

Education policy and curricula differ among England, Scotland, Wales, and Northern Ireland. These differences affect educational structures, curricula, and qualifications.

Further information is available at https://commonslibrary.parliament.uk/research-briefings/cbp-9834/.

For example, England uses year groups and Key Stages in its school structure, while Scotland has its own Curriculum for Excellence and distinct arrangements.

Implication for Optimus Orien

The platform should not treat the United Kingdom as one indivisible educational configuration.

Where the relevant rules differ, the system must allow the school to select the appropriate jurisdiction and educational arrangement.

A school located in England should not automatically receive Scottish curriculum settings merely because both schools are in the United Kingdom.

This is a general principle that applies to other countries with regional educational differences.

7. The Most Important Software Design Principle So Far

We have now seen examples of several educational arrangements. The next question is how Optimus Orien should support them.

Consider three schools. Greenfield School is in Pakistan and follows a Pakistani examination-board pathway, using its own school year groups. Horizon International School is also in Pakistan but offers Cambridge International qualifications alongside its school-defined year groups. Northstar Academy is in the United States and follows an American-style programme with a grade-based structure.

These schools should be able to use the same core platform. They need not share every educational rule.

Their records may have common features, such as students, teachers, attendance, and academic results. At the same time, their educational arrangements may differ.

This leads to a central design principle:

Common foundation, configurable educational arrangements.

Optimus Orien should have a shared foundation for managing schools and analysing performance. Within that foundation, schools must be able to configure the educational arrangements they actually use.

The platform should not require a completely separate application for every country. Nor should it force all countries into one rigid educational model.

What This Means for the Developer

The implementation must distinguish between shared platform capabilities and configurable educational arrangements.

Shared platform capabilities include user accounts, schools and campuses, students and staff, academic records, attendance, reporting, and permissions.

Configurable educational arrangements include school year and stage structures, programmes and pathways, subjects and course definitions, assessment methods, grading rules, and progression requirements.

Some features will interact with both categories. For example, an assessment is a common platform capability, but the way a particular school configures that assessment may depend on its educational arrangement.

The detailed technical architecture will be specified later.

8. What Must Be Configurable?

Not every piece of information should be freely configurable.

For example, the basic concept of a student record is common to the platform. A school should not be able to redefine what a user account is.

However, many educational arrangements legitimately differ.

The school may need to configure its educational structure, including year groups, levels, and stages. It may need to define which programmes it offers, which qualifications are relevant, which subjects and courses are taught, which curriculum information applies, and how its academic calendar is organised.

The school may also need to configure assessment methods, grading systems, progression requirements, and reporting preferences.

For example, one school may use Grade 8, another may use Year 9, and another may use a custom year-group structure. One school may use three academic terms, while another uses semesters.

The word configure does not mean that a school must manually create every item from scratch.

The platform may offer templates for supported arrangements. A school could select an appropriate template and adjust the settings permitted by that template.

However, the platform must distinguish between a verified official template and a school-defined configuration.

9. How School Setup Should Work

Now let's translate the educational concepts into an actual user experience.

Imagine a new school subscribes to Optimus Orien. The principal signs in for the first time.

The principal is not an education-system expert, and we should not expect them to understand software architecture. The setup process should guide the principal through a sequence of simple questions.

Step 1: Tell Us About Your School

The system asks for basic information, including the school name, country, relevant region or jurisdiction where applicable, and whether the institution is independent or part of a larger organisation.

These questions establish the school's context. They must not automatically determine every educational setting.

Step 2: Choose How Your School Organises Education

The interface explains the available options in plain English.

For example, it might ask, "Which educational programme or arrangement does your school follow?"

The school might select a supported national pathway, a Cambridge International programme, an IB programme, an American-style structure, or a custom arrangement.

The exact options depend on the reference information available in the platform.

The system must allow for schools that combine arrangements or use their own structure.

Step 3: Set Up School Levels and Year Groups

The system asks the school how it organises its students.

For example, the school might use Grade 1 through Grade 12, Year 1 through Year 13, or a custom set of year groups.

The platform should not assume that these labels are universal or directly interchangeable.

Step 4: Select the Programmes the School Offers

A school may offer one programme or several.

For example, it might use one arrangement in primary school and another for secondary students.

The setup process should let the school configure each relevant offering without forcing every student into the same pathway.

Step 5: Set Up Subjects and Courses

The school chooses the subjects it teaches and associates them with the appropriate year groups or programmes.

Where the platform has reliable reference data, it can suggest relevant subjects. The school should also be able to create appropriate school-defined subjects or courses.

Step 6: Configure Assessments and Grading

The school selects or defines how it records assessment results.

For example, it might use marks out of a maximum value, percentage marks, letter grades, or a supported qualification's grading arrangement.

The exact supported options and validation rules will be defined in Part 3.

The platform must not invent official grade boundaries or assume that the same percentage means the same thing for every qualification.

Step 7: Review and Confirm

The system displays a summary of the school's configuration.

For example:

Your School Setup

Country: Pakistan
Educational arrangement: Cambridge International
School year groups: Year 1–Year 13
Programmes offered: As configured
Subjects: As configured
Assessment configuration: As selected

The actual summary must display the school's saved choices rather than fixed example values.

The school confirms the setup and can later make authorised changes through the appropriate settings.

Developer Requirement: Guided Configuration

The setup process must be conditional.

If a school chooses an arrangement that requires additional configuration, the system should ask the relevant follow-up questions. It should not present every possible educational setting to every school.

It must also validate that required information has been provided before allowing the school to publish or activate the relevant configuration.

10. What Happens When a School Does Not Follow a Prebuilt Model?

This is a critical requirement for international expansion.

We cannot reasonably assume that we will have a complete template for every school at launch.

Some schools may use a national educational framework, a recognised international programme, a combination of educational arrangements, a locally defined curriculum, an internal grading policy, or a specialised educational structure.

The system needs a controlled way to accommodate arrangements that are not available as prebuilt templates.

10.1 Start With a Supported Template

A school selects an available educational arrangement. The system loads the relevant supported structure and settings. The school then completes any required institution-specific configuration.

10.2 Create a Custom Arrangement

A school defines the structure it actually uses, within the capabilities supported by Optimus Orien.

It may configure year groups, subjects, programmes, assessment methods, grading rules, and relevant academic periods.

However, custom configuration must not mean that the school can bypass data validation or create arbitrary system behaviour.

For example, the system must still validate that a recorded assessment result conforms to its configured input rules.

A custom arrangement should also be labelled appropriately. The platform must not present a school-defined configuration as an officially verified educational model.

Important Limitation

A custom configuration cannot automatically reproduce every possible education system.

If a particular educational model requires a feature the software does not support, the system should identify that limitation rather than pretending the model has been implemented completely.

For example, a school may require a specialised progression rule or a complex qualification structure that the current platform does not yet support.

We will need a formal process for identifying and adding such capabilities.

11. How Optimus Orien Should Handle New Educational Models

Suppose Optimus Orien launches with support for a defined set of educational arrangements. Later, a school in another country wants to use the platform, but its educational model is not yet available.

The product should not require the developer to rewrite the entire application.

Instead, the process should begin by identifying the school's actual educational requirements and determining which existing platform capabilities already support those requirements. The team should then identify genuine gaps, verify the relevant educational rules using reliable sources, design any necessary additional configuration, implement the new capabilities, and test the new arrangement against realistic scenarios.

Once verified, the new arrangement may be made available as an approved template or supported configuration where appropriate.

This process may require product development. It should not be assumed that every new educational model can be supported by adding a few database records.

Developer Requirement

The platform must separate educational reference data from application logic wherever practical.

For example, adding a new supported subject or grading scale should not normally require rewriting unrelated parts of the application.

However, genuinely new educational behaviour may require code changes, reviewed rules, and additional tests.

The goal is extensibility without pretending that all educational differences are merely data-entry differences.

12. How the Platform Should Represent Official Information

A platform that supports international education must be careful about the source of its educational information.

Imagine Optimus Orien contains a template for a qualification. The template might include the qualification name, the relevant organisation, the subjects available, the applicable syllabus versions, the assessment arrangements, the grading rules, and the relevant effective dates.

If any of this information is incorrect or outdated, the platform could produce misleading reports.

Therefore, official educational information must be treated as maintained reference data, not as arbitrary text created by the application.

12.1 Proposed Requirements

Where an official educational template is offered, the system should retain appropriate metadata, including the organisation responsible for the information, the official source or reference, the version of the information where applicable, the date the information was checked, its applicable effective period where known, its status such as draft, verified, or retired, and any relevant implementation limitations.

The exact metadata fields will be finalised in a later specification.

12.2 What If a Template Changes?

Suppose an awarding organisation updates a syllabus.

Optimus Orien must not silently replace the historical syllabus associated with every student record.

The platform should support versioned educational configurations so that previous records remain interpretable.

This does not mean every minor wording change requires a separate database version. The versioning rules must be appropriate to the type of information.

However, changes that affect the interpretation of student results must be handled carefully.

13. What Optimus Orien Must Not Assume

This section establishes explicit constraints for the developer and AI coding tools.

13.1 One Country Means One Education System

This assumption is incorrect. A country can contain multiple educational arrangements. The platform must support this variation.

13.2 One School Follows Only One Programme

This assumption is incorrect. A school may offer multiple programmes or qualifications. The platform must support that possibility.

13.3 One Student Always Belongs to One Educational Pathway

This assumption is too restrictive. A student may participate in different programmes or courses over time and may study subjects associated with different arrangements.

The system must represent the student's actual enrolments rather than imposing a universal one-programme rule. The precise rules for concurrent pathways will be defined later.

13.4 A Grade Label Means the Same Thing Everywhere

This assumption is incorrect. Grade 8, Year 9, and a particular qualification level are not automatically equivalent. The system must preserve the original labels and their context.

13.5 All Subjects With the Same Name Are Identical

This assumption is incorrect. Two subjects called Mathematics may have different specifications, content, levels, or assessment arrangements. The system must allow them to be distinguished when necessary.

13.6 All Assessments Use Marks Out of 100

This assumption is incorrect. Assessment structures and result formats vary. The system must support explicitly configured result types and rules.

13.7 All Grading Systems Are Directly Comparable

This assumption is incorrect. Two grading scales may differ in meaning, structure, and intended use. The platform must not assume that an identical numerical result represents identical achievement across different assessments.

13.8 Every School Uses an Official Curriculum Template

This assumption is incorrect. Some schools use custom or combined arrangements. The system must support appropriate custom configurations.

13.9 Selecting an Educational Model Automatically Configures Everything Correctly

This assumption is incorrect. A template may require additional choices, school-specific settings, or validation. The setup process must make these requirements clear.

13.10 Every New Educational Model Can Be Supported Without Code Changes

This assumption is incorrect. Some differences may require additional software functionality. The system should be designed to minimise unnecessary changes, not to promise impossible flexibility.

14. Part 2 — Developer Requirements

The following requirements translate this document into a preliminary implementation baseline. They are intended to guide the later technical specifications and AI-assisted development.

They are not permission to create a production database without completing the subsequent design work.

EDUC-001 — Country and Educational Arrangement Must Be Separate

The system must store a school's country independently of its educational programmes, qualifications, and assessment arrangements. Selecting a country must not automatically force the school into one educational model.

Acceptance test: A school in Pakistan can be configured for a supported Cambridge International pathway without being forced into a Pakistani examination-board pathway.

EDUC-002 — Support Multiple Programmes per Institution

The system must allow an institution to offer multiple supported educational programmes or pathways.

Acceptance test: A school can configure different programmes for different sections of its institution without creating separate, unrelated school records.

EDUC-003 — Support Custom Educational Configurations

The system must provide a controlled way to represent school-defined educational arrangements when an appropriate supported template is unavailable.

Acceptance test: A school can define supported custom year groups and subjects without claiming that its configuration is an officially verified educational model.

EDUC-004 — Distinguish Subjects From Qualifications

The system must not represent a qualification as though it were simply a subject or school year.

Acceptance test: A student can study multiple subjects associated with a qualification while retaining the school's separate year-group information.

EDUC-005 — Support Subject-Specific Configurations

The system must allow relevant subjects to have different courses, specifications, levels, or assessment arrangements where required.

Acceptance test: Two Mathematics courses can be represented separately when their specifications or assessment requirements differ.

EDUC-006 — Preserve School-Defined Labels

The system must support school-defined year-group and course labels where appropriate. It must not assume that every institution uses the same naming convention.

Acceptance test: A school using Year 8 and a school using Grade 8 can retain those labels without the system silently treating them as equivalent.

EDUC-007 — Support Multiple Assessment Configurations

The system must allow assessment arrangements to be configured according to the supported educational model and the school's needs.

Acceptance test: The system can represent assessments with different maximum marks and result formats without forcing all results into one universal structure.

Detailed assessment requirements will be defined in Part 3.

EDUC-008 — Maintain Educational Reference Information

Official educational templates must have appropriate source, version, and status information.

Acceptance test: An administrator can distinguish an approved, verified template from a school-defined configuration.

EDUC-009 — Preserve Relevant Historical Context

Changes to future educational configurations must not silently rewrite the meaning of historical student results.

Acceptance test: A change to a future syllabus or grading configuration does not automatically replace the configuration associated with previously finalised results.

EDUC-010 — Validate Educational Configurations

The system must validate required fields and relationships before allowing a configuration to become active.

Acceptance test: A configuration missing required information cannot be published until the missing information is supplied or an authorised exception is applied under documented rules.

EDUC-011 — Support Controlled Template Expansion

The platform must allow new supported educational templates to be introduced without requiring an unrelated rewrite of the application.

Acceptance test: Adding a new supported configuration does not require rewriting the general user-account, school-management, and permissions systems unless a genuine new capability is required.

EDUC-012 — Avoid Unsupported Equivalence Claims

The system must not automatically treat different school years, qualifications, subjects, or grades as equivalent merely because their labels look similar.

Acceptance test: Any cross-model equivalence used in reporting must be supported by an explicitly defined and documented mapping or methodology.

EDUC-013 — Preserve Configuration Provenance

Where a result depends on a particular educational or grading configuration, the system must retain sufficient information to explain which configuration applied.

Acceptance test: An authorised user can determine which applicable configuration was used to calculate a historical finalised result.

EDUC-014 — Keep User-Facing Terminology Simple

The school setup process must use clear labels, contextual explanations, and relevant follow-up questions.

Acceptance test: A school administrator can configure a supported educational arrangement without needing to understand database or software architecture terminology.

EDUC-015 — Do Not Invent Unsupported Official Rules

The application must not silently fabricate official curricula, examination-board requirements, qualification rules, or grading boundaries.

Acceptance test: If a required official configuration is unavailable or unverified, the system clearly identifies that limitation and offers only supported alternatives.

15. What Part 2 Has Established

We have established the basic vocabulary and relationships needed to understand international education.

A country is where the school operates. An education system is the broader arrangement through which education is organised. An educational programme or pathway is a more specific arrangement followed by students. An examination board or awarding organisation is responsible for specified examinations, qualifications, assessments, or awards. A qualification is an educational award or credential.

A curriculum describes the intended learning students should experience. A syllabus describes more specific subject or course content and requirements in the context of the relevant educational arrangement. A subject is an area of study. A school level or year group describes a student's place within the school's structure. An assessment collects evidence of learning. A grading system represents or classifies the student's performance.

We have also established that schools can combine educational arrangements and that Optimus Orien must preserve meaningful differences instead of forcing every school into a single structure.

Most importantly, we have begun translating those concepts into explicit software requirements.

16. What We Have Not Yet Solved

We have deliberately not finalised the complete structure of Pakistani provincial and federal board arrangements, the exact requirements for every supported Cambridge qualification, the detailed structures of the International Baccalaureate programmes, the full relationship between curricula, syllabuses, subjects, courses, and learning outcomes, the detailed assessment and grading models, the precise rules for calculating final results, the database structure required to represent all these relationships, or the full list of supported international templates for launch.

These are not minor details. They are separate areas of work that must be handled systematically.

We should not pretend to support a particular educational model fully until we have identified its requirements and tested the implementation.

Part 3 will explain **assessment and grading from zero**, including how marks, percentages, grades, class averages, examinations, skills, and student progress work across different educational systems. I'll write it so you and your developer can understand the education concepts and translate them into software requirements without guessing how schools operate.

Part 3:

Understanding Assessment and Grading from Zero

1. Purpose of This Document

This document explains how schools measure student performance and how Optimus Orien should record, calculate, display, and compare that performance.

The goal is to help the development team understand what a student's result actually means, how different schools calculate results, and why the software must not assume that every school follows the same grading rules.

By the end of this document, the development team should understand the difference between marks, percentages, grades, assessments, examinations, class averages, subject performance, skill ratings, and progress over time.

The document also establishes the software requirements for handling these concepts correctly.

2. What Is Assessment?

An assessment is any method a school uses to find out what a student knows, understands, or can do.

For example, imagine that a student is learning mathematics. The teacher wants to determine whether the student understands multiplication.

The teacher could give the student a short test, ask the student to solve a problem in class, assign homework, observe the student completing an activity, or ask the student to explain the answer verbally.

Each of these activities can provide information about the student's learning.

The school uses this information to understand the student's current performance and decide whether additional teaching, practice, or support is needed.

In simple terms, **assessment is the process of collecting evidence about student learning**.

For Optimus Orien, assessment data is important because it helps the platform identify performance patterns. For example, a student might perform well in homework but poorly in examinations. Another student might have good examination marks but regularly fail to complete assignments.

These patterns can help teachers and school leaders understand where a student may need support.

However, the platform must distinguish between a recorded result and an explanation for that result. Low marks alone do not prove why a student is struggling.

3. The Main Types of Assessment

Schools use different types of assessment for different purposes. Optimus Orien must be able to represent these differences without forcing every school into one model.

3.1 Diagnostic Assessment

A diagnostic assessment is used to understand what a student already knows and where learning difficulties may exist before or at the beginning of a learning period.

For example, a mathematics teacher might give students a short test before beginning algebra. The results help the teacher identify which students already understand the necessary mathematical concepts and which students need additional preparation.

A diagnostic assessment is not necessarily used to calculate a student's final grade.

For Optimus Orien, diagnostic results should be recordable as a distinct assessment type. The school should be able to decide whether these results contribute to its formal academic calculations.

3.2 Formative Assessment

A formative assessment helps teachers monitor learning while teaching is still taking place.

Examples include classroom quizzes, practice questions, homework, classroom questioning, short assignments, and teacher observations.

Suppose a teacher teaches fractions over three weeks. During the first week, students complete a short quiz. The teacher notices that many students are making mistakes when adding fractions with different denominators.

The teacher can address the problem before the final examination.

Formative assessment is especially useful for identifying learning difficulties early.

Optimus Orien should allow schools to record formative assessments and examine their results over time. Whether these assessments contribute to a formal subject grade must be configurable.

3.3 Summative Assessment

A summative assessment measures what a student has learned after a particular period of instruction.

Examples include term examinations, final examinations, end-of-unit tests, and major projects.

Suppose a student completes a mathematics examination at the end of a term and receives 78 out of 100 marks.

That result provides a measure of performance in that particular examination. It does not automatically establish that the student understands every mathematics topic equally well.

For Optimus Orien, summative assessments may contribute to term results, annual results, or final grades, depending on the school's rules.

The software must not assume that every school uses the same combination of formative and summative assessments.

3.4 Standardised Assessments

A standardised assessment follows a defined set of administration and scoring rules so that results can be interpreted using a consistent framework.

Some standardised assessments are used across many schools, regions, or countries. Others are used to assess particular skills or compare performance against an established reference group.

A standardised assessment is not automatically the same as a school examination. Its purpose, scoring system, and interpretation may differ.

Optimus Orien should be able to store results from standardised assessments when a school needs them. It should not assume that the marks from such assessments can be directly compared with ordinary classroom examination marks.

3.5 Assessment Is Not the Same as an Examination

An examination is one type of assessment.

Assessment is the broader concept. It includes examinations, quizzes, assignments, projects, practical demonstrations, oral assessments, and other methods of measuring learning.

The system should therefore use a general assessment model rather than building its entire academic structure around examinations.

4. Understanding Marks

Marks are numerical values awarded for a student's performance in an assessment.

Suppose a student takes a science test containing 40 marks and receives 32 marks.

The result is recorded as:

Marks obtained: 32

Maximum possible marks: 40

The student has earned 32 of the 40 available marks.

This is different from saying that the student received 32 out of 100. The maximum possible marks matter because they establish the scale on which the result was recorded.

A student might receive 32 out of 40 in one test and 75 out of 100 in another. The raw marks cannot be compared directly because the assessments have different maximum scores.

The software must store both the marks obtained and the maximum possible marks.

It should also retain the original result rather than replacing it with a percentage.

4.1 Calculating a Percentage

A percentage expresses a result on a scale where the maximum is 100.

The formula is:

Percentage = (Marks Obtained ÷ Maximum Possible Marks) × 100

Consider the science test:

Marks obtained = 32

Maximum possible marks = 40

Percentage = (32 ÷ 40) × 100

Percentage = 80%

The student has achieved 80% of the available marks.

Another student might receive 45 out of 60 marks.

Percentage = (45 ÷ 60) × 100

Percentage = 75%

Although the second student obtained more raw marks than the first, the first student achieved a higher percentage.

This illustrates why the software must retain both raw marks and calculated percentages.

4.2 What Happens When the Maximum Marks Are Different?

Schools and teachers may set assessments with different maximum marks.

One mathematics quiz might be worth 10 marks, another 25 marks, and a final examination 100 marks.

Optimus Orien must allow the maximum marks to be specified for each assessment.

It must validate that the maximum possible marks are greater than zero and that a student's recorded result is within the permitted range.

For a conventional assessment, a result should not exceed the maximum possible marks or fall below zero. If a school uses a different scoring method, that method must be explicitly supported rather than silently accepting invalid values.

The system must also distinguish between a score of zero and a missing result.

A student who completed an assessment and received zero marks is not in the same situation as a student whose result has not yet been entered.

5. Missing Results, Absences, and Exemptions

A missing result means that the system does not currently have a recorded mark for that assessment.

There can be several reasons.

The teacher may not have entered the result yet. The student may have been absent. The student may have been exempted from the assessment. The assessment may have been cancelled. The student may be required to take a replacement assessment.

These situations should not be treated as equivalent.

For example, imagine that a class contains 30 students. Twenty-eight students have completed a test, while two students were absent.

The system must not automatically assign zero marks to the absent students unless the school's applicable rules explicitly require that outcome and an authorised user records the appropriate result.

The system must not calculate a class average by treating every missing result as zero.

Instead, each assessment record should have a result status, such as recorded, not yet entered, absent, exempt, cancelled, or awaiting a replacement assessment.

The available statuses should be configurable where school policies differ.

When a result is missing, the system should explain why it is excluded from a calculation. It should not quietly turn missing data into a numerical score.

This distinction is important because incorrect handling of missing results can distort student rankings, class averages, school comparisons, and performance indicators.

6. Understanding Grades

A grade is a category used to describe performance.

Examples include A, B, C, D, and F, or categories such as Excellent, Very Good, Good, Satisfactory, and Needs Improvement.

Some systems use numerical grades, such as 1 to 7. Other systems use grade-point scales, percentage bands, descriptive categories, or combinations of these methods.

A grade is not the same as a mark.

A mark is a numerical score. A grade is a category assigned according to a set of rules.

Suppose a school uses the following illustrative grading scale:

90–100% = A

80–89% = B

70–79% = C

60–69% = D

Below 60% = F

Under this particular scale, a student who achieves 85% receives a B.

However, this scale is only an example. It must not be treated as an official grading system or applied automatically to every school.

Another school might award an A at 85%. A different school might use grades from 1 to 9. Another might use standards-based descriptions instead of letter grades.

Optimus Orien must not hard-code one grading scale for all schools.

Each school must be able to use an appropriate grading configuration.

6.1 Grade Boundaries

A grade boundary is the score at which one grade changes into another.

For example, if a school's rules state that a B begins at 80%, then 80% is the lower boundary for a B.

The system must store grade boundaries as part of a grading configuration.

It must also handle boundary values correctly.

If the configured rule says that a B is awarded from 80% up to, but not including, 90%, a score of exactly 80% should receive a B and a score of exactly 90% should receive the grade assigned to the next band.

The software should not leave these cases ambiguous.

The rules must clearly define whether each boundary is inclusive or exclusive.

6.2 Grades May Be Based on More Than Percentages

Not every grading system converts percentages directly into letter grades.

Some systems use achievement levels, learning standards, proficiency descriptions, grade points, or criteria-based judgments.

For example, a teacher might evaluate whether a student can independently write a structured paragraph, explain a scientific concept, or solve a particular type of problem.

The teacher may assign a proficiency level such as Beginning, Developing, Proficient, or Advanced.

These categories describe performance against a defined standard rather than simply converting a percentage into a letter.

Optimus Orien should support both numerical and categorical results. It should not require every assessment to produce a percentage.

7. Understanding Weighted and Unweighted Results

Schools may combine several assessments to produce a term result or final subject result.

The method used to combine them matters.

There are two common approaches: weighted calculation and unweighted calculation.

7.1 Unweighted Average of Assessment Percentages

Suppose a student receives these percentages in three assessments:

Quiz 1: 80%

Quiz 2: 90%

Final examination: 70%

If the school gives all three assessments equal importance, the average is:

Average = (80 + 90 + 70) ÷ 3

Average = 80%

Each assessment contributes one-third of the final calculation, even though the assessments may have different maximum marks.

This is an unweighted average of the three assessment percentages.

7.2 Weighted Average

A school may decide that different assessments should contribute different amounts to the final result.

For example, a school might configure:

Homework: 20%

Quizzes: 30%

Final examination: 50%

These values are illustrative. They are not universal school rules.

Suppose a student achieves:

Homework average: 90%

Quiz average: 80%

Final examination: 70%

The weighted result is calculated as follows:

Homework contribution = 90 × 0.20 = 18

Quiz contribution = 80 × 0.30 = 24

Examination contribution = 70 × 0.50 = 35

Final result = 18 + 24 + 35

Final result = 77%

The student receives a weighted result of 77%, assuming the school's configured rules permit this calculation and all three components are available.

7.3 Why Optimus Orien Must Support Different Calculation Methods

A school might calculate its final result using the weighted method above.

Another school might use an average of examination marks.

A third school might calculate the final result by adding all marks obtained and dividing by all marks available.

These approaches can produce different outcomes.

For example, suppose a student receives 8 out of 10 on a short quiz and 60 out of 100 on a final examination.

The unweighted average of the percentages is:

(80% + 60%) ÷ 2 = 70%

The combined marks percentage is:

(8 + 60) ÷ (10 + 100) × 100 = approximately 61.82%

Neither calculation is universally correct for every school. They answer different questions.

The first gives equal importance to the two assessment percentages. The second gives greater influence to the examination because it has more available marks.

Optimus Orien must use the school's configured calculation method and make the method understandable to authorised users.

It must never silently choose a calculation method merely because it is easier to implement.

8. Missing Components in Weighted Calculations

Consider a subject in which homework contributes 20%, quizzes contribute 30%, and the final examination contributes 50%.

A student has completed the homework and quizzes, but the final examination result has not been entered.

The system cannot safely assume that the student's final result is zero.

It also cannot automatically treat the available 50% of the assessment structure as though it represents the entire course, unless the school's configured policy specifically permits that approach.

There are different possible policies.

The school might require all components before calculating a final result. It might display a provisional result based on the available components. It might redistribute weights under a defined rule when an assessment is formally excluded.

These are different policies and can produce different results.

Optimus Orien must allow the school to define how incomplete assessment sets are handled.

Until a final result is valid under the configured rules, the interface should label it as provisional, incomplete, or another suitable status rather than presenting it as a confirmed final result.

The system should also display which components are missing.

9. What Is a Term?

A term is a defined period within a school's academic year.

Some schools divide the year into two semesters. Others use three terms, four quarters, or different arrangements.

A school might use the following structure:

Academic year: 2026–2027

Term 1: August to December

Term 2: January to March

Term 3: April to June

This is only an example. Academic calendars differ by country, school, and educational programme.

A term may contain several assessments. A subject result may be calculated at the end of the term using those assessments.

Optimus Orien must allow each school to configure its academic calendar, including the number of terms, their names, their dates, and their relationship to the academic year.

It must not assume that every school has three terms or that every term follows the same calendar dates.

10. What Is an Academic Year?

An academic year is the period a school uses to organise its educational activities and records.

It may cross two calendar years.

For example, an academic year named 2026–2027 might begin in August 2026 and end in June 2027.

A school using a different calendar might begin and end its academic year at different times.

Optimus Orien must treat the academic year as a configurable entity rather than assuming that it begins in January.

Every assessment and result should be associated with the appropriate academic year and, where applicable, the relevant term or reporting period.

This is necessary to make historical performance reports accurate.

For example, when a principal views a student's performance in the academic year 2025–2026, the system must not accidentally include assessments from 2026–2027.

11. Understanding Subject Results

A subject is an area of study, such as mathematics, English, biology, chemistry, physics, history, or computer science.

A student may have a separate result for each subject.

Consider this example:

Mathematics: 82%

English: 74%

Science: 91%

History: 68%

These results provide more information than a single overall score.

The student performs particularly well in science but has a lower result in history.

This does not automatically mean that the student is weak in every aspect of history or strong in every aspect of science. The assessment methods and topics matter.

Nevertheless, subject-level results can help schools identify areas where a student may benefit from additional attention.

Optimus Orien must maintain subject results separately before calculating any overall performance measure.

The system must not combine every mark from every subject without a defined calculation rule.

12. What Is an Overall Student Result?

An overall student result is a summary of performance across multiple subjects or assessment components.

There are different ways to calculate it.

Suppose a student receives the following subject percentages:

Mathematics: 80%

English: 70%

Science: 90%

History: 60%

If every subject receives equal importance, the overall average is:

(80 + 70 + 90 + 60) ÷ 4 = 75%

The student's equal-weight subject average is 75%.

However, a school might give different subjects different weights. It might also use the total marks obtained across subjects.

These methods are not necessarily equivalent.

For example, suppose mathematics has a maximum of 100 marks and English has a maximum of 50 marks. Combining the raw marks directly gives English a different influence than it would have if the school first converted both results to percentages and gave them equal weight.

Therefore, Optimus Orien must not assume that an overall result always means the arithmetic average of subject percentages.

The calculation method must be explicitly defined.

The system should show which subjects contributed to the result, what weights were used, and whether any results were excluded.

13. Understanding Class Averages

A class average summarises the performance of students in a class under a defined calculation method.

Suppose five students receive these mathematics percentages:

Student A: 80%

Student B: 70%

Student C: 90%

Student D: 60%

Student E: 100%

The class average is:

(80 + 70 + 90 + 60 + 100) ÷ 5 = 80%

The class average is 80%.

This provides a summary of the class's performance on the relevant measure.

However, the average does not describe every student.

Two classes can have the same average but very different distributions of performance. One class might have most students close to 80%, while another might contain several students with very high results and several with very low results.

For this reason, Optimus Orien should display class averages alongside other useful information, such as score distributions and the number of students whose results are included.

13.1 Class Average Must Have a Clear Meaning

The phrase "class average" can refer to different calculations.

It might mean the average mathematics result for one class. It might mean the average across all subjects for a class. It might mean the average examination score for a particular assessment.

These values must not be mixed together.

Every class average displayed by Optimus Orien should identify what it measures.

For example:

"Class 7A — Mathematics — Term 1 average: 76.4%"

This is more informative than simply displaying "Class Average: 76.4%."

The system should also identify the reporting period and the calculation method when necessary.

13.2 How Optimus Orien Should Calculate Class Performance

Optimus Orien's class-level performance reporting must follow a defined calculation policy.

If the intended measure is the average subject performance for a class, the system should first calculate each eligible student's subject result using the school's configured rules. It should then calculate the average of those eligible student results for that subject.

If the intended measure is the class's overall performance across subjects, the system should first calculate each student's overall result using the configured subject-weighting rules. It should then calculate the class average from those eligible student results.

These are two different measures.

The software must not treat the average of all raw marks in a class as interchangeable with the average of student percentages or the average of subject results.

The selected measure must be clearly named in reports and charts.

14. Understanding School Averages

A school average summarises a defined set of student or class results across a school.

For example, a principal may want to know the average mathematics performance across all Year 8 classes.

Suppose the school has three Year 8 classes.

Class 8A average: 80%

Class 8B average: 70%

Class 8C average: 90%

If each class contains the same number of eligible students, the average of the three class averages is 80%.

However, suppose Class 8A contains 30 students, Class 8B contains 20 students, and Class 8C contains 10 students.

The class averages alone are no longer sufficient to calculate the average across all 60 students accurately if each student is intended to receive equal weight.

The correct student-weighted average would be:

(30 × 80 + 20 × 70 + 10 × 90) ÷ 60 = approximately 78.33%.

The simple average of the three class averages would still be 80%.

Both numbers can be useful, but they describe different things.

The first represents the average result across all eligible students. The second represents the average of the three class averages, giving each class equal importance regardless of size.

Optimus Orien must distinguish between these calculations.

For school-wide academic performance, the platform should normally calculate the measure from the underlying eligible student results when the intended metric is average student performance. If the school wants an average of class averages, the interface must label it accordingly.

The system must not calculate school performance by averaging class averages without considering the intended meaning of the measure.

15. Understanding Student Rankings

A ranking places students in an order according to a selected performance measure.

For example, if five students receive mathematics percentages of 95%, 88%, 82%, 76%, and 65%, they can be ordered from highest to lowest.

However, ranking is meaningful only when the comparison is defined.

A student might rank first in mathematics but fifth in English. Another student might rank highly in one class but have a different position when compared across the entire year group.

Optimus Orien should allow rankings to be generated for defined groups and measures, such as:

A particular subject within one class.

Overall performance within a class.

A subject within a year group.

Overall performance within a year group.

A wider school-wide comparison where the students are being assessed on a sufficiently comparable basis.

The platform must clearly state the measure, subject where relevant, reporting period, and comparison group.

It must not compare students from different assessment systems as though their results were automatically equivalent.

15.1 What Happens When Students Have the Same Result?

Suppose three students receive 90%, 85%, and 85%.

Two students have the same score.

The platform must have a defined method for handling ties.

One option is competition ranking, in which the positions are 1, 2, and 2. The next position is 4.

Another option is dense ranking, in which the positions are 1, 2, and 2. The next position is 3.

Both methods exist, but they produce different rank numbers.

Optimus Orien should use a clearly defined ranking method and apply it consistently. If the ranking method is configurable, the school must be able to identify which method it uses.

The interface should not suggest that a tied score represents a meaningful difference in ability merely because a ranking number is displayed.

15.2 Ranking Is Not the Same as Learning Progress

A student can improve their own performance while moving down a ranking if other students improve more quickly.

For example, a student might increase their mathematics result from 65% to 75%. This is an improvement of 10 percentage points.

However, if other students improve by even larger amounts, the student's rank may decrease.

Therefore, Optimus Orien must treat ranking and individual progress as separate indicators.

Ranking describes relative position within a defined comparison group. Progress describes change in a student's performance over time.

16. Understanding Percentage Points and Percentage Change

These two concepts are often confused.

Suppose a student improves from 60% to 70%.

The increase is 10 percentage points.

The relative percentage increase is:

((70 − 60) ÷ 60) × 100 = approximately 16.67%.

The student improved by 10 percentage points, which represents an approximately 16.67% increase relative to the original score.

These statements are not interchangeable.

Optimus Orien should use the term "percentage-point change" when subtracting one percentage score from another.

For example:

Previous result: 60%

Current result: 70%

Change: +10 percentage points

This is clearer than saying the student improved by 10% when the intended calculation is simply the difference between the two percentage scores.

If the platform displays relative percentage change, it should label that calculation separately.

The system must also handle cases where the original value is zero, because relative percentage change from zero cannot be calculated using the usual formula.

17. Understanding Student Progress Over Time

A single assessment gives information about performance at one point in time.

Progress requires comparing results across multiple points in time.

Suppose a student receives these mathematics results:

Term 1: 58%

Term 2: 64%

Term 3: 72%

The results show an upward pattern.

The student improved by 6 percentage points from Term 1 to Term 2 and by another 8 percentage points from Term 2 to Term 3.

This suggests that the student's recorded performance improved across these reporting periods.

However, the platform should not automatically conclude that the student has mastered mathematics or that a particular teaching intervention caused the improvement.

The assessments may have differed in difficulty, content, or scoring rules.

A meaningful progress comparison requires enough context to understand what is being compared.

Optimus Orien should therefore store the assessment or reporting-period context for every result.

When displaying a trend, the platform should indicate the subject, reporting periods, and relevant score scale.

Where assessment methods are not sufficiently comparable, the platform should warn users that the trend must be interpreted cautiously.

18. Understanding Skills and Competencies

A subject result does not always reveal a student's strengths and weaknesses in enough detail.

For example, a student might receive 75% in English. That score provides a general indication of performance, but it does not necessarily show whether the student struggles with reading, writing, speaking, listening, grammar, or comprehension.

Schools may therefore assess particular skills or competencies.

A skill is a specific ability that a student is expected to demonstrate. A competency is a broader ability to apply knowledge and skills appropriately in a particular context.

Examples of skills include reading fluency, written expression, oral communication, mathematical reasoning, problem-solving, scientific observation, and analytical thinking.

The exact skills associated with a subject depend on the educational programme and the school's chosen framework.

Optimus Orien should allow a school to define the skills it wants to track and associate them with appropriate subjects, year groups, and educational programmes.

The system should not assume that every school uses the same skill list.

18.1 Skill Ratings

A school may choose to record skill ratings using a numerical scale.

For example, a school might use a scale from 1 to 10.

Under a hypothetical scale, a teacher might rate a student's presentation skills as 7 out of 10.

However, a rating of 7 does not automatically mean the same thing in every school.

One school might use 1 for "Beginning" and 10 for "Highly Advanced." Another might use a different set of descriptions.

For skill ratings to be meaningful, the school must define what the scale represents.

Optimus Orien should allow authorised users to configure the scale and its descriptive levels.

If the school uses descriptive categories instead of numbers, the platform should support those categories rather than forcing the school to invent numerical scores.

18.2 Who Can Record Skill Ratings?

A school may assign particular skills to teachers who are qualified or responsible for assessing them.

For example, the English teacher may assess reading and writing, while the mathematics teacher may assess mathematical reasoning.

The platform should allow authorised users to record skill assessments only for the students and skills they are permitted to assess.

The system should record who entered or changed a rating and when the change occurred.

Where the school requires a review or approval process, the rating should remain distinguishable from a reviewed or approved rating.

18.3 Skill Ratings Must Not Be Confused With Examination Marks

A score of 80% in a science examination and a skill rating of 8 out of 10 are not necessarily equivalent.

The examination score may reflect performance on a particular test. The skill rating may represent a teacher's judgment based on observations across several activities.

The platform must preserve the type and context of each measure.

It must not combine skill ratings and examination percentages into a single overall score unless the school explicitly configures a justified calculation method.

19. Understanding Attendance and Homework Completion as Separate Indicators

Optimus Orien is intended to provide school performance intelligence, so academic results are not the only information that may matter.

Attendance records indicate whether a student attended school or a particular class according to the school's attendance rules.

Homework records indicate whether an assigned task was completed, submitted, checked, or confirmed according to the school's workflow.

These indicators may help users identify patterns that deserve attention.

For example, a student might have declining academic results and a recent increase in absences. Another might attend regularly but frequently fail to submit homework.

However, attendance, homework completion, and academic performance measure different things.

A student who attends every class does not necessarily achieve high marks. A student who misses school does not automatically perform poorly.

Optimus Orien must keep these indicators distinct and allow users to examine how they relate over time.

Any relationship displayed between these indicators and academic results should be presented as an observed pattern rather than proof of cause and effect.

The rules for recording attendance and homework completion should be defined separately from the rules for calculating marks and grades.

20. Understanding Assessment Comparability

Comparability means that two results can reasonably be compared for a particular purpose.

Two students might both receive 80%, but their assessments may differ in important ways.

One student might have completed a mathematics test covering basic arithmetic. Another might have completed an advanced test covering algebra and geometry.

Both results are 80%, but they do not necessarily represent the same level of subject knowledge.

Similarly, two schools may use different grading scales or assessment methods.

Optimus Orien must not assume that matching percentages prove equivalent achievement.

The system should retain information about the educational programme, subject, year group, assessment, maximum marks, grading configuration, and reporting period.

This information helps users understand the context of each result.

Where the system compares results across classes, year groups, branches, schools, or educational systems, it should make the comparison rules clear.

For example, a comparison might be appropriate because all students completed the same assessment under the same scoring rules. Another comparison might be limited to descriptive benchmarking because the assessments differ.

The platform should not describe results as directly equivalent when the necessary evidence for comparability is missing.

21. How Results Should Appear in Optimus Orien

Optimus Orien should display academic results in ways that are understandable to school leaders, teachers, parents, and students.

A result should not appear as an unexplained number.

For example, displaying "78.4" alone does not tell the user whether the number represents marks, a percentage, a grade point, a skill rating, or a class average.

A clearer presentation would be:

Mathematics — Term 1

Marks: 78 out of 100

Percentage: 78%

Grade: B, according to the school's configured grading scale

Class average: 74%

Difference from class average: +4 percentage points

This example assumes a school configuration that awards a B for 78%. The actual grade must come from the school's own grading rules.

The system should display only the measures that are applicable to the particular result.

If the school does not use letter grades, the interface should not invent one.

If the maximum marks are unknown, the system should not calculate a percentage.

If the class average cannot be calculated because no eligible results exist, the interface should communicate that the value is unavailable rather than displaying zero.

22. Historical Results and Changing Rules

School assessment rules may change over time.

A school might introduce a new grading scale, change the weighting of examinations, or revise the subjects and skills it assesses.

The software must not silently apply a new configuration to historical results.

For example, suppose a school uses one grading scale in 2025–2026 and changes to a different scale in 2026–2027.

When a user opens a result from 2025–2026, the system should preserve the grade assigned under the applicable historical rules.

If the school needs to recalculate a historical result using a new rule, that should be a deliberate operation with a recorded reason, appropriate permissions, and a clear indication that the result has been recalculated.

The platform should maintain a history of relevant configuration changes.

This principle is essential for trust, auditing, and accurate longitudinal reporting.

23. Developer Requirements

The following requirements translate the concepts in this document into software behaviour.

EDUC-016: Configurable Assessment Types

The system must support different assessment types, including diagnostic, formative, summative, and standardised assessments.

A school must be able to configure which assessment types it uses.

Acceptance test: An administrator creates a quiz, a term examination, and a diagnostic assessment. Each assessment is stored with its correct type and can be distinguished in reports.

EDUC-017: Assessment Metadata

Every assessment must support a name, subject, relevant class or student group, assessment type, maximum marks where applicable, assessment date, reporting period, academic year, and responsible teacher or authorised creator.

Additional fields may be required for particular assessment methods.

Acceptance test: An authorised teacher creates a mathematics quiz for Class 7A, specifies a maximum of 25 marks, assigns it to Term 1, and saves it. The assessment appears under the correct subject, class, and reporting period.

EDUC-018: Marks and Percentage Calculation

The system must store marks obtained and maximum possible marks separately.

For conventional numerical assessments, the system must calculate the percentage using the formula defined in this document.

Acceptance test: A student receives 18 out of 20. The system retains both values and calculates 90%. It must not replace the original marks with the percentage.

EDUC-019: Result Validation

The system must validate conventional assessment results and prevent invalid marks from being accepted without an explicit, supported exception.

Acceptance test: For an assessment worth 20 marks, the system accepts a result of 18 and rejects a result of 25. It also rejects a negative result unless a supported scoring model explicitly permits such a value.

EDUC-020: Missing Results and Result Statuses

The system must distinguish a missing result from a recorded zero and must support relevant result statuses.

Acceptance test: One student receives zero marks, another has not yet received a recorded mark, and a third is marked absent. Reports display these as different states. The absent and missing results are not silently treated as zero.

EDUC-021: Configurable Grading Scales

The system must support configurable grading scales, including numerical bands, letter grades, and descriptive categories where applicable.

The configuration must define grade boundaries and their inclusion rules.

Acceptance test: An administrator creates a grading scale with a B beginning at 80%. A result of exactly 80% receives a B, provided the configured boundary rules specify that 80% is included.

EDUC-022: Configurable Calculation Methods

The system must support different methods of calculating combined results, including an unweighted average of percentages, a weighted average, and a combined marks percentage.

The system must not select a calculation method silently.

Acceptance test: A developer or tester configures the same assessment data under two different calculation methods. The system produces the appropriate result for each method and identifies which method was used.

EDUC-023: Configurable Assessment Weights

The system must allow authorised users to configure assessment weights when a school uses weighted results.

The configuration must define the contribution of each component and validate the applicable weight rules.

Acceptance test: A school configures homework at 20%, quizzes at 30%, and the final examination at 50%. The system calculates the weighted result correctly when all required component results are present.

EDUC-024: Incomplete Assessment Sets

The system must handle missing components in combined results according to a school-configured policy.

It must not automatically treat missing results as zero or redistribute assessment weights without an applicable rule.

Acceptance test: A student's examination result is missing. The system follows the configured incomplete-result policy and does not display an unqualified final result when the required components are unavailable.

EDUC-025: Academic Calendar Configuration

The system must support configurable academic years, terms, semesters, quarters, or other reporting periods.

Acceptance test: An administrator configures an academic year running from August 2026 to June 2027, divided into two terms. The system stores assessments under the correct academic year and term without assuming a January start.

EDUC-026: Subject-Level Results

The system must maintain results separately for each subject before calculating overall performance.

Acceptance test: A student has results in mathematics, English, and science. The system displays each subject result independently and does not merge the marks into a single result without a defined calculation rule.

EDUC-027: Configurable Overall Results

The system must support a configured method for calculating overall student performance across subjects.

The calculation method must be identifiable and must handle missing subject results according to defined rules.

Acceptance test: An administrator configures an equal-weight average across four subjects. The system calculates the student's overall result using the four eligible subject percentages. If one subject result is missing, the system applies the configured missing-data policy.

EDUC-028: Class Average Calculation

The system must calculate class averages according to an explicitly defined measure and method.

It must distinguish the average of student results from the average of class averages.

Acceptance test: Two classes have different enrolments and different average scores. The system calculates the average across all eligible student results correctly when student-weighted performance is requested. It produces a separately labelled result if an average of class averages is requested.

EDUC-029: School Average Calculation

The system must support school-level averages calculated from the appropriate underlying results.

The measure must identify the subjects, reporting period, comparison population, and calculation method.

Acceptance test: A principal views the school's mathematics performance for Term 1. The system uses eligible student results for the intended measure and does not substitute an unweighted average of class averages without identifying that method.

EDUC-030: Ranking and Ties

The system must support rankings based on defined comparison groups and performance measures.

The ranking method must be consistent and must handle tied results according to the configured policy.

Acceptance test: Two students receive the same result. The system assigns their positions according to the configured ranking method and clearly identifies the ranking scope.

EDUC-031: Progress Over Time

The system must support historical result views and comparisons across reporting periods.

It must distinguish changes in percentage points from relative percentage changes.

Acceptance test: A student's result increases from 60% to 70%. The system displays a change of +10 percentage points. If relative percentage change is also shown, it is separately labelled and calculated correctly.

EDUC-032: Configurable Skills and Competencies

The system must allow schools to define skills or competencies and associate them with relevant subjects, year groups, or educational programmes.

Acceptance test: An administrator creates a reading-comprehension skill for a specified year group and associates it with English. An authorised teacher can record an assessment of that skill for eligible students.

EDUC-033: Configurable Skill Rating Scales

The system must support numerical and descriptive skill-rating scales.

Acceptance test: One school uses a numerical scale from 1 to 10, while another uses Beginning, Developing, Proficient, and Advanced. Each school can record and view skill assessments using its own configured scale.

EDUC-034: Skill Assessment Permissions

The system must enforce permissions for recording and changing skill assessments.

It must retain a record of who created or changed a rating and when the change occurred.

Acceptance test: A teacher can record a skill rating for an authorised student and skill but cannot change a rating outside the teacher's permitted scope. An authorised administrator can inspect the change history.

EDUC-035: Separation of Performance Measures

The system must keep examination marks, subject grades, skill ratings, attendance, and homework completion as distinct measures.

Acceptance test: A student has an examination percentage of 80%, a skill rating of 8 out of 10, and attendance of 95%. The system displays each measure correctly and does not combine them into one score unless a specifically configured and authorised calculation requires it.

EDUC-036: Result Context and Comparability

The system must preserve the context necessary to interpret and compare results, including the assessment method, subject, educational programme where relevant, academic year, reporting period, grading configuration, and applicable scoring rules.

Acceptance test: A user compares results from two academic years in which the grading configuration changed. The system retains the historical context and does not silently recalculate old grades using the new configuration.

EDUC-037: Historical Configuration and Audit Trail

The system must preserve the applicable historical grading and calculation rules and record significant configuration changes.

Acceptance test: An administrator changes a grading boundary for a new academic year. Historical results remain associated with the earlier configuration unless an authorised recalculation is deliberately performed and recorded.

EDUC-038: Explainable Results

The system must provide enough information for authorised users to understand how a displayed result was calculated.

Where relevant, this includes the original marks, maximum marks, percentage, grade, component weights, included assessments, excluded assessments, and calculation method.

Acceptance test: A principal opens a student's weighted subject result and can identify the component results and weights that produced it.

24. Final Principles

Assessment is the process of collecting evidence about learning. Marks are numerical scores. Percentages express results on a scale of 100. Grades classify performance according to defined rules. Averages summarise a selected group of results. Rankings

describe relative positions within a comparison group. Progress describes changes over time. Skill ratings represent assessments of specific abilities or competencies.

These concepts are related, but they are not interchangeable.

Optimus Orien must preserve these distinctions in its database, calculations, permissions, and interface.

The system should be configurable enough to support different educational arrangements without requiring the developer to rewrite the software every time a school uses a different grading scale or reporting structure.

At the same time, configurability must not mean that the system accepts unexplained or invalid data. Each configuration should have defined rules, validation, and a clear explanation of how results are produced.

The platform's purpose is to help schools understand student and school performance accurately. Its calculations should therefore be transparent, reproducible, and appropriate to the educational context.

25. What This Part Establishes

This document establishes the foundations for recording and interpreting assessment results in Optimus Orien.

It defines how the platform should handle marks, percentages, grades, weighted results, missing data, academic years, reporting periods, subject results, overall performance, class averages, school averages, rankings, progress, skills, and historical calculation rules.

It also establishes the requirement that the platform must never assume that every school follows the same assessment or grading system.

Part 4 builds on the first three documents. It explains how a school is organised, how multiple schools can belong to one organisation, how students and teachers connect to classes and subjects, and how all of this should be represented in Optimus Orien's database.

I’ve kept it in simple English, with implementation requirements and acceptance tests so your developer can use it as a guide when coding.

Part 4:
How Schools Are Organised and How Their Data Connects

1. Purpose of This Document

Optimus Orien needs to understand how schools are organised before it can accurately calculate school performance.

Consider three different situations.

In the first situation, a school has one campus, one principal, 30 teachers, and 500 students.

In the second situation, a school organisation operates five campuses in different parts of a city. Each campus has its own principal, teachers, students, and classes, but the organisation wants to see performance across all five campuses.

In the third situation, an education group operates several schools in different countries. Each school may follow a different curriculum, academic calendar, grading system, and educational structure.

All three situations should be possible within Optimus Orien.

The platform must not assume that one organisation always means one school, that one school always means one campus, or that all campuses follow identical academic rules.

This document establishes how Optimus Orien should represent organisations, schools, campuses, academic years, classes, subjects, students, teachers, parents, and enrolments.

It also explains how these entities connect to one another, how their data should be separated, and how users should access the appropriate information.

The objective is to build a system that can start with a single school in Pakistan and expand to larger organisations without requiring its entire structure to be redesigned.

2. Begin With a Simple Example

Imagine a school called Bright Future School.

Bright Future School has one principal, 20 teachers, 300 students, and several classes.

The school teaches mathematics, English, science, and social studies.

Students are assigned to classes. Teachers teach particular subjects to particular classes. Students receive assessment results in the subjects they study.

The principal needs to see the performance of the entire school.

A mathematics teacher needs to see the students and results relevant to the teacher's assigned classes.

A parent needs to see information about their own child.

A student needs to see their own results.

These users are looking at different parts of the same underlying school information.

The software must store the information in a connected structure so that it does not need to create separate copies of the same student record for the principal, teacher, student, and parent portals.

Instead, the system should maintain the appropriate records and provide each user with the information they are authorised to access.

This is the foundation of the Optimus Orien data model.

3. Understanding an Organisation

An organisation is an entity that owns, operates, manages, or governs one or more educational institutions.

For example, a company may operate several private schools. A charitable organisation may operate schools in different regions. A school group may operate institutions under one brand.

The organisation is the higher-level entity responsible for the schools it operates.

In Optimus Orien, an organisation should be able to manage its authorised schools and view the information available at the organisation level.

For example, suppose the Bright Future Education Group operates three schools:

Bright Future School — North Campus.

Bright Future School — Central Campus.

Bright Future School — South Campus.

The organisation may want to compare the performance of all three schools.

However, the principal of the North Campus should not automatically receive access to the private records of the Central and South campuses.

The organisation-level administrator may have broader access, while each principal has access limited to their assigned school or campus.

The software must distinguish between the organisation and the individual institutions it manages.

An organisation can have multiple schools, but it does not have to. A single independent school can also be represented within an organisation structure.

4. Understanding a School

A school is an educational institution that provides education to students.

For Optimus Orien, a school is an important unit of academic management and reporting.

A school may have its own principal, students, teachers, subjects, classes, academic calendar, educational programmes, and grading configuration.

Some schools have only one campus. Others operate multiple campuses.

The software must support both arrangements.

A school should have a unique internal identifier that remains stable even if its name changes.

For example, Bright Future School might change its public name to Bright Future Academy. The software should update the displayed name without creating an entirely new school or losing its historical records.

The system should also retain relevant historical information when organisational changes occur.

For example, if a campus changes its name or moves under a different organisation, the system must not automatically erase its earlier results or associations.

The exact handling of ownership transfers, mergers, and closures should be controlled by authorised administrative procedures.

5. Understanding a Campus or Branch

A campus or branch is a physical or operational location associated with a school or school organisation.

The words "campus" and "branch" are often used interchangeably, but different organisations may use them differently.

For example, a school group might call each location a campus. Another organisation might call each location a branch.

Optimus Orien should allow the organisation to configure the terminology displayed in the interface.

Internally, the platform should maintain a consistent structure for representing school locations.

Consider an organisation operating three campuses:

North Campus has 600 students.

Central Campus has 450 students.

South Campus has 350 students.

The organisation has 1,400 students across all three campuses.

Each campus may have its own principal, teachers, classes, subjects, and academic performance records.

The organisation-level administrator should be able to view the combined total and examine the individual campuses.

The principal of North Campus should normally see only the records belonging to North Campus, subject to any additional permissions explicitly granted.

5.1 Must Every School Have Multiple Campuses?

No.

The platform must support a school with one campus without forcing administrators to create unnecessary additional campuses.

It should also support organisations that operate several schools, each of which has its own locations.

The system should not require every organisation to follow a complex hierarchy if it does not need one.

5.2 How Campus Information Should Be Displayed

An authorised organisation administrator should be able to view a list or map of the organisation's campuses.

For each campus, the platform may display its name, principal, student count, teacher count, attendance information, academic performance indicators, and school health score.

The meaning of each indicator must be defined.

For example, a school health score is not a universal educational measurement. Optimus Orien must calculate it using an explicitly defined model rather than presenting an unexplained number as an established standard.

The map should show the locations of campuses only when suitable location information is available and the user is authorised to see it.

The system must not invent locations for campuses whose addresses or coordinates have not been supplied.

6. Understanding the Relationship Between Organisation, School, and Campus

The relationship between these entities must be defined carefully.

For a simple organisation, the structure may be:

Organisation → School → Campus.

For example:

Bright Future Education Group → Bright Future School → North Campus.

However, a real education group might operate several separately named schools, each with multiple campuses.

In that case, the structure may be:

Organisation → School A → Campus A1 and Campus A2.

Organisation → School B → Campus B1 and Campus B2.

The platform should support these arrangements.

Some institutions may also operate without a meaningful distinction between the school and its single location.

The data model must support that case without forcing unnecessary duplication.

Before development, the team must decide whether each campus belongs to exactly one school, whether schools may share physical locations, and whether an organisation can manage schools through more than one administrative level.

For the initial version, a practical default is that an organisation can manage multiple schools, and each school can have one or more operational locations. A single-location school can have one campus record or use a clearly defined single-campus arrangement.

These rules should be consistent across the application.

The platform must not create different interpretations of a school or campus in different modules.

7. Understanding an Academic Year Within a School

An academic year is the period a school uses to organise teaching, assessment, and reporting.

As established in Part 3, an academic year does not necessarily begin in January.

For example, a school might operate from August 2026 to June 2027 and call this academic year 2026–2027.

Another school might follow a different calendar.

Optimus Orien must associate academic years with the appropriate school and preserve the dates and configuration applicable to each year.

This is important because schools may change their calendars, assessment structures, and grading policies over time.

A student might belong to Class 7 during one academic year and Class 8 during the next.

The system must preserve both records rather than replacing the student's old class assignment with the new one.

When a principal views historical performance, the platform must use the correct academic year and the relevant configuration for that period.

8. Understanding Year Groups, Grades, and Stages

Students are usually organised according to their educational stage or level.

Depending on the country and educational system, schools may use terms such as Grade 5, Year 5, Class 5, Standard 5, Form 5, or another local designation.

These terms do not always mean the same thing across countries.

For example, a school may use the word "Grade" to refer to the student's year group. Another may use it to describe an achievement level or examination result.

Optimus Orien must therefore distinguish between a student's educational year group and the student's academic grade or mark.

The system should use an internally consistent concept for the year group while allowing each school to configure the label shown to users.

For example, the same internal concept could be displayed as "Grade 7" in one school and "Year 7" in another.

The system must not assume that two identically named year groups in different educational systems are academically equivalent.

Where a school uses stages, phases, key stages, or other groupings, the platform should be able to associate year groups with those structures.

For example, a school might group several year groups under a primary stage and others under a secondary stage.

The exact stage names and relationships must be configurable.

9. Understanding a Class

A class is a defined group of students who are taught together for a particular purpose or period.

In many schools, students belong to a main class or homeroom. They may also attend different subject classes.

For example, Class 7A may contain 30 students who share a main classroom teacher.

These students may attend mathematics, English, and science lessons together. However, some schools divide students into different groups for particular subjects.

The system must distinguish between a general student group and a specific teaching group where necessary.

For a first implementation, Optimus Orien should support a class or student group associated with a school, academic year, and year group.

The class should have a name, such as 7A, 7B, or Year 8 Blue.

Students should be associated with the class through an enrolment or membership record rather than having their historical class assignment permanently overwritten.

This allows the platform to preserve the student's class history.

For example, a student might belong to Class 7A in 2025–2026 and Class 8B in 2026–2027.

Both assignments must remain available for historical reporting.

10. Understanding Subject Classes and Teaching Assignments

A student may belong to a general class but attend a different group for a particular subject.

Consider a school in which Class 7A has 30 students.

For mathematics, the school divides these students into two groups.

Mathematics Group 1 contains 15 students.

Mathematics Group 2 contains the other 15 students.

The English teacher teaches all 30 students together.

The mathematics teacher teaches the two groups separately.

If Optimus Orien assumes that every student has exactly one class for every subject, it will not correctly represent this arrangement.

The system should therefore distinguish between a student's general class membership and their participation in a specific teaching group or subject offering.

A subject offering represents a subject being taught to a defined group of students during a particular academic period.

For example, "Mathematics — Class 7A — Term 1" might represent a teaching assignment for a specific group, subject, and reporting period.

The exact data model can be refined during technical design, but the software must support the underlying relationships.

This structure also allows a teacher to teach mathematics to Class 7A and science to Class 8B without creating duplicate student records.

11. Understanding a Subject

A subject is an area of study, such as mathematics, English, biology, history, or computer science.

A subject may exist across several year groups.

For example, a school may teach mathematics to students from Grade 1 through Grade 12.

However, the topics, assessment methods, and learning expectations may differ at each level.

The platform should distinguish between the general subject and the way that subject is taught or assessed in a particular educational programme or year group.

For example, mathematics in one programme may have a different curriculum and assessment structure from mathematics in another programme.

The platform must not assume that every subject with the same name has identical content or grading rules.

For the initial implementation, schools should be able to create or configure subjects, assign them to relevant year groups or programmes, and associate them with the appropriate teachers and student groups.

Subject names should be editable by authorised users, while stable internal identifiers should preserve the relationships between records.

12. Understanding a Teacher

A teacher is a person responsible for teaching, assessing, or supporting students.

In Optimus Orien, a teacher should have an individual record that is separate from the teacher's account used to sign in.

This distinction matters because the teacher's account provides access to the software, while the teacher's record describes their role within the school.

A teacher may teach several subjects, several classes, or several student groups.

A teacher may also change schools or teach at more than one campus.

The platform must not assume that one teacher always belongs to one class or teaches only one subject.

A teacher's assignments should be recorded for the relevant academic period.

For example, a teacher may teach mathematics to Class 7A and Class 7B in 2026–2027. The following year, the same teacher may teach different classes.

Historical teaching assignments should remain available when required for auditing and interpreting past results.

The teacher's permissions should be based on their authorised role and assignments.

A mathematics teacher should not automatically receive permission to change every student's results across the entire organisation.

The platform must define which records the teacher can view, create, edit, or approve.

13. Understanding a Student

A student is an individual enrolled in an educational institution.

Each student should have a stable internal student identifier.

The identifier should not depend on the student's name, class, year group, or school location.

Names can change, students can move between classes, and schools can reorganise their structures. A stable identifier helps the system preserve the student's history.

The student record may contain the information necessary to identify the student and manage their school participation.

The exact personal information collected should be limited to what the platform genuinely requires.

For example, the platform may need the student's name, internal identifier, enrolment status, and relevant educational assignments. Additional information should be collected only when justified by a defined purpose.

The system must protect student information and restrict access according to role and permission.

A student should not automatically gain access to another student's private academic record simply because both students belong to the same class.

14. Understanding Enrolment

Enrolment is the formal association between a student and an educational institution or educational programme for a defined period.

It is not the same as the student's permanent identity record.

Suppose a student named Sara joins Bright Future School in August 2025.

The system creates or identifies Sara's student record and records her enrolment in the school for the relevant academic period.

In the next academic year, Sara moves into a different year group.

The system should create or update the appropriate enrolment and class-assignment records while preserving the previous year's information.

If Sara transfers to another school, the platform should preserve her historical records according to the organisation's policies and applicable privacy requirements.

It should not automatically treat the transfer as a new person if the platform can reliably establish that the incoming record belongs to the same student.

However, the system must not automatically merge two student records based only on matching names. Two different students can have the same name.

Any record matching or merging process must use appropriate evidence, permissions, and safeguards.

For the initial implementation, the platform should maintain a student record and separate enrolment records for school participation over time.

This allows the system to represent admission, active enrolment, transfer, withdrawal, completion, and other relevant status changes without destroying historical data.

15. Understanding Parents and Guardians

A parent or guardian is a person authorised to receive relevant information about a student.

A student may have more than one parent or guardian. One parent or guardian may also be associated with more than one student.

For example, a parent may have two children attending the same school.

The platform should represent these relationships without creating a separate parent identity for each child.

A parent account should be associated with the relevant student records through an explicit relationship.

The relationship must include appropriate permissions governing which information the parent can access.

The platform must not assume that every adult associated with a student automatically has full access to the student's records.

The school may need to verify the relationship and determine the appropriate access rights.

The exact handling of custody arrangements, disputed access, or other sensitive circumstances requires a clearly defined administrative process.

For the initial version, the important software principle is that parent access must be based on an authorised relationship, not merely a matching surname, email address, or shared household.

A parent who has two children should be able to switch between those children's authorised views without gaining access to unrelated students.

16. Understanding User Accounts and Roles

A user account allows someone to sign in to Optimus Orien.

The account is separate from the real-world person or organisation represented in the platform.

For example, a teacher has a staff record and a user account. The staff record describes the teacher's relationship with the school. The user account authenticates the teacher and provides access to authorised features.

Similarly, a principal has a staff record, a user account, and a principal role within a defined school or campus.

The system must not assume that creating an account automatically gives the user unrestricted access to school information.

Access must depend on assigned roles, the scope of those roles, and the permissions associated with them.

A single person might have more than one role.

For example, a person may be a teacher at one school and an administrator at another.

The platform should be able to represent multiple authorised assignments without confusing the access rights associated with each one.

17. Understanding Role-Based Access

Role-based access control means that permissions are granted according to the user's role and the scope of their responsibility.

In Optimus Orien, different roles should have different default access.

An organisation administrator may need to manage several schools.

A principal may need to manage the school or campus assigned to them.

A teacher may need to view and enter results for assigned students and subjects.

A student may need access to their own permitted academic information.

A parent may need access to the records of their authorised children.

These roles must not be treated as interchangeable.

For example, a teacher should not automatically be able to edit another teacher's assessments. A parent should not be able to access every student in their child's class. A principal should not automatically be able to access every school operated by the organisation.

Permissions should specify both the action and the scope.

An action might be viewing results, entering marks, editing assessments, approving a skill rating, managing a teacher, or exporting a report.

The scope might be a particular student, class, subject, campus, school, or organisation.

The system must check both when deciding whether to allow an operation.

18. Understanding Data Isolation

Data isolation means ensuring that a user or organisation cannot access records outside their authorised scope.

This is especially important when Optimus Orien serves multiple schools.

Suppose School A and School B both use the platform.

A teacher at School A should not be able to access School B's student records simply by changing a URL, modifying a request, or guessing a record identifier.

The software must enforce access restrictions on the server side. Hiding a button in the interface is not sufficient protection.

Every request involving protected records must verify that the user is authenticated and authorised to access those records.

The same principle applies to reports, exports, charts, search results, and summary calculations.

For example, if a principal requests the average mathematics result for their school, the system must ensure that the calculation includes only the intended school's eligible results.

It must not include results from another school because the underlying query omitted a required scope restriction.

The developer must design data access rules consistently rather than implementing separate, incomplete checks in each page.

19. How the Main Entities Connect

The platform needs a consistent way to connect its records.

An organisation manages or operates schools.

A school has one or more campuses or operational locations.

A school has academic years.

An academic year contains reporting periods and academic arrangements applicable to that year.

A year group belongs to the relevant educational structure.

A class or student group is associated with a school, year group, and academic period.

A student has a stable identity and may have multiple enrolments over time.

An enrolment associates the student with a school or programme for a defined period.

A student may belong to a general class and may also participate in several subject-specific teaching groups.

A subject offering connects a subject, a teaching group, the relevant teacher assignments, and the applicable academic period.

Assessments belong to the appropriate subject offering or another explicitly defined assessment context.

Assessment results connect individual students to the assessments they completed.

Parents and guardians connect to students through authorised relationships.

Staff members connect to schools and campuses through employment or assignment records.

User accounts connect to the relevant people or organisations and receive explicitly defined roles and permissions.

This structure allows the system to answer questions such as:

How many students are enrolled at a particular campus?

Which classes does a particular teacher teach this academic year?

Which mathematics assessments belong to Class 7A?

What was a student's mathematics result in the previous academic year?

Which parent or guardian is authorised to view a particular student's information?

What is the average mathematics performance across all eligible students at a school?

Which schools belong to a particular organisation?

These questions should be answered from the connected data rather than by maintaining separate, manually synchronised copies of the same information.

20. Why Historical Records Must Be Preserved

School structures change over time.

A student moves to another class. A teacher changes subjects. A principal leaves. A school introduces a new year group. A campus changes its name. An organisation acquires another school.

The platform must preserve enough history to understand what was true at the time a result was recorded.

For example, suppose Sara studied in Class 7A during 2025–2026 and received 85% in mathematics.

In 2026–2027, she moves to Class 8B.

When the principal views Sara's mathematics result from 2025–2026, the system must retain the historical context showing that she was in Class 7A during that period.

The result must not be moved into Class 8B merely because that is her current class.

Similarly, a teacher who changes schools should not have their historical teaching assignments rewritten as though they always belonged to the new school.

The developer should use dated assignments and enrolment records where historical changes matter.

Records should be deactivated or closed when appropriate rather than indiscriminately deleted.

The exact retention and deletion policies must comply with applicable law and the organisation's requirements, particularly for personal and student information.

21. Handling Students Who Change Schools or Campuses

A student may transfer from one campus to another while remaining within the same organisation.

A student may also leave one school and join another.

These events must be represented explicitly.

For example, Sara may transfer from the North Campus to the Central Campus in January 2027.

The system should preserve her earlier enrolment and class membership, close them when appropriate, and establish the new enrolment and class membership.

The system must not simply replace the campus field on a student record and thereby erase the history of where Sara studied.

Historical academic results should remain associated with the original school, campus, subject, academic year, and assessment context.

Whether those historical results can be viewed by the receiving campus depends on the organisation's access rules, the purpose of access, and applicable privacy requirements.

A transfer should not automatically grant every employee at the receiving campus unrestricted access to all of the student's historical information.

22. Handling Teachers Who Teach at Multiple Locations

Some teachers work across more than one campus.

For example, a science teacher might teach at the North Campus on Monday and the Central Campus on Wednesday.

The system must support this arrangement.

The teacher should have one staff identity where the organisation can reliably establish that it is the same person, with separate assignments for the relevant schools, campuses, subjects, and periods.

The teacher's access must reflect the assignments that have been authorised.

The teacher should not automatically gain access to every student at every campus simply because the teacher works at more than one location.

This structure also makes it easier to understand historical teaching responsibilities when the teacher's assignments change.

23. Handling Multiple Educational Programmes Within One School

A school may teach more than one educational programme.

For example, it may offer a national curriculum to some students and an international programme to others.

The students may share the same campus and some facilities but follow different curricula, assessment methods, and grading rules.

Optimus Orien must not assume that all students at one campus follow the same educational structure.

The platform should allow relevant classes, subjects, assessments, and student enrolments to be associated with the appropriate educational programme.

The applicable grading and calculation configuration must be selected according to the school's defined rules.

For example, the mathematics result of a student following one programme should not automatically be compared with that of a student following another programme merely because both are labelled Grade 8.

A comparison may still be useful, but the platform must explain the comparison basis and must not imply equivalence without justification.

24. Understanding the Difference Between Organisation-Wide Rules and School-Specific Rules

Some settings may be managed centrally by an organisation.

For example, an organisation might define its preferred report format, shared naming conventions, or default user roles.

Other settings may need to be managed separately by each school.

For example, two schools in the same group might follow different curricula, use different academic calendars, or operate different grading systems.

Optimus Orien should support central defaults without forcing every school to use identical academic rules.

A central administrator may be allowed to establish a default configuration. An authorised school administrator may then be permitted to override particular settings where the organisation allows it.

The system must make it clear which settings are centrally controlled and which can be changed at school level.

A change to an organisation-wide default should not silently overwrite a school's existing configuration unless the change has been explicitly approved and the intended effect is clear.

The platform should record significant configuration changes and preserve historical context where required.

25. What the Developer Should Build First

Before building complex dashboards, the development team should establish the underlying data relationships.

The first step is to define stable records for organisations, schools, campuses, academic years, year groups, classes, subjects, staff members, students, user accounts, and parent or guardian relationships.

The next step is to define the dated relationships connecting these records.

For example, a student should be connected to a school through an enrolment record, and their class assignment should apply to a defined academic period.

A teacher should be connected to a subject offering through an authorised teaching assignment.

The platform should then define the relationships between assessments, student results, and the relevant academic structures.

Access control must be designed alongside these relationships, not added as an afterthought.

The developer should establish the rules for who can read, create, update, approve, and export each type of record.

The next priority is to create validation and automated tests that prevent one school's data from appearing in another school's portal without authorisation.

Only after these foundations are stable should the team build the full organisation map, principal dashboard, teacher portal, parent portal, student portal, and organisation-level performance reports.

This order reduces the risk of building attractive interfaces on top of an inconsistent data model.

26. Developer Requirements

The following requirements convert this document into implementation expectations.

ORG-001: Organisation Records

The system must support an organisation that manages one or more schools.

Each organisation must have a stable internal identifier and an appropriate administrative record.

Acceptance test: An authorised administrator creates an organisation and associates two schools with it. Both schools appear under the correct organisation.

ORG-002: School Records

The system must represent each school as a distinct entity with a stable internal identifier.

Acceptance test: An authorised administrator renames a school. Its historical results and existing relationships remain associated with the same school record.

ORG-003: Campus Records

The system must support schools or organisations with one or more campuses or operational locations.

Acceptance test: An organisation has three campuses. An authorised administrator can view all three and open each campus's authorised information separately.

ORG-004: Configurable Terminology

The system should allow appropriate interface terminology, such as campus or branch, to be configured without changing the underlying identity of existing records.

Acceptance test: An organisation changes the displayed term from "Campus" to "Branch." Existing campus records and relationships continue to work.

ORG-005: School and Campus Relationships

The system must enforce the configured relationship between schools and campuses.

Acceptance test: An administrator cannot associate a campus with an invalid or unauthorised school relationship. A single-campus school can operate without unnecessary additional school records.

ORG-006: Academic Year Records

The system must support academic years with configurable start dates, end dates, and reporting periods.

Acceptance test: An administrator creates an academic year running from August 2026 to June 2027 and associates its assessments with the correct period.

ORG-007: Historical Academic Context

The system must preserve the academic year and relevant class, subject, and assessment context associated with historical results.

Acceptance test: A student moves from Class 7A to Class 8B. The student's previous year's results remain associated with Class 7A and the original academic year.

ORG-008: Configurable Year Groups and Stages

The system must support configurable year-group labels and optional stage or phase groupings.

Acceptance test: One school uses "Grade 7" while another uses "Year 7." Both can use the platform without the system assuming that the two groups are academically equivalent.

ORG-009: Class Records

The system must support classes or student groups associated with the relevant school, academic year, and year group.

Acceptance test: An administrator creates Class 7A for a particular academic year. The class can be associated with eligible student enrolments and teaching assignments.

ORG-010: Historical Class Membership

The system must preserve student class membership over time.

Acceptance test: A student belongs to Class 7A in one academic year and Class 8B in the next. Both memberships remain available for historical reporting.

ORG-011: Subject Records

The system must support configurable subjects with stable internal identifiers.

Acceptance test: An administrator creates mathematics and English as subjects and associates them with appropriate year groups or educational programmes.

ORG-012: Subject Teaching Groups

The system must support teaching groups or subject offerings that connect a subject to a defined group of students and an academic period.

Acceptance test: Students in Class 7A are divided into two mathematics groups but remain part of the same general class. The system can represent both arrangements.

ORG-013: Staff Records

The system must maintain staff records separately from user accounts.

Acceptance test: An authorised administrator creates a teacher's staff record and associates an authorised user account with it. Updating the account's sign-in details does not create a duplicate teacher record.

ORG-014: Multiple Teaching Assignments

The system must allow a teacher to have multiple authorised assignments across subjects, classes, campuses, or academic periods.

Acceptance test: A teacher teaches mathematics to Class 7A and science to Class 8B. The system records both assignments and grants access according to the relevant permissions.

ORG-015: Student Records

The system must maintain a stable student identity independent of the student's current class or campus.

Acceptance test: A student's class changes and their name is updated. The student's historical assessment records remain connected to the correct student identity.

ORG-016: Enrolment Records

The system must maintain student enrolments separately from the permanent student identity record.

Acceptance test: A student leaves a school and later joins another school. The system can preserve the earlier enrolment and establish the new enrolment without automatically creating a duplicate student identity.

ORG-017: Student Transfers

The system must support transfers between classes, campuses, and schools through explicit, authorised changes to enrolment and assignment records.

Acceptance test: A student transfers from one campus to another. The system preserves the earlier campus association and applies the new association from the appropriate effective date.

ORG-018: Parent and Guardian Relationships

The system must support multiple authorised parents or guardians for a student and multiple student relationships for a parent or guardian.

Acceptance test: One parent has two children at the school. The parent can access both children's permitted records through the same authorised account without gaining access to unrelated students.

ORG-019: Role-Based Access

The system must enforce role-based permissions for actions such as viewing, creating, editing, approving, and exporting records.

Acceptance test: A teacher can enter marks for assigned students but cannot edit unrelated students' results. A principal can access records within the principal's authorised scope.

ORG-020: Scope-Based Access

Permissions must specify the scope within which a user may perform an action.

Acceptance test: A principal assigned to the North Campus cannot access the Central Campus's private student records unless an additional authorised permission grants that access.

ORG-021: Server-Side Authorisation

The system must enforce access restrictions on the server side for every protected operation.

Acceptance test: A user who is not authorised to access a student record cannot retrieve it by manually changing a URL, identifier, or request parameter.

ORG-022: Data Isolation Between Organisations

The system must prevent unauthorised access to information belonging to another organisation.

Acceptance test: A teacher belonging to Organisation A attempts to retrieve Organisation B's student results. The request is denied, and no protected data is returned.

ORG-023: Correctly Scoped Reports

All reports, dashboards, searches, exports, and calculations must apply the appropriate organisation, school, campus, class, and academic-period filters.

Acceptance test: A principal requests the average mathematics result for their school. The calculation excludes unauthorised records from other schools.

ORG-024: Historical Staff Assignments

The system must preserve relevant historical staff assignments and their effective periods.

Acceptance test: A teacher changes campuses. Historical teaching assignments remain available for interpreting previous assessments and auditing responsibilities.

ORG-025: Multiple Educational Programmes

The system must support multiple educational programmes within the same school or campus.

Acceptance test: A school offers two programmes with different grading configurations. The system associates students and assessments with the appropriate programme and does not automatically merge the grading rules.

ORG-026: Organisation Defaults and School Overrides

The system must support centrally managed defaults and authorised school-level configuration where permitted.

Acceptance test: An organisation defines a default report format while allowing one school to use a different authorised academic configuration. Changing the default does not silently overwrite the school's existing academic rules.

ORG-027: Record Lifecycle Management

The system must support appropriate status changes and historical preservation for schools, campuses, staff, students, enrolments, and assignments.

Acceptance test: An enrolment is closed when a student leaves a school. Historical results remain available to authorised users, and the system does not treat the closed enrolment as a current one.

ORG-028: Audit Trail

The system must record significant administrative changes to school structures, enrolments, staff assignments, access permissions, and academic configurations.

Acceptance test: An authorised administrator changes a student's campus assignment. The system records the change, the responsible user, and the applicable date or timestamp.

ORG-029: Minimum Necessary Personal Data

The system should collect and expose only the personal information necessary for its defined functions and authorised purposes.

Acceptance test: A teacher's portal displays the information needed for the teacher's authorised work but does not expose unrelated personal or administrative records.

ORG-030: Consistent Entity Relationships

All modules must use the same underlying records and relationship rules for organisations, schools, campuses, academic periods, students, teachers, classes, subjects, and enrolments.

Acceptance test: A student shown in the principal's dashboard is the same underlying student record used by the teacher portal and the authorised parent portal. The application does not create separate student identities for each portal.

27. Final Principles

Optimus Orien must model the real structure of an educational organisation.

An organisation may manage multiple schools. A school may operate multiple campuses. Students have stable identities and dated enrolments. Teachers have dated assignments. Classes, subjects, and teaching groups represent different educational relationships. Parents and guardians connect to students through authorised relationships. User accounts provide access, but an account does not automatically grant unrestricted access to school information.

These distinctions allow the platform to serve independent schools, school groups, and larger educational organisations.

The software must preserve historical information so that past results remain connected to the school, class, subject, academic year, and teaching arrangements that applied when those results were recorded.

It must also enforce permissions consistently, ensuring that users see only the records they are authorised to access.

The underlying structure should remain consistent even when the names, educational models, academic calendars, and organisational arrangements differ.

28. What This Part Establishes

This document establishes the organisational and relational foundation of Optimus Orien.

It explains how organisations, schools, campuses, academic years, year groups, classes, subjects, teachers, students, parents, enrolments, user accounts, and permissions connect to one another.

It also establishes that historical assignments must be preserved and that school data must be separated according to authorised access.

Part 5 is particularly important because it defines **who can access Optimus Orien, what each person can see, what they can change, and how the system protects student and school information**.

This document connects the educational structure from Parts 1–4 to the actual software. Your developer should be able to use it to build the portals, navigation, permissions, and security rules without having to guess what each role means.

Part 5:

User Roles, Portals, Permissions, and Access Control

1. Purpose of This Document

Optimus Orien will be used by different people who have different responsibilities within an educational organisation.

An organisation administrator needs to understand performance across the schools they manage. A principal needs to understand the performance of their school. A teacher needs to record and review information about assigned students. A student needs to understand their own progress. A parent needs to understand their child's performance.

These users should not see the same information or have the same permissions.

For example, a teacher should be able to enter marks for an assigned class but should not automatically be able to change the results of students at every school in the organisation.

A parent should be able to see their child's authorised academic information but should not be able to open another parent's child's private records.

A principal should be able to manage their school but should not automatically gain access to every school operated by the organisation.

This document explains the difference between a user, an account, a role, a permission, and an access scope. It then defines the intended portals and establishes the rules the developer must implement.

The goal is to make Optimus Orien secure, understandable, and suitable for organisations that operate one or many schools.

2. Understanding a User

A user is a person or authorised organisational representative who interacts with Optimus Orien.

Examples include a principal, teacher, student, parent, and organisation administrator.

However, the word "user" does not automatically tell us what the person is allowed to do.

Two users might both work at the same school, but one may be a principal and the other may be a teacher. Their responsibilities and permissions differ.

Similarly, two teachers at the same school might teach different classes and subjects. They should not necessarily have identical access to student records.

The system must identify the person and determine their authorised responsibilities before allowing access to protected information.

3. Understanding a User Account

A user account allows a person to sign in to Optimus Orien.

The account is responsible for authentication. Authentication is the process of verifying that someone is the person they claim to be when they attempt to sign in.

For example, a teacher may sign in using an email address and password or another supported authentication method.

After the person signs in, the system must determine what the person is authorised to access.

Creating an account must not automatically grant access to all platform data.

The account should be connected to the appropriate person or organisational record.

For example, a teacher's account should be associated with the teacher's staff record. A parent's account should be associated with the parent's verified relationship to their child.

This distinction allows the platform to manage sign-in credentials separately from educational records.

It also makes it possible to disable an account without deleting the underlying student, parent, or staff record.

4. Understanding a Role

A role describes a user's responsibilities within the platform.

Examples include Organisation Administrator, School Administrator, Principal, Teacher, Student, and Parent or Guardian.

A role determines the general category of work a person is permitted to perform.

For example, a teacher role may permit the user to view assigned classes and enter assessment results. A principal role may permit the user to review school performance and manage certain school records.

However, a role alone is not sufficient to determine every permission.

A teacher may be assigned to Class 7A but not Class 8B. A principal may manage the North Campus but not the Central Campus.

The system must therefore consider both the user's role and the scope of their assignment.

5. Understanding a Permission

A permission is an explicit authorisation to perform a particular action.

Examples include viewing a student's academic results, creating an assessment, entering marks, editing a class, approving a skill rating, managing staff assignments, or exporting a report.

It is useful to think of a permission as answering a specific question:

What action is this user allowed to perform?

For example, a teacher might have permission to view student results and enter marks but not permission to change the school's grading configuration.

A principal might have permission to view all student results in their school and manage teacher assignments but not permission to change organisation-wide security settings.

The software should not rely on vague rules such as "teachers can access academic data."

Instead, permissions should define the relevant actions precisely.

6. Understanding Access Scope

Access scope defines the records or organisational area to which a permission applies.

For example, a permission to view student results might apply to one student, one class, one campus, one school, or an entire organisation.

Consider two principals.

The first principal manages North Campus. The second manages South Campus.

Both have a principal role, but their access scopes differ.

The first principal should be able to view authorised records belonging to North Campus. The second should be able to view authorised records belonging to South Campus.

Neither should automatically be able to access the other's private school records.

For Optimus Orien, every protected action must be checked against both the user's permission and the relevant access scope.

This principle applies to pages, search results, dashboards, reports, charts, exports, and direct requests to the server.

7. Understanding the Difference Between Hiding a Button and Enforcing Permission

Suppose a teacher is not permitted to edit a particular student's marks.

The interface should not display an editing button to that teacher.

However, hiding the button is not enough.

A technically knowledgeable user might attempt to send a direct request to the server to change the marks.

If the server accepts that request without checking permission, the student record is still vulnerable.

The developer must therefore enforce permissions on the server side.

The interface should help users understand what they can do, but the server must make the final decision about whether an action is authorised.

Every operation involving protected information must verify the user's identity, permission, and access scope.

8. The Main User Roles in Optimus Orien

The initial platform should support six principal user categories.

The first is the Organisation Administrator, who manages the organisation and its authorised schools.

The second is the School Administrator, who manages administrative records within an assigned school or campus. This role is useful for schools that delegate administrative work to staff members who are not principals.

The third is the Principal, who oversees the academic and operational performance of the school or campus.

The fourth is the Teacher, who teaches students and records or reviews authorised academic information.

The fifth is the Student, who accesses their own permitted educational information.

The sixth is the Parent or Guardian, who accesses the information they are authorised to receive about their child or children.

These are logical roles rather than necessarily six separate applications. Optimus Orien should use one consistent platform with different interfaces and permissions for different users.

A future version may introduce additional roles, such as government education officers, auditors, or other authorised oversight personnel. These roles should not be granted access until their responsibilities, data scope, and permissions have been explicitly defined.

9. Organisation Administrator Portal

The Organisation Administrator manages the schools and organisational information within their authorised scope.

For example, an organisation might operate five schools across two cities. The organisation administrator needs to understand the performance of the group and examine the individual schools.

The portal should provide an organisation-level dashboard showing relevant school performance indicators, attendance summaries, student and teacher counts, and configured school health indicators.

The administrator should be able to select a school or campus to examine more detailed information.

The portal should also provide access to organisation management functions, such as creating school records, assigning school administrators, managing campus records, and configuring permitted organisation-wide settings.

The organisation administrator should be able to compare schools using clearly defined measures.

For example, the administrator might compare attendance rates or mathematics performance across three schools.

However, the administrator must understand what is being compared. If the schools use different assessment methods or grading systems, the platform should not present their results as directly equivalent without an appropriate comparison basis.

The portal should distinguish organisation-level summaries from school-level details.

It should also identify missing data and incomplete reporting where these affect the reliability of a comparison.

9.1 Organisation Administrator Permissions

The organisation administrator may be permitted to create and manage schools, campuses, and school-level administrator assignments within the organisation.

The administrator may view the organisation's authorised student, teacher, attendance, and academic performance summaries.

Access to individual student records should be explicitly defined by the organisation's policy and applicable privacy requirements rather than assumed to be unrestricted merely because the user is an organisation administrator.

The administrator may manage permitted organisation-wide settings but should not automatically be able to change every historical assessment result.

Changes to grading rules, permissions, and other important settings should be subject to appropriate controls and audit records.

9.2 Organisation Administrator Restrictions

An organisation administrator must not automatically gain access to data belonging to an unrelated organisation.

The administrator must not be able to bypass record protections by manually changing a school identifier in a request.

The administrator must not silently alter historical results or overwrite school-specific academic settings without an authorised process.

10. School Administrator Portal

A School Administrator helps manage the administrative structure of an assigned school or campus.

This role is distinct from the principal role.

For example, a school might employ an administrator responsible for maintaining student enrolments, updating class lists, managing staff assignments, and helping prepare reports.

The administrator may not be responsible for the school's academic leadership.

The portal should provide the functions required for the administrator's assigned responsibilities.

These may include managing student records, updating class membership, maintaining staff assignments, supporting academic-year setup, and managing certain school settings.

The available functions should be determined by explicit permissions.

A school administrator should not automatically be allowed to approve academic results, change grading rules, or view sensitive information outside their responsibilities.

An organisation may choose not to use this role at all. In that case, the principal or another authorised user may perform the necessary administrative tasks.

The system must not force every school to create a separate School Administrator account.

11. Principal Portal

The principal is responsible for overseeing the performance and operation of the assigned school or campus.

The principal portal should focus on understanding what is happening within that school.

It should provide access to relevant information about students, teachers, classes, subjects, attendance, academic results, skills, and performance patterns.

The principal should be able to view class averages, subject averages, and school-level academic summaries.

For example, the principal might notice that mathematics performance is declining across several Year 8 classes.

The principal should be able to open the relevant subject or class report, examine the underlying results, and identify where additional review may be appropriate.

The portal should also support the identification of students whose recorded results or other configured indicators suggest that they may need additional attention.

The platform should describe these students using neutral language such as "Student Requiring Review" or "Student at Risk" where the school has defined and validated the relevant criteria.

A flag must not be treated as a diagnosis or a definitive explanation of why a student is struggling.

The principal should be able to view relevant information, examine the evidence behind a flag, and take appropriate administrative action within their authority.

11.1 Principal Permissions

The principal should be able to view authorised student, class, subject, teacher, attendance, and academic performance information within the assigned school or campus.

The principal may be permitted to manage class assignments, review staff assignments, approve particular records, and oversee school reporting.

The principal's permissions should be configurable according to the organisation's governance model.

For example, one organisation may permit principals to create classes. Another may require the central administrator to create classes.

These differences should be handled through permissions and settings rather than separate versions of the software.

11.2 Principal Restrictions

A principal assigned to one school must not automatically be able to access another school's private records.

The principal should not be able to change historical assessment results without an authorised procedure.

The principal should not be able to change organisation-wide permissions unless explicitly granted that authority.

12. Teacher Portal

The teacher portal is designed for teachers who need to manage their assigned classes, students, subjects, and assessments.

A teacher should be able to view the students they are authorised to teach.

The teacher should be able to create assessments where permitted, enter marks, view relevant student results, and record skill assessments where the school uses them.

The portal should show the teacher the information needed to complete these tasks without overwhelming the interface with unrelated organisational information.

For example, a mathematics teacher assigned to Class 7A should be able to open Class 7A, select mathematics, create a quiz, enter student marks, and view the resulting performance summary.

The teacher should also be able to identify students whose recorded results show a decline or a repeated difficulty in the relevant subject.

The platform should make it easy to review the evidence behind a performance indicator rather than displaying an unexplained warning.

12.1 Teacher Permissions

A teacher may be permitted to view assigned classes and student records, create assessments, enter marks, correct their own entries under the applicable rules, and record relevant skill assessments.

The teacher may be able to view class averages for their assigned classes and subjects.

The teacher should not automatically be able to change the school's academic-year structure, grading configuration, or organisation settings.

12.2 Teacher Restrictions

A teacher must not automatically have access to every student in the school.

A teacher must not be able to edit another teacher's assessments merely because both teachers work at the same school.

If a teacher is assigned to multiple classes or campuses, access must follow the relevant assignments.

The system must also distinguish between entering a result and approving a result if the school uses an approval process.

13. Student Portal

The student portal should help students understand their own academic performance.

A student should be able to view permitted results for their subjects, assessments, skills, and reporting periods.

The student may also view attendance and homework information where the school has enabled those features and authorised the student to access them.

The interface should make results understandable.

For example, a student viewing a mathematics result should be able to identify the assessment, marks obtained, maximum marks, percentage, and grade where applicable.

The student should also be able to view progress across reporting periods when the underlying results can be meaningfully compared.

The platform should explain relevant indicators in simple language.

For example, if a student's mathematics result falls from 78% to 65%, the portal might indicate that the recorded result decreased by 13 percentage points.

It should not automatically state that the student has become less intelligent or that a particular cause explains the change.

The student portal should encourage understanding of performance rather than create unnecessary shame or pressure.

13.1 Student Permissions

A student should be able to view their own permitted educational information.

If the school allows it, the student may also view assignments or homework statuses relevant to them.

13.2 Student Restrictions

A student must not be able to view another student's private academic results.

The student must not be able to change marks, grades, attendance records, teacher assessments, or official skill ratings unless a separate, explicitly defined function grants an appropriate permission.

A student should not be able to alter a record by modifying a request or using another student's identifier.

14. Parent or Guardian Portal

The parent or guardian portal helps authorised adults understand the educational performance of their child or children.

A parent may have more than one child enrolled in the same school or across different schools that use Optimus Orien.

The portal should allow the parent to select the relevant child and view the information authorised for that relationship.

This may include academic results, subject performance, progress across reporting periods, attendance summaries, homework status, and relevant skill assessments.

The platform should present this information in understandable language.

For example, a parent might see that their child is performing well in science but has declining results in English.

The platform may identify the pattern and show the relevant results. It should not automatically claim that the child has a particular learning disorder or that a parent or teacher caused the decline.

The portal should help parents understand what the recorded information shows and where further discussion with the school may be useful.

14.1 Parent Permissions

A parent or guardian may view the records of students for whom the school has authorised the relationship.

The exact information available should follow the school's policy and applicable privacy requirements.

14.2 Parent Restrictions

A parent must not be able to access another student's private records merely because that student is in the same class as their child.

A shared surname, email address, or home address must not be treated as sufficient proof of the parent-child relationship.

The platform must use an authorised relationship established through the school's process.

15. Future Government or Oversight Portal

A future version of Optimus Orien may include a portal for authorised government education officers or other oversight bodies.

Such a portal might support analysis across districts, regions, or schools.

However, this role requires a separate definition of responsibilities, legal authority, and data access.

A government officer should not automatically receive unrestricted access to every student's personal information simply because they represent a public authority.

The system should distinguish between aggregate reporting and access to individual records.

Aggregate reporting summarises groups of records without necessarily revealing the identity of individual students.

For example, a report might show average attendance across a district or the proportion of students whose mathematics performance falls below a configured threshold.

Individual-level information should be available only where there is a defined purpose, appropriate authorisation, and a lawful basis for access.

This portal should be treated as a future capability until its requirements and governance rules have been approved.

16. Understanding the Public Homepage and Leaderboards

Optimus Orien may provide a public homepage displaying selected school performance information.

A public homepage is different from an authenticated portal.

Authenticated portals require users to sign in and provide access to protected information according to their permissions.

Public pages may be accessible without an account, so they must contain only information explicitly approved for public disclosure.

For example, the platform might display a school leaderboard or a publicly approved school health indicator.

However, a school leaderboard must not expose student names, individual marks, private attendance information, or other protected student records.

The platform must distinguish between public school-level information and private student-level information.

Before publishing a school ranking, the platform must define the calculation method, reporting period, eligibility rules, and meaning of the displayed score.

It should also identify missing or incomplete data when this affects the interpretation of the ranking.

A school with incomplete data should not automatically be treated as having poor performance simply because the system has fewer recorded results.

The public homepage must never provide a route around the authenticated access controls used by the private portals.

17. Understanding Multi-Role Users

One person may have more than one role.

For example, a principal may also teach a class. A school administrator may also be a parent of a student enrolled at the school. A teacher may have children attending another school in the same organisation.

The platform should support these situations without creating unnecessary duplicate identities.

The person's account should be connected to the appropriate records and authorised roles.

When the person uses the platform, the system must determine which role and scope apply to the requested action.

For example, a principal who also teaches mathematics may need access to principal-level school reporting and teacher-level assessment functions.

The system should make the active context clear.

A principal performing a teacher task should not accidentally create a record under the wrong school or class.

A parent who is also a teacher must not receive access to a student's private record through their teacher role unless that access is independently authorised.

The developer must ensure that permissions are checked for the specific action and relevant scope, not merely by checking whether the account has any role associated with the school.

18. Understanding Invitations and Account Activation

Schools need a controlled way to provide access to their staff, students, and parents.

The system should not allow anyone to create an account and automatically become a principal, teacher, or parent with access to school records.

An authorised administrator should be able to initiate the appropriate account creation or invitation process.

For example, a principal might invite a teacher to activate an account associated with an existing staff record.

A school might initiate a parent invitation after verifying the parent's relationship to a student.

The exact activation method may involve email, a secure invitation link, a verified contact method, or another supported process.

Invitation links should be protected against unauthorised reuse and should expire according to a defined policy.

An invitation should establish only the role and scope for which it was created.

For example, a teacher invitation for North Campus must not allow the recipient to become an organisation administrator.

If an invitation is revoked, expires, or is accepted by the wrong person, the system should handle the situation without exposing protected records.

19. Understanding Account Suspension and Role Changes

People change jobs, students leave schools, and parents' authorised relationships may change.

The platform must be able to update access without deleting important historical records.

If a teacher leaves a school, an authorised administrator should be able to revoke or end the teacher's access.

The teacher's historical assessments should remain available to authorised users when required.

If a student leaves a school, the student's account and enrolment should be handled according to the school's policies. Closing an enrolment should not automatically erase the student's historical academic records.

If a principal moves to another campus, their previous permissions should end or change according to the effective date of the new assignment.

The system must not continue granting access to a former school merely because the user's account still exists.

Role changes and access revocations should take effect reliably.

The platform should record significant changes so that administrators can investigate who had access and when, to the extent supported by the system's audit records.

20. Understanding Sensitive Actions and Approval

Not every action should be treated as equally important.

Viewing a class average is different from changing a student's final examination result.

Creating a draft assessment is different from publishing an official result.

Optimus Orien should distinguish between ordinary data entry and actions that can significantly affect official records.

Depending on the school's policy, sensitive actions may require additional permission, approval, or confirmation.

For example, a school might require a teacher to submit final marks for review before the marks become official.

Another school might allow teachers to publish their own results but require a record of subsequent changes.

The platform should support the chosen workflow rather than assuming that every school has the same approval process.

Where an approval process exists, the system should clearly distinguish between draft, submitted, approved, and published states as appropriate.

It should record who performed the action and when.

An unauthorised user must not be able to bypass the approval process by directly modifying the underlying record.

21. Understanding the Audit Trail

An audit trail is a record of significant actions and changes performed in the system.

It helps an organisation understand what happened, who performed an action, and when it occurred.

For example, if a student's mathematics result changes from 65 to 75, an authorised administrator may need to know who changed it and why.

The audit record should identify the relevant record, the responsible user, the action performed, the time of the action, and other appropriate information needed to understand the change.

Where appropriate, it may record the previous and updated values.

The audit trail must be protected from ordinary users who might otherwise modify or erase evidence of their own actions.

Not every event requires the same level of detail. The developer should define an appropriate audit policy for sensitive records and administrative actions.

Audit records themselves may contain sensitive information and must also be protected by access controls and retention rules.

22. Understanding Security for Student Information

Optimus Orien will handle information about children and their education.

The platform must therefore be designed to protect student information from unauthorised access, accidental disclosure, and inappropriate use.

Security should be built into the system rather than treated as a final feature.

At a minimum, the developer must implement server-side authorisation, secure authentication, appropriate session handling, protected password storage where passwords are used, secure communication, and validation of inputs.

The system should collect only the information necessary for its intended functions.

Administrative permissions should follow the principle of least privilege. This means users receive only the access they need to perform their authorised responsibilities.

Sensitive information should not appear in public pages, unprotected exports, error messages, or application logs without a justified and controlled purpose.

The developer should also establish secure backup, recovery, and incident-handling procedures.

Because Optimus Orien may operate internationally, the organisation must assess the privacy and education-data requirements that apply in each jurisdiction where the service is offered.

This document establishes software design principles; it does not replace a jurisdiction-specific legal review.

23. Recommended Portal Navigation

The interface should present each user with navigation suited to their role.

The organisation administrator should have access to organisation overview, schools and campuses, performance reports, user and role management, and permitted organisation settings.

The school administrator should have access to the assigned school's records and the administrative tasks they are authorised to perform.

The principal should have access to the school overview, students, teachers, classes, subjects, attendance, academic performance, skills, reporting periods, and relevant reports.

The teacher should have access to assigned classes, students, subjects, assessments, marks, skill assessments, and permitted performance summaries.

The student should have access to their own dashboard, subject results, assessments, progress, and other enabled information relevant to them.

The parent or guardian should have access to their authorised children, academic summaries, subject results, progress, attendance, and other information enabled by the school.

These are functional navigation requirements rather than a mandate to create completely separate applications.

The developer should build reusable interface components and apply role-aware navigation consistently.

Pages that are not relevant to a user's role should normally not appear in that user's navigation.

However, hiding a page does not replace the need for server-side permission checks.

24. Developer Requirements

The following requirements translate this document into implementation expectations.

AUTH-001: Separate User Accounts from Educational Records

The system must maintain user accounts separately from the relevant student, parent, staff, and organisational records.

Acceptance test: A teacher's account is disabled. The teacher's historical assessment records remain associated with the correct staff identity.

AUTH-002: Configurable Roles

The system must support the defined initial roles: Organisation Administrator, School Administrator, Principal, Teacher, Student, and Parent or Guardian.

The role model should permit future authorised roles without requiring the entire permission system to be redesigned.

Acceptance test: An administrator assigns a teacher role to a user. The system grants only the permissions associated with that role and its assigned scope.

AUTH-003: Explicit Permissions

The system must define permissions for protected actions rather than relying solely on role names.

Acceptance test: A teacher has permission to view and enter marks but does not have permission to change organisation-wide grading settings.

AUTH-004: Scope-Based Permissions

The system must apply permissions to the relevant organisation, school, campus, class, subject, student, or other defined scope.

Acceptance test: A principal assigned to North Campus cannot retrieve private student records from South Campus unless an additional authorised permission grants that access.

AUTH-005: Server-Side Access Enforcement

Every protected operation must verify the user's authentication, permission, and access scope on the server side.

Acceptance test: A user manually changes a request to reference an unauthorised student's identifier. The server denies access and does not return the protected record.

AUTH-006: Organisation Administrator Portal

The system must provide organisation-level views and administrative functions within the organisation administrator's authorised scope.

Acceptance test: An organisation administrator can view the schools they are authorised to manage and examine their permitted performance summaries without accessing unrelated organisations.

AUTH-007: School Administrator Portal

The system must support a school administrator role with configurable permissions within the assigned school or campus.

Acceptance test: A school administrator can update permitted student enrolment information but cannot approve official marks unless that permission has been granted.

AUTH-008: Principal Portal

The system must provide school-level performance summaries and authorised management functions for the principal's assigned school or campus.

Acceptance test: A principal can view class averages and student performance within the assigned school but cannot access another school's private records without explicit authorisation.

AUTH-009: Teacher Portal

The system must provide teachers with access to their authorised classes, students, subjects, assessments, and results.

Acceptance test: A teacher can create an assessment for an assigned class but cannot create or modify assessments for an unrelated class without permission.

AUTH-010: Student Portal

The system must allow students to view their own authorised academic information.

Acceptance test: A student can view their mathematics results but cannot retrieve another student's results by changing a record identifier.

AUTH-011: Parent or Guardian Portal

The system must provide parents or guardians with access to the records of students linked to them through an authorised relationship.

Acceptance test: A parent with two authorised children can switch between their children's permitted views but cannot access a third student's records.

AUTH-012: Public and Private Data Separation

The system must distinguish explicitly approved public information from private authenticated information.

Acceptance test: A public school leaderboard displays only approved school-level indicators and never exposes individual student marks or private attendance records.

AUTH-013: Multi-Role Accounts

The system must support one person having multiple authorised roles or assignments.

Acceptance test: A person who is both a teacher and a parent can access teacher functions and their own children's permitted information without those roles granting unintended access to unrelated records.

AUTH-014: Account Invitations

The system must support controlled account invitation and activation processes for authorised users.

Acceptance test: An authorised administrator invites a teacher for one school. Accepting the invitation does not grant the recipient access to other schools or an organisation administrator role.

AUTH-015: Invitation Security

Invitation links or equivalent activation mechanisms must follow defined security and expiry rules.

Acceptance test: An expired or revoked invitation cannot be used to activate an account with the intended permissions.

AUTH-016: Role and Assignment Changes

The system must support authorised changes to roles, assignments, and access scopes.

Acceptance test: A principal transfers from one campus to another. The principal's previous campus access ends according to the configured effective date, while authorised historical records remain intact.

AUTH-017: Account Suspension and Revocation

The system must support account suspension and access revocation without unnecessarily deleting educational history.

Acceptance test: A teacher's account is disabled. The teacher can no longer sign in, but authorised school users can still access relevant historical assessment records.

AUTH-018: Approval Workflows

The system must support configurable approval or publication workflows for sensitive academic records where required by the school.

Acceptance test: A school requires principal approval for final results. A teacher can submit marks but cannot publish the official result without the required approval.

AUTH-019: Audit Trail for Sensitive Actions

The system must record significant changes to academic results, permissions, enrolments, staff assignments, and administrative configurations.

Acceptance test: An authorised user changes a student's recorded marks. The audit trail records the user, the time, the affected record, and the change details permitted by the audit policy.

AUTH-020: Audit Record Protection

Audit records must be protected against unauthorised alteration or deletion.

Acceptance test: A teacher cannot edit or delete the audit entry created when the teacher changes an assessment result.

AUTH-021: Least-Privilege Access

Users should receive only the permissions necessary for their authorised responsibilities.

Acceptance test: A parent account receives access to the parent's authorised children but does not receive teacher, principal, or school administrator permissions.

AUTH-022: Secure Authentication and Sessions

The system must implement appropriate authentication and session security for the selected authentication method.

Acceptance test: An unauthenticated user cannot access a protected portal, and a disabled account cannot continue using an otherwise valid session beyond the system's defined revocation behaviour.

AUTH-023: Secure Data Handling

The system must protect sensitive information in application responses, exports, logs, and error messages.

Acceptance test: A request for an unauthorised student record does not expose the student's marks, personal details, or other protected information through the error response.

AUTH-024: Consistent Permissions Across Features

The same authorisation rules must apply across pages, dashboards, searches, charts, reports, exports, and direct server requests.

Acceptance test: A teacher cannot access an unauthorised student's marks through a report export when access to the same marks is denied in the student detail page.

AUTH-025: Controlled Government or Oversight Access

Any future government or oversight role must have explicitly defined permissions and an authorised data scope.

Acceptance test: A future oversight account can access approved aggregate reports within its assigned scope but cannot access individual student records unless a separate permission explicitly authorises that access.

AUTH-026: Role-Aware Navigation

The interface must display navigation appropriate to the user's roles and authorised responsibilities.

Acceptance test: A student does not see organisation management or teacher assessment-entry pages in the normal navigation. Direct attempts to access protected functions are still denied by the server.

AUTH-027: Privacy and Data Minimisation

The system must collect and expose only the information necessary for its defined functions and authorised purposes.

Acceptance test: A teacher sees the student information needed for authorised teaching tasks but cannot browse unrelated sensitive administrative information.

AUTH-028: International Privacy Readiness

The system architecture must support configurable privacy and data-handling requirements where different jurisdictions require different treatment.

Acceptance test: The implementation can apply a defined school or jurisdictional policy without granting users additional access by default. The applicable legal requirements must be confirmed before the relevant deployment.

25. Final Principles

The most important principle in this document is that **a user's role, permissions, and access scope are separate concepts**.

A role describes the user's responsibilities. A permission defines an allowed action. The access scope determines which records that action applies to.

Optimus Orien must evaluate all three when protecting private information.

The organisation administrator, school administrator, principal, teacher, student, and parent portals should be different views of one connected platform, not separate systems containing duplicate records.

Public pages must remain separate from private student information. Account invitations must be controlled. Role changes must update access correctly. Sensitive changes must be auditable.

The system must enforce security on the server side and must not rely only on what the interface displays.

These principles should guide the database design, backend services, interface, reports, and future integrations.

26. What This Part Establishes

This document establishes the initial user-role and access-control model for Optimus Orien.

It defines the intended responsibilities of the organisation administrator, school administrator, principal, teacher, student, and parent or guardian. It also explains how accounts, roles, permissions, access scopes, invitations, approval workflows, and audit records should work together.

It establishes the requirement that private student information must remain protected across every portal and feature.

Part 6 brings together the concepts from the previous documents and explains how Optimus Orien should build a complete student performance record, identify students who may need support, and help schools take appropriate action.

This document is written for both you and your developer. It explains the educational concepts first, then translates them into concrete software requirements and acceptance tests.

PART 6

THE COMPLETE STUDENT PERFORMANCE RECORD AND EARLY INTERVENTION

Project: Optimus Orien
Document type: Educational Foundations and Software Implementation Requirements
Audience: Product Owner, Developer, and Future Development Team
Purpose: Define how Optimus Orien records, analyses, displays, and monitors student performance while helping schools identify and support students who may be experiencing difficulties.

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1. Why This Document Exists

In Part 2, we explained how education systems differ between countries and why Optimus Orien must support configurable educational arrangements.

In Part 3, we discussed assessment and grading: how schools assess students, record marks, calculate results, and assign grades.

In Part 4, we explained how schools, branches, classes, subjects, teachers, students, and enrolments are connected.

In Part 5, we explained user roles, portals, permissions, and access control.

Part 6 combines these foundations into one important question:

How can Optimus Orien build a reliable picture of a student's progress over time and help a school recognise when that student may need additional support?

Consider a student who receives 82% in an examination. That result alone does not tell us everything about the student's educational situation.

The student might have excellent attendance but struggle with homework. Another student might achieve good marks while their performance has declined significantly over the last three reporting periods. A third student might have low marks because they recently transferred schools and missed several weeks of instruction.

These situations require different interpretations and potentially different responses.

Optimus Orien must therefore avoid treating one mark, one percentage, or one overall number as a complete description of a student.

Instead, the platform must bring together several types of information, show what the available evidence actually says, identify patterns that deserve attention, and allow authorised school staff to investigate and respond.

The goal is not to label students as failures. The goal is to help schools notice potential difficulties early enough to respond constructively.

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2. What Is a Student Performance Record?

A student performance record is a connected collection of information about a student's educational progress within a particular school, programme, academic year, and reporting period.

It includes both the original information entered into the system and the indicators calculated from that information.

For example, a student record might contain examination marks, attendance, homework completion, classroom participation, subject-specific skills, and changes in performance over several terms.

These records must be connected to the correct student, subject, class, school, academic year, and reporting period.

This connection matters because a mark without context can be misleading.

A score of 70 out of 100 means something different from a score of 70 out of 80. A student's result in one subject should not be treated as their result in another subject. A student's current school record should not accidentally include marks belonging to another student with a similar name.

The system must also distinguish between information that belongs to the student personally and information that belongs to a particular period of their education.

For example, the student's date of birth is a personal record. Their Mathematics mark of 74% belongs to a particular assessment and reporting period. Their attendance percentage belongs to a defined attendance period and calculation method.

These different types of information should be connected through the data model rather than stored as one large, unstructured student profile.

2.1 A Simple Example

Imagine that Student A attends Grade 8 at a school.

During Term 1, the system records the following:

Mathematics examination: 72%.

English examination: 84%.

Science examination: 76%.

Attendance: 93%.

Homework completion: 88%.

English communication skill rating: 8 out of 10.

Science analytical skill rating: 6 out of 10.

These figures describe different aspects of the student's educational experience.

The examination results describe performance on specific assessments. Attendance describes presence during the relevant attendance opportunities. Homework completion describes the recorded completion of assigned work. Skill ratings describe a teacher's assessment of particular skills using the school's chosen rating method.

The system should display these measures together, but it should not automatically combine them into one score.

A student with 72% in Mathematics, 93% attendance, and an English communication rating of 8 out of 10 does not have a meaningful overall score of 57.7 merely because those three numbers were averaged.

They measure different things and use different scales.

Core principle: Connected information should create a clearer picture of performance, not an artificial number that hides important differences.

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3. The Six Main Categories of Student Performance Information

Optimus Orien's initial student performance record should organise information into six main categories.

These categories are academic results, attendance, homework completion, classroom participation, subject-linked skills, and historical progress.

Each category must have its own definitions, calculation rules, data requirements, and visual presentation.

3.1 Academic Results

Academic results describe how a student performed in assessments.

An assessment may be an examination, test, quiz, assignment, practical activity, oral examination, project, or another assessment type supported by the school's educational arrangements.

The exact types of assessments available should be configurable because different programmes and schools use different assessment methods.

For every recorded result, Optimus Orien should retain the student, subject, assessment, assessment type, maximum possible marks, marks obtained, assessment date, reporting period, academic year, and the identity of the person who entered or changed the result.

The system should also retain the applicable grading rules when grades or other derived results are calculated.

For example, if a student obtains 36 marks out of 45, the percentage is:

$36 \div 45 \times 100 = 80\%$.

The system may display 80% as the percentage result, but it must also retain the original score of 36 out of 45.

The percentage is calculated information. The original marks are the source information.

If the school later corrects the original score, the system should recalculate the dependent indicators according to the applicable rules.

It must not require staff to edit the percentage manually.

What the student performance record should display

The academic results section should show individual assessment results and appropriate summaries.

An assessment result should display the marks obtained, maximum possible marks, calculated percentage where meaningful, assessment name, subject, date, reporting period, and applicable grade where the school uses a grading system.

A subject summary may show the student's performance across the assessments included in a particular reporting period.

The system should also be able to display a comparison with the relevant class average when the comparison is valid and the user has permission to see it.

However, the developer must not assume that every assessment contributes equally to the final subject result. Some schools use weighted assessments, while others use different calculation methods.

The assessment and grading rules established in Part 3 must determine how these summaries are calculated.

3.2 Attendance

Attendance describes whether a student was present for the relevant opportunities to attend school or instruction.

Attendance can provide useful context for understanding a student's educational experience. However, attendance alone does not establish why a student is struggling academically.

A student may have low attendance and strong marks. Another student may attend regularly but experience difficulties with a subject.

Optimus Orien must show these measures separately.

Understanding attendance percentage

Suppose a student was present for 45 out of 50 eligible attendance days.

The attendance percentage is:

45 ÷ 50 × 100 = 90%.

The denominator is important. It represents the number of attendance opportunities included in the calculation.

The system must not automatically count days before a student's enrolment began, days after the enrolment ended, or days when the school was officially closed as absences.

It must also handle incomplete attendance records carefully.

For example, if a school has recorded attendance for only 30 out of 50 eligible days, the system must not silently treat the remaining 20 days as absences.

It should identify the missing records and communicate that the attendance information is incomplete.

Where supported by the school's policy, attendance records may distinguish authorised absences, unauthorised absences, late arrivals, and partial attendance.

The system must not invent these categories or impose a universal definition. Their availability and calculation must be configurable.

What the attendance section should display

The student record should display attendance percentage, days or sessions present, recorded absences, the eligible attendance period, and a clear indication of whether the data is complete.

Where the school records different attendance statuses, the record should display the relevant breakdown.

The interface should also allow authorised staff to inspect the underlying attendance records when they need to understand how a percentage was calculated.

A low attendance percentage may contribute to a rule-based alert, but the system must not automatically claim that the student is disengaged, irresponsible, or academically weak.

Those conclusions cannot be established from attendance percentage alone.

3.3 Homework Completion

Homework completion describes whether assigned work has been recorded as completed according to the school's defined workflow.

It does not automatically tell us whether the student understood the work or performed well on it.

A student may complete every assignment but misunderstand the material. Another student may understand the subject but fail to submit some assignments.

Optimus Orien must distinguish between completing work and demonstrating academic mastery.

Homework statuses

The platform should support a clear homework workflow.

A teacher creates an assignment and defines its due date and relevant class or students.

The assignment becomes available to the assigned students.

The system records the relevant status of each student's work.

Depending on the school's configuration, statuses may include assigned, submitted, awaiting confirmation, completed, incomplete, or overdue.

The exact status names can vary, but their meanings must remain clear and consistent.

In Optimus Orien's intended workflow, a student or teacher may indicate that work has been completed, but where teacher confirmation is required by the school's policy, the final confirmed-completion status must not be awarded until the teacher confirms it.

The interface should make that distinction visible.

For example, a student marking homework as finished should not automatically cause the record to display it as teacher-confirmed completed work.

Calculating homework completion

Suppose a student has 20 homework assignments that are eligible for inclusion in a reporting period.

The system records 16 as completed according to the applicable confirmation rules.

The completion rate is:

16 ÷ 20 × 100 = 80%.

The calculation must define which assignments are included in the denominator.

Cancelled assignments, assignments created before enrolment, and assignments not applicable to the student must not automatically be counted as incomplete.

Missing records must not be silently converted into completed or incomplete work.

The system should explain the period covered, the number of eligible assignments, the number completed, and any excluded or unconfirmed records that materially affect interpretation.

3.4 Classroom Participation

Classroom participation describes a teacher's assessment of a student's participation in learning activities.

Depending on the school's policy, this may include contribution to discussions, engagement with classroom activities, asking questions, collaboration, or other defined behaviours.

Participation is not the same as intelligence, knowledge, academic achievement, or personality.

A quiet student may understand the material very well. A student who speaks frequently may still misunderstand it.

The system must therefore treat participation as a separate measure rather than a substitute for academic results.

If Optimus Orien uses a numerical participation scale, such as 1 to 10, the school must define what the scale means.

For example, a school might define what a rating of 2, 5, or 9 represents through a rubric. A rubric is a set of criteria that helps people make ratings consistently.

Without clear criteria, one teacher's rating of 7 may not mean the same thing as another teacher's rating of 7.

Required information

Every participation rating should be associated with the relevant student, subject or class, teacher, date or reporting period, rating scale, and applicable criteria.

The system should retain enough information to determine who submitted the rating and when it was recorded or changed.

A rating must not be displayed as if it were an objective measurement independent of the teacher's judgement.

The interface should identify it as a teacher-assessed measure and explain the scale when necessary.

Participation ratings may help staff identify patterns worth reviewing, but they must not independently establish that a student is academically at risk.

3.5 Subject-Linked Skills

Subject-linked skills describe particular abilities that a student demonstrates in relation to a subject or defined learning area.

Examples may include English communication, reading, speaking, listening, creativity, analytical thinking, or other skills selected by the school.

The important distinction is that a skill should be connected to a defined context.

For example, an English teacher may assess communication in English. A science teacher may assess analytical thinking within science activities.

The system must not assume that one rating in one subject proves the student's ability in every other subject.

Skill ratings

If a school uses a rating scale from 1 to 10, the platform should allow the school to define the meaning of the scale and, where appropriate, the criteria for each rating.

A rating of 8 out of 10 should be connected to the skill, subject, assessment period, and person who assigned it.

The interface should allow the school to explain what the skill means and how it is assessed.

For example, the school may define English communication through clarity of expression, comprehension, and appropriate use of language.

That is an example of a possible rubric, not a universal definition that every school must adopt.

Skills may be assessed at different times during the year. The system should therefore support a history of ratings rather than storing only the most recent value.

If a student's analytical-thinking rating changes from 5 to 7, the system should show when each rating was recorded and the applicable assessment criteria.

It must not automatically interpret the change as proof that a particular teaching intervention caused improvement.

The change is an observed difference. Explaining its cause requires additional evidence.

3.6 Historical Progress

Historical progress describes how a student's recorded performance changes across assessments, reporting periods, and academic years.

This is one of the most valuable functions Optimus Orien can provide because a single result often reveals less than a series of results.

Suppose a student receives the following Mathematics results:

Term 1: 82%.

Term 2: 76%.

Term 3: 69%.

The results show a decline of 13 percentage points between Term 1 and Term 3.

This is a useful pattern to investigate.

However, it does not automatically establish why the decline occurred.

Possible explanations may include changes in assessment difficulty, gaps in prerequisite knowledge, illness, changes in attendance, differences in the content assessed, or other factors.

Optimus Orien should present the pattern and allow authorised staff to investigate it rather than choosing an explanation without evidence.

Percentage change versus percentage-point change

These terms must not be confused.

If a result changes from 82% to 69%, the difference is:

$69 - 82 = -13$ percentage points.

The relative percentage change is different:

$(69 - 82) \div 82 \times 100 \approx -15.85\%$.

The first figure describes the difference between two percentages. The second describes the change relative to the original value.

For ordinary school performance dashboards, displaying the percentage-point difference is often easier to interpret. If the platform also displays relative percentage change, it must label it clearly.

Comparing results fairly

Not every historical result can be compared directly.

A student's Mathematics result from one year may use a different assessment structure from the previous year. A school may change its grading system, subject curriculum, or assessment weighting.

The system should therefore preserve the context of historical results, including the applicable grading rules, assessment definitions, programme, subject, and reporting period.

When the comparison is not valid or the information is insufficient, Optimus Orien should show a warning or withhold the comparison rather than manufacture a misleading trend.

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4. Raw Data, Calculated Indicators, and Interpretations

The developer must understand three different layers of information.

They are raw data, calculated indicators, and interpretations or flags.

These layers must not be treated as interchangeable.

4.1 Raw Data

Raw data is the original information recorded in the system.

Examples include 36 marks out of 45, an attendance record marked present, a homework assignment marked submitted, or a teacher's skill rating of 7 out of 10.

Raw data should be traceable to its source and should not be overwritten by a calculated summary.

4.2 Calculated Indicators

A calculated indicator is produced by applying a defined calculation to the underlying data.

Examples include an assessment percentage, an attendance percentage, a homework completion rate, a subject average, or a change between reporting periods.

Every indicator must have a defined calculation method.

The platform should know which records were included, which records were excluded, which configuration was used, and when the indicator was last calculated.

Where relevant, the interface should let an authorised user inspect the records contributing to a result.

4.3 Interpretations and Flags

An interpretation or flag indicates that a defined pattern may deserve attention.

For example, a student might receive a review flag because attendance has fallen below a school's configured threshold or because performance has declined across several comparable reporting periods.

A flag is not the original data. It is not the same as a diagnosis. It is not proof that a student will fail.

It is a signal produced by applying an explicit rule to available information.

The platform must make the reason for each flag visible to authorised staff.

The fundamental rule is that Optimus Orien must never present an interpretation as if it were an unquestionable fact about the student.

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5. Individual Subject Performance and Overall Student Performance

These are related concepts, but they are not identical.

5.1 Subject Performance

Subject performance describes the student's recorded results within one subject and a defined period.

For example, a student's Mathematics summary might show an assessment average of 74%, an applicable class average of 79%, and a change of −6 percentage points from the previous comparable period.

The interface should make clear whether the 74% represents a simple average of assessment percentages, a weighted result, or another school-defined calculation.

These methods are not interchangeable.

If a school uses weighted assessments, the subject summary must follow the configured weighting rules rather than calculating an unweighted average by default.

5.2 Overall Student Performance

Overall student performance provides a broader view across the subjects and other indicators relevant to the student's educational record.

However, there is no single universally correct formula for combining Mathematics marks, English marks, attendance, homework completion, participation, and skill ratings into one number.

The measures represent different dimensions of educational experience.

For example, a school may choose to display an academic summary alongside separate attendance and homework indicators. This is often more informative than combining everything into one score.

Optimus Orien should therefore use a multidimensional student overview as its default.

It may display subject results, attendance, homework completion, participation, skills, and historical trends together while preserving their individual meanings.

If a school later chooses to configure a combined indicator, the formula, purpose, limitations, and validation requirements must be explicitly defined. The platform must not silently create such a score.

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6. How Class Averages Should Work

Class averages help users understand how a student's result compares with an appropriate group.

However, the term "class average" can refer to different calculations.

The system must define exactly what is being averaged.

Suppose five students receive the following Mathematics percentages:

60%, 70%, 80%, 90%, and 100%.

The arithmetic mean is:

(60 + 70 + 80 + 90 + 100) ÷ 5 = 80%.

This is the average of the five students' percentages.

However, if one student has not completed the assessment, the system must not automatically include that student's result as zero unless the assessment rules explicitly define that treatment.

A missing result and a score of zero are different situations.

The class average should also use an appropriate comparison group. A student in one programme should not automatically be compared with students taking a different assessment under a different grading system.

6.1 Averaging Across Subjects

Optimus Orien's intended reporting model includes class performance calculated from average subject performance. That calculation must be defined carefully.

Suppose a class has an average of 80% in Mathematics and 70% in English.

The arithmetic mean of the two subject averages is 75%.

However, this does not necessarily mean that the average student achieved 75% across their subjects. The result may differ if the subjects contain different numbers of assessments, different weighting rules, or different numbers of eligible students.

The dashboard must therefore label the method accurately.

If the platform displays the mean of subject averages, it should say so. If it displays a student-level average across comparable subject results, it should explain how that measure is calculated.

The developer must not assume that two differently calculated indicators are equivalent simply because both are expressed as percentages.

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7. Identifying Students Who May Need Support

A central purpose of Optimus Orien is to help schools identify students whose recorded circumstances or performance patterns may require attention.

This is the basis of early intervention.

In education, early intervention means responding to a potential difficulty before it becomes more serious or continues unnoticed.

For example, if a student's results decline over several reporting periods, a teacher or school leader may decide to review the student's work and determine whether additional support is appropriate.

The platform can help staff notice the pattern. It cannot automatically know the cause.

7.1 What Does "Student at Risk" Mean?

The phrase "student at risk" can be misleading if it is treated as a permanent description of the student.

For Optimus Orien, it should mean that **available information meets a defined review rule suggesting that authorised staff should examine the student's situation**.

The phrase must not mean that the student is certain to fail, has a particular psychological condition, lacks ability, or is personally responsible for the difficulty.

A student who receives a review flag is still an individual with circumstances that may not be fully represented by the data.

The platform should use constructive labels such as:

"Review recommended."

"Attendance requires attention."

"Academic decline detected."

"Insufficient information to assess this pattern."

These labels describe the situation or the required action rather than defining the student.

7.2 Rule-Based Review Flags

The initial version of Optimus Orien should use transparent, configurable rules.

It does not need artificial intelligence to identify straightforward patterns in recorded information.

A rule-based flag is created when data meets a condition defined by the school or an authorised administrator.

For example, a school may configure a rule that requests a review when a student's attendance falls below 85% during a specified reporting period.

Another school may use a different threshold because its policies and circumstances differ.

The figure of 85% is an example only. It must not become a universal default applied to every school without explicit product and school-policy decisions.

Other possible rules include a sustained decline across comparable assessment periods, several results below a configured academic threshold, or repeated incomplete homework records.

The platform must define the conditions precisely and record the rule version used to create each flag.

7.3 Combining Multiple Indicators

A school may want to identify situations in which several concerns appear together.

For example, a student may have declining Mathematics results, falling attendance, and several incomplete homework assignments.

Together, these observations may justify a closer review.

However, the system must not assume that every combination has the same meaning or that the factors caused one another.

The platform should show each contributing indicator separately.

It should also distinguish between a flag based on one rule and a flag based on multiple rules.

A staff member should be able to understand why the system recommended a review without reverse-engineering an unexplained score.

7.4 Data Completeness Must Affect Flags

Suppose a student's recorded attendance is 60%, but the school has entered attendance for only half of the relevant days.

That percentage may not provide a reliable picture of the student's attendance for the full period.

Similarly, a student may appear to have declining marks because results from one reporting period have not yet been entered.

The platform must consider data completeness when evaluating rules.

If information is insufficient, the system should identify the limitation.

It must not treat missing records as negative evidence by default.

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8. Designing a Responsible Early Intervention Workflow

Identifying a possible concern is only the first step.

A useful early intervention system also helps authorised staff record what they reviewed, what they decided, and what happened afterward.

Optimus Orien should therefore distinguish between a detected flag and an actual intervention.

8.1 Stage One: A Review Rule Is Triggered

The platform detects that available information meets a configured rule.

For example, a student's Mathematics results have declined across three comparable reporting periods.

The system creates a review flag containing the relevant rule, supporting evidence, detection date, and current status.

The flag should not immediately be interpreted as a confirmed problem.

8.2 Stage Two: An Authorised Staff Member Reviews the Evidence

A teacher, principal, or other authorised staff member opens the flag.

The interface displays the underlying results and explains which conditions triggered the rule.

The reviewer may determine that the pattern requires support, that the evidence is insufficient, or that the rule has identified a situation that does not require further action.

For example, the decline might be explained by an assessment change that makes the results unsuitable for direct comparison.

The system should allow the reviewer to record an appropriate decision.

8.3 Stage Three: Support Is Planned

If the concern is confirmed and support is appropriate, the reviewer may record an intervention or support action.

Examples might include additional subject instruction, a discussion with the family, a review of missed work, or another school-approved action.

These examples are possibilities, not universal requirements.

The platform should allow the school to define the types of support it records.

Each action should identify the responsible staff member, the action date, the intended purpose, and any scheduled follow-up date.

Where appropriate, the school may record a short note describing the planned support.

Sensitive personal or family information should not be collected unnecessarily.

8.4 Stage Four: Follow-Up

At the follow-up date, authorised staff should be able to review the student's updated information and record what happened.

Possible outcomes include continued support, successful resolution of the concern, further review, or insufficient information to determine an outcome.

The platform must not automatically mark an intervention as successful simply because the student's next mark increased.

Likewise, a student whose marks do not immediately improve should not automatically be considered a failure.

The reviewer should consider the purpose of the intervention, the period available for change, the quality of the evidence, and the student's circumstances.

8.5 Stage Five: Close or Continue the Review

The review can be closed when authorised staff determine that no further action is currently required.

It can also remain open, be reassigned, or be scheduled for another review.

The system must preserve the history of the flag and associated actions.

Closing a review should not delete the underlying evidence or erase the fact that a review previously occurred.

A new concern may also arise later. The platform should create a new review event or appropriately update the existing case according to a clearly defined workflow.

The system must avoid creating endless duplicate alerts every time a background calculation runs.

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9. The Difference Between a Flag and an Intervention

These concepts must remain separate in the database and user interface.

A **flag** is a system-generated or authorised review indicator based on defined conditions.

An **intervention** is a deliberate action taken by school staff in response to a concern or identified need.

For example, the platform may detect that attendance has fallen below a configured threshold.

That is a flag.

A principal may review the records, speak with the appropriate staff, and arrange a follow-up meeting with the family.

Those are intervention actions.

The platform must not assume that the flag itself means the school has taken action.

It must also not assume that creating an intervention means the underlying concern has been resolved.

These are separate events and should have separate records and statuses.

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10. Who Should See Which Information?

The complete student performance record must respect the access-control rules established in Part 5.

Different users require different views of the same underlying information.

The platform should not create separate, inconsistent copies of student performance data for every portal. It should use the same authorised source records and present appropriate views according to the user's permissions.

10.1 Student Portal

Students should be able to see their own authorised academic results, attendance information, homework records, skill ratings, and historical progress.

The interface should explain indicators in understandable language.

A student should be able to distinguish between an actual assessment result and a calculated summary.

The platform should not display internal staff-only notes or sensitive intervention information merely because the student can view their own academic results.

The school must define appropriate student visibility for different record types.

10.2 Parent or Guardian Portal

Parents and guardians should be able to view information about the student or students they are authorised to access.

The portal may display academic progress, attendance, homework completion, skill development, and relevant school-approved notifications.

A parent must not be able to access another family's student records by changing a URL, guessing an identifier, or manipulating a request.

Parents should see enough information to understand their child's progress without automatically receiving confidential staff notes or sensitive information about other people.

10.3 Teacher Portal

Teachers should be able to see the students and classes assigned to them, subject results relevant to their responsibilities, and any review flags they are authorised to access.

A teacher should be able to investigate a flag by inspecting relevant evidence.

However, teachers must not automatically have access to every student's complete record across the school.

Access should be based on the teacher's assignment, role, school policy, and explicitly granted permissions.

10.4 Principal Portal

Principals should be able to review student performance within their authorised school or campus scope.

They may need to see students requiring review, patterns across subjects, attendance concerns, intervention statuses, and school-level performance summaries.

A principal's authority over one school must not automatically grant access to another school's private student information.

Organisation-level administrators may have broader access according to their permissions.

10.5 Organisation Administrator Portal

An organisation administrator may need to compare performance indicators across multiple schools or branches.

The organisation-level dashboard can display aggregated indicators, such as attendance rates, subject averages, and counts of open review flags, for the schools within the administrator's scope.

However, aggregated information and identifiable student records are not the same thing.

Access to a school-level summary should not automatically grant unrestricted access to every underlying student record.

The platform must apply the relevant permissions when a user drills down from an organisation-wide chart into a school, class, or student.

10.6 Public Homepage

The public homepage may display approved school-level indicators and leaderboards according to Optimus Orien's product rules.

It must not expose identifiable student-level academic records, attendance histories, homework records, skill ratings, or intervention details.

A student-level record must never become public merely because the system uses that information to calculate an aggregate.

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11. How the Student Performance Dashboard Should Be Organised

The student performance dashboard should provide a clear overview before showing detailed records.

The design should be modern, professional, and easy to understand. It should not overwhelm users with dozens of numbers and charts on the first screen.

11.1 Student Overview

The first section should identify the student within the appropriate school and academic context.

It should display the student's name, relevant student identifier, school, class or year group, programme where applicable, and selected academic year or reporting period.

The interface should make it obvious which period the displayed results represent.

If the user changes the reporting period, the data should update consistently across the dashboard.

11.2 Academic Summary

The academic summary should show the relevant subject results for the selected period.

Where valid, it may show assessment percentages, applicable grades, class averages, and changes from comparable previous periods.

The dashboard must label each figure clearly.

For example, it should distinguish a student's assessment average from the final subject result calculated under the school's grading rules.

The interface should not use the word "average" without making the calculation method clear when different methods are possible.

11.3 Attendance Summary

The attendance section should display the attendance percentage, the period covered, and the number of eligible attendance opportunities.

It should identify incomplete records where that affects interpretation.

Where the school records different absence types or late arrivals, those details may be available in a breakdown.

11.4 Homework Summary

The homework section should show the number of eligible assignments, completed assignments, and the calculated completion rate.

It should distinguish completed work from submitted work and work awaiting teacher confirmation wherever those statuses apply.

The interface should not treat homework completion as a direct measurement of subject mastery.

11.5 Participation and Skills

The participation and skills section should display the configured ratings, their definitions, the applicable subjects, and the relevant reporting periods.

Historical ratings should remain available so authorised users can understand changes over time.

Where a rubric exists, the interface should provide access to its definitions.

11.6 Historical Progress

The historical section should show how comparable results have changed over time.

Charts may display subject results across terms, attendance over successive periods, homework completion rates, and skill ratings where the scale and assessment method remain comparable.

A trend chart should identify the period represented by each data point.

If the system cannot make a fair comparison, it should explain the limitation rather than drawing a misleading continuous trend.

11.7 Reviews and Support

Authorised users should see open review flags and relevant intervention statuses according to their permissions.

The section should explain why a review was triggered, who is responsible for the next action, and whether a follow-up date exists.

A student-facing or parent-facing dashboard should not automatically expose every internal review record.

The information shown must follow the school's visibility policy and the user's permissions.

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12. Data Quality and Corrections

A student performance dashboard is only useful when the underlying information is sufficiently accurate.

Schools may enter incorrect marks, record attendance late, change class assignments, or discover that a homework status was entered incorrectly.

Optimus Orien must support correction without destroying the history of important changes.

12.1 Corrections to Academic Results

Suppose a teacher enters 68 marks out of 100 but later discovers that the correct result is 78.

The system should allow an authorised correction.

Any dependent calculations should update according to the applicable calculation rules.

The system should retain an audit record showing who changed the mark, when the change occurred, and the relevant before-and-after values, subject to the platform's data-retention and privacy policies.

12.2 Late Data Entry

A result may be entered after a reporting period has ended.

The system should identify the correct assessment and reporting period rather than assigning the result to the current period merely because it was entered today.

The record should distinguish the date of the assessment from the date the result was entered.

12.3 Data Completeness

The system should identify situations in which an indicator cannot be calculated reliably because required information is missing.

For example, it should not display an attendance percentage of zero when no attendance records exist.

Zero percent means that the applicable attendance records indicate no attendance. No records means the system lacks the information needed to calculate the result.

Those situations must remain distinct.

12.4 Source of Truth

Each data category should have a clear source of truth.

Academic results should come from authorised assessment records. Attendance indicators should come from the relevant attendance records. Homework completion should come from the assignment workflow. Skill ratings should come from authorised ratings.

Calculated summaries should use these source records rather than independently maintained, manually edited copies.

This reduces inconsistencies across portals.

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13. Historical Records and Changing Rules

School policies may change over time.

A school may introduce a new grading scale, change an assessment weighting, modify its attendance policy, or revise its definition of a skill-rating rubric.

The platform must not silently apply a new rule to old results in a way that changes their meaning.

Where a calculation depends on a particular configuration, the system should preserve the applicable rule version or sufficient historical information to reproduce the result.

For example, if a subject result was calculated under one assessment-weighting scheme, the historical record should retain the context needed to understand that calculation.

If the school later changes the weighting scheme, the system should apply the new configuration to the appropriate records without silently rewriting the meaning of previous reports.

The same principle applies to review rules.

If an attendance review threshold changes from 85% to 80%, the system should preserve which rule version generated a historical flag.

The school should be able to distinguish a flag that was triggered under the previous rule from one generated under the new rule.

Historical data is not merely a collection of old numbers. It is a record of what those numbers meant under the rules in effect at the time.

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14. Student Transfers and Changes in Enrolment

Students may join a school after the academic year has started, transfer between schools, change classes, or leave a programme.

These events affect how performance indicators should be calculated.

Suppose a student joins a school halfway through a term.

The student should not automatically be marked absent for days before enrolment began.

Similarly, the school should not automatically treat assignments created before the student's enrolment as incomplete homework.

The system should calculate indicators using the student's applicable enrolment period and the relevant school policies.

Historical results from a previous school may be recorded when the school has appropriate authority and sufficient information. However, those results must retain their source and context.

The system must not merge previous-school results into the current school's class averages without an explicit, valid rule.

If a student changes classes, previous results should remain associated with the appropriate historical class, subject, and reporting period.

The current class assignment should not overwrite the historical record.

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15. Avoiding Unsupported Conclusions

Optimus Orien must be careful about the difference between identifying a pattern and explaining its cause.

This is especially important because schools may use the platform to make decisions that affect students.

Consider these examples.

A student has low attendance and declining marks. The system may report that both indicators have changed. It must not automatically conclude that low attendance caused the decline.

A student's homework completion rate increases, followed by an improvement in marks. The system may show both changes. It must not automatically claim that completing more homework caused the improvement.

A student receives a low skill rating. The system may show the rating and its defined criteria. It must not conclude that the student lacks intelligence or has a psychological disorder.

A student has repeated low results. The system may recommend a review under the school's configured rules. It must not claim that failure is certain.

These distinctions are essential.

The initial Optimus Orien platform should provide transparent calculations, descriptive trends, and rule-based review flags. It should not claim to predict a student's future with certainty, diagnose a condition, or establish causal relationships from simple correlations.

The platform should help people ask better questions, not pretend that incomplete data provides all the answers.

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16. Privacy and Responsible Handling of Intervention Information

Intervention records can be more sensitive than ordinary academic summaries.

A record might contain a concern raised by a teacher, an agreed support action, or a note about a meeting with a family.

Such information should be collected only when it serves a legitimate educational purpose.

The platform should avoid encouraging staff to record unnecessary personal details.

For example, a short note that a follow-up meeting is scheduled may be sufficient. The system should not require a teacher to document private family circumstances that are irrelevant to the educational decision.

Access to intervention records should be controlled separately from general academic record access where necessary.

The school should define which roles can create, view, edit, close, and reopen intervention records.

Sensitive notes should not automatically appear in parent, student, teacher, organisation-wide, or public views.

The platform should also maintain an audit trail for access and changes to sensitive records where required by its security design and applicable policies.

Data retention, deletion, export, and access requirements must be reviewed for the jurisdictions in which Optimus Orien operates.

The platform must not assume that one country's privacy rules apply identically everywhere.

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17. The Correct Technical Architecture

The developer should implement student performance as a connected set of records rather than a single large table containing every possible field.

The following is a conceptual model. The developer may choose different physical database tables or implementation details, but the relationships and meanings must be preserved.

A student record identifies the person.

An enrolment record connects that student to a school, academic period, programme, and class assignment for a defined period.

Assessment records describe the assessments that were created.

Assessment-result records connect individual students to the marks recorded for those assessments.

Attendance records describe attendance events or statuses for the relevant student and attendance opportunity.

Homework records describe assignments and each student's workflow status.

Participation ratings describe teacher-assessed participation within a defined subject or class and period.

Skill definitions describe the skills a school has chosen to assess.

Skill-rating records connect a student to a defined skill, applicable context, rating scale, period, and authorised assessor.

Calculated indicators store or produce summaries from the underlying records according to explicit rules.

Review flags record that a configured condition was met and identify the supporting evidence.

Intervention records describe actions taken by authorised staff in response to a review.

Audit records preserve the history of important changes and actions.

The developer should not store a manually edited attendance percentage as if it were the original attendance evidence. The percentage should be calculated from the appropriate records, with any stored result linked to its calculation method and relevant data version.

Likewise, an overall student performance indicator should not be stored without a defined formula, data scope, and explanation.

18. Calculation and Display Rules

The following rules should guide implementation.

First, every percentage must have a defined numerator and denominator. A percentage should not appear merely because two numbers were available.

Second, missing data must remain distinguishable from zero.

Third, calculations must use the correct academic period, enrolment, subject, school, and programme.

Fourth, class averages must use a clearly defined eligible population and comparison method.

Fifth, indicators based on different scales must not be combined without an explicitly defined and justified method.

Sixth, trends should compare results that are sufficiently comparable. Where comparability is uncertain, the interface should disclose the limitation.

Seventh, changes to source records should update dependent calculations according to defined rules.

Eighth, calculations must be reproducible. The system should be able to identify the records and configuration used to produce an indicator.

Ninth, the same underlying indicator should not produce conflicting values in different portals merely because each portal implements its own formula.

Tenth, users should be able to understand what an indicator means without needing to know how the database works.

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19. Developer Implementation Requirements

The following requirements translate the educational model into testable software behaviour.

They are proposed requirements for the initial version of Optimus Orien. They should be implemented alongside the organisation, education-configuration, and access-control requirements from Parts 2–5.

PERF-001 — Complete Student Performance Overview

The system must provide a student performance overview that brings together academic results, attendance, homework completion, participation, subject-linked skills, and historical progress where those records are available.

Acceptance test: An authorised user opens a student's record and can navigate to each supported category without the system mixing information from unrelated students or reporting periods.

PERF-002 — Preserve Original Results

The system must retain the original assessment marks and maximum possible marks separately from calculated percentages and summaries.

Acceptance test: A result of 36 out of 45 displays as 80% while retaining the original values of 36 and 45.

PERF-003 — Use Configured Assessment Rules

Subject summaries and academic indicators must follow the applicable assessment and grading configuration.

Acceptance test: When a school uses weighted assessments, the summary follows the configured weighting instead of silently calculating an unweighted mean.

PERF-004 — Correct Attendance Calculation

Attendance percentages must use the defined eligible attendance opportunities and the school's configured attendance rules.

Acceptance test: Days before a student's enrolment and official school-closure days are not counted as absences unless an explicitly defined and valid policy requires a different treatment.

PERF-005 — Distinguish Missing Attendance

The system must distinguish between zero attendance and unavailable attendance data.

Acceptance test: If no attendance records exist for the selected period, the dashboard displays an unavailable or incomplete-data state rather than automatically displaying 0%.

PERF-006 — Handle Homework Confirmation

The homework workflow must distinguish student-reported completion from teacher-confirmed completion wherever teacher confirmation is required by policy.

Acceptance test: A student marks an assignment as completed, but the assignment remains awaiting confirmation until an authorised teacher confirms it.

PERF-007 — Define Homework Denominator

Homework completion rates must use only assignments eligible under the applicable rules.

Acceptance test: A cancelled assignment or an assignment that predates the student's enrolment is not silently counted as incomplete.

PERF-008 — Store Participation Context

Participation ratings must identify the relevant student, assessor, subject or class, rating scale, and applicable reporting period.

Acceptance test: An authorised user can determine who submitted a rating and which defined scale was used.

PERF-009 — Store Skill-Rating Context

Skill ratings must be linked to a defined skill, subject or context, rating scale, reporting period, and assessor.

Acceptance test: A Mathematics analytical-thinking rating cannot accidentally appear as the student's English communication rating.

PERF-010 — Preserve Historical Ratings

The system must retain previous skill and participation ratings according to the platform's retention rules.

Acceptance test: When a new skill rating is entered, the previous rating remains available in the appropriate historical view rather than being overwritten without a trace.

PERF-011 — Separate Data Layers

The system must distinguish original records, calculated indicators, and review flags.

Acceptance test: Changing an authorised source assessment result triggers the required recalculation process without changing the source record into a calculated summary.

PERF-012 — Explain Calculated Indicators

Every major indicator must have a defined calculation method and an understandable label.

Acceptance test: An authorised user can determine how the displayed attendance percentage or homework completion rate was calculated, including the relevant period and eligible record count.

PERF-013 — Do Not Create an Unspecified Overall Score

The system must not automatically combine marks, attendance, homework, participation, and skills into an unexplained overall performance score.

Acceptance test: A student's dashboard displays the categories separately unless an explicitly approved, configured, and documented combined indicator has been implemented.

PERF-014 — Define Class-Average Method

Each class-average indicator must have a documented calculation method and comparison population.

Acceptance test: The system can explain whether an indicator represents the average of student percentages, a mean of subject averages, or another defined calculation.

PERF-015 — Exclude Ineligible Records

Class and subject summaries must follow the configured rules for eligible students, assessments, and reporting periods.

Acceptance test: A student without a result for an assessment is not silently assigned a zero and included in the average unless the applicable assessment policy explicitly defines that treatment.

PERF-016 — Calculate Percentage-Point Differences Correctly

When the platform compares two percentages, it must calculate and label the difference correctly.

Acceptance test: A change from 82% to 69% is displayed as a decrease of 13 percentage points, not as a decrease of 13% unless a separately calculated relative percentage change is intended and labelled.

PERF-017 — Preserve Historical Context

Historical indicators must retain sufficient context to explain the rules and source records used in their calculation.

Acceptance test: A user reviewing an older result can identify the relevant reporting period and grading configuration, even if the school later changes its grading rules.

PERF-018 — Compare Only Suitable Results

The system must not automatically treat results as directly comparable when their definitions, assessment structures, or grading contexts make the comparison invalid.

Acceptance test: If two historical results use incompatible grading scales and no valid comparison method is configured, the platform withholds the misleading comparison or displays an appropriate limitation.

PERF-019 — Configurable Review Rules

Authorised administrators must be able to configure supported review thresholds and conditions within the platform's approved configuration model.

Acceptance test: A school can change its configured attendance-review threshold without requiring a developer to hard-code a new number into the application.

PERF-020 — Version Review Rules

The system must retain the rule version associated with a review flag.

Acceptance test: If an attendance threshold changes, an older flag still identifies the rule that originally triggered it.

PERF-021 — Explain Every Review Flag

Every system-generated review flag must identify the rule that triggered it and the relevant supporting evidence.

Acceptance test: An authorised user opens a flag and can see the recorded conditions that caused it to be created.

PERF-022 — Consider Data Completeness

Review rules must account for required information being missing or incomplete.

Acceptance test: A rule that requires a complete attendance period does not treat missing attendance records as confirmed absences.

PERF-023 — No Unsupported Causal Claims

The system must present observed changes and configured rule outcomes without automatically asserting that one factor caused another.

Acceptance test: When attendance and academic performance both decline, the dashboard may show both patterns but does not automatically state that attendance caused the academic decline.

PERF-024 — Separate Flags and Interventions

The system must maintain distinct records for review flags and intervention actions.

Acceptance test: Creating a review flag does not automatically mark an intervention as completed or create a fictional record of staff action.

PERF-025 — Record Review Decisions

Authorised staff must be able to record the outcome of reviewing a flag.

Acceptance test: A reviewer can mark a flag as requiring support, requiring more information, or not requiring further action according to the configured workflow.

PERF-026 — Record Support Actions

Where the school enables intervention tracking, authorised staff must be able to record an action, its responsible person, relevant date, and follow-up information.

Acceptance test: A principal records a support action and assigns responsibility to an authorised staff member. The action remains linked to the relevant review and student.

PERF-027 — Track Follow-Up

The system must support follow-up dates and intervention status changes.

Acceptance test: An authorised user can identify overdue follow-ups and record whether the support should continue, be reviewed, or be closed.

PERF-028 — Preserve Intervention History

Closing or resolving a review must not silently delete its history.

Acceptance test: After a review is closed, an authorised user can still inspect the original flag, recorded decision, and associated support actions according to the retention and access rules.

PERF-029 — Avoid Duplicate Flags

The system must not create uncontrolled duplicate review flags every time the same unchanged rule is evaluated.

Acceptance test: Recalculating the same rule against unchanged records does not repeatedly create identical open flags. The system follows a documented rule for updating, reopening, or creating a new review event.

PERF-030 — Enforce Role and Scope Restrictions

All student performance and intervention data must be protected by server-side authorisation.

Acceptance test: A teacher cannot access another teacher's restricted student records by changing a URL, record identifier, or application request.

PERF-031 — Protect Parent and Student Views

Students and parents must see only the records authorised for their account and relationship.

Acceptance test: A parent cannot retrieve another family's student records through direct navigation or a modified request.

PERF-032 — Protect Sensitive Intervention Records

Intervention notes and actions must follow explicit access permissions.

Acceptance test: A user who can view ordinary academic results but lacks permission to view internal intervention notes cannot retrieve those notes through the interface or an API request.

PERF-033 — Use One Authorised Source of Truth

Different portals must derive their displayed indicators from the same defined calculation logic and authorised source records.

Acceptance test: When the same authorised user views the same indicator, student, and reporting period through two permitted views, the values and calculation definitions agree.

PERF-034 — Support Corrections and Audit

The system must allow authorised corrections while preserving the required history of important changes.

Acceptance test: A teacher corrects a mark, the applicable summaries update, and the audit record identifies the correction without losing the required history.

PERF-035 — Distinguish Assessment Date from Entry Date

The system must retain the date an assessment occurred and the date its result was entered when both are relevant.

Acceptance test: A result entered after a term ends remains associated with the correct assessment and reporting period.

PERF-036 — Handle Student Transfers

Performance indicators must respect the student's historical enrolment periods and applicable school policies.

Acceptance test: A student joining halfway through a term is not automatically assigned absences or incomplete homework for periods before their applicable enrolment began.

PERF-037 — Preserve Historical Class Assignments

Changes to a student's class must not overwrite historical class membership.

Acceptance test: A student who changes classes can still be viewed in the appropriate historical class record for earlier reporting periods.

PERF-038 — Show Reporting-Period Context

The dashboard must clearly identify the academic year, reporting period, and relevant educational context for the displayed indicators.

Acceptance test: When a user changes the selected reporting period, all relevant dashboard sections update consistently and display the newly selected period.

PERF-039 — Handle Insufficient Data Clearly

The system must provide a clear state when a requested indicator cannot be calculated reliably.

Acceptance test: If a subject has no eligible assessment results, the dashboard displays an appropriate unavailable state rather than inventing a zero result.

PERF-040 — Keep Public Data Separate

Public leaderboards must not expose private student-level information.

Acceptance test: A public user cannot retrieve identifiable student marks, attendance histories, skill ratings, homework records, or intervention details through a public page or API.

PERF-041 — Explain Rating Scales

Configured participation and skill-rating scales must have accessible definitions.

Acceptance test: An authorised user can inspect the meaning of a displayed rating and identify the applicable scale or rubric.

PERF-042 — Preserve Data Provenance

The system must identify the relevant source records, calculation configuration, and rule version used for important derived indicators and flags.

Acceptance test: An authorised user can trace a review flag to the supporting indicator and the underlying records that contributed to it.

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20. Recommended Development Order

The developer should not attempt to build every dashboard, calculation, and intervention feature simultaneously.

The components depend on one another and should be implemented in a logical order.

First, complete the underlying student, enrolment, class, subject, academic-period, and permission structures from the earlier parts.

Second, ensure that assessment results, attendance records, homework statuses, participation ratings, and skill ratings can be stored accurately with their required context.

Third, implement the calculation services for percentages, subject summaries, attendance, homework completion, and historical comparisons.

Fourth, implement data-completeness checks and ensure that missing values remain distinct from zero.

Fifth, build the student performance overview and detailed views for authorised users.

Sixth, implement configurable rule-based review flags and the supporting evidence needed to explain them.

Seventh, implement review decisions, intervention actions, follow-up dates, and intervention history.

Eighth, test role permissions, historical records, student transfers, data corrections, and cross-school isolation.

Ninth, validate calculations using known example datasets and test cases before using the indicators in real school decisions.

The user interface should be built on top of stable data definitions and calculation rules. A visually attractive dashboard cannot compensate for incorrect underlying calculations.

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21. Example End-to-End Scenario

Consider a student named Student A who is enrolled in Grade 8 Mathematics.

During the first reporting period, the student receives 82%.

During the second reporting period, the student receives 76%.

During the third reporting period, the student receives 69%.

The system also records that attendance has declined during the same general period.

A school has configured a rule that requests a review when a comparable subject result declines across three reporting periods.

The rule is evaluated against the available records.

The platform checks that the results belong to the correct student, subject, and reporting periods. It checks that the results are comparable under the school's configuration. It checks that the required data is present.

The rule is met, so Optimus Orien creates a review flag.

The flag states that Mathematics performance has declined by 13 percentage points from the first to the third reporting period. It separately identifies the recorded attendance trend.

It does not state that attendance caused the decline.

An authorised teacher or principal opens the flag and examines the underlying results. The reviewer determines whether additional information is needed.

If the school decides to provide support, the authorised staff member records an appropriate action and follow-up date.

At the follow-up, the reviewer examines the updated information and records the outcome.

The system preserves the original results, the calculation method, the flag, the review decision, and the support history.

This is the complete process:

Record data → Calculate indicators → Detect a defined pattern → Explain the evidence → Review the situation → Provide support if appropriate → Follow up → Preserve the history.

That process is the foundation of responsible early intervention in Optimus Orien.

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22. Final Principles for the Product Owner and Developer

Optimus Orien should provide a reliable, understandable picture of student performance without pretending that a few numbers explain everything about a learner.

Academic results, attendance, homework completion, participation, and skills must remain distinct measures with clearly defined meanings.

Historical trends should help staff notice changes, but the platform must not automatically invent explanations for those changes.

Review flags should be transparent, configurable, and based on available evidence. They should prompt authorised staff to investigate rather than label a student as destined to fail.

Intervention records should document real actions taken by school staff, not merely the existence of a system-generated alert.

The same underlying records and calculation logic should support all authorised portals, while access permissions determine which information each user can see.

Historical results must retain the educational and calculation context needed to understand them.

Missing information must not silently become zero. Different rating scales must not be treated as equivalent. Comparisons must be made only when the underlying measures are suitable for comparison.

Most importantly, the platform should be designed to support better educational decisions, not replace professional judgement.

A dashboard can show that a student's results have declined. It cannot, by itself, explain the student's circumstances or determine the most appropriate response.

Optimus Orien's purpose is to make meaningful patterns visible, make the evidence understandable, and help schools respond responsibly.

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23. What Part 6 Establishes

After implementing the requirements in this document, Optimus Orien should have a clear foundation for creating a complete student performance record.

The platform should be able to organise different categories of educational information, calculate transparent indicators, compare appropriate historical results, identify defined patterns that warrant review, and support the recording of follow-up actions.

It should do this without confusing missing information with poor performance, treating subjective ratings as unquestionable facts, making unsupported causal claims, or exposing private student information.

Part 6 establishes the foundation for student-level performance intelligence and early intervention.

Part 7 focuses on how Optimus Orien should turn educational data into clear, meaningful visualisations. It will explain the principles to you first, then give your developer practical implementation requirements and acceptance tests.

PART 7

INFORMATION VISUALISATION, CHARTS, DASHBOARDS, AND

DATA INTERPRETATION

Project: Optimus Orien
Document type: Educational Foundations and Software Implementation Requirements
Audience: Product Owner, Developer, and Future Development Team
Purpose: Define how Optimus Orien should transform school data into accurate, understandable, accessible, and actionable visual information.

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1. Why This Document Exists

In Part 6, we established how Optimus Orien should record student performance, calculate indicators, identify patterns that require attention, and support early intervention.

However, calculating accurate information is only half the problem.

Imagine that a principal opens a dashboard and sees 47 numbers, 12 charts, 8 colours, and several tables. The information may be accurate, but if the principal cannot quickly understand what matters, the dashboard has failed its purpose.

Now imagine a different dashboard. It shows the school's attendance trend, academic performance by subject, the number of students requiring review, and a clear comparison with the previous reporting period. Each chart has a meaningful title, its figures are easy to understand, and users can open the underlying records when they need more detail.

The second dashboard is more useful because it helps the user understand the information.

This is the purpose of information visualisation.

Optimus Orien must not merely display data. It must make educational information easier to understand and use.

This document explains what information visualisation means, why different charts serve different purposes, how to choose the right visual representation, how to avoid misleading users, and how the developer should implement dashboards throughout the platform.

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2. What Is Information Visualisation?

Information visualisation is the presentation of information in a visual form so that people can recognise patterns, compare values, understand relationships, and identify changes.

A chart is one example of information visualisation. A dashboard, progress indicator, comparison table, or map can also be part of an information visualisation system.

Consider two ways to present the same attendance information.

The first method displays a paragraph:

"Attendance was 96% in Term 1, 92% in Term 2, 88% in Term 3, and 91% in Term 4."

The second method displays a line chart with four points representing those reporting periods.

Both methods communicate the same values. However, the line chart makes the decline and subsequent improvement easier to recognise quickly.

The visual representation helps the user identify a pattern without reading every value individually.

This does not mean that charts are always better than text or tables. Each method has a different purpose.

A chart may be better for understanding trends. A table may be better for inspecting exact values. A short explanatory sentence may be best for communicating the meaning of an indicator.

Optimus Orien should use the presentation method that best supports the user's task.

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3. What Is a Dashboard?

A dashboard is a screen that brings together a selected group of important indicators and visualisations.

It is designed to help a particular user understand a situation without searching through many separate records.

A dashboard is not simply a page containing every available chart.

It should answer a defined set of questions.

For example, an organisation administrator may need to know which branches require attention, whether attendance is improving across the organisation, and how school-level performance differs.

A principal may need to know which classes have declining results, whether attendance is changing, and which students require a review.

A teacher may need to understand the performance of assigned students in a particular subject.

These users need different dashboards because they make different decisions.

The organisation administrator needs an organisation-wide view. The principal needs a school-wide view. The teacher needs a view relevant to assigned classes and subjects.

A single universal dashboard would either provide too little information to some users or overwhelm others with information they do not need.

Every dashboard should be designed around the decisions its intended user needs to make.

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4. Begin with a Question, Not a Chart

One of the most important principles in dashboard design is that a chart should exist for a reason.

The developer should not begin by asking, "Which charts can we add?"

Instead, the developer should ask, "What question should this visualisation help the user answer?"

Consider the following examples.

If the question is, "How has attendance changed over the academic year?", a line chart may be suitable.

If the question is, "Which subjects have the highest and lowest average results?", a bar chart may be suitable.

If the question is, "How many students are in each configured review category?", a bar chart or clearly labelled summary cards may work.

If the question is, "Which schools are located in areas where attendance requires attention?", a map may be useful, provided the location and underlying data are sufficiently reliable.

If the question is, "What exact marks did this student receive in each assessment?", a table may be more useful than a chart.

If the question is, "Why did this student receive a review flag?", the interface should show the triggering rule and underlying evidence rather than presenting a decorative chart that does not explain the decision.

A visualisation should support a meaningful task, not exist merely to make a page look sophisticated.

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5. The Main Types of Charts Optimus Orien Needs

Different charts communicate different kinds of information. Choosing the wrong chart can make accurate data difficult to interpret or even misleading.

The following sections establish which chart types are useful for Optimus Orien and how they should be implemented.

5.1 Bar Charts

A bar chart compares values across categories.

For example, a school may have the following average subject results:

Mathematics: 72%.

English: 84%.

Science: 78%.

History: 81%.

A bar chart makes it easy to compare the subjects and identify which averages are higher or lower.

When Optimus Orien should use bar charts

Bar charts are suitable for comparing subject averages, class averages, branch-level attendance rates, school-level indicators, and other categorical values.

They are particularly useful when users need to compare several groups.

Horizontal versus vertical bars

Horizontal bar charts are often suitable when category names are long.

For example, a school may have subjects with lengthy names. Horizontal bars allow those names to remain readable without excessive rotation or abbreviation.

Vertical bars may be suitable when the number of categories is small and their labels fit comfortably.

The developer should choose the orientation that makes the information easiest to read.

Ordering bars

For comparison dashboards, Optimus Orien should generally order categories in a meaningful way.

For example, a principal may want to see subjects ranked from lowest to highest average performance when investigating areas that may require attention.

A chart may also use alphabetical ordering when users need to locate categories predictably.

The ordering should support the chart's purpose.

It must not change unpredictably every time the page loads.

Important implementation rule

Bar lengths must represent values accurately.

For a standard bar chart, the numerical axis should generally begin at zero. Truncating the axis can exaggerate small differences between bars.

If a non-zero baseline is genuinely necessary for a specialised analytical view, it must be clearly indicated and must not create a misleading comparison.

5.2 Line Charts

A line chart shows how a value changes across an ordered sequence, usually time.

For example, a student's Mathematics results may be:

Term 1: 82%.

Term 2: 76%.

Term 3: 69%.

A line chart helps the user recognise the downward trend.

If the next result is 74%, the chart shows that the result increased from the previous period.

When Optimus Orien should use line charts

Line charts are useful for showing academic progress across reporting periods, attendance trends, changes in homework completion, and changes in other consistently measured indicators.

They can also support school-level and organisation-level monitoring over time.

Time must be ordered correctly

The horizontal axis must represent the actual order of the reporting periods.

The developer must not sort terms alphabetically or order dates as plain text if doing so produces the wrong sequence.

The system should use the configured academic calendar and reporting-period definitions.

For example, a school's academic year may not follow the January-to-December calendar year. The chart should follow the school's configured sequence.

Missing results must remain missing

Suppose a student has results for Term 1 and Term 3 but no recorded result for Term 2.

The system must not invent a Term 2 value merely to create a smooth line.

Depending on the charting approach, the line should break at the missing value or the visualisation should otherwise make the missing information clear.

The interface should not imply that the platform knows what happened during a period for which it has no data.

Comparable measures are required

A line chart can visually suggest that every point represents the same measure.

Therefore, Optimus Orien must avoid drawing a continuous trend when the underlying results are not meaningfully comparable.

If the grading system or assessment definition changes, the chart should show the change in context or separate the incompatible periods.

5.3 Tables

A table presents information in rows and columns.

Tables are particularly useful when users need exact values, multiple attributes, or detailed records.

For example, a teacher may need to see each student's assessment marks, attendance percentage, and current review status.

A table allows the teacher to inspect individual records more precisely than a chart usually can.

When Optimus Orien should use tables

Tables are appropriate for assessment results, student lists, homework records, review flags, intervention records, and detailed school or branch comparisons.

They are also useful when users need sorting, filtering, pagination, or controlled export.

Table design principles

Column names should be understandable.

Numbers should use consistent formatting.

Percentages should be labelled as percentages rather than displayed as unexplained decimal values.

Dates should use a clear format appropriate to the user's locale and school settings.

The table should make it possible to distinguish missing values from zero.

Where a value depends on a selected academic period, the table should make that context clear.

The developer should also avoid presenting too many columns at once. Important information should remain visible, while less frequently needed details may be available through a record's detail view.

Tables must support accessible keyboard navigation and readable layouts on smaller screens.

5.4 Summary Cards

A summary card displays one important indicator with enough context to understand it.

For example:

Attendance

91%

Selected reporting period

Previous comparable period: 88%

Change: +3 percentage points

A summary card allows a user to understand an important value quickly.

Appropriate uses

Optimus Orien may use summary cards for current attendance, a clearly defined academic summary, homework completion, the number of open review flags, or the number of branches within the user's scope.

However, a card should not contain a number without explaining what it represents.

A card displaying "78" is ambiguous. It could represent marks, a percentage, a number of students, or a rating.

The card should display the indicator's name, unit, applicable period, and any relevant comparison.

Avoid meaningless summary cards

A dashboard should not contain dozens of cards simply because they look modern.

Every card occupies attention.

The developer should prioritise indicators that help the user understand the situation or decide what to investigate next.

5.5 Progress Indicators

A progress indicator shows the current value relative to a meaningful scale or target.

For example, a progress indicator may show homework completion at 80%.

It may also show attendance against a school-defined target if the target has a clear meaning.

The developer must distinguish between a measured value and a target.

A student's attendance of 90% is an observed indicator. A school target of 95% is a policy or goal. These are different kinds of information.

The interface should not imply that a target is a universal educational standard when it is merely a local policy.

Progress indicators should also make incomplete data visible. An attendance indicator calculated from insufficient records should not appear as a fully reliable result.

5.6 Stacked Bar Charts

A stacked bar chart divides each bar into component parts.

For example, a school dashboard might show the number of students whose review status is open, resolved, or awaiting more information.

A stacked bar can show the total and the composition of the total at the same time.

However, stacked bars become difficult to interpret when there are many categories or when users need to compare small components across groups.

Optimus Orien should use them only when the component relationships matter and the categories are clearly defined.

The system must not combine incompatible categories merely to produce an attractive visual.

For example, academic performance, attendance, and skill ratings should not be stacked together as if they were parts of one total. They are separate measures, not components of a shared quantity.

5.7 Scatter Plots

A scatter plot displays the relationship between two numerical variables.

For example, a school may want to explore whether students with higher attendance percentages also tend to have higher Mathematics results.

Each point represents an observation, such as one student's attendance and Mathematics result.

A scatter plot can help reveal patterns, clusters, and unusual observations.

However, the presence of a relationship does not establish causation.

If students with higher attendance also have higher marks, the chart does not prove that attendance alone caused the difference.

Other factors may be involved.

Scatter plots should therefore be used only when the comparison has a clear purpose and the data is sufficiently reliable.

They should not be part of the default dashboard simply because the chart type is available.

The system must also protect student privacy. In small classes, individual points may reveal a student's identity even when names are omitted. Access and aggregation rules must account for this risk.

5.8 Maps

Maps can help organisation administrators understand geographic patterns across branches or schools.

Optimus Orien's organisation-level view may use a map to display the locations of branches and selected school-level indicators.

A branch map may include the branch name, principal, number of students, number of teachers, attendance indicator, and school health indicator, subject to the administrator's permissions.

A geographic view can help the user understand where branches are located and how indicators differ across locations.

However, a map should not be treated as proof that a geographic location caused a particular educational outcome.

A map showing lower attendance in one district does not explain why attendance is lower.

The platform should also distinguish between the physical location of a school and the geographic area from which its students come.

These are different data points.

Map requirements

The map should use reliable location records and clearly distinguish missing location information from a school that does not exist.

It should support searching, zooming, panning, and appropriate filters where those capabilities are part of the product requirements.

Users should be able to open an authorised branch detail view from a map marker.

The system must not expose private student locations or other sensitive geographic information on public pages.

Maps should supplement the underlying data, not replace accessible tables or other ways of exploring it.

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6. Choosing the Correct Chart

The following table gives the developer a practical starting point.

User's question	Recommended presentation	Reason
How did attendance change over the year?	Line chart	Shows change over time.
Which subjects have the highest or lowest averages?	Bar chart	Makes category comparisons clear.
What exact marks did each student receive?	Table	Preserves precise values and details.
How many review flags are open?	Summary card or bar chart	Makes the current count easy to identify.
What proportion of homework was completed?	Percentage card or progress indicator	Communicates a defined proportion.
How are review statuses distributed across branches?	Stacked bar chart or grouped bars	Shows the composition of defined categories.
Do attendance and marks appear related?	Scatter plot	Shows the relationship between two numerical measures.
Where are the organisation's branches located?	Map with a supporting list or table	Shows geographic distribution.
Why did a review flag appear?	Evidence panel with relevant records	Explains the actual rule and supporting data.

This table is a guide, not a rigid law.

The developer should consider the data type, the user's question, the number of categories, the reliability of the data, and the consequences of misunderstanding the result.

A simple table is often better than an elaborate chart.

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7. Understanding Axes, Scales, and Units

Charts can mislead users even when the underlying values are correct.

The developer must understand how axes and scales affect interpretation.

7.1 The Horizontal Axis

In many charts, the horizontal axis represents categories or time.

For example, a line chart might place reporting periods along the horizontal axis.

The labels must be readable and appear in the correct order.

The platform must not assume that all schools have the same academic calendar.

7.2 The Vertical Axis

The vertical axis represents the numerical value being measured.

It may represent marks, percentages, student counts, attendance rates, or another defined measure.

The axis label should explain the unit.

A chart displaying attendance should not have an unexplained axis that could be mistaken for academic marks.

7.3 Percentage Scales

Percentages generally range from 0% to 100% when they represent a proportion with a maximum of 100%.

The chart must not display values outside the valid range unless the measure has a different defined meaning.

For a conventional bar chart of percentages, starting the vertical axis at zero generally provides a fair comparison.

A line chart may use a narrower scale when the purpose is to examine small changes, but the scale must be clearly visible so users do not mistake a small change for a dramatic one.

7.4 Counts Versus Percentages

A count and a percentage answer different questions.

Suppose School A has 40 students requiring review out of 1,000 students. School B has 12 students requiring review out of 100 students.

School A has the larger count.

School B has the larger proportion.

School A's proportion is 4%.

School B's proportion is 12%.

If an organisation administrator sees only counts, School A may appear to have the more serious situation. If the administrator sees only percentages, they may overlook the difference in the number of students affected.

For organisation-wide monitoring, the dashboard may need to show both the count and the proportion, clearly labelled.

The appropriate choice depends on the decision the user needs to make.

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8. Colour Should Communicate Meaning, Not Decoration

Colour can help people recognise categories and patterns quickly.

However, colour must be used consistently and must not carry meaning that is impossible to understand without seeing the colour itself.

Optimus Orien should use a controlled colour system.

For example, a school may configure a review indicator as green, yellow, or red to communicate defined school-health categories.

The meaning of each category must be documented.

The platform should not automatically assume that every indicator uses the same thresholds.

A school's attendance category may use one set of rules, while an academic review category uses another.

The colour should be generated from the applicable indicator's configuration.

8.1 Green, Yellow, and Red

If Optimus Orien uses a green/yellow/red system, the interface must define what each colour means in that particular context.

For example, green may indicate that a measure is within the configured range, yellow may indicate that it warrants attention, and red may indicate that it meets a more serious review condition.

These meanings are examples, not universal rules.

The system must not use red simply to make a chart look dramatic.

Nor should a red category automatically imply that a student is a problem.

A colour should communicate the status of an indicator, not the worth of a person.

8.2 Never Use Colour Alone

A user may have colour-vision differences, use a screen reader, or view the platform on a display where colours are difficult to distinguish.

The system should pair colour with a text label, icon, pattern, or another accessible indicator.

For example, a red status should also display text such as "Review required" or the relevant configured status.

The user should not have to guess what a colour means.

8.3 Use Colour Consistently

If red means "requires review" in one part of the platform and "excellent performance" in another, users may become confused.

The design system should define the meaning of semantic colours and use them consistently where the meaning is the same.

Different colours may be used to distinguish separate subjects or categories, but the system should avoid introducing too many colours without a clear purpose.

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9. Avoiding Misleading Visualisations

A chart can be technically correct and still create a false impression.

Optimus Orien must treat accurate visual communication as a product requirement.

9.1 Do Not Exaggerate Small Differences

Suppose two branches have attendance rates of 92% and 93%.

The difference is one percentage point.

A chart with a truncated axis may make the difference appear enormous.

For a standard bar chart, the numerical scale should generally begin at zero. If another scale is necessary for a specific analytical purpose, the chart must make that choice clear.

9.2 Do Not Hide the Denominator

A percentage is easier to interpret when the user knows how many observations contributed to it.

Suppose one class has a homework completion rate of 100% based on two assignments, while another has 94% based on 500 assignments.

The first rate is numerically higher, but the amount of supporting information differs substantially.

The dashboard should provide the relevant counts where they materially affect interpretation.

9.3 Do Not Compare Incompatible Groups

A school's average result should not be compared directly with another school's average if the measures are not sufficiently comparable.

Differences in curricula, assessment structures, grading rules, reporting periods, or student populations may affect the interpretation.

The platform should provide the comparison context and withhold misleading comparisons when necessary.

9.4 Do Not Connect Unrelated Data Points

A line chart implies an ordered progression.

The developer must not connect points merely because several values exist.

For example, connecting the Mathematics result from one academic year to an unrelated English result from another year may suggest a trend that does not exist.

The system should use the correct subject and reporting-period definitions.

9.5 Do Not Treat Correlation as Causation

Two indicators may change together without one causing the other.

A chart may show that attendance and academic results both declined during a particular period.

It cannot, by itself, prove that one decline caused the other.

The interface should describe the observed pattern rather than automatically assigning a cause.

9.6 Do Not Make Missing Values Look Like Zero

If a school has not entered data for a period, the chart should show a gap, unavailable state, or another appropriate indication.

The system must not silently plot zero.

Zero is a value. Missing information is a data-quality condition.

9.7 Do Not Use Decorative Three-Dimensional Charts

Three-dimensional bars and pie charts can make values harder to compare because perspective distorts apparent size.

Optimus Orien should prefer clear, two-dimensional visualisations.

The purpose is to help users understand the data, not to create a visually impressive but less accurate representation.

---

10. Accessibility and Cognitive Load

Cognitive load refers to the mental effort required to process information.

A dashboard with too many charts, colours, labels, and controls places unnecessary demands on the user.

Optimus Orien should reduce that burden.

The interface should help users understand the most important information first, while allowing them to open detailed views when needed.

10.1 Clear Titles

Every chart should have a title that describes the information being displayed.

"Attendance Trend Across Reporting Periods" is more informative than "Chart 4".

The title should make it clear what the chart measures and, where relevant, which students, school, or reporting period it covers.

10.2 Clear Labels

Axes, units, legends, and categories should be labelled.

A chart displaying 78 should identify whether the value represents 78 students, 78 marks, or 78%.

If a chart contains several series, its legend should explain what each series represents.

10.3 Tooltips

A tooltip is a small information box that appears when a user points to or focuses on an element.

For example, a user may hover over a point on an attendance chart and see:

Reporting period: Term 2.

Attendance: 92%.

Present: 46 eligible days recorded as present.

Eligible attendance days: 50.

The actual content should depend on the available data and the user's permissions.

Tooltips should supplement visible information rather than contain essential explanations that are unavailable anywhere else.

They should also be accessible through keyboard interaction where appropriate.

10.4 Readable Text

The platform should use readable font sizes, clear headings, sufficient spacing, and strong contrast between text and its background.

Small labels should not become unreadable merely because the developer wants to fit more information into one chart.

The interface should work on common desktop and mobile screen sizes, even if some complex analytical views are primarily designed for desktop use.

10.5 Accessible Alternatives

Charts should have appropriate accessible names and textual summaries.

Users who cannot interpret a visual chart should still be able to understand its key message.

Where appropriate, a chart should have a supporting data table or another accessible representation.

The interface must not rely entirely on colour, shape, or mouse interaction to communicate important information.

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11. The Organisation Administrator Dashboard

The organisation administrator oversees the schools or branches within their authorised scope.

This dashboard should provide a broad overview while allowing authorised users to investigate specific branches.

The first screen should prioritise a small number of meaningful organisation-level indicators.

These may include the number of active branches, total enrolled students, total teachers, an appropriately calculated attendance indicator, a defined school-health summary, and the number of open review flags across the organisation.

The exact indicators should reflect the final product requirements and the quality of available data.

Each indicator should have a clear definition and a visible reporting period.

11.1 Organisation-Wide Comparisons

The dashboard may display branch comparisons using bar charts and tables.

For example, the administrator may compare attendance rates across branches or examine subject-level performance summaries.

The comparison must use a consistent, documented calculation method.

If a branch has incomplete data, the dashboard should disclose that limitation.

It must not silently rank branches using values calculated from incompatible periods.

11.2 Branch Map

The organisation administrator's map may show each branch as a marker.

Selecting a marker should reveal authorised branch information, such as branch name, principal, number of students, number of teachers, attendance indicator, and school-health status.

The map should support the navigation and filtering functions established in the Optimus Orien product requirements.

It should not expose private student-level information.

11.3 Drill-Down Navigation

Drill-down means moving from a broad summary to a more detailed view.

For example, the administrator selects a branch, opens its school dashboard, and then inspects a particular subject or class.

The information should become more detailed at each step without changing the meaning of the original indicator.

Access permissions must be enforced at every level.

A user must not obtain student-level access merely because the interface provides a drill-down button.

---

12. The Principal Dashboard

The principal's dashboard should focus on the educational situation within the principal's authorised school or campus.

It should help the principal understand where attention may be needed.

The dashboard may display academic summaries by subject, class-level comparisons, attendance trends, homework completion, student review flags, and intervention follow-up status.

The exact layout should prioritise the school's most important monitoring tasks.

For example, a principal may first see attendance across classes, then open a subject comparison to inspect differences in academic results.

The principal should be able to move from a summary to the supporting data without losing the reporting-period context.

The dashboard must not encourage a principal to treat a single chart as a complete explanation of a school's educational performance.

It should show the evidence needed to investigate a pattern.

For example, a class with a lower average may warrant examination, but the interface should not automatically conclude that the teacher is ineffective.

The difference may reflect many factors, including assessment structure, student intake, missing records, or other circumstances.

The dashboard should make patterns visible without assigning unsupported blame.

---

13. The Teacher Dashboard

Teachers need a more focused dashboard than organisation administrators or principals.

A teacher should be able to select an assigned class, subject, and reporting period and inspect the relevant student records.

The teacher dashboard may show assessment summaries, class averages, individual student results, attendance information where authorised, homework completion, participation and skill ratings, and review flags relevant to the teacher's responsibilities.

The teacher should be able to move from a class-level summary to an individual student record.

The interface should make it clear which students and records are within the teacher's authorised scope.

The platform should not expose other teachers' restricted performance records simply because those records exist in the same school.

A teacher's dashboard should support practical educational tasks rather than replicate every feature of the principal's dashboard.

---

14. The Student and Parent Dashboards

Students and parents require understandable information about progress.

Their dashboards should avoid excessive technical terminology and unnecessary complexity.

The interface should show relevant academic results, attendance, homework records, skill ratings where appropriate, and historical progress.

Where a comparison with the class average is permitted and appropriate, the platform should explain what the comparison means.

It should not use a class comparison to shame or humiliate an individual student.

A student who sees a lower result should be able to understand the actual result and relevant context rather than being reduced to a rank or colour.

A parent should be able to identify the reporting period and distinguish between recorded results and calculated indicators.

The interface should not automatically display confidential staff notes or every internal review flag.

Student and parent views must follow the school's visibility policies and the platform's permission model.

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15. Public Leaderboards and Rankings

Optimus Orien has considered public school leaderboards and school-level health indicators.

These features require particular care because rankings can influence how people perceive schools.

A ranking is a relative position within a defined comparison group.

For example, a school ranked third among 20 schools is third within that particular ranking method and group.

It does not necessarily mean that the school provides the third-best education in every respect.

15.1 Define the Ranking Method

Every ranking must use a clearly defined method.

The platform should specify which indicators contribute to the ranking, how they are calculated, which schools are eligible, what period is covered, and how missing data is handled.

If the system uses a school-health indicator, the formula and meaning of that indicator must be documented.

The platform must not combine unrelated indicators using arbitrary weights simply to generate a leaderboard.

15.2 Compare Suitable Schools

A fair comparison may require grouping schools according to relevant characteristics or providing clear contextual information.

Schools may differ in programmes, assessment systems, student populations, and available data.

The platform must not automatically assume that every school is directly comparable.

Where valid comparisons cannot be established, the ranking should be restricted, qualified, or withheld.

15.3 Protect Student Privacy

Public rankings must not reveal individual student marks, attendance, skill ratings, homework records, or intervention histories.

A school-level summary should be generated in a way that protects private records.

Small groups require particular care because an aggregate can sometimes reveal information about an identifiable individual.

Public reporting should therefore be governed by explicit publication rules, privacy review, and appropriate aggregation thresholds.

15.4 Avoid False Precision

A ranking difference of one place may not represent a meaningful difference in educational quality.

The interface should avoid suggesting that every difference in rank is important.

Where useful, the platform may show the underlying indicator values and explain how the ranking was calculated.

The ranking should support understanding rather than replace it.

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16. Filtering, Sorting, and Drill-Down

A dashboard becomes more useful when users can explore relevant information without losing control of the context.

Filters allow users to restrict the data being displayed.

For example, a principal may filter by academic year, reporting period, class, subject, or review status, subject to permissions.

Sorting allows users to order a list by a particular value, such as attendance percentage or subject average.

Drill-down allows users to move from a summary to the relevant underlying records.

These functions must work consistently.

If a principal selects Term 2, the dashboard should not continue displaying a Term 1 chart beside a Term 2 summary card without clearly identifying the difference.

If a user filters a class list, the associated counts and charts should update according to the same defined filter context.

The platform should make active filters visible so users understand which records are included.

It should also provide a clear way to reset filters.

Filters must not bypass access permissions. A user must not be able to reveal unauthorised records by manipulating a filter parameter.

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17. Responsive Design

Responsive design means that the interface adjusts to different screen sizes.

Optimus Orien should be designed primarily around clear information hierarchy rather than assuming that every user has a large desktop monitor.

On a wide screen, a dashboard may display several charts in a carefully arranged grid.

On a narrower screen, those charts may move into a vertical layout.

Tables may require horizontal scrolling or a more compact record view.

Navigation should remain understandable, and important controls should not disappear outside the visible area.

The developer should avoid shrinking every chart until its labels become unreadable.

Some complex views, such as a detailed organisation map, may be more convenient on desktop. Even then, the platform should provide a usable alternative for smaller screens.

Responsive design should preserve the meaning and accessibility of the information, not merely make the page fit.

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18. Visual Design Principles for Optimus Orien

The platform should have a consistent visual identity across portals.

The design should be minimal, modern, professional, and appropriate for schools, parents, and education leaders.

The default interface should use clear surfaces, readable typography, deliberate spacing, and a controlled colour palette.

The platform should not place every available chart inside a decorative card with unnecessary shadows, borders, gradients, and animations.

Visual styling should support information hierarchy.

For example, a primary indicator may receive more emphasis than a secondary explanatory label.

A chart title should be easier to identify than an optional technical detail.

The dashboard should not force users to read long explanations before seeing the main information, but it should make definitions and calculation details available when needed.

Tooltips, help text, and explanatory panels can help users understand technical terms without overcrowding the main view.

The interface should use consistent chart styles and labels so users do not need to relearn the meaning of each element on every page.

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19. Chart Data and Calculation Architecture

The developer must separate data calculation from chart rendering.

The charting component should not independently invent its own formula for attendance, class averages, or historical change.

Instead, the platform should provide validated data and explicit metadata to the visualisation layer.

For example, an attendance chart should receive the appropriate reporting periods, calculated attendance values, completeness information, and labels.

It should not decide by itself which days count as eligible attendance opportunities.

That logic belongs in the attendance calculation service established in Part 6.

Similarly, a class-average chart should receive values calculated using the platform's defined comparison method.

The chart should display the values and communicate their meaning.

This separation has several advantages.

It helps ensure that different portals use the same calculations. It allows the calculation logic to be tested independently of the interface. It also reduces the risk that one chart displays a different result from a summary card representing the same indicator.

19.1 Required Information for a Visualisation

Each visualisation should have a clear definition of the indicator being displayed.

It should also know the selected school, class, subject, academic year, reporting period, and other applicable filters.

Where relevant, it should receive the unit, calculation method, comparison group, data-completeness status, and information about whether historical comparisons are valid.

The exact fields will depend on the chart type.

The developer should define a consistent data contract between the calculation services and the visualisation components.

19.2 Loading, Error, and Empty States

A chart must not appear as a blank box without explanation when data is unavailable.

The interface should distinguish among loading, no eligible records, incomplete information, permission restrictions, and genuine errors.

For example, if no Mathematics results exist for the selected reporting period, the interface should explain that there are no eligible results to display.

It should not display an empty chart that looks like a software failure.

If a request fails, the interface should show an appropriate error state and allow the user to retry when suitable.

19.3 Data Freshness

The dashboard should make it clear when an indicator was last updated if that information is important to the user's decision.

If a teacher has just entered a new result, the relevant summaries should update according to the platform's recalculation rules.

The interface should not continue displaying an outdated chart while implying that it represents the latest data.

Where recalculation takes time, the system should communicate that the indicator is being updated.

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20. Performance and Scalability

Optimus Orien may eventually manage many schools, branches, classes, students, and historical records.

The dashboard architecture must therefore account for performance from the beginning.

The developer should avoid loading every historical student record into the browser simply to display a small summary card.

The server should calculate or retrieve appropriately scoped results, apply authorisation, and return the information needed for the selected view.

Large tables should support pagination or another appropriate way of retrieving manageable amounts of data.

Expensive organisation-wide calculations may require caching or precomputed summaries, provided that the system has a reliable method for keeping them accurate and respecting access permissions.

A cached value must not be displayed as current if its underlying data has changed and the platform has not refreshed it according to its defined policy.

The implementation should be tested with realistic data volumes rather than only a handful of demonstration students.

Performance optimisation must not weaken access control or calculation accuracy.

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21. Recommended Dashboard Structure

The following is a proposed information hierarchy, not a requirement that every portal must use identical screens.

An organisation administrator's dashboard should begin with an organisation-level overview, followed by branch comparisons, the branch map, and access to more detailed school records.

A principal's dashboard should begin with the school's current overview, followed by class and subject comparisons, attendance trends, and students or reviews requiring attention.

A teacher's dashboard should begin with assigned classes and subjects, followed by relevant student summaries, assessment information, and review indicators.

A student dashboard should begin with the student's own current reporting-period overview, followed by subject results, attendance, homework, skills, and historical progress.

A parent dashboard should provide a clear view of the linked student's progress, attendance, homework, and other school-approved information.

In every portal, the user should be able to identify the selected period, understand the visible indicators, and open more detailed information where authorised.

The interface should prioritise a small number of important measures and reveal further details through navigation rather than displaying everything at once.

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22. Developer Implementation Requirements

The following requirements define expected software behaviour for the initial visualisation system.

They should be implemented alongside the educational calculations, access controls, and data-quality rules established in Parts 2–6.

VIS-001 — Purposeful Visualisations

Every dashboard chart must support a defined user question or task.

Acceptance test: The product documentation identifies what each chart measures, why it exists, and which user decision it supports.

VIS-002 — Appropriate Chart Selection

The platform must select chart types appropriate to the data and the user's task.

Acceptance test: Historical trends are not presented as unrelated category comparisons, and exact assessment records remain accessible in a table or equivalent detailed view.

VIS-003 — Accurate Bar Charts

Bar charts used for ordinary numerical comparisons must represent values proportionally and generally use a zero baseline.

Acceptance test: A chart comparing attendance percentages does not exaggerate small differences through an unexplained truncated bar axis.

VIS-004 — Correct Time Ordering

Time-based charts must follow the configured academic calendar and reporting-period sequence.

Acceptance test: Terms are displayed in the school's configured order rather than alphabetical order.

VIS-005 — Missing Values in Charts

Visualisations must distinguish missing values from actual zero values.

Acceptance test: A missing Term 2 result is shown as unavailable or as a gap, not automatically plotted as 0%.

VIS-006 — Comparable Historical Data

The platform must avoid presenting incompatible historical measures as a valid continuous trend.

Acceptance test: When a grading change makes historical results unsuitable for direct comparison, the chart communicates the limitation or separates the incompatible periods.

VIS-007 — Clear Chart Titles

Every chart must have an understandable title that describes the measure being displayed.

Acceptance test: A user can identify the purpose of a chart without relying on an internal component name or undocumented abbreviation.

VIS-008 — Label Units and Axes

Charts must identify their numerical units and relevant axes.

Acceptance test: A user can distinguish a percentage from a count without guessing from the numerical value.

VIS-009 — Meaningful Legends

Charts with multiple data series must explain what each series represents.

Acceptance test: A comparison chart with several series provides a visible or otherwise accessible legend identifying each one.

VIS-010 — Summary Card Context

Summary cards must display the indicator name, relevant unit, and reporting-period context where applicable.

Acceptance test: A card displaying attendance identifies it as an attendance percentage and makes its selected period clear.

VIS-011 — Count and Percentage Distinction

The platform must distinguish between counts and percentages in school and organisation comparisons.

Acceptance test: A dashboard displaying students requiring review can show both the count and proportion where both are relevant, without presenting them as the same measure.

VIS-012 — Defined Progress Targets

Progress indicators must distinguish measured values from configured targets.

Acceptance test: An attendance percentage and the school's attendance target are labelled separately.

VIS-013 — Consistent Semantic Colours

The design system must define the meaning of status colours and use them consistently where the meaning is the same.

Acceptance test: The same configured review status has the same documented meaning across permitted dashboard views.

VIS-014 — Do Not Use Colour Alone

Important statuses must include text or another accessible indicator in addition to colour.

Acceptance test: A user can identify a review status without relying solely on whether a chart element appears red, yellow, or green.

VIS-015 — Accessible Chart Information

Charts must provide appropriate accessible names and a textual summary or data alternative where needed.

Acceptance test: A user navigating with assistive technology can identify the chart's purpose and obtain its essential information without depending exclusively on visual interpretation.

VIS-016 — Accessible Tooltips

Tooltips must be usable through appropriate pointer and keyboard interactions.

Acceptance test: Essential tooltip information can be accessed through keyboard focus when the chart supports interactive data points.

VIS-017 — Clear Empty States

Charts and tables must distinguish an empty result from a loading state, permission restriction, or system error.

Acceptance test: Selecting a period with no eligible results displays a clear explanation instead of an unexplained blank chart.

VIS-018 — Clear Error States

Failed data retrieval must produce an appropriate error message and a suitable recovery option where possible.

Acceptance test: If a chart request fails, the user sees an error state rather than a misleading zero-value chart.

VIS-019 — Consistent Filter Context

Charts, summary cards, and tables within the same filtered view must use consistent applicable filters.

Acceptance test: Selecting Term 2 updates all relevant components to the selected period and prevents one component from silently displaying Term 1.

VIS-020 — Visible Active Filters

The interface must make active filters understandable and provide a suitable way to reset them.

Acceptance test: A user can identify the selected subject, class, and reporting period and return to the default view.

VIS-021 — Enforce Access During Drill-Down

Every drill-down request must enforce the user's permissions and data scope.

Acceptance test: A user cannot access an unauthorised student's record by changing a URL or manipulating a drill-down request.

VIS-022 — Correct Class and Subject Comparisons

Comparison charts must use the documented calculation methods and eligible records.

Acceptance test: The same class average displayed in a chart and a summary table agrees when both use the same definition, data scope, and reporting period.

VIS-023 — Data Completeness Disclosure

Visualisations must communicate material data-quality limitations.

Acceptance test: A school attendance chart identifies incomplete source records when their absence affects the reliability or interpretation of the indicator.

VIS-024 — Fair Comparison Context

The platform must identify or restrict comparisons that use incompatible educational contexts.

Acceptance test: Schools using incompatible assessment measures are not silently ranked together under a comparison presented as directly equivalent.

VIS-025 — Separate Chart Logic from Calculation Logic

Visualisation components must consume defined, validated indicators rather than independently implementing educational calculation formulas.

Acceptance test: Changing the approved attendance calculation updates the shared indicator source and produces consistent results across every portal using that indicator.

VIS-026 — Consistent Data Contracts

The calculation and visualisation layers must use documented data structures.

Acceptance test: The chart receives the defined value, unit, period, and relevant completeness metadata needed to display the indicator correctly.

VIS-027 — Respect Reporting Periods

Every relevant dashboard must identify the academic year or reporting period represented.

Acceptance test: A user can determine which academic period the visible values belong to without inferring it from the chart's position on the page.

VIS-028 — Branch Map Interaction

Where the branch map is implemented, authorised users must be able to inspect branch information through its markers.

Acceptance test: Selecting a branch marker opens the relevant branch information without exposing unauthorised student-level records.

VIS-029 — Map and Table Alternatives

Important branch information must remain accessible through a list, table, or equivalent non-map representation.

Acceptance test: A user can find a branch and inspect its authorised summary without having to interact with the map.

VIS-030 — Responsive Dashboards

Dashboards must adapt to supported screen sizes while preserving readable information.

Acceptance test: On a narrow screen, charts and cards reflow appropriately and do not require the user to interpret unreadable labels caused by forced compression.

VIS-031 — Accessible Tables

Tables must support readable labels and appropriate keyboard navigation.

Acceptance test: A user can navigate the relevant table controls and identify column headings without depending exclusively on a mouse.

VIS-032 — Avoid Misleading Three-Dimensional Charts

The default visualisation system should use clear two-dimensional charts.

Acceptance test: Standard academic and attendance comparisons do not use three-dimensional perspective effects that distort apparent values.

VIS-033 — Public Data Protection

Public visualisations must not reveal private student-level information.

Acceptance test: A public user cannot obtain individual student results or intervention records by interacting with a leaderboard, chart, map, or underlying request.

VIS-034 — Ranking Method Transparency

Every public school ranking must use a defined and documented calculation method.

Acceptance test: The platform can identify the indicators, reporting period, eligibility rules, and calculation method used for a published ranking.

VIS-035 — Appropriate Ranking Comparisons

The platform must apply defined eligibility and comparability rules before ranking schools.

Acceptance test: A school with incompatible or insufficient data is not silently assigned a comparable rank without following the approved ranking policy.

VIS-036 — Keep Visualisations Updated

Charts and summary indicators must reflect source-data changes according to the platform's defined recalculation and refresh rules.

Acceptance test: After an authorised result correction has been processed, the affected charts and summaries display the updated indicator rather than an outdated value.

VIS-037 — Communicate Data Freshness

Where freshness materially affects interpretation, the interface must communicate when the data was last updated or whether it is still being recalculated.

Acceptance test: A user can distinguish a current indicator from one that is waiting for an update when the distinction matters.

VIS-038 — Scalable Data Retrieval

Large dashboards must retrieve appropriately scoped information rather than unnecessarily loading all underlying records into the browser.

Acceptance test: A large student table can be navigated through the platform's approved pagination or retrieval approach without loading the entire historical dataset at once.

VIS-039 — Consistent Indicator Definitions

The same indicator must retain the same documented meaning wherever it appears.

Acceptance test: Attendance shown in two permitted dashboard views uses the same approved calculation definition when the student population and reporting period are the same.

VIS-040 — Avoid Unsupported Causal Claims

Chart titles, labels, summaries, and automatically generated explanations must not assert causal relationships unsupported by the available evidence.

Acceptance test: A chart showing attendance and marks declining together does not automatically state that one decline caused the other.

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23. Recommended Testing Strategy

The developer should test visualisations using known datasets before connecting them to live school records.

A known dataset is a small collection of records for which the correct expected result has already been calculated.

For example, a test dataset might contain five students with known Mathematics percentages and a known class average.

The developer can verify that the chart displays the expected values and that the summary card and table agree.

Testing should cover ordinary cases, missing information, zero values, incomplete reporting periods, changed filters, and historical comparisons.

It should also cover large datasets and users with different permissions.

The developer should inspect the interface on wide and narrow screens, using keyboard navigation and accessibility checks where appropriate.

Finally, visualisation tests should verify meaning, not just appearance.

A chart that renders successfully but displays the wrong reporting period has failed.

A chart that uses the correct values but exaggerates their differences has also failed.

The acceptance criteria should therefore test both the underlying data and how the interface communicates it.

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24. What Part 7 Establishes

After implementing Part 7, Optimus Orien should have a consistent foundation for presenting educational information visually.

The platform should use appropriate charts, clear labels, meaningful comparisons, accessible colours, readable tables, and useful dashboard layouts.

Users should be able to move from broad summaries to detailed records while retaining the correct reporting-period context and respecting their permissions.

The system should distinguish counts from percentages, missing information from zero, measured values from targets, and observed patterns from unsupported explanations.

The developer should also have a clear separation between calculation services and visualisation components, allowing the same validated indicators to be used consistently throughout the platform.

Most importantly, the dashboard should help users understand educational information rather than overwhelm them with it.

A successful Optimus Orien dashboard is not the one with the most charts. It is the one that helps the right person understand the right information and make a better-informed decision.

Part 8 explains how Optimus Orien should compare student performance, calculate rankings, measure school health, and present differences between schools fairly. This is particularly important because rankings and school-health indicators could influence how principals, parents, education leaders, and government organisations judge a school.

The document distinguishes what the platform can calculate objectively from what requires context and professional judgement. It also includes implementation requirements and acceptance tests for your developer.

PART 8

ACADEMIC PROGRESS, STUDENT RANKINGS, SCHOOL HEALTH INDICATORS, AND FAIR COMPARISON

Project: Optimus Orien
Document type: Educational Foundations and Software Implementation Requirements
Audience: Product Owner, Developer, and Future Development Team
Purpose: Define how Optimus Orien should measure progress, compare performance, calculate rankings, and communicate school-health indicators without misleading users or creating unfair comparisons.

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1. Why This Document Exists

In Part 6, we established how Optimus Orien should build student performance records and identify patterns that may require early intervention.

In Part 7, we explained how the platform should visualise educational data through dashboards, charts, tables, and indicators.

Part 8 addresses the next question:

How should Optimus Orien compare students, classes, subjects, and schools in a way that is mathematically accurate, educationally meaningful, and fair?

Imagine that Optimus Orien displays two schools.

School A has an average examination result of 88%.

School B has an average examination result of 74%.

At first glance, School A appears to perform better.

But suppose School A uses a grading system with relatively accessible assessments, while School B uses a different curriculum and a more demanding assessment structure.

The two percentages may not represent directly comparable levels of achievement.

Now imagine that School A has excellent examination results but declining attendance, while School B has moderate examination results, improving attendance, and substantial improvement among previously struggling students.

Which school is performing better?

There is no universally correct answer without defining what "better" means and examining the evidence.

This is why Optimus Orien must distinguish academic results, progress over time, attendance, student support, and other measures rather than treating them as interchangeable.

The platform can calculate results, differences, averages, and rankings according to explicit rules. It must not pretend that a single number completely describes the quality of a school.

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2. The Difference Between Performance, Progress, and Quality

Before building comparisons, the developer must understand three important concepts.

They are related, but they are not identical.

2.1 Performance

Performance describes the result or level recorded during a defined period.

For example, a student receives 82% in Mathematics during Term 2.

That is a measure of performance for a particular subject, assessment context, and period.

A school may also have an average examination result of 78% across an eligible group of students.

That is a school-level performance indicator, provided the calculation method and eligible group are defined.

Performance tells us what was recorded. It does not necessarily tell us how much improvement occurred or why the result was achieved.

2.2 Progress

Progress describes how a measure changes over time.

Suppose a student's Mathematics result changes from 62% in Term 1 to 74% in Term 2.

The difference is:

$74 - 62 = 12$ percentage points.

The student has a higher recorded result in the second period.

However, the comparison is meaningful only if the results are sufficiently comparable.

The assessments may have differed in difficulty or content. The school may have changed the grading rules. The student may have moved to a different programme.

Progress indicators should therefore preserve the context of the underlying results.

Progress also differs from final performance.

A student who improves from 40% to 60% has improved by 20 percentage points but still records 60% in the later period.

Another student who moves from 90% to 92% has a higher later result but a smaller change.

Neither number alone tells the complete story.

2.3 Educational Quality

Educational quality is broader than a single performance or progress indicator.

It may involve academic achievement, learning opportunities, student wellbeing, teaching quality, inclusion, safety, leadership, resources, and other factors.

Optimus Orien does not automatically measure every aspect of educational quality.

Its initial scope focuses on the indicators the platform can record and calculate reliably.

Therefore, a high academic average should not automatically be presented as proof that a school is excellent in every respect.

Likewise, a lower average should not automatically be treated as proof that the school is ineffective.

Product rule: Optimus Orien must clearly identify what it measures and avoid presenting a limited set of indicators as a complete judgement of educational quality.

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3. The Four Main Levels of Comparison

Optimus Orien should support four broad levels of comparison.

These are individual students, subjects and classes, schools or branches, and organisation-wide performance.

Each level answers different questions.

3.1 Individual Student Comparison

An individual student can be compared with their own previous results, an appropriate class average, or another valid reference group.

The most useful comparison depends on the purpose.

Comparing a student's current Mathematics result with their previous Mathematics result may help identify a change in performance.

Comparing the same result with the class average may help the teacher understand how the student's result relates to the group's results.

However, the class average is not a perfect measure of the student's ability.

The student may have different educational needs, may have recently enrolled, or may have been assessed under different circumstances.

The dashboard should therefore present comparisons as context rather than as a complete judgement about the student.

3.2 Subject and Class Comparison

A school may want to compare Mathematics and English results across classes or examine differences between reporting periods.

These comparisons require consistent definitions.

If the platform compares two classes, it should identify the subject, assessment or summary measure, reporting period, and eligible student population.

A comparison between a Mathematics average and an English average may be useful for school monitoring, but the platform should not assume that the difference directly measures the relative quality of teaching in the two subjects.

The assessments and grading structures may differ.

3.3 School and Branch Comparison

An organisation administrator may want to compare attendance, academic results, or review indicators across branches.

The system should provide a defined comparison method and identify the reporting period.

Where appropriate, it should display both counts and rates.

For example, one branch may have 30 students requiring review out of 1,000 students, while another has 12 students requiring review out of 100 students.

The first branch has the larger number of students requiring review.

The second branch has the larger proportion.

These are different findings, and both may be useful.

3.4 Organisation-Wide Comparison

An organisation-wide dashboard should summarise the schools within the administrator's authorised scope.

However, the organisation's overall figures must be calculated using clearly defined methods.

The platform must not simply average every branch's percentage unless that is the intended and documented calculation.

Averaging branch percentages can produce a different result from calculating the percentage across all eligible students or attendance opportunities.

The dashboard must identify which method it uses.

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4. Understanding Averages Before Building Rankings

Averages are among the most common calculations in school reporting.

They are also a common source of misleading comparisons.

The developer must understand the difference between a student-level average, a class average, a subject average, and an organisation-wide average.

4.1 Arithmetic Mean

The arithmetic mean is calculated by adding a set of values and dividing the total by the number of values included.

Suppose five students receive the following Mathematics results:

60%, 70%, 80%, 90%, and 100%.

The mean is:

(60 + 70 + 80 + 90 + 100) ÷ 5 = 80%.

The class average is 80% if these are the five eligible results and the platform's defined method is the arithmetic mean of student percentages.

The system must identify which students are included and how missing results are treated.

4.2 Why Averaging School Percentages Can Be Misleading

Suppose an organisation has two branches.

Branch A has 100 eligible students and an average result of 80%.

Branch B has 20 eligible students and an average result of 60%.

The simple average of the two branch averages is:

(80 + 60) ÷ 2 = 70%.

However, this calculation gives each branch equal weight, even though one branch has five times as many students.

If every student has one eligible result and the intended organisation-level indicator is the mean of all student percentages, the calculation should be:

((100 × 80) + (20 × 60)) ÷ 120

= 7,200 ÷ 120

= 60? No. The correct weighted total is 8,000 + 1,200 = 9,200.

9,200 ÷ 120 ≈ 76.67%.

The organisation-level average is therefore approximately 76.67% under these assumptions.

The difference between 70% and 76.67% arises because the calculations answer different questions.

The simple mean of branch averages treats the two branches equally.

The student-weighted mean gives each eligible student equal weight.

Neither method is universally correct for every purpose. The product must choose and label the method appropriate to the indicator.

In this example, the arithmetic is based on the assumption that each branch average represents the same kind of student-level percentage and that the underlying results are comparable.

If the branches use different assessment methods, even the weighted calculation may not support a valid educational comparison.

4.3 The Denominator Must Be Explicit

A denominator is the number or quantity by which the relevant total is divided.

For an average, it is the number of eligible values included in the calculation.

For attendance, the denominator may be the total eligible attendance opportunities.

For homework completion, it may be the number of eligible assignments.

For a school-health indicator, the denominator depends on the specific measure being calculated.

The developer must define the denominator for every rate, percentage, or average.

The system should not quietly change the denominator when data is missing or when a student joins or leaves a school.

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5. Measuring Academic Progress Fairly

Progress indicators can help schools identify improvement, decline, and stability.

However, a progress calculation must compare the right things.

5.1 Compare the Same Subject

A student's Mathematics result should not automatically be compared with their English result to calculate an academic trend.

The two subjects measure different things.

If the platform provides a combined academic summary, it must define how that summary is calculated and explain its limitations.

5.2 Compare Appropriate Periods

A result from Term 1 should normally be compared with a suitably defined later period for the same subject.

However, not all schools use identical reporting periods.

The platform must use the school's configured academic calendar rather than assuming that every school has four terms or follows the same sequence.

5.3 Understand Percentage-Point Change

Suppose a student's attendance changes from 95% to 90%.

The difference is:

90 − 95 = −5 percentage points.

The system should display a decrease of 5 percentage points.

It should not automatically call this a 5% decrease because relative percentage change is a different calculation.

5.4 Distinguish Improvement from Final Achievement

Suppose Student A improves from 45% to 65%.

Student B improves from 82% to 86%.

Student A improves by 20 percentage points.

Student B improves by 4 percentage points.

Student A shows the larger numerical change, while Student B has the higher later result.

A school may want to understand both achievement and progress.

Optimus Orien should not assume that one measure replaces the other.

5.5 Handle Different Assessment Difficulties

A result may change because an assessment was easier or harder, not because the student's underlying knowledge changed by the same amount.

The platform should avoid presenting raw score changes as definitive evidence of learning growth when assessment comparability has not been established.

Where schools use a validated method for comparing achievement across assessments, the platform may use that method if its requirements are defined and supported.

Otherwise, it should display the recorded changes without claiming more than the data supports.

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6. Student Rankings

A ranking assigns positions to students according to a defined measure.

For example, students may be ordered by their Mathematics result in a particular assessment.

Ranking is mathematically straightforward when the eligible results and sorting rule are clear.

The difficulty is deciding what the ranking means and whether it is appropriate for the intended use.

6.1 Ranking by a Specific Measure

Optimus Orien may support rankings for a clearly defined measure, such as results in a particular assessment.

For example:

Student A: 92%.

Student B: 88%.

Student C: 81%.

If higher percentages are better under the applicable assessment rules, the order is Student A, Student B, and Student C.

The ranking must identify the assessment or summary measure, subject, reporting period, and eligible comparison group.

A ranking of Mathematics results should not be presented as an overall ranking of student worth or ability.

6.2 Ranking by Overall Academic Performance

An overall academic ranking requires a carefully defined calculation.

Different subjects may use different assessment structures, maximum marks, weighting rules, and grading systems.

The platform must not simply average all raw marks because those marks may use different scales.

If the school chooses to rank students using an overall academic measure, the platform must define which subject results are included, how they are made comparable, how subject weighting works, and how missing results are treated.

The school must also understand the limitations of the measure.

If the required information is unavailable, the platform should not invent an overall ranking.

6.3 Ties

Two students may have the same result.

For example:

Student A: 88%.

Student B: 88%.

Student C: 84%.

The platform must have a documented tie-handling method.

One option is competition ranking, which assigns both students rank 1 and assigns Student C rank 3.

Another option is dense ranking, which assigns both students rank 1 and Student C rank 2.

Both methods can be mathematically valid. The platform should use a defined method and apply it consistently.

The interface should not break ties arbitrarily based on database order, record identifiers, or name sorting while presenting the resulting order as a meaningful performance distinction.

6.4 Incomplete Results

Suppose one student has completed all required assessments while another has completed only two of five.

The student with fewer results may have an apparently higher average because the missing assessments are not included.

The platform must apply the configured eligibility rules before assigning rankings.

It may exclude students with insufficient results from a particular ranking, or it may display an incomplete-results status where the ranking method supports it.

The method must be documented.

Missing assessments must not automatically become zero unless the relevant assessment rules explicitly require that treatment.

6.5 Student Privacy and Public Rankings

Optimus Orien may display authorised rankings inside private school portals.

However, public student-level rankings create privacy, safeguarding, and reputational risks.

The initial public homepage should focus on approved school-level information rather than identifiable student-level marks or rankings.

Private student records must not become public merely because a school wants to celebrate high performers.

Any future public student ranking feature would require a separate product decision, privacy assessment, appropriate legal review, and explicit publication rules.

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7. School Rankings

A school ranking orders schools according to a defined measure or composite indicator.

For example, a ranking might order schools by their attendance percentage during a particular academic year.

Such a ranking is only meaningful if the comparison group and calculation method are appropriate.

7.1 A Ranking Is Not a Universal Judgement

Suppose School A ranks first in attendance among a defined group of schools.

This means it has the highest attendance indicator under the specified method and period.

It does not establish that School A is the best school overall.

The ranking does not automatically measure curriculum quality, student wellbeing, teaching quality, inclusion, facilities, or every other aspect of education.

The platform should communicate exactly what the ranking represents.

7.2 Define Eligible Schools

Before ranking schools, the platform must determine which schools qualify for inclusion.

The eligibility rules may depend on the availability of required data, the reporting period, the selected education model, and the indicator being ranked.

For example, a school without sufficient attendance records should not automatically be ranked alongside schools with complete records if the ranking is intended to compare attendance reliably.

The platform should explain exclusions where appropriate.

7.3 Compare Suitable Educational Contexts

Different schools may use different curricula, assessment methods, grading systems, and reporting periods.

A ranking based on raw examination percentages may be unsuitable when the underlying assessments are not comparable.

Where a valid comparison method is unavailable, the platform should restrict the ranking to an appropriate group or avoid publishing the ranking.

The developer must not solve an educational comparability problem by simply converting every result to a percentage.

A percentage normalises the scale of a result, but it does not automatically make different assessments equivalent.

7.4 Show the Underlying Indicator

Whenever possible, a school ranking should display the underlying measure alongside the position.

For example:

Rank: 4 of 25 eligible schools.

Attendance: 94%.

Reporting period: Academic year 2026–2027.

Comparison group: Schools meeting the configured eligibility requirements.

The exact details depend on the ranking.

The user should be able to understand what the position means without treating the rank as a complete assessment of the school.

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8. What Is a School Health Indicator?

Optimus Orien has considered a school-level health indicator or School Health Score.

This indicator could help school leaders understand whether the school's recorded performance measures fall within defined ranges or whether some measures require attention.

However, the term "school health" can be interpreted broadly.

It must not imply that the platform measures every aspect of a school's educational quality or wellbeing.

For the initial product, the indicator should be understood as a **defined summary of selected, measurable school-performance information**.

It is not a medical measure, a complete inspection result, or proof of the school's overall quality.

8.1 Define the Purpose

Before implementing a School Health Score, the product owner must decide what question the score is intended to answer.

For example, it might summarise selected indicators related to academic performance, attendance, and homework completion.

Alternatively, it might represent a rule-based status indicating whether selected school-level measures meet their configured thresholds.

These are different designs.

The platform should not mix them without defining the method.

8.2 Option A: A Composite Score

A composite score combines several indicators into one calculated value.

For example, a school might want to combine attendance, academic performance, and homework completion into a single numerical indicator.

However, these measures have different meanings.

Attendance measures presence. Academic results measure performance on assessments. Homework completion measures recorded completion of assigned work.

A composite score requires an explicit method for converting these different measures into a defined combined indicator.

The product owner would need to decide which indicators are included, how each indicator is scaled, whether some indicators receive more weight, how missing information is handled, and what the final score is intended to represent.

Any weights or thresholds must have a documented rationale. They must not be invented simply to make the dashboard produce one convenient number.

A composite score can hide important differences.

Two schools may receive the same score while having very different underlying situations.

For example, one school might have strong academic results and weaker attendance, while another has moderate academic results and strong attendance.

The composite score may obscure these differences.

If Optimus Orien introduces a composite School Health Score, the dashboard must display its underlying indicators and explain the calculation method.

The score must not be presented as a validated measure of educational quality unless it has been appropriately developed and validated for that purpose.

8.3 Option B: A Rule-Based School Status

An alternative is to use a status derived from explicit conditions rather than combining everything into one numerical score.

For example, a school may be classified as requiring review if a configured attendance rule is triggered or if the school has a significant data-completeness problem.

The status should identify the specific condition that triggered it.

This method may be easier to explain and audit because the user can see the reason for the status.

However, a rule-based status also requires careful design.

The platform must define which rules apply, how they are configured, how multiple conditions interact, and what each status means.

A status must not imply that the school is failing simply because a particular indicator requires review.

8.4 Recommended Initial Approach

For the initial version of Optimus Orien, the safest approach is to display the principal school-level indicators separately and use clearly defined review statuses where necessary.

A single composite score should not be treated as an essential requirement until its purpose, calculation method, and limitations have been defined.

The platform can still display a school-health section with separate attendance, academic, homework, and data-completeness indicators.

This provides useful information without pretending that unrelated measures can be reduced to one universally meaningful number.

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9. Designing a Transparent School Health Framework

If Optimus Orien uses a School Health Score or a green/yellow/red school-health status, the framework must be documented.

The following questions must be answered before implementation.

What does the indicator measure?

The platform must specify the exact educational or operational dimensions represented.

Which source records are used?

The platform must identify whether the indicator uses attendance records, academic results, homework completion, or other approved measures.

What is the calculation method?

The method must be explicit, reproducible, and testable.

Which reporting period is covered?

A school-health indicator for one term must not be confused with an indicator covering an entire academic year.

How is missing data handled?

The system must distinguish between poor performance and insufficient information.

How are thresholds determined?

The thresholds must come from an approved configuration or documented framework rather than arbitrary numbers embedded in the code.

Who can change the rules?

Only authorised users should be able to modify the applicable settings.

How are changes recorded?

The system should retain the applicable configuration version so historical results remain understandable.

What does each status mean?

The platform must explain whether a status indicates a threshold was met, information is incomplete, or a human review is required.

These definitions should exist before the score becomes visible to users.

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10. Fair Comparison and Context

A fair comparison does not mean that every student or school must receive the same result.

It means that the comparison uses an appropriate method and provides enough context to interpret the result.

10.1 Different Curricula

Schools may follow different educational programmes.

One school may use a local national curriculum, while another uses an international programme.

Their assessments may differ in content, structure, grading rules, and reporting periods.

The platform must preserve the programme context and avoid assuming that raw percentages establish equivalence.

If a comparison method is available and appropriate, the system may use it according to its defined requirements.

Otherwise, it should present the results separately or clearly qualify the comparison.

10.2 Different Student Populations

Schools may enrol students with different educational histories, needs, and circumstances.

The platform's initial data may not include enough information to explain these differences.

It should not automatically assume that the school with the highest average has the most effective teaching or that a school with lower averages is ineffective.

Comparisons should be presented as evidence about the recorded indicators, not as automatic judgements about staff or students.

10.3 Different Reporting Periods

A comparison between one school's Term 1 and another school's full academic year is not a fair comparison of the same period.

The platform should ensure that the selected reporting periods are appropriate for the indicator.

If the periods differ, the interface should identify the difference or prevent the comparison.

10.4 Data Completeness

A school with incomplete data may appear better or worse than it really is.

For example, a school may have recorded results only for students who completed an assessment.

If students without results are excluded, the average may not represent the entire eligible student population.

The platform must disclose relevant completeness limitations and apply the approved eligibility rules consistently.

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11. Ranking Versus Benchmarking

Ranking and benchmarking are related, but they answer different questions.

A ranking tells the user where a school or student stands relative to other members of a defined comparison group.

Benchmarking compares a measure with a reference value, target, or defined standard.

For example, a school may rank fourth among 25 eligible schools for attendance.

The same school may also have attendance of 91%, compared with its own configured target of 95%.

The ranking describes relative position.

The target comparison describes the difference between the observed value and the target.

A school can rank highly while still missing its target if the other schools have similar or lower results.

It can also rank lower while meeting its own target.

Optimus Orien should support these concepts separately.

The interface must not use ranking position as a substitute for a defined target or standard.

11.1 Types of Benchmarks

A benchmark may be an internal school target, an organisation-wide reference, a historical baseline, or an externally established standard.

These are not interchangeable.

An internal target is selected by the school or organisation.

A historical baseline describes a previous result.

An external standard comes from an identifiable external framework or authority and must be verified before it is represented as official.

The system should identify the source and meaning of the benchmark.

It must not label an arbitrary administrator-entered target as a national or international standard.

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12. Student Improvement and Recognition

Rankings often focus on who achieved the highest result.

However, educational progress can also be important.

Optimus Orien should be able to display improvement indicators where the underlying data supports valid comparisons.

For example, a school may want to identify students whose Mathematics results improved across comparable reporting periods.

A principal may want to identify classes with improving attendance.

An organisation administrator may want to examine branches that have improved a particular indicator over the academic year.

These views can provide a broader perspective than rankings based only on current achievement.

However, improvement must be calculated consistently.

The system should not reward a large percentage-point change without considering whether the underlying results are comparable.

It should also avoid presenting a small numerical improvement as definitive proof that a particular intervention caused the change.

12.1 Improvement Is Not the Same as Final Achievement

Suppose Student A improves from 40% to 60%.

Student B improves from 80% to 85%.

Student A improves by 20 percentage points.

Student B improves by 5 percentage points.

The first student shows a larger numerical change. The second has the higher final result.

Both facts may be relevant.

The platform should not automatically combine them into a single ranking unless a defined and appropriate method has been approved.

12.2 Recognising Improvement Without Shaming Students

Optimus Orien should use language that recognises progress without labelling other students as failures.

For example, an authorised dashboard might show that a student's recorded Mathematics result improved by 12 percentage points across comparable periods.

It should not publicly identify lower-performing students or turn private intervention flags into public rankings.

Recognition features must follow the platform's privacy rules and the school's policies.

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13. How the Main Portals Should Use Comparisons

Different users need different comparison tools.

13.1 Organisation Administrator

The organisation administrator may compare branch-level indicators, inspect organisation-wide trends, and identify branches requiring review.

The dashboard should provide the relevant counts, rates, comparison periods, and data-completeness information.

The administrator should be able to move from an organisation-wide summary to a branch view when authorised.

13.2 Principal

The principal may compare classes, subjects, and reporting periods within the authorised school.

The principal should be able to identify areas that may need further investigation.

The interface should not automatically treat a lower class average as evidence of poor teaching.

It should make the relevant assessment and student-population context available.

13.3 Teacher

Teachers may compare assigned students' results with appropriate class summaries and historical results.

The platform should help teachers understand student progress without creating unnecessary public competition.

Teachers should not receive access to restricted student records outside their authorised scope.

13.4 Student

Students may view their own results and appropriate historical comparisons.

Where class averages are displayed, the interface should explain what they represent.

The platform should not automatically show every student's identity and individual result in a public or unrestricted ranking.

13.5 Parent or Guardian

Parents may view their linked student's progress, relevant targets, and historical changes.

They should be able to understand what the figures mean without needing to know the technical details of the calculation.

Parents should not be able to access other families' records.

13.6 Public Visitors

Public visitors may view approved school-level information and leaderboards.

The public interface must use only data that has been approved for publication.

It must not expose private student records or confidential intervention information.

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14. Handling Rankings When Data Changes

Rankings can change when new results are entered, data is corrected, a school joins the comparison group, or the calculation method changes.

Optimus Orien must define how these changes affect published rankings.

Suppose a school initially ranks fifth because some other schools have not yet submitted their data.

When those schools submit results, the ranking may change.

The system should not imply that the original ranking represented a complete comparison if important data was missing.

The platform should define whether rankings are provisional or final for the selected reporting period.

If a published ranking is recalculated after an authorised correction, the system should follow the configured update policy and preserve the information needed to understand the change.

Where historical rankings are retained, the platform should distinguish a historical snapshot from a newly recalculated ranking.

The developer must not silently rewrite historical published reports if doing so would misrepresent what was previously reported.

The applicable retention and publication rules should determine how historical versions are stored and presented.

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15. Recommended Initial Product Decisions

The following recommendations are intended to keep Optimus Orien transparent and manageable during its initial development.

First, prioritise clearly defined academic, attendance, and homework indicators before building a composite school-health score.

Second, allow rankings only where the measure, comparison group, eligibility rules, and reporting period are explicitly defined.

Third, keep student-level rankings private and do not include identifiable student information on public leaderboards.

Fourth, display both counts and percentages when they answer materially different questions.

Fifth, distinguish progress from current achievement.

Sixth, use configured targets and thresholds rather than assuming that one value applies to every country, programme, or school.

Seventh, preserve historical calculation and configuration context.

Eighth, provide explanations for missing data and unsuitable comparisons.

Ninth, treat school-health indicators as summaries of selected measures rather than complete judgements of educational quality.

Tenth, do not introduce an overall student or school score simply because it makes the interface look simpler.

These decisions are consistent with Optimus Orien's purpose: providing transparent school performance intelligence that supports informed decisions.

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16. Developer Implementation Requirements

The following requirements translate the concepts in this document into testable software behaviour.

They should be implemented alongside the calculation, access-control, and visualisation requirements established in Parts 2–7.

COMP-001 — Define Every Comparison

Every comparison must identify the measure, eligible population, reporting period, and calculation method.

Acceptance test: An authorised user can determine which students, classes, or schools contributed to a displayed comparison and which method was used.

COMP-002 — Preserve the Difference Between Performance and Progress

The system must represent current results and changes over time as distinct indicators.

Acceptance test: A student with a higher final result but a smaller improvement is not automatically described as having made less progress than a student whose result increased by a larger amount.

COMP-003 — Define Average Calculations

Every average must use a documented formula and a defined set of eligible values.

Acceptance test: Given a known set of five student results, the system calculates the arithmetic mean correctly and identifies which results were included.

COMP-004 — Define Organisation-Level Aggregation

Organisation-wide indicators must use the approved aggregation method.

Acceptance test: The platform distinguishes a simple mean of branch averages from a student-weighted average and produces the correct result for each defined calculation.

COMP-005 — Define Attendance Denominators

Attendance rates must use the configured eligible attendance opportunities.

Acceptance test: The attendance calculation excludes periods outside the student's applicable enrolment or another explicitly defined eligibility rule.

COMP-006 — Distinguish Counts from Rates

The system must retain and display counts and percentages as separate measures.

Acceptance test: A branch comparison can display both the number of students requiring review and the proportion of eligible students represented by that count.

COMP-007 — Calculate Percentage-Point Changes

Changes between percentages must be calculated and labelled correctly.

Acceptance test: A change from 95% to 90% is shown as a decrease of 5 percentage points.

COMP-008 — Restrict Incompatible Comparisons

The platform must apply defined comparability rules before directly comparing educational results.

Acceptance test: Results from incompatible grading contexts are not presented as directly equivalent unless an approved comparison method supports that use.

COMP-009 — Preserve Educational Context

Comparison records must retain the relevant school, programme, subject, assessment, and reporting-period context.

Acceptance test: A user can identify the context of a historical result used in a comparison.

COMP-010 — Define Student Ranking Measures

Every student ranking must identify the measure used to determine the order.

Acceptance test: A ranking based on Mathematics results is labelled as a Mathematics ranking for the specified assessment or reporting period.

COMP-011 — Define Overall Academic Rankings

An overall academic ranking must use an explicitly configured and documented method.

Acceptance test: The system does not rank students by averaging unrelated raw marks with different maximum values or assessment meanings.

COMP-012 — Handle Ties Consistently

The system must use a documented tie-handling method.

Acceptance test: Students with identical ranking values receive the configured tie treatment, and the system does not create arbitrary distinctions based on database ordering.

COMP-013 — Handle Incomplete Ranking Data

Ranking eligibility must account for missing or incomplete results.

Acceptance test: A student with insufficient eligible results is handled according to the documented policy rather than silently receiving an artificially high or low rank.

COMP-014 — Protect Student Rankings

Student-level ranking data must follow the platform's access and publication policies.

Acceptance test: An unauthorised public visitor cannot retrieve identifiable student rankings or underlying private assessment results.

COMP-015 — Define School Ranking Eligibility

Every school ranking must specify which schools qualify for inclusion.

Acceptance test: A school with insufficient data is handled according to the documented eligibility policy rather than automatically receiving a misleading rank.

COMP-016 — Define School Ranking Method

Every school ranking must use a documented indicator and calculation method.

Acceptance test: An authorised user can identify the measure, reporting period, eligible comparison group, and method used to produce a school's position.

COMP-017 — Display Underlying Ranking Values

Where appropriate, rankings must display the underlying indicator alongside the position.

Acceptance test: A school ranking shows the relevant indicator and reporting period so the user does not have to infer what the rank represents.

COMP-018 — Avoid Unsupported Quality Claims

The platform must not present one performance indicator as proof of complete educational quality.

Acceptance test: A high attendance or academic average is not automatically labelled as proof that the school is excellent in every aspect of education.

COMP-019 — Define School Health Indicators

Every School Health Score or status must have a documented purpose and definition before implementation.

Acceptance test: The product configuration identifies what the indicator represents and which source measures contribute to it.

COMP-020 — Document Composite Score Formulas

If a composite School Health Score is implemented, its formula, input indicators, weighting rules, and limitations must be documented.

Acceptance test: An authorised user can inspect the documented method and identify the source measures contributing to the score.

COMP-021 — Do Not Invent Weights

The platform must not silently assign arbitrary weights to academic results, attendance, homework completion, or other measures.

Acceptance test: A composite indicator cannot be activated with undocumented or undefined weighting rules.

COMP-022 — Prefer Separate Indicators When Appropriate

The initial dashboard must support separate display of major school-level indicators without requiring a composite score.

Acceptance test: Attendance, academic performance, and homework completion can be displayed independently even if no combined School Health Score is configured.

COMP-023 — Define Status Thresholds

Green, yellow, red, or other school-health statuses must use explicit, configurable rules.

Acceptance test: A status is generated from the approved configuration rather than an undocumented threshold embedded in the user interface.

COMP-024 — Distinguish Poor Results from Missing Data

School-health calculations must distinguish an unfavourable measured result from insufficient information.

Acceptance test: A branch with missing attendance records is not automatically treated as having zero attendance or assigned an unfavourable status solely because the records are absent.

COMP-025 — Version Ranking and Health Rules

The platform must preserve the applicable rule or calculation version for historical rankings and school-health indicators where required.

Acceptance test: An authorised user can identify which approved method generated a historical indicator after the organisation changes its configuration.

COMP-026 — Separate Ranking from Benchmarking

The platform must implement relative rankings and target comparisons as separate concepts.

Acceptance test: A school can rank highly relative to other schools while still being shown as below its own configured attendance target.

COMP-027 — Identify Benchmark Sources

The system must identify whether a benchmark is an internal target, historical baseline, organisation-wide reference, or verified external standard.

Acceptance test: An administrator-entered target is not labelled as an official national standard without verified supporting configuration.

COMP-028 — Support Progress Indicators

The system must calculate supported progress indicators using comparable records and defined methods.

Acceptance test: A valid change in Mathematics performance is calculated from the appropriate Mathematics records and does not accidentally compare an unrelated subject.

COMP-029 — Avoid Causal Claims

The platform must not automatically claim that one indicator caused another indicator to change.

Acceptance test: A simultaneous change in attendance and academic results is presented as an observed pattern, not as proof of a causal relationship.

COMP-030 — Define Publication Status

Rankings intended for publication must have an explicit publication state or equivalent control.

Acceptance test: A provisional ranking is not presented as final when the approved publication rules require it to be marked provisional.

COMP-031 — Control Ranking Recalculation

The system must follow a documented policy when new data or corrections change a ranking.

Acceptance test: An authorised correction triggers the required recalculation, and the platform preserves relevant historical information according to the configured policy.

COMP-032 — Prevent Duplicate Ranking Records

The system must avoid uncontrolled duplicate ranking entries when a ranking is recalculated for the same scope and reporting period.

Acceptance test: Recalculating a ranking updates or versions the appropriate result according to the documented method instead of creating multiple indistinguishable published rankings.

COMP-033 — Respect Permissions in Aggregated Views

School and organisation comparisons must enforce the user's authorised data scope.

Acceptance test: A principal cannot retrieve another school's restricted comparison records by manipulating a filter or API request.

COMP-034 — Protect Small-Group Privacy

Public aggregate indicators must be assessed for the risk of exposing identifiable student information in small groups.

Acceptance test: A public reporting rule prevents disclosure when the configured privacy policy determines that an aggregate could reveal sensitive student-level information.

COMP-035 — Keep Comparison Methods Consistent

The same indicator must use the same documented calculation method across permitted views.

Acceptance test: A school attendance rate shown in a school dashboard and an organisation comparison agrees when both use the same eligible records, period, and definition.

COMP-036 — Explain Ranking Limitations

Where a ranking has important limitations, the interface must make the relevant context available.

Acceptance test: A user can identify the ranking measure and comparison group and access relevant warnings about incomplete or non-comparable data.

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17. Recommended Development Order

The developer should implement comparisons in a controlled sequence.

First, establish reliable calculations for academic results, attendance, homework completion, and other supported indicators.

Second, define how eligible records and denominators are determined.

Third, implement class-level and subject-level summaries.

Fourth, implement historical comparisons using the relevant educational context and comparability rules.

Fifth, implement organisation-wide aggregation and ensure that the chosen calculation method is documented.

Sixth, introduce narrowly defined rankings for measures that support valid comparisons.

Seventh, implement school-health statuses using explicit rules.

Eighth, consider a composite School Health Score only if the product owner approves a defined purpose, calculation method, and validation approach.

Ninth, implement ranking publication controls and historical versioning.

Tenth, test missing data, ties, incompatible grading contexts, student transfers, school transfers, and corrected results.

Finally, review the public-facing language and interface to ensure that the platform does not turn limited indicators into unsupported judgements about students or schools.

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18. Example: Comparing Three Schools

Consider an organisation that wants to review three branches.

The following figures are illustrative and assume that the academic results are comparable under the organisation's approved method.

Indicator	School A	School B	School C
Eligible students	1,000	100	500
Academic average	82%	76%	79%
Attendance	91%	96%	93%
Homework completion	88%	92%	85%
Students requiring review	40	12	15

These values describe different aspects of the schools' recorded information.

School A has the highest academic average.

School B has the highest attendance and homework completion rates.

School C has the fewest students requiring review in absolute numbers among the three schools with the given counts? No. School B has 12, while School C has 15 and School A has 40. School B has the smallest count.

However, comparing only counts can hide differences in school size.

The proportion of students requiring review is:

School A: 40 ÷ 1,000 × 100 = 4%.

School B: 12 ÷ 100 × 100 = 12%.

School C: 15 ÷ 500 × 100 = 3%.

School A has 40 students requiring review, but 4% of its eligible student population is represented by that count.

School B has 12 students requiring review, representing 12% of its eligible student population.

School C has 15 students requiring review, representing 3% of its eligible student population.

These indicators provide different perspectives.

The organisation administrator may want to investigate School A because it has the largest number of students requiring review. The administrator may also want to investigate School B because it has the highest proportion.

Neither observation alone establishes the cause of the pattern.

The dashboard should make the evidence visible and allow the administrator to investigate further.

It should not automatically label one school as the best or worst school.

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19. Final Principles

Optimus Orien should compare information only after defining what is being compared, why the comparison matters, and which records are eligible.

Performance and progress must remain separate concepts.

Counts and percentages must not be confused.

Rankings must use defined measures, eligible populations, and consistent tie-handling rules.

School-health indicators must have a clear purpose and transparent calculation method.

A composite score must not be introduced simply because one number appears convenient.

Different curricula, assessment systems, reporting periods, and student populations must be considered when deciding whether comparisons are meaningful.

Missing information must not be silently converted into zero or interpreted as evidence of poor performance.

Public rankings must respect privacy and must not expose individual student records.

Most importantly, the platform must communicate what its indicators actually establish.

Optimus Orien should make comparison more informative, not turn comparison into an unsupported judgement.

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20. What Part 8 Establishes

Part 8 establishes the foundation for academic progress indicators, student and school rankings, benchmarking, and school-health reporting.

After implementing its requirements, Optimus Orien should be able to calculate clearly defined averages, distinguish counts from rates, compare suitable historical results, and present rankings with the necessary context.

The platform should also have a defined approach to school-health indicators that prioritises transparent individual measures before introducing a composite score.

These foundations are essential because a school performance intelligence platform must do more than calculate numbers. It must help people understand what the numbers mean and where their limitations lie.

The next stage is to define the attendance system in greater detail: attendance events, registers, statuses, calculation rules, corrections, reporting, and the way attendance information should connect to the wider student performance record.

Part 9 will cover the complete attendance system for Optimus Orien, from the educational concepts behind attendance to the calculations, workflows, privacy rules, and software requirements your developer needs to implement it correctly.

PART 9
THE COMPLETE ATTENDANCE SYSTEM

Understanding Attendance, Recording It Correctly, Calculating Attendance Rates, and Connecting Attendance to Student Performance

Project: Optimus Orien
Document purpose: Education-domain learning and software implementation specification

Target audience: Product owner and software developer
Implementation principle: Attendance must be transparent, configurable, auditable, and appropriate for different educational systems.

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1. Why This Document Exists

Attendance is one of the most important types of information a school collects about its students.

However, an attendance system is more complicated than a table containing student names and a Present or Absent button.

To calculate attendance correctly, the system must understand when a student was enrolled, when the school was open, when the student was expected to attend, whether the student arrived late, how absences are classified, and which attendance rules apply to that school.

Consider three students.

Student A attends school every day they are expected to attend.

Student B joins the school halfway through the academic year.

Student C is absent for several days, but some of those absences are officially excused under the school's attendance policy.

If Optimus Orien treats every calendar day as an attendance opportunity, it may unfairly count weekends, holidays, and days before Student B enrolled as absences. If it treats every absence as identical, it may also hide important distinctions in the school's records.

These are not merely presentation problems. They are problems with the underlying data and calculation rules.

The central rule is that Optimus Orien must calculate attendance using the student's actual eligible attendance opportunities and the attendance policy that applies to those opportunities.

This document defines how that should work.

2. What Attendance Means in Education

Attendance describes whether a student participated in an educational activity they were expected to attend.

The relevant activity depends on the school's system.

In one school, attendance may be recorded once per school day. In another, attendance may be recorded separately for every lesson. A school may also record attendance for morning registration, practical sessions, examinations, or approved school activities.

These approaches are related, but they are not identical.

2.1 Daily Attendance

Daily attendance records whether a student attended school on a particular school day.

For example, a register might show that a student was present on Monday, absent on Tuesday, and late on Wednesday.

Daily attendance is useful for understanding whether a student regularly comes to school.

2.2 Lesson or Period Attendance

Lesson attendance records whether a student attended a particular scheduled lesson or period.

For example, a student may attend Mathematics, miss English, and arrive late to Science on the same day.

A daily register could show that this student attended school. A lesson-level register provides more detail about which learning sessions the student attended.

Lesson attendance is useful when a school needs subject-level or session-level attendance information.

2.3 Why the Difference Matters

Suppose a student attends four out of five scheduled lessons.

The student's lesson attendance rate is:

$4 \div 5 \times 100 = 80\%$.

However, it would be incorrect to automatically conclude that the student attended 80% of school days. The school day is a different unit of measurement.

Optimus Orien must identify whether a calculation concerns school days, lessons, periods, or another configured attendance unit.

The system must never combine these different units into one percentage without a clearly defined and appropriate calculation method.

3. The Main Concepts Your Developer Must Understand

Before building the attendance database, your developer needs to understand the following concepts.

3.1 Attendance Opportunity

An attendance opportunity is a scheduled occasion when a particular student is expected to attend.

For daily attendance, one attendance opportunity may be one eligible school day.

For lesson attendance, one attendance opportunity may be one scheduled lesson.

Imagine that a school is open from Monday to Friday. Saturday and Sunday are non-school days, and the school closes for a public holiday on Wednesday.

If a student is enrolled throughout that week, the school may have three eligible school-day attendance opportunities: Monday, Tuesday, and Thursday, assuming the other days are ordinary school days.

The weekend and the school holiday should not automatically count as absences.

The system must calculate eligible opportunities from the school's calendar, schedule, attendance model, and the student's enrolment.

3.2 Attendance Register

An attendance register is a record used to mark attendance for a defined group of students at a particular time or during a particular activity.

For example, a teacher might open the attendance register for Grade 8, Section A, for the Mathematics lesson on Monday.

The register identifies the relevant students and allows the authorised teacher to record their attendance.

A register is not the same thing as an individual attendance record. One register can contain many individual student attendance records.

3.3 Attendance Session

An attendance session identifies the activity or time period for which attendance is recorded.

Examples include:

- A full school day.
- A morning registration session.
- A scheduled Mathematics lesson.
- An examination session.

A daily attendance session and a Mathematics lesson session must remain distinguishable in the data model.

3.4 School Calendar

A school calendar identifies dates and periods relevant to the school's operation.

It may include the start and end of the academic year, teaching days, weekends, public holidays, school holidays, examination periods, and exceptional closures.

Different schools can follow different calendars, including schools within the same organisation.

The calendar must therefore be configurable for the relevant school or academic programme.

3.5 Enrolment Period

A student's enrolment period identifies when that student belongs to a particular school, programme, or class.

Suppose a student joins a school on 1 September.

Attendance opportunities before 1 September must not be attributed to that student as absences for that enrolment.

If the student transfers to another school, the system must preserve the attendance history of the previous enrolment while recording attendance under the new enrolment.

3.6 Attendance Status

An attendance status describes what happened during a particular attendance opportunity.

Examples include Present, Absent, Late, and Partially Present.

The exact statuses and their meanings must be configurable because schools and educational jurisdictions may follow different rules.

The software must store both the recorded status and the policy information needed to interpret it.

4. Attendance Statuses and Their Meanings

Optimus Orien should support configurable attendance statuses rather than assuming that every school uses an identical set of codes.

The following statuses are reasonable starting examples. They are not a claim that every school or jurisdiction officially recognises the same categories.

4.1 Present

Present means the student attended the relevant session according to the school's attendance rules.

The system should record which session the status refers to.

A student marked Present for the morning session is not automatically Present for every lesson later that day unless the school's attendance policy explicitly supports that interpretation.

4.2 Absent

Absent means the student did not attend the relevant session.

The system may distinguish authorised or excused absences from unauthorised or unexcused absences.

The distinction is important because schools may need to report both the total amount of missed attendance and the circumstances under which the absence was classified.

4.3 Authorised or Excused Absence

An authorised or excused absence is an absence that has been accepted under the applicable school policy.

For example, a school may accept a documented absence under its rules.

This status does not mean the student attended. It means the absence has a particular administrative classification.

Whether authorised absences count towards an attendance-rate calculation depends on the applicable policy. Some calculations may count them as missed attendance, while other reports may show authorised and unauthorised absences separately.

Optimus Orien must not silently treat authorised absence as Present.

4.4 Unauthorised or Unexcused Absence

An unauthorised or unexcused absence is an absence that has not been accepted as authorised under the applicable rules.

The platform should store the classification without making assumptions about the student's motivation or circumstances.

An unauthorised absence does not, by itself, prove that a student is careless, disengaged, or experiencing a particular problem.

4.5 Late

Late means the student arrived after the relevant expected start time, according to the school's policy.

For example, a student might arrive ten minutes after the school day begins.

A school may count a late student as present for its attendance-rate calculation while separately counting late arrivals. Another school may use a different policy.

Optimus Orien should therefore allow a school to define how Late affects each applicable metric.

Where the school records arrival times, the system may calculate lateness from the scheduled start time and the recorded arrival time. However, the calculation must respect the school's configured grace period and rules.

4.6 Partially Present

Partially Present may be useful when a student attends only part of a school day or session.

For example, a student may arrive after the morning lessons have started and attend the remainder of the day.

The school must define how partial attendance is recorded and whether it contributes to a daily or session-level attendance calculation.

The system must not automatically count a partially attended day as a full present day unless that is the school's configured policy.

4.7 Approved School Activity

A student may be participating in an activity officially organised or approved by the school outside the usual classroom setting.

Depending on the school's policy, the activity might count as attendance for the relevant session.

The system should support this category when required, while keeping the recorded status and calculation treatment explicit.

4.8 Other Configurable Statuses

A school may need additional statuses for its circumstances.

Optimus Orien should allow authorised administrators to configure appropriate statuses, descriptions, reporting categories, and calculation behaviour.

Each status should have a stable internal identifier. The displayed name may change without breaking historical records.

The system should also retain the meaning and calculation treatment that applied when a record was created or finalised.

Important: A configurable status must not automatically be included in attendance calculations merely because its name contains the word Present or Absent. Its calculation behaviour must be defined explicitly.

5. How the Attendance Data Model Should Work

The attendance system must store enough information to establish who was expected to attend, what was recorded, which rules applied, and whether the record was subsequently corrected.

A practical starting model includes the following entities.

5.1 Student

The student is the person whose attendance is being recorded.

Each student must have a stable internal identifier that does not change when their name, class, or contact information changes.

5.2 Enrolment

The enrolment connects the student to a particular school or programme during a defined period.

It should contain the relevant start date, end date where applicable, and other information required to determine which attendance opportunities belong to that enrolment.

The system must not rely only on the student's current class to interpret old attendance records.

5.3 School Calendar and Schedule

The calendar defines the relevant school dates. The schedule identifies the expected sessions, lessons, or periods.

Together, these determine which attendance opportunities should exist.

A schedule may differ by class, student, programme, or day of the week.

For example, a part-time student may not be expected to attend on every day that the school is open. The system must respect the student's actual schedule.

5.4 Attendance Session

The attendance session identifies the specific occasion for which attendance is being recorded.

It should identify the relevant school, date, session type, and scheduled time where applicable.

For lesson-level attendance, it should also connect to the relevant class, subject, or scheduled lesson.

5.5 Attendance Record

An individual attendance record should contain, as appropriate:

1.

A unique record identifier.

2.
3.

The student identifier.

4.
5.

The relevant enrolment identifier.

6.
7.

The attendance session identifier.

8.
9.

The recorded attendance status.

10.
11.

Arrival and departure times, if supported.

12.
13.

The user who recorded the attendance.

14.
15.

The time the record was created.

16.
17.

The time of the most recent permitted change.

18.
19.

The applicable attendance-policy version or an equivalent historical reference.

20.
21.

A reason category or restricted note, where permitted.

22.
23.

The source of the record, such as manual entry or an authorised import.

24.

Not every school will need every field. Nevertheless, the data model should support the information required by the configured attendance model.

The developer should not store attendance as only a student's name, date, and a text value. That structure would make transfers, lesson-level attendance, policy changes, corrections, and historical reporting difficult to handle safely.

5.6 Attendance Register Submission

The register should have its own workflow state, such as Draft, Submitted, Finalised, or Reopened.

These states are different from the student's attendance status.

For example, a register can be submitted even though it contains absent students. A register marked Draft may contain preliminary attendance records that are not yet approved for official reporting.

The system must not confuse a register's workflow status with the student's attendance status.

6. Calculating Attendance Correctly

This section is particularly important because incorrect attendance calculations could make a school's data misleading.

6.1 The Basic Attendance-Rate Formula

A common attendance-rate formula is:

Attendance rate = Eligible attended opportunities ÷ Eligible scheduled opportunities × 100

However, the words *eligible attended opportunities* require a precise definition.

The school must define which statuses count as attended, whether partial attendance is supported, how late arrivals are treated, and whether approved activities count as attendance.

The denominator must include only attendance opportunities eligible under the applicable calendar, schedule, and enrolment rules.

The formula should never count every calendar day in the year.

6.2 Example: Basic Attendance Rate

Suppose a student has 100 eligible school-day attendance opportunities.

The student's recorded statuses are:

Present: 80 days.

Late: 5 days.

Authorised absence: 5 days.

Unauthorised absence: 10 days.

Assume the school's policy counts Late as attended for the basic attendance rate, while authorised and unauthorised absences both count as missed attendance.

The calculation is:

Attended opportunities = 80 + 5 = 85.

Eligible opportunities = 100.

Attendance rate = 85 ÷ 100 × 100 = 85%.

The school may also report five late arrivals, five authorised absences, and ten unauthorised absences.

These additional figures explain the composition of the attendance record. They do not replace the attendance rate.

If another school defines Late differently, its attendance-rate result may differ. That is why the applicable calculation policy must be recorded.

6.3 Do Not Treat Missing Records as Absences Automatically

Suppose a school should have recorded attendance for 100 school days, but the register is missing for 10 days.

The database contains records for only 90 days.

The system must not silently assume that the student was absent on all ten unrecorded days.

The missing records may indicate incomplete administrative work rather than student absence.

Optimus Orien should distinguish at least three conditions:

Recorded attendance: A valid attendance status exists for the relevant opportunity.

Recorded absence: A valid attendance record explicitly indicates absence.

Missing attendance data: No valid attendance record exists for the relevant opportunity.

If 80 days are recorded as attended, 10 days are recorded as absent, and 10 days have no attendance record, the system must display the incomplete data.

It may calculate a provisional rate over the 90 recorded opportunities if that is permitted by the reporting policy, but it must clearly label the calculation and disclose the missing ten opportunities.

It must not present the resulting percentage as if all 100 opportunities had been fully verified.

6.4 Attendance Rate and Absence Rate

The attendance rate and absence rate may be related, but the system must define their relationship explicitly.

Under a simple policy in which every eligible opportunity is classified as attended or absent, with no missing records or special partial-attendance rules:

Attendance rate + Absence rate = 100%.

However, that relationship may not hold when there are missing records, partial attendance, or categories whose calculation treatment differs.

The system should therefore calculate each measure from its own documented rules rather than assuming that one can always be obtained by subtracting the other from 100%.

6.5 On-Time Attendance Rate

A school may want to understand how often students arrive on time.

One possible formula is:

On-time attendance rate = Eligible on-time arrivals ÷ Eligible opportunities requiring arrival × 100

The precise denominator depends on the school's policy and the data available.

For example, a school might measure whether students arrived on time for morning registration. A lesson-level system could measure whether students arrived on time for each scheduled lesson.

Optimus Orien should not calculate an on-time rate unless the required schedule and attendance information are available.

6.6 Partial Attendance

Suppose a student attends half of a school day.

The system must not automatically decide that this represents exactly half a day of attendance unless the school has adopted that method.

A school may record partial attendance through lesson-level records, time-based calculations, or a configured partial-day category.

The selected approach must be documented, consistently applied, and reflected in reports.

6.7 The Denominator Must Be Visible

A percentage alone does not tell the whole story.

Compare these two records:

Student A: 90% attendance across 100 eligible school days.

Student B: 90% attendance across 10 eligible school days.

Both percentages are 90%, but they represent different amounts of evidence.

Optimus Orien should display the relevant numerator and denominator alongside the percentage wherever practical.

For example:

Attendance: 90% — 90 of 100 eligible days recorded as attended.

The wording must accurately reflect the school's calculation policy and the actual data.

7. Determining Which Days Count

The system must determine eligible attendance opportunities before calculating attendance rates.

This requires more than checking whether a date falls between the start and end of an academic year.

7.1 Weekends and Non-Teaching Days

If a school does not normally operate on Saturday and Sunday, those dates should not automatically count as attendance opportunities.

However, another school may schedule classes on weekends.

The system must use the relevant school calendar and schedule rather than hardcoding Monday through Friday as universal teaching days.

7.2 Public Holidays and School Holidays

Dates marked as holidays under the applicable school calendar should not automatically count as attendance opportunities.

If a school conducts a special examination or educational activity during a holiday, that event should be handled according to its actual schedule and attendance policy.

7.3 Unexpected School Closures

A school may close because of severe weather, a local emergency, maintenance, or another reason.

An authorised closure should be recorded in the calendar or relevant schedule.

The system must recalculate eligible opportunities according to the applicable rules without creating false student absences.

If a closure is entered after attendance records have already been finalised, the system should identify affected reports and recalculate them according to a controlled process.

It must preserve the history of the change.

7.4 Students Who Join or Leave Mid-Year

A student joins on 1 September and leaves on 31 December.

Only the attendance opportunities belonging to the relevant enrolment should contribute to that enrolment's attendance calculations.

The system must not count days before the student joined as absences.

It must also avoid assigning attendance at a new school to the old school's enrolment.

7.5 Students With Different Schedules

Some students may follow a part-time, modified, or otherwise different timetable.

The system should use the attendance opportunities applicable to that student rather than assuming that all students in the same school have identical schedules.

Where the software cannot establish the correct schedule, it should flag the data for review instead of inventing attendance opportunities.

8. Daily Attendance and Lesson Attendance Must Remain Separate

Optimus Orien should support a clearly configured attendance model.

The initial school setup should establish whether attendance is recorded primarily by school day, by lesson or period, or through another defined arrangement.

The system may later support more than one attendance model where a school has a genuine need.

However, the calculations must remain distinct.

Consider a school with six lessons each day.

If a student attends five lessons and misses one, a lesson-level calculation may show five attended opportunities out of six.

It would be incorrect to count the student as having attended five school days.

Similarly, the system must not add daily attendance records and lesson attendance records together as if they were equivalent events.

If both forms of attendance are stored, the system must define which is authoritative for each report and how conflicting information is handled.

A school should not receive an inflated attendance rate simply because the same day is represented by multiple lesson records.

9. Attendance Register Workflow

An attendance system needs a controlled workflow so that records can be entered, checked, corrected, and finalised.

9.1 Opening a Register

An authorised teacher or administrator opens the register for the relevant school, class, date, and session.

The system loads the students who are eligible for that session according to their enrolments and schedules.

Students who do not belong to the relevant class or session must not appear as ordinary attendees.

The register should identify the date, class, session, and responsible user.

9.2 Recording Attendance

The authorised user records the appropriate status for each student.

The interface should make it easy to identify students whose attendance has not yet been recorded.

The system should distinguish an unmarked record from an explicitly recorded absence.

It should not automatically mark every unselected student Absent unless the school has a carefully defined and approved workflow that explicitly requires such behaviour. Even then, the resulting records must be distinguishable from verified attendance decisions.

9.3 Submitting a Register

When the user submits the register, the system validates it.

Possible checks include:

-

Are all expected students represented?

-
-

Are any statuses invalid?

-
-

Does the session actually exist?

-

-

Is the user authorised to submit the register?

-
-

Are there duplicate attendance records?

-
-

Is the register already finalised?

-
-

Are any students no longer enrolled for the relevant date?

-

The system should warn about missing information and block invalid submissions where necessary.

9.4 Finalising a Register

Finalisation indicates that the register has reached the school's defined state for official reporting.

A finalised register should not be freely editable by every user.

The system must define who can finalise registers and under what conditions.

The finalised state must not imply that every record is necessarily correct forever. Errors may still need correction through an authorised process.

9.5 Correcting Attendance

Suppose a teacher accidentally marks a student Absent when the student was Present.

The system must allow an authorised correction under the school's policy.

However, it should not silently overwrite the previous record without preserving an audit trail.

The audit history should record the old value, new value, user making the correction, time of correction, and reason where required.

This allows authorised school staff to investigate what changed and why.

9.6 Reopening a Register

A school may need to reopen a submitted or finalised register.

This action should be restricted to authorised users.

The system should record who reopened it, when, and why.

If the change affects an attendance report or a student performance indicator, the affected calculations must be updated in a controlled manner.

The platform should make it possible to distinguish a current corrected report from an earlier report that was produced before the correction.

10. Attendance Data Quality

A reliable attendance system must detect incorrect, incomplete, or conflicting information.

The following checks should be part of the implementation.

10.1 Duplicate Attendance Records

A student should not accidentally receive two conflicting attendance records for the same attendance opportunity.

The database should enforce an appropriate uniqueness rule based on the student's enrolment and session.

The exact key will depend on the attendance model.

For example, a daily register may permit one primary attendance record per student per eligible school day, while a lesson-level register requires a separate record for each relevant lesson.

Legitimate corrections must update the existing record through the authorised workflow rather than creating duplicate attendance events.

10.2 Missing Registers

If attendance is expected for a session but the register has not been submitted, the system should identify it as missing or incomplete.

It must not interpret the missing register as proof that every student was absent.

10.3 Conflicting Statuses

If the system detects incompatible records for the same attendance opportunity, it should flag the conflict for review.

For example, a student should not simultaneously have two finalised records marking the same lesson as Present and Absent.

10.4 Attendance Outside the Enrolment Period

The system should flag attendance records that fall outside the applicable enrolment period.

The system should not silently attach those records to another enrolment or alter the student's enrolment dates to make the records appear valid.

10.5 Attendance on a Closed Day

If an attendance record exists for a date marked as a school closure, the system should check whether an approved activity or special session explains the record.

If no valid explanation exists, the record should be flagged.

10.6 Late or Incomplete Submission

Schools may want to know which registers have not been submitted by the expected deadline.

The platform should be able to identify overdue or incomplete registers using configurable deadlines where this functionality is required.

The deadline is an administrative setting, not a universal educational rule.

11. Connecting Attendance to the Student Performance Record

Attendance is one part of a student's educational record. It must be interpreted alongside other information without being confused with those other measures.

A student may have high attendance and low academic results. Another may have lower attendance but strong assessment results.

These situations demonstrate why attendance and academic achievement must remain separate indicators.

Attendance can provide useful context for school staff, but it does not establish the cause of a student's academic performance.

11.1 Separate Indicators

Optimus Orien should show attendance as its own measure.

A student profile might display:

-

Attendance rate for the selected reporting period.

-
-

Number of eligible attendance opportunities.

-
-

Number of attended opportunities.

-
-

Number of recorded absences.

-
-

Authorised and unauthorised absence counts, where supported.

-
-

Number of late arrivals.

-
-

Attendance-data completeness.

-
-

Attendance trend over comparable reporting periods.

-

The precise set of indicators should reflect the school's configured attendance model and available data.

11.2 Do Not Automatically Combine Attendance and Marks

Suppose a student has 60% attendance and 85% academic performance.

The platform must not automatically calculate an overall score of 72.5% by averaging those numbers.

Attendance percentage and academic performance percentage measure different things. Their denominators, meanings, and educational implications are different.

A combined indicator would require a separately defined, justified, and validated formula. It should not be introduced merely because both values are percentages.

For the current Optimus Orien scope, the recommended approach is to display attendance and academic performance as separate indicators.

11.3 Attendance Trends

The system should support comparisons between compatible reporting periods.

For example, a student's attendance may increase from 75% in one term to 88% in the next.

That is an increase of 13 percentage points.

The system must not describe this as a 13% relative increase unless it explicitly calculates the relative change.

A relative increase from 75% to 88% would be approximately 17.3%, calculated as:

$(88 - 75) \div 75 \times 100 \approx 17.3\%$.

The interface should use the appropriate term for the calculation it displays.

11.4 Early Intervention and Attendance

Optimus Orien is intended to help schools identify students who may require review or support.

Attendance can contribute to this process through transparent, configurable rules.

For example, a school may decide that a student should be reviewed when their attendance rate falls below a specified threshold, when several absences occur consecutively, or when attendance declines significantly compared with an earlier period.

These are examples of rule-based review criteria, not universal definitions of a student being at risk.

The school must configure and approve the criteria.

When a rule is triggered, the system should explain the evidence behind the flag.

For example:

Attendance review suggested: The student's recorded attendance is 68% across 50 eligible school days, which is below the school's configured review threshold of 80%.

This is more transparent than displaying an unexplained warning.

The flag must not automatically diagnose the student, assign blame, or claim that absence caused poor academic performance.

A qualified member of school staff should review the circumstances before deciding what support, if any, is appropriate.

11.5 Human Review and Follow-Up

If the platform includes an attendance-related intervention workflow, the authorised staff member should be able to record that the case was reviewed and document the agreed follow-up.

The workflow may include the review date, responsible staff member, action taken, and follow-up date.

Sensitive information should be restricted to users who need it for their responsibilities.

The platform should not assume that every attendance warning requires the same response.

12. Attendance Reports and Visualisations

Attendance data should be presented in a way that helps school staff understand the situation without misleading them.

12.1 Student-Level Report

The student's own attendance report should show the relevant reporting period, attendance rate, numerator and denominator, absence counts, late arrivals, and trend where supported.

It should also make incomplete attendance data visible.

Parents or guardians should see information only for students to whom they have an authorised relationship.

Students should see their own records according to the school's configured permissions.

12.2 Teacher-Level Report

A teacher should be able to access attendance registers for the classes and sessions assigned to them.

The teacher should not automatically receive access to every student's full attendance history across the organisation.

Any broader access must be explicitly authorised.

12.3 Principal-Level Report

The principal should be able to review attendance within the school they are authorised to manage.

Useful views may include:

-

Attendance rate by class or year group.

-
-

Attendance rate by reporting period.

-
-

Counts of authorised and unauthorised absences, where supported.

-
-

Late-arrival patterns.

-
-

Missing or incomplete registers.

-

-

Students who meet configured review criteria.

-
-

Attendance-data completeness.

-

These views must respect the school's data permissions and the applicable attendance policies.

12.4 Organisation-Level Report

An organisation administrator responsible for several schools should be able to review school-level attendance indicators across authorised branches.

The administrator should be able to compare attendance rates only when the measures are sufficiently compatible.

If one school reports daily attendance and another reports lesson attendance, the platform must not present their raw percentages as directly equivalent without a clearly justified comparison method.

The organisation view should identify the underlying attendance model and any limitations that affect comparisons.

12.5 Public Information

Individual student attendance records are private educational information.

Optimus Orien must not expose a student's attendance history, absence reasons, or attendance-related review status through public leaderboards.

Any public school-level attendance statistic must be explicitly approved for publication and reviewed for privacy risks.

13. Student Attendance and Employee Attendance Are Different Systems

Optimus Orien has previously considered attendance information for students and school staff.

However, student attendance and employee attendance should not be treated as the same data model with different labels.

Student attendance concerns participation in scheduled educational activities.

Employee attendance concerns staff work schedules, presence, leave, and employment-related rules.

These systems may share technical components, such as calendars and timestamps, but their permissions, calculations, policies, and privacy requirements differ.

For example, a student's absence may be recorded against a school lesson. An employee's absence may need to be recorded against a scheduled working day or shift.

The initial student attendance implementation should therefore remain clearly separated from any future employee attendance module.

If employee attendance is included in the same product release, it requires its own explicit requirements and acceptance tests.

14. Privacy, Security, and Sensitive Information

Attendance data can reveal information about a student's routine, circumstances, and participation in education.

The system must protect it accordingly.

14.1 Limit Access

A teacher should access only the attendance information needed for their assigned responsibilities.

A principal may have broader access within their authorised school.

An organisation administrator may see branch-level summaries and other information permitted by their role.

Parents or guardians should see only records for students they are authorised to access.

A student should not be able to access another student's attendance history by changing a URL, request parameter, or record identifier.

All permissions must be enforced by the server, not merely by hiding interface elements.

14.2 Restrict Absence Reasons

A school may need to record a reason category for an absence.

The system should collect only information necessary for the school's legitimate purpose.

Detailed medical, family, or other sensitive information should not be placed in a general-purpose attendance comment that every teacher can read.

If the product supports sensitive notes, it must define separate permissions, retention rules, and access logging.

14.3 Protect Public Displays

Public pages must not expose individual attendance records or identify students as absent, frequently late, or requiring intervention.

School-level statistics should be checked for disclosure risks, particularly when groups are very small.

14.4 Record Access and Changes

The platform should maintain appropriate logs for attendance changes and sensitive administrative actions.

The logs should identify who performed the action, when it occurred, and what changed.

Audit records must not be editable by ordinary users.

14.5 Retention and Deletion

Attendance retention requirements can vary by school policy and jurisdiction.

The product must support a documented retention approach rather than assuming that every attendance record should be kept forever or deleted after the same period.

The applicable rules should be reviewed before deployment in each jurisdiction.

15. Configurable Attendance Policies

Optimus Orien is intended to support schools in multiple countries. Attendance behaviour must therefore be configurable.

The system should not hardcode one country's calendar, school week, absence categories, or reporting formula as universal.

At minimum, authorised configuration should cover the following areas.

Attendance model: Whether the school uses daily attendance, lesson-level attendance, or another supported arrangement.

Calendar: Teaching days, non-teaching days, holidays, closures, academic-year boundaries, and relevant timezone.

Schedules: Scheduled sessions and, where applicable, different timetables for classes or individual students.

Attendance statuses: Names, descriptions, categories, and calculation behaviour.

Calculation rules: How late arrivals, authorised absences, unauthorised absences, partial attendance, and approved activities affect each metric.

Register workflow: Who may create, submit, finalise, reopen, and correct attendance records.

Review criteria: Configurable rules for identifying attendance patterns that warrant human review.

Reporting periods: The periods used for term, year, or other school-level reports.

Permissions: Which roles may view, enter, correct, approve, export, or administer attendance information.

Historical versions: The ability to preserve the policies and calculation rules that applied to earlier attendance records.

These settings should have clear administrative interfaces and sensible validation.

A school must not be allowed to change a calculation rule accidentally without understanding which reports and historical records could be affected.

16. Historical Records and Changes to Rules

Attendance rules may change over time.

For example, a school may revise its late-arrival policy, change its timetable, or adopt a different definition for an attendance indicator.

The system must not allow a new rule to silently rewrite the meaning of every historical report.

Each attendance record and report should remain interpretable using the relevant historical context.

This requires the system to preserve or reference the applicable calendar, schedule, enrolment, attendance-status definitions, and calculation-policy versions.

When a policy change requires recalculation of historical information, the system should follow an authorised process.

It should identify the affected records and reports, preserve the change history, and make the updated calculation method clear.

The developer should distinguish between correcting an inaccurate historical record and changing the policy used to calculate a historical indicator. These are different operations.

17. Data Import and Export

A school may already maintain attendance information in spreadsheets or another school management system.

Optimus Orien may need to support controlled import and export.

An import process should validate the file before creating records.

It should check that students, enrolments, dates, sessions, and attendance statuses can be matched correctly.

It should also identify duplicate records, invalid dates, unrecognised status codes, and records outside the applicable enrolment period.

The system should provide a summary of accepted and rejected records and explain the errors that require correction.

An import must not silently create new student identities because a student's name does not match exactly.

Where matching is uncertain, the process should request authorised review.

Exports should be restricted by role and should include only the data necessary for the user's authorised purpose.

Any import or export of sensitive attendance data should be logged appropriately.

18. Software Implementation Requirements and Acceptance Tests

The following requirements define the minimum expected behaviour of the attendance system.

The developer should treat them as testable product requirements rather than general suggestions.

18.1 Attendance Configuration

ATT-001 — Configurable Attendance Model

The system must support a clearly configured attendance model, including daily attendance and lesson-level attendance where enabled.

Acceptance test: An administrator configures a school to use daily attendance. The system creates daily attendance opportunities and does not treat lesson records as additional school days. A second school can use lesson-level attendance without changing the first school's configuration.

ATT-002 — Configurable Attendance Statuses

The system must allow authorised administrators to configure attendance statuses and their calculation behaviour.

Acceptance test: An administrator creates a new status called Approved School Activity and defines how it affects the relevant attendance metrics. The system applies that rule consistently and does not assume that every new status automatically counts as Present.

ATT-003 — Versioned Policies

The system must preserve historical references to attendance policies and status definitions.

Acceptance test: An administrator changes the treatment of Late for future reporting. Previously finalised reports remain identifiable as having used the earlier policy unless an authorised recalculation is performed.

ATT-004 — School Calendar

The system must support configurable school calendars, including teaching days, holidays, closures, and academic-year boundaries.

Acceptance test: An administrator marks a date as a school closure. That date is excluded from eligible attendance opportunities where the applicable policy requires it.

ATT-005 — Configurable Schedules

The system must support schedules relevant to the selected attendance model.

Acceptance test: A part-time student has no scheduled attendance opportunity on a particular weekday. The system does not automatically count that weekday as an absence for the student.

ATT-006 — Timezone Handling

The system must use the appropriate configured school timezone when interpreting local dates and attendance times.

Acceptance test: An attendance event recorded shortly after midnight in the school's local timezone is assigned to the correct local date, rather than the date incorrectly inferred from a different timezone.

18.2 Attendance Records

ATT-007 — Stable Record Identifiers

Every attendance record must have a stable unique identifier.

Acceptance test: A student's name changes. Their existing attendance records remain connected to the correct student.

ATT-008 — Enrolment Connection

Every attendance record must be connected to the appropriate student enrolment and attendance opportunity.

Acceptance test: A student transfers to another school. Attendance recorded at the previous school remains associated with the previous enrolment, and later attendance belongs to the new enrolment.

ATT-009 — No False Pre-Enrolment Absences

The system must not create absence records for dates before the relevant enrolment begins.

Acceptance test: A student enrols on 1 September. A report for that enrolment does not count eligible dates in August as absences.

ATT-010 — Missing Data Is Not Absence

The system must distinguish missing attendance data from an explicitly recorded absence.

Acceptance test: A register is never submitted for a scheduled session. The report identifies missing attendance data rather than marking every student absent.

ATT-011 — Session-Specific Records

The system must associate attendance records with the correct daily or lesson-level session.

Acceptance test: A student is Present for Mathematics and Absent for English. The system retains both records and does not overwrite one with the other.

ATT-012 — Record Source

Where applicable, the system must record whether attendance was entered manually, imported, or created through another authorised process.

Acceptance test: An administrator reviews a record created through an import and can identify its recorded source.

18.3 Register Workflow

ATT-013 — Open and Submit Registers

The system must support the configured register workflow.

Acceptance test: An authorised teacher opens a register, records attendance, and submits it. An unauthorised user cannot submit that register.

ATT-014 — Unmarked Students

The system must identify students whose attendance has not been recorded.

Acceptance test: A register contains 30 expected students, but only 28 have attendance statuses. The system identifies the two unmarked students and does not silently treat them as verified absences.

ATT-015 — Register Validation

The system must validate registers before submission or finalisation according to configured rules.

Acceptance test: A register contains an invalid attendance status. The system explains the validation error and prevents finalisation until the issue is resolved or handled through an authorised exception process.

ATT-016 — Finalised Register Protection

The system must restrict changes to finalised registers.

Acceptance test: A teacher attempts to edit a finalised register without the required permission. The server rejects the operation.

ATT-017 — Controlled Reopening

The system must restrict reopening of finalised registers to authorised users.

Acceptance test: A principal with the appropriate permission reopens a register and provides the required reason. The system records the action in the audit history.

ATT-018 — Audited Corrections

The system must preserve the history of attendance corrections.

Acceptance test: An authorised user changes a student's status from Absent to Present. The audit history records the previous status, new status, user, time, and reason where required.

ATT-019 — Duplicate Prevention

The system must prevent conflicting duplicate attendance records for the same student and attendance opportunity.

Acceptance test: A user submits a second record for the same student and session. The system prevents an unintended duplicate or routes the action through the authorised correction workflow.

18.4 Calculation Rules

ATT-020 — Eligible Opportunities

The system must calculate eligible attendance opportunities from the applicable calendar, schedule, enrolment, and attendance model.

Acceptance test: A student is enrolled only for part of the academic year. The denominator includes only eligible opportunities within that enrolment period.

ATT-021 — Explicit Attendance Formula

Every attendance metric must have a documented calculation formula.

Acceptance test: An administrator reviews the attendance-rate definition and can identify which statuses count as attended, how partial attendance is handled, and which opportunities are included in the denominator.

ATT-022 — Visible Numerator and Denominator

Reports must show the relevant counts behind an attendance percentage wherever practical.

Acceptance test: A student has 85 attended opportunities out of 100 eligible opportunities. The report displays 85% and the corresponding counts.

ATT-023 — Configurable Late Treatment

The system must apply the configured treatment of Late to each relevant attendance metric.

Acceptance test: One school counts Late as attended for its attendance rate, while another reports Late separately under a different policy. Each school receives results consistent with its own settings.

ATT-024 — Separate Absence Categories

Where authorised and unauthorised absences are supported, the system must report them according to the configured classification rules.

Acceptance test: A student has five authorised and three unauthorised absences. The report preserves the separate counts and does not silently combine them into a single unexplained category.

ATT-025 — Partial Attendance

The system must apply the configured method for partial attendance.

Acceptance test: A student attends part of a school day. The system uses the configured partial-attendance rule and does not automatically count the student as fully present unless the rule explicitly requires it.

ATT-026 — No Mixed Units

The system must not combine daily attendance and lesson attendance as if they were the same unit.

Acceptance test: A student attends five out of six lessons. The lesson attendance report displays the lesson-based result and does not report five school days attended.

ATT-027 — Missing Data Disclosure

The system must disclose incomplete attendance data when calculating or presenting attendance metrics.

Acceptance test: A report contains 80 attended days, 10 absent days, and 10 days with missing records. The system identifies the missing ten days and does not present the 80% calculation as a complete, verified rate over all 100 days.

ATT-028 — No Unexplained Zero

The system must distinguish zero attendance from the absence of attendance data.

Acceptance test: A student has no recorded attendance opportunities because the schedule has not been configured. The system displays unavailable or incomplete data rather than claiming that the student's attendance rate is 0%.

ATT-029 — Separate On-Time Metric

If on-time attendance is enabled, the system must calculate it using a documented rule and appropriate data.

Acceptance test: The school enables on-time reporting with a configured start time and grace period. The report follows those rules and does not infer punctuality where the required timestamps or status records are missing.

ATT-030 — Reproducible Calculations

The system must produce consistent results when the same records and policy version are used.

Acceptance test: Recalculating the same attendance report with unchanged inputs and rules produces the same result.

18.5 Reporting and Early Intervention

ATT-031 — Student Attendance Report

The system must provide authorised access to individual attendance reports.

Acceptance test: A student can view their own permitted attendance information. They cannot access another student's report by changing a record identifier.

ATT-032 — Teacher Scope

Teachers must access attendance records only for classes and sessions within their authorised scope.

Acceptance test: A teacher assigned to Grade 8, Section A cannot retrieve another teacher's restricted register by directly calling the relevant server endpoint.

ATT-033 — Principal Scope

Principals must be restricted to the school or schools they are authorised to manage.

Acceptance test: A principal from School A cannot access School B's individual attendance records unless a separate explicit permission authorises that access.

ATT-034 — Organisation Scope

Organisation administrators must be restricted to the branches and records assigned to their organisational permissions.

Acceptance test: An organisation administrator can review permitted branch summaries but cannot retrieve data belonging to an unrelated organisation.

ATT-035 — Compatible Comparisons

The system must identify the attendance model and calculation rules used for school comparisons.

Acceptance test: Two schools use different attendance units. The system does not present their percentages as directly comparable without an explicitly justified comparison method.

ATT-036 — Attendance Review Rules

The system must support transparent, configurable attendance-review criteria where early intervention is enabled.

Acceptance test: An authorised administrator sets a review threshold of 80%. A student with a valid recorded rate of 68% is flagged with an explanation of the applicable rule and supporting evidence.

ATT-037 — Human Review

An attendance-related flag must not automatically be presented as a diagnosis or proof of the cause of an academic problem.

Acceptance test: A student triggers an attendance-review rule. The platform displays the relevant evidence and requests staff review without claiming that absence caused a low mark.

ATT-038 — Review History

Where attendance-review workflows are enabled, the system must preserve the review and follow-up history.

Acceptance test: An authorised staff member records that a case was reviewed and schedules a follow-up. The system retains the responsible staff member, review date, and follow-up information under the configured access rules.

ATT-039 — Reporting Filters

Attendance reports must support the relevant configured filters, such as school, class, student, date range, and reporting period.

Acceptance test: A principal filters a report to one class and one term. The displayed counts and rates reflect only the eligible records within that selection.

ATT-040 — Historical Reporting

The system must preserve enough historical context to interpret earlier attendance reports.

Acceptance test: A school's attendance policy changes in a new academic year. An authorised user can identify which policy was used for a report from the previous year.

18.6 Privacy, Security, and Data Quality

ATT-041 — Server-Side Permissions

All attendance access and modification permissions must be enforced on the server.

Acceptance test: A user without permission attempts to access a restricted attendance record through a direct request. The server denies access regardless of what the interface displays.

ATT-042 — Parent and Guardian Scope

Parents and guardians must access only students connected to their authorised relationship.

Acceptance test: A parent linked to Student A cannot access Student B's attendance report by changing the student identifier.

ATT-043 — Public Privacy

Public pages must not expose individual student attendance records or restricted attendance-review information.

Acceptance test: A visitor who is not authenticated cannot access an individual student's attendance history through a public page or direct endpoint.

ATT-044 — Sensitive Notes

Sensitive attendance notes must be protected by appropriate permissions and data-minimisation rules.

Acceptance test: A teacher without permission to view restricted notes cannot retrieve them through the interface, an export, or a direct server request.

ATT-045 — Auditable Imports

Attendance imports must be validated and traceable.

Acceptance test: An imported file contains invalid dates and unrecognised statuses. The system identifies rejected records and explains the errors instead of silently creating inaccurate attendance entries.

ATT-046 — Controlled Exports

Attendance exports must respect the requesting user's permissions.

Acceptance test: A teacher exports attendance data for an assigned class. The exported file does not contain restricted records from other classes or schools.

ATT-047 — Closure and Schedule Validation

The system must detect attendance records that conflict with the calendar or schedule.

Acceptance test: An attendance record exists for a closed day without an authorised scheduled activity. The system flags the record for review.

ATT-048 — Employee Attendance Separation

If employee attendance is implemented, its data and permissions must be separated from student attendance.

Acceptance test: A student attendance report cannot accidentally include staff attendance records, and a teacher's access to student registers does not automatically grant access to confidential employee attendance records.

19. Recommended Development Order

The developer should implement attendance in a sequence that protects calculation accuracy.

Stage 1: Establish the Rules

Build the school calendar, attendance model, schedule structure, configurable statuses, and policy-versioning foundation.

Before creating dashboards, ensure that the system can determine which attendance opportunities are eligible.

Stage 2: Build the Core Data Model

Implement student enrolments, attendance sessions, attendance records, and the relationships between them.

Add appropriate database constraints to prevent duplicate or invalid records.

Stage 3: Build the Register Workflow

Implement opening registers, recording attendance, validation, submission, finalisation, reopening, and audited corrections.

The system should clearly distinguish unmarked records from recorded absences.

Stage 4: Implement and Test Calculations

Implement attendance rates, absence counts, late-arrival measures, partial-attendance treatment, and missing-data handling.

Create automated tests using small, manually verified examples.

Do not rely only on checking whether a percentage looks plausible.

Stage 5: Enforce Access Control

Implement role- and scope-based access on the server.

Test student, parent, teacher, principal, and organisation administrator permissions separately.

Include tests that attempt to access unauthorised records directly rather than relying only on normal interface navigation.

Stage 6: Build Reports

Create student-level, teacher-level, principal-level, and organisation-level attendance reports.

Ensure each report identifies its reporting period and uses the correct attendance model and calculation rules.

Stage 7: Add Review Criteria

Implement configurable attendance-review rules after the underlying attendance calculations are reliable.

Each flag must show the rule and evidence that triggered it.

Stage 8: Test Historical Changes

Test enrolment transfers, schedule changes, school closures, attendance corrections, and policy changes.

Verify that historical reports remain interpretable and that recalculation does not silently erase the previous record of what happened.

Stage 9: Add Imports and Exports

Implement validated imports and permission-controlled exports once the core data model and calculations are stable.

Test duplicate handling, student matching, invalid dates, and audit records.

20. Example: A Complete Attendance Scenario

Imagine that a school uses daily attendance.

Its policy states that Present and Late count as attended for the main attendance rate. Authorised and unauthorised absences count as missed attendance. Missing registers are excluded from a provisional calculation and must be disclosed.

A student has the following records for a selected reporting period:

Present: 70 days.

Late: 5 days.

Authorised absence: 8 days.

Unauthorised absence: 7 days.

Missing attendance data: 10 days.

The total number of eligible attendance opportunities is 100.

The number of opportunities with recorded statuses is 90.

The number of attended opportunities under this policy is 70 + 5 = 75.

The provisional attendance rate over recorded opportunities is:

$75 \div 90 \times 100 \approx 83.3\%$.

The system must not report this as a fully verified 75% attendance rate across all 100 opportunities, because ten days have no recorded attendance status.

It must also not treat the missing ten days as confirmed absences.

A suitable report would explain:

Provisional attendance rate: 83.3% of recorded eligible opportunities.

Recorded attended opportunities: 75.

Recorded absent opportunities: 15.

Missing attendance records: 10.

Total eligible opportunities: 100.

The school can then address the missing registers before producing a final report.

This example demonstrates the relationship between attendance status, eligible opportunities, calculation policy, and data completeness.

The correct result depends not only on arithmetic but also on whether the underlying records are complete and correctly classified.

21. Decisions Optimus Orien Should Make Before Release

The following decisions should be documented rather than left to the developer's assumptions.

Decision 1: Initial attendance model. Decide whether the first release supports daily attendance only or both daily and lesson-level attendance. The simplest reliable option is to implement one clearly defined model first while keeping the architecture capable of supporting additional models later.

Decision 2: Initial status set. Define the default statuses offered during school setup. Allow authorised schools to configure additional statuses without changing the software code.

Decision 3: Attendance-rate policy. Define how each supported status affects each attendance metric. Do not assume that the same rule is suitable for every school.

Decision 4: Register finalisation. Define who may submit, finalise, reopen, and correct registers, including the audit information required for each action.

Decision 5: Missing-data policy. Decide how provisional reports are calculated when records are incomplete, how missing information is displayed, and what conditions must be met before a report is treated as final.

Decision 6: Attendance-review rules. Define whether the initial release includes configurable review thresholds. If included, make the rules transparent and require human review.

Decision 7: Historical reporting. Decide how attendance policies, enrolments, schedules, and calendars are versioned so historical reports remain interpretable.

Decision 8: Staff attendance. Decide whether employee attendance is outside the initial release. If included, define it as a separate module with its own requirements.

Decision 9: Data retention. Establish a policy for retaining, exporting, correcting, and deleting attendance information in accordance with the applicable legal and operational requirements.

Decision 10: Notifications. Decide whether attendance notifications are part of the current release. Do not assume that parent notifications, automated messages, or messaging integrations already exist merely because attendance data is available.

22. Final Principles

The attendance system must be built around accurate records and explicit rules.

It must know when a student was expected to attend, what was recorded, and which calculation policy applies.

It must distinguish an absence from missing data, a daily attendance opportunity from a lesson, and a student's current enrolment from their historical enrolments.

It must protect student information, prevent unauthorised changes, preserve correction histories, and explain how every attendance metric was calculated.

It must support different school calendars, schedules, attendance statuses, and reporting requirements without pretending that all educational systems operate identically.

Most importantly, attendance should be treated as an educational indicator that can inform a human decision, not as automatic proof of a student's character, motivation, or academic ability.

For Optimus Orien, a trustworthy attendance system is not merely one that displays percentages. It is one that can explain where those percentages came from, which records were included, what information is missing, and which rules were applied.

That transparency is what makes attendance information useful for school leaders, teachers, students, and parents.

Part 10 covers the complete assessment and marks system for Optimus Orien. This is one of the most important foundations of the platform because inaccurate assessment calculations would make student profiles, class averages, academic trends, and school comparisons unreliable.

The document explains the educational concepts from the beginning, then translates them into requirements and acceptance tests your developer can use while building the software with AI coding tools.

PART 10

THE COMPLETE ASSESSMENT AND MARKS SYSTEM

Understanding How Schools Measure Learning and How Optimus Orien Must Calculate, Store, and Present Academic Performance

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner and software developer
Primary objective: Build an accurate, transparent, configurable, and internationally adaptable academic assessment system.

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1. Why This Document Exists

Optimus Orien is a School Performance Intelligence and Early Intervention platform.

One of its central responsibilities is helping schools understand how students are performing academically. To do that reliably, it needs to collect assessment results and turn them into meaningful indicators.

However, academic assessment is more complicated than asking a teacher to enter a mark out of 100.

Consider these examples.

A student scores 18 out of 20 on a quiz.

Another scores 72 out of 100 on an examination.

A third receives an A grade rather than a numerical mark.

A fourth transfers from a school that uses letter grades to one that uses percentages.

A fifth misses an examination because of an approved absence.

Although all these situations relate to academic assessment, they cannot all be handled using the same simplistic calculation.

The software must understand what each result represents, what the student was assessed on, how the result was calculated, and which rules apply.

The central principle is that Optimus Orien must preserve the original assessment result and apply explicit, configurable rules when calculating academic-performance indicators.

It must not assume that every school uses the same examinations, grading scales, reporting periods, or assessment weights.

This document establishes the foundation for doing that correctly.

2. What Is Assessment in Education?

An assessment is a method of gathering evidence about what a student knows, understands, or can do.

Teachers and schools use assessments to understand student learning and make educational decisions.

Assessments can measure different things.

A Mathematics examination may measure a student's ability to solve problems. An English speaking assessment may measure oral communication. A science practical may measure the student's ability to perform an experiment and interpret the results.

These assessments may all contribute to an understanding of student performance, but they do not necessarily measure the same skill.

Optimus Orien should therefore preserve the identity and purpose of each assessment rather than treating every mark as an interchangeable number.

2.1 Assessment, Examination, Test, Quiz, and Assignment

These terms are related, but their meanings can vary between schools.

An **assessment** is the broad category. It includes any organised activity used to gather evidence about learning.

An **examination** is usually a formal assessment conducted under specified conditions. It may cover a substantial portion of a subject or course.

A **test** is an assessment used to measure knowledge or performance in a particular area. It may be formal or informal.

A **quiz** is often a shorter assessment covering a smaller amount of material.

An **assignment** is a task completed by a student, sometimes independently and sometimes collaboratively.

A **practical assessment** measures performance through an activity, demonstration, experiment, or other applied task.

A **project** may assess the ability to combine knowledge, research, problem-solving, and practical application.

These are common uses, not universal definitions. A school may use the terms differently.

The software should allow schools to classify assessments according to their own academic arrangements.

2.2 Formative and Summative Assessment

Two important concepts in educational assessment are formative assessment and summative assessment.

Formative assessment gathers evidence that helps a teacher and student understand learning progress and identify what needs improvement.

Examples may include a short quiz used to identify a misunderstanding, feedback on a draft, or a classroom task used to check understanding.

Summative assessment evaluates learning at a particular point, often at the end of a unit, term, course, or programme.

Examples may include a final examination, an end-of-term test, or a final project grade.

The distinction concerns the purpose and use of the assessment, not simply its size.

A short quiz can contribute to a final grade, and a large examination can also reveal gaps that teachers use to improve future instruction.

For Optimus Orien, the assessment type and its reporting purpose should be configurable.

The platform must not automatically assume that every quiz is formative or that every examination is summative.

3. The Difference Between an Assessment and an Assessment Result

This distinction is essential for the database design.

An assessment is the activity that students are expected to complete.

An assessment result is the outcome recorded for a particular student.

Imagine that a teacher creates a Mathematics examination worth 50 marks.

The examination itself has properties such as its name, subject, date, maximum marks, and reporting period.

Student A receives 42 marks.

Student B receives 35 marks.

Student C is recorded as absent.

The examination is one assessment, but it has three different student outcomes.

The system should therefore distinguish the assessment definition from the result for each student.

3.1 Assessment Definition

The assessment definition should identify the assessment and its academic context.

Depending on the school, this may include:

-

Assessment name.

-
-

Assessment type.

-
-

School and academic programme.

-
-

Academic year and reporting period.

-
-

Subject.

-
-

Class or group.

-
-

Assessment date.

-
-

Maximum possible marks.

-
-

Assessment weighting, if applicable.

-
-

Grading rules, if applicable.

-
-

Whether the assessment is draft, published, or finalised.

-

Not every assessment will require every field. The system should support the information needed by the school's configured assessment model.

3.2 Student Assessment Result

A student assessment result should identify the student's outcome for that particular assessment.

It may contain:

-

Student identifier.

-
-

Enrolment identifier.

-
-

Assessment identifier.

-
-

Marks obtained, where applicable.

-
-

Maximum marks applicable to the result, if the system permits individual variations.

-
-

Grade or rubric outcome, where applicable.

-
-

Result status.

-
-

Teacher feedback, if supported.

-
-

Recorded-by user.

-
-

Date recorded.

-
-

Correction history.

-
-

Applicable grading or calculation rules.

-

The developer must not assume that every result is a number.

Some assessment systems use marks. Others use letter grades, performance bands, rubric levels, pass/fail outcomes, or combinations of these.

Optimus Orien should preserve the original outcome and support the result types that the school has configured.

4. Marks Obtained and Maximum Marks

A mark is a numerical value awarded for an assessment.

For example, a student may receive 18 marks out of a possible 20.

In this case:

Marks obtained = 18

Maximum marks = 20

The maximum marks represent the total available score for the assessment or applicable result.

A raw mark cannot be interpreted properly without knowing the maximum possible marks.

For example, 18 marks could represent an excellent result in an assessment worth 20 marks or a poor result in an assessment worth 100 marks.

The system must store the marks obtained and the applicable maximum marks separately.

4.1 Converting Marks to a Percentage

When the assessment is numerical and the maximum marks are positive, a simple percentage can be calculated as:

Percentage = Marks obtained ÷ Maximum marks × 100

Example:

A student receives 18 out of 20.

18 ÷ 20 × 100 = 90%.

Another student receives 72 out of 100.

72 ÷ 100 × 100 = 72%.

Percentages help users compare results from assessments with different maximum marks, but only within appropriate limits.

4.2 What Percentages Do Not Tell Us

A percentage does not automatically tell us how difficult an assessment was, whether two assessments measured the same learning outcomes, or whether a student has mastered an entire subject.

For example, 90% on a short quiz covering one topic is not necessarily equivalent to 90% on a comprehensive final examination.

The software may display both percentages, but it should preserve the assessment context and avoid claiming that the results are educationally equivalent.

4.3 Validation of Marks

The system must validate marks against the applicable assessment rules.

For a conventional numerical assessment with a maximum of 50 marks, a mark of 42 may be valid, while a mark of 55 would normally be invalid.

However, some assessment systems may allow negative marking, decimal marks, adjusted scores, or other specialised rules.

The system should support the configurations that Optimus Orien explicitly implements. It must not silently assume every school uses whole-number marks between zero and the maximum.

Where a result violates the configured rules, the system should explain the problem and prevent invalid publication.

5. Assessment Result Statuses

A result is not always a completed numerical mark.

Students may be absent, awaiting a make-up assessment, exempted under an approved policy, or still waiting for a teacher to enter their result.

The system must distinguish these situations.

5.1 Recorded Result

A recorded result means that a valid outcome has been entered for the student.

The outcome may be a numerical mark, grade, rubric level, or another supported result type.

5.2 Not Yet Marked

Not Yet Marked means the assessment result has not been entered or finalised.

It does not mean that the student scored zero.

For example, a teacher may have created a final examination but not yet graded every student's paper.

The platform must display the missing result as incomplete rather than interpreting it as a zero.

5.3 Absent

Absent means the student did not participate in the assessment under the applicable attendance or examination rules.

It does not automatically determine the mark that should be entered.

The school may have a separate policy for make-up examinations, authorised absences, or missed assessments.

The platform must record the result status and apply the configured academic policy.

5.4 Exempt

Exempt means the student is not required to complete the assessment under an applicable rule or authorised decision.

An exemption must not automatically be treated as a mark of zero.

The system must define whether exempt assessments are excluded from the calculation, handled through an alternative assessment, or treated under another explicit policy.

5.5 Incomplete

Incomplete means that the result or assessment process is not yet complete.

For example, a student may have completed one component of a multi-part assessment while another component remains outstanding.

The system must not automatically treat incomplete results as final.

5.6 Withdrawn or Not Applicable

Depending on the school's academic arrangements, an assessment may no longer apply to a student, or a student may have withdrawn from the relevant course.

These situations should be represented through explicit statuses or enrolment changes rather than by inventing a zero mark.

5.7 Why Statuses Matter

Suppose five students are expected to complete an examination.

Three have marks recorded.

One is absent.

One has not yet been graded.

If the system automatically assigns zero to the absent student and the ungraded student, the calculated class average will be misleading.

Optimus Orien must distinguish actual results from missing, incomplete, and non-applicable outcomes.

6. Subjects, Assessments, and Academic Reporting Periods

Assessment results must be connected to the correct subject and academic context.

A student may study Mathematics, English, Science, and History. Each subject may have several assessments during a term.

The system must understand which assessments belong to which subjects, classes, academic years, and reporting periods.

6.1 Academic Year

An academic year is the defined period in which a school organises its educational activities.

It may follow a calendar-year arrangement or another schedule.

Optimus Orien must not assume that every academic year begins in January or ends in December.

6.2 Term or Reporting Period

A reporting period is the period for which a school collects or presents academic results.

It may be a term, semester, quarter, trimester, module, or another configured period.

For example, a school might report results at the end of three terms.

Another may use two semesters.

The system should allow the relevant academic structure to be configured.

6.3 Assessment-to-Subject Relationship

Every subject-related assessment should connect to the relevant subject offering and student enrolment.

If a school changes its subject name, the historical assessment records must remain connected to the correct subject identity.

If a student transfers to another class or school, the previous results must not be overwritten with the new class assignment.

6.4 Final Results and Historical Records

A student may have several results for the same subject across multiple reporting periods.

The platform should preserve these records so that users can understand the student's progress over time.

A new academic year must not overwrite the previous year's results.

The system must be able to distinguish the student's current academic record from their historical records.

7. Understanding Assessment Weighting

Weighting is one of the most important concepts in academic calculations.

A weight determines how much a particular assessment contributes to a larger result.

Schools may use different methods to combine assessment results.

The developer must understand these methods before implementing subject averages or final grades.

7.1 Method A: Simple Average of Percentages

Suppose a student receives the following results:

Quiz 1: 80%.

Quiz 2: 60%.

Quiz 3: 100%.

If all three percentages receive equal weight, the simple average is:

(80 + 60 + 100) ÷ 3 = 80%.

Each quiz contributes equally to the final average, regardless of its maximum marks.

This method can be appropriate when the school intentionally gives each assessment equal weight.

However, it is not the same as adding all marks together and dividing by all available marks.

7.2 Method B: Combine Marks Before Converting to a Percentage

Now suppose the student has these results:

Assessment A: 8 out of 10.

Assessment B: 30 out of 50.

Assessment C: 18 out of 20.

The combined marks obtained are:

8 + 30 + 18 = 56.

The combined maximum marks are:

10 + 50 + 20 = 80.

The combined percentage is:

56 ÷ 80 × 100 = 70%.

This method gives more influence to assessments with larger maximum marks because each raw mark contributes to the combined total.

That may or may not be the intended academic policy.

The developer must not assume that this method is always correct simply because it is easy to calculate.

7.3 Method C: Weighted Average

A weighted average assigns an explicit weight to each assessment or assessment category.

Suppose a school defines the following assessment structure for a subject:

Classwork: 20%.

Midterm examination: 30%.

Final examination: 50%.

A student receives:

Classwork: 80%.

Midterm: 70%.

Final examination: 90%.

The weighted result is:

$(80 \times 0.20) + (70 \times 0.30) + (90 \times 0.50)$

$= 16 + 21 + 45$

$= 82\%$.

The student's weighted result is 82%.

The weights total 100%, so the calculation represents the intended contribution of each component.

This method is appropriate only when the school has actually defined the weights and the results are eligible for inclusion.

7.4 Why These Methods Cannot Be Interchanged

The three methods can produce different results from the same assessment data.

A school may intentionally use equal assessment weights, combine raw marks, or apply category weights.

The system must preserve the chosen calculation method rather than selecting one automatically.

Optimus Orien must not calculate a final subject result until it knows which assessment method and grading rules apply.

8. Assessment Categories and Category Weighting

Schools may organise assessments into categories.

For example:

- Classwork.
- Homework.
- Quizzes.
- Practical work.
- Midterm examinations.
- Final examinations.

A school may assign weights to these categories rather than giving every individual assessment a separate weight.

Consider a subject where the school assigns 30% to coursework and 70% to examinations.

The coursework category may contain several assignments and quizzes. The examination category may contain a midterm and a final examination.

The system must know how results are combined inside each category and how the category contributes to the overall subject result.

These are two separate calculations.

8.1 Calculating a Category Result

Suppose a coursework category contains three assessments:

Assignment 1: 80%.

Assignment 2: 90%.

Quiz: 70%.

If the school gives these assessments equal weight within the category, the category result is:

(80 + 90 + 70) ÷ 3 = 80%.

If the school instead assigns individual weights, the category result must use those weights.

The system must not assume that category weighting and individual assessment weighting are interchangeable.

8.2 Calculating the Overall Subject Result

Suppose coursework contributes 30% and examinations contribute 70%.

The coursework result is 80%.

The examination result is 75%.

The overall result is:

(80 × 0.30) + (75 × 0.70)

= 24 + 52.5

= 76.5%.

The result is 76.5%, assuming the school permits decimal results and the assessment categories are complete and eligible.

If the school requires rounding, the platform should apply the configured rounding rule at the appropriate stage.

8.3 Weight Validation

If a school requires category weights to total 100%, the software must validate that condition.

For example, weights of 30%, 30%, and 40% total 100%.

Weights of 30%, 30%, and 30% total only 90%.

The system must not silently assume that the missing 10% should be assigned to a particular category.

If a school uses a valid scheme in which weights do not directly total 100%, the system must support that scheme only through an explicitly defined calculation method.

9. Missing Results and Incomplete Assessment Categories

One of the most common calculation problems occurs when a student has not completed every assessment.

Suppose a subject has three assessments:

Quiz: 80%.

Midterm: 70%.

Final examination: Not yet marked.

The quiz contributes 20%, the midterm contributes 30%, and the final examination contributes 50%.

The system cannot automatically calculate the final subject result by treating the missing final examination as zero.

Doing so would turn an incomplete academic record into an apparently poor result.

It also cannot silently redistribute the final examination's weight unless the applicable policy explicitly permits that behaviour.

The software should distinguish between a provisional result and a final result.

9.1 Provisional Results

A provisional result is calculated using an explicitly defined rule while some information remains incomplete.

For example, a school may choose to show the weighted contribution of completed assessments while making it clear that the remaining assessment is outstanding.

In the example above, the completed contributions are:

Quiz contribution: 80 × 0.20 = 16.

Midterm contribution: 70 × 0.30 = 21.

Completed contribution total: 37 percentage points.

The missing final examination contributes nothing to the current recorded total because no final result exists. This does not mean the student received zero on the final examination.

The system could display:

Recorded contribution so far: 37 out of the 100 possible weighted percentage points.

It should not present 37% as the student's final subject grade.

If the school supports a provisional average based only on completed assessments, it must use a separately defined formula and label that result clearly.

9.2 Missing Results Must Be Visible

The platform should identify outstanding assessments and incomplete result sets.

Teachers and school administrators should be able to distinguish between:

-

A result that has been recorded.

-
-

A result that is awaiting marking.

-
-

A student who was absent.

-
-

An exempt assessment.

-
-

A result that is incomplete.

-
-

An assessment that does not apply to the student.

-

These conditions must not be merged into a generic zero.

9.3 Excluded Results and Renormalisation

Some academic policies permit the removal of an assessment from a calculation under specified conditions.

If an assessment is excluded, the system may need to redistribute the remaining weights.

For example, if a 20%-weighted assessment is formally excluded and the policy requires the remaining 80% of weights to be renormalised to 100%, the system may do so.

However, the system must not renormalise weights automatically whenever a result is missing.

Exclusion and renormalisation must be explicitly permitted by the applicable academic policy.

10. Grading Scales: Turning Numerical Results into Grades

A grading scale maps a numerical result or another assessment outcome to a grade or performance category.

Schools may use percentages, letters, grade points, performance bands, pass/fail categories, or other systems.

Optimus Orien must support the grading arrangements it explicitly implements and must not assume one scale is universal.

10.1 Example of a Percentage-Based Grading Scale

A hypothetical school might use the following scale:

Percentage	Grade
90–100	A
80–89.99	B
70–79.99	C
60–69.99	D
Below 60	F

This is only an illustrative example. It must not be presented as an official grading scale for Pakistan, Cambridge International, the International Baccalaureate, or any other educational system.

A school must configure its own authorised grading rules or use a verified configuration appropriate to its programme.

10.2 Boundary Rules

Grading boundaries must be explicit.

For example, a grade may begin at 80%.

The system must define whether 80% belongs to that grade and how decimal values are handled.

It must also define whether the percentage is rounded before applying the grading scale.

A result of 79.96% could receive a different grade depending on the rounding policy.

The developer must not decide this arbitrarily.

10.3 Rounding

Rounding changes the displayed or calculated value according to a defined rule.

For example, 76.46% might be displayed as 76.5% when one decimal place is used.

However, the underlying value should be preserved with sufficient precision for reliable calculations.

The platform must define when rounding occurs:

-

When calculating an individual assessment percentage.

-
-

When calculating a category result.

-
-

When calculating a final subject result.

-
-

When displaying the result.

-
-

When assigning a grade.

-

The system should not repeatedly round intermediate results if doing so changes the final outcome unexpectedly.

10.4 Grades Are Not Always Equivalent Across Schools

A grade of A at one school may not represent exactly the same percentage or performance standard as an A at another school.

Even two programmes that use letter grades may define them differently.

Optimus Orien should preserve the grading scale and academic context that produced each grade.

It must not assume that identical grade labels automatically indicate equivalent achievement.

11. Grade Points, GPA, and CGPA

Some educational systems use grade points or grade-point averages.

These require separate explanations because they cannot be calculated reliably by simply averaging every visible grade.

11.1 Grade Point

A grade point is a numerical value associated with a grade under a particular grading system.

For example, a hypothetical grading system might assign 4 points to A, 3 points to B, 2 points to C, and 1 point to D.

These are illustrative values, not a universal standard.

The system must know the relevant grade-point mapping before using it.

11.2 Grade Point Average

A Grade Point Average (GPA) summarises grade points over a defined set of courses or subjects under a specified calculation method.

Some systems calculate a simple average of grade points. Others use course credits or another weighting method.

For a credit-weighted GPA, the formula is:

GPA = Sum of (Grade Point × Course Credits) ÷ Sum of Course Credits

Consider three courses in a hypothetical system.

Course A: Grade point 4.0, worth 3 credits.

Course B: Grade point 3.0, worth 2 credits.

Course C: Grade point 2.0, worth 1 credit.

The calculation is:

$(4.0 \times 3 + 3.0 \times 2 + 2.0 \times 1) \div (3 + 2 + 1)$

$= (12 + 6 + 2) \div 6$

$= 20 \div 6$

≈ 3.33.

This example assumes that the institution uses a credit-weighted GPA and the stated grade-point values.

The system must not apply this formula where the institution uses a different method.

11.3 Cumulative Grade Point Average

Cumulative Grade Point Average (CGPA) summarises grade-point performance across multiple eligible courses or academic periods under the applicable institutional rules.

It may depend on course credits, repeated courses, withdrawals, transferred credits, and other academic policies.

The platform should not calculate CGPA merely by averaging displayed term GPAs unless that method is explicitly valid for the institution.

For example, if one term contains 12 credits and another contains 24 credits, a simple average of the two term GPAs may not produce the correct cumulative result under a credit-weighted system.

11.4 Should Optimus Orien Implement GPA Immediately?

The product should prioritise the academic calculation methods required by its initial customers.

If the initial implementation focuses on numerical marks, percentages, subject averages, and school-defined grades, it should implement those reliably before expanding into every possible GPA or credit-based model.

However, the data model should avoid making future support impossible.

GPA and CGPA must be treated as separate, configurable academic calculations rather than as synonyms for a percentage average.

12. Rubric-Based Assessment

Not every assessment uses numerical marks.

A rubric is a structured assessment tool that describes criteria and performance levels.

For example, a speaking assessment may evaluate pronunciation, vocabulary, fluency, and organisation.

Each criterion may have several performance levels, each with its own description.

A student may perform strongly in vocabulary but need improvement in fluency.

A rubric can preserve that detail rather than reducing the entire assessment to one unexplained mark.

12.1 Rubric Criteria

A criterion is the specific aspect being assessed.

Examples include:

-

Accuracy of reasoning.

-
-

Clarity of communication.

-
-

Use of evidence.

-
-

Practical technique.

-
-

Organisation of an argument.

-

The criteria should be defined by the school or assessment designer.

12.2 Performance Levels

A performance level describes the quality of work demonstrated for a criterion.

A school might use levels such as Beginning, Developing, Proficient, and Advanced.

These labels are examples only. The school must configure the labels and descriptions it uses.

12.3 Rubric Results

The system should preserve the student's result for each criterion where rubric assessment is supported.

If the rubric includes numerical values, the system may calculate a combined result using the defined rubric rules.

If the rubric uses descriptive levels only, the platform should not invent numerical values simply to force the assessment into a percentage.

A rubric result should retain its meaning and structure.

12.4 Relationship to Subject Skills

Optimus Orien also intends to track subject-linked skills, potentially using ratings on a scale such as 1–10.

A rubric-based assessment and a teacher-entered skill rating may be related, but they are not automatically the same data type.

The platform should distinguish between:

-

A result from a specific assessment.

-
-

A teacher's rating of a student's skill.

-
-

A calculated academic-performance indicator.

-

If a rubric result contributes to a skill rating, the relationship must be explicitly defined.

The platform should not automatically convert every rubric level into a 1–10 skill score without an approved mapping.

13. Subject Performance and Overall Academic Performance

This section explains how assessment results can be aggregated into the academic indicators displayed by Optimus Orien.

The central challenge is that a school may have many assessments for each subject and many subjects for each student.

The platform must calculate each level correctly.

13.1 Individual Assessment Performance

This is the result of one student on one assessment.

For a numerical assessment, the system may display marks obtained, maximum marks, percentage, and applicable grade.

The original mark should remain available even when a percentage or grade is also displayed.

13.2 Subject Performance

Subject performance summarises a student's results in a particular subject over a defined reporting period.

It may be calculated using:

-

A school-defined weighted assessment formula.

-
-

A simple average of assessment percentages.

-
-

Combined marks, where appropriate.

-
-

Another explicitly defined grading method.

-

The applicable method must be stored or referenced.

For example, a student's English subject result for Term 1 should be calculated using the English assessment rules applicable to Term 1.

The platform must not assume that the same assessment weights apply to every subject or every academic year.

13.3 Overall Academic Performance

Overall academic performance summarises performance across multiple subjects for a defined period.

A simple method is to calculate the arithmetic mean of eligible subject percentages.

Suppose a student has these subject results:

Mathematics: 80%.

English: 70%.

Science: 90%.

The simple subject average is:

(80 + 70 + 90) ÷ 3 = 80%.

This calculation gives each subject equal weight.

That may be suitable for a school that intentionally uses equal subject weighting.

However, other schools may use credits, subject weights, or different aggregation rules.

The system must therefore identify the method used.

13.4 Missing Subjects

Suppose a student studies six subjects, but only four subject results are finalised.

The platform must not automatically interpret the two missing results as zero.

It should identify the incomplete record and apply the school's defined policy for provisional reporting.

If the system displays an average across the four available subjects, it must label the result as an average of the four available subject results rather than presenting it as a final average across all six subjects.

13.5 Class Average

A class average summarises the performance of a defined class or group over a specified period.

However, the system must define exactly what it is averaging.

A useful option for Optimus Orien is:

1.

Calculate each student's eligible subject result using the applicable assessment rules.

2.
3.

Calculate the class average for that subject using the eligible student results.

4.
5.

Calculate a broader class-level indicator from subject averages only when the aggregation method is explicitly defined.

6.

This allows the platform to show subject-level comparisons and class-level performance without confusing different units.

The class average must identify the eligible students and reporting period.

13.6 Class Average Is Not the Same as Total Class Marks

Suppose one class has 20 students and another has 35.

Adding all marks together would generally produce a larger total for the larger class even if the students' average performance were identical.

A total mark is therefore not a suitable substitute for an average when the purpose is to compare performance across classes of different sizes.

The system should use an appropriate average or rate for the intended comparison and display the relevant denominator.

14. Comparing Students Fairly

Academic comparisons can help schools identify patterns, but only when the compared results are sufficiently compatible.

A percentage is not a guarantee that two students completed equivalent assessments.

14.1 Compare the Same Subject Where Possible

Comparing two students' Mathematics results is generally more interpretable than directly comparing one student's Mathematics percentage with another student's English percentage.

Even within the same subject, the assessments and grading rules may differ.

The platform should therefore identify the subject, reporting period, assessment context, and grading rules where relevant.

14.2 Compare the Same Reporting Period

A student's Term 1 result should not be directly compared with another student's full-year result as if they covered the same period.

The platform should filter or label comparisons to make their reporting periods clear.

14.3 Distinguish Percentage Points From Relative Change

Suppose a student's Mathematics result increases from 60% to 75%.

The increase is 15 percentage points.

The relative increase is:

$(75 - 60) \div 60 \times 100 = 25\%$.

Both figures are mathematically valid, but they describe different things.

The interface must use the correct label for the selected calculation.

14.4 Preserve Context

A student's result may be affected by changes in curriculum, assessment design, maximum marks, grading scales, or school programme.

The system should preserve these contextual details where available.

If the system cannot establish that two results are sufficiently comparable, it should avoid presenting the comparison as definitive.

14.5 Avoid Unsupported Conclusions

A lower mark does not, by itself, explain why a student performed poorly.

Similarly, a higher mark does not prove that a particular teaching method caused improvement.

Optimus Orien should show the evidence available in the records without claiming to know causes that have not been established.

15. Historical Academic Records

Academic data becomes especially valuable when the platform preserves it across multiple years.

However, historical comparison introduces additional complexity.

A student may move between schools, programmes, curricula, subjects, or grading systems.

The system must preserve the context of the original result.

15.1 Do Not Overwrite Previous Results

Suppose a student's Mathematics result was 72% in one academic year and 84% in the next.

The platform should retain both records.

A new result must not overwrite the previous year's value.

15.2 Preserve the Original Result

If a teacher records 42 out of 50, the system should preserve that original result.

A later grading-policy change may affect the grade displayed for the result, but it should not silently change the original mark.

If an authorised correction changes 42 to 44, the system should retain the correction history.

15.3 Preserve Grading Rules

Suppose a school changes its grade boundaries between academic years.

The system must preserve enough information to establish which boundaries applied when a historical result was finalised.

If a report is recalculated under a new policy, the system should make that change explicit.

15.4 Student Transfers

A student's academic history may include results from different schools.

The platform should preserve the source school, subject, programme, reporting period, and applicable grading context for each record.

A transferred result should not automatically be converted into a local grade unless the receiving school has an approved conversion rule.

Where the original grading scale is incompatible with the receiving school's scale, the system should preserve the original result and clearly identify any authorised conversion.

16. Result Corrections and Academic Integrity

Assessment records may contain mistakes.

A teacher may enter the wrong mark, select the wrong student, or publish a result before completing the marking process.

The software must support correction without erasing the history of the record.

16.1 Draft and Published Results

The system should distinguish between results that are being entered and results that have been approved or published under the school's workflow.

A draft result should not automatically appear as a final academic result in every student, parent, principal, or organisation dashboard.

The school's configuration should determine who can see draft, provisional, and final results.

16.2 Correcting a Mark

If a mark changes from 32 to 42, the system should record the old mark, new mark, user making the correction, timestamp, and reason where required.

The correction must trigger recalculation of affected subject results, overall averages, and reports.

The platform should not leave the student dashboard displaying a new mark while the class average still uses the old mark.

16.3 Audit History

The audit history should allow authorised users to understand what changed and when.

Ordinary users must not be able to alter or delete audit records.

16.4 Prevent Unauthorised Result Changes

A teacher should be permitted to enter or correct marks only within their assigned scope and the school's correction rules.

A teacher assigned to one class should not be able to alter another teacher's restricted assessment results by changing a record identifier in a request.

Permissions must be enforced on the server.

17. Recommended Data Model

The following is a conceptual starting point for the database. It is not a requirement to create exactly these database tables if the implementation uses a different architecture. The important point is that the relationships and historical meaning must be preserved.

17.1 Assessment

Represents the assessment itself.

It should identify the assessment, subject, academic context, reporting period, maximum marks where applicable, assessment type, and applicable calculation rules.

17.2 Assessment Result

Represents one student's result for one assessment.

It should preserve the original result, result type, status, applicable maximum marks, and correction history.

17.3 Assessment Category

Represents a group of assessments, such as coursework, quizzes, or examinations, when category-based weighting is used.

17.4 Assessment Weighting Rule

Represents the weights and aggregation method that determine how results contribute to a larger academic result.

It should preserve the relevant effective dates or version so that historical calculations remain interpretable.

17.5 Grading Scale

Represents a configured system for translating eligible numerical results into grades or performance bands.

It should contain the boundaries, labels, applicable rules, and version information.

17.6 Rubric and Rubric Result

If rubric assessments are supported, the system should store the rubric definition, criteria, performance levels, and the student's result for each relevant criterion.

17.7 Subject Result

Represents the calculated subject-level result for a student over a defined academic period.

It should identify the calculation method and the underlying results or rule version used.

Where the system calculates results dynamically rather than storing them, it must still preserve enough information to reproduce the historical result.

17.8 Overall Academic Result

Represents an aggregate result across eligible subjects for a defined period.

It must identify the aggregation method, included subjects, denominator, completeness, and applicable rules.

17.9 Audit Record

Records authorised changes to assessments, results, weights, grading rules, and published calculations.

The audit record should identify the action, user, time, and relevant before-and-after values.

18. Software Implementation Requirements and Acceptance Tests

The following requirements define the minimum expected behaviour for Optimus Orien's assessment system.

The developer should implement and test these requirements before building complex academic dashboards.

18.1 Assessment Configuration

ASM-001 — Assessment Definitions

The system must allow authorised users to create assessments with the required academic context.

Acceptance test: A teacher creates a Mathematics quiz for an assigned class, links it to the correct academic period, defines its maximum marks, and saves it without creating results for unrelated students.

ASM-002 — Configurable Assessment Types

The system must support configurable assessment categories or types.

Acceptance test: An administrator creates a Practical Assessment type. The system stores the type and makes it available according to the relevant permissions without requiring a code change.

ASM-003 — Subject and Class Connections

Every subject-related assessment must be connected to the appropriate subject and class or student group.

Acceptance test: A Mathematics assessment cannot accidentally appear as an English assessment because a display label or class name changes.

ASM-004 — Academic Periods

Assessments must be linked to the relevant academic year and reporting period.

Acceptance test: A Term 1 assessment is not included in a Term 2 report unless an explicit rule permits it.

ASM-005 — Assessment Status

The system must distinguish draft, published, and finalised assessment states where those states are supported.

Acceptance test: A draft examination is not treated as a published final result in the student dashboard.

ASM-006 — Assessment Maximum Marks

The system must preserve the applicable maximum marks for numerical assessments.

Acceptance test: A teacher creates an assessment worth 50 marks. A result of 42 is interpreted as 42 out of 50, not 42 out of 100.

18.2 Student Results

ASM-007 — Separate Assessment and Result Records

The system must distinguish assessment definitions from individual student results.

Acceptance test: One examination is created for 30 students. Each student can have a separate result without creating 30 duplicate assessment definitions.

ASM-008 — Original Marks

The system must preserve marks obtained and the applicable maximum marks separately.

Acceptance test: A student receives 18 out of 20. The system retains both values and calculates the percentage as 90%.

ASM-009 — Result Validation

The system must validate results against the configured marking rules.

Acceptance test: Under a conventional 0–50 marking rule, a mark of 55 is rejected with a clear explanation unless an explicitly supported rule permits that value.

ASM-010 — Decimal Marks

The system must support decimal marks where enabled by the school's assessment policy.

Acceptance test: A school permits half marks. A result of 17.5 out of 20 is stored and calculated without being silently rounded to 18.

ASM-011 — Non-Numerical Results

The system must support configured non-numerical outcomes where implemented.

Acceptance test: An assessment using a configured rubric stores the relevant rubric level rather than forcing the teacher to enter an invented percentage.

ASM-012 — Result Statuses

The system must distinguish recorded, unmarked, absent, exempt, and incomplete outcomes where those statuses are supported.

Acceptance test: A student with an unmarked examination has no fabricated zero mark, and the result is displayed as incomplete or awaiting marking.

ASM-013 — Student Enrolment Context

Results must remain connected to the appropriate student enrolment and academic context.

Acceptance test: A student transfers to another school. Historical results remain connected to the original school and enrolment.

ASM-014 — Duplicate Prevention

The system must prevent unintended duplicate results for the same student and assessment.

Acceptance test: A user attempts to create a second result for the same student and assessment. The system blocks the duplicate or routes the action through an authorised correction workflow.

18.3 Percentages, Weights, and Subject Results

ASM-015 — Percentage Calculation

The system must calculate a numerical assessment percentage using the applicable maximum marks.

Acceptance test: A student receives 18 out of 20. The displayed percentage is 90%, subject to the configured display precision.

ASM-016 — Explicit Aggregation Method

The system must identify the method used to combine assessment results.

Acceptance test: One subject uses equal assessment weights and another uses category weights. Each subject is calculated using its own configured rules.

ASM-017 — Weighted Assessment Calculation

The system must correctly calculate weighted results when weights are configured.

Acceptance test: A subject has coursework weighted at 20%, a midterm at 30%, and a final examination at 50%. Results of 80%, 70%, and 90% produce a weighted result of 82%.

ASM-018 — Weight Validation

The system must validate weights according to the selected calculation method.

Acceptance test: A calculation method requires weights totalling 100%. The system rejects weights totalling 90% and explains the problem instead of silently redistributing the difference.

ASM-019 — Category Calculations

The system must calculate category results according to the configured rules.

Acceptance test: A coursework category contains three equally weighted assessments with results of 80%, 90%, and 70%. Its category result is 80%.

ASM-020 — Incomplete Assessments

The system must not automatically treat missing results as zero.

Acceptance test: A final examination has not been marked. The student has no fabricated final examination mark, and the subject result is labelled incomplete or provisional according to the applicable policy.

ASM-021 — Exempted Assessments

The system must apply the configured policy for exempt assessments.

Acceptance test: An assessment is marked Exempt under an approved rule. The system excludes or otherwise handles it according to the rule instead of automatically assigning zero.

ASM-022 — Provisional Results

The system must distinguish provisional results from final results.

Acceptance test: A subject has completed assessments and one outstanding assessment. The dashboard labels the calculated contribution or permitted provisional average clearly and does not misrepresent it as a final grade.

ASM-023 — Exclusion and Renormalisation

The system must not automatically redistribute weights when results are missing.

Acceptance test: A required assessment is unmarked. The system preserves the original weighting scheme unless an authorised rule explicitly permits exclusion and renormalisation.

ASM-024 — Subject Result Versioning

The system must preserve or identify the calculation rules used for a subject result.

Acceptance test: Assessment weights change for the next academic year. Historical subject results remain identifiable as having used the previous weights.

18.4 Grading Scales and Grade Points

ASM-025 — Configurable Grading Scales

The system must support configurable grading scales.

Acceptance test: An authorised administrator defines a percentage-to-grade mapping. The system applies it to eligible results within the configured academic context.

ASM-026 — Boundary Validation

The system must validate grading boundaries and prevent unintended overlaps or gaps where the chosen scale requires complete coverage.

Acceptance test: An administrator defines two conflicting grade ranges. The system identifies the conflict and prevents publication of the invalid scale.

ASM-027 — Rounding Rules

The system must apply configured rounding rules consistently.

Acceptance test: A school defines one-decimal display precision and a specified grading-boundary rule. The system displays the result at the correct precision and assigns the grade according to the configured sequence of operations.

ASM-028 — Grade Point Mapping

If grade points are supported, the system must use the configured mapping between grades and grade points.

Acceptance test: A school configures a grade-point mapping. The system uses that mapping rather than assuming that a letter grade has the same numerical value in every institution.

ASM-029 — GPA Calculation

If GPA is implemented, the system must use the institution's configured calculation method.

Acceptance test: An institution uses credit-weighted GPA. The system multiplies grade points by the applicable course credits and divides by the total eligible credits.

ASM-030 — CGPA Calculation

If CGPA is implemented, the system must apply the institution's rules across the relevant academic history.

Acceptance test: Two terms have different total credit values. The system does not automatically use a simple average of the term GPAs when the configured CGPA method requires credit weighting.

18.5 Academic Averages and Comparisons

ASM-031 — Subject Average

The system must calculate subject results using the applicable assessment rules.

Acceptance test: A student's subject result uses the correct configured aggregation method and excludes results that are not eligible under the applicable policy.

ASM-032 — Overall Academic Average

The system must identify how multiple subject results contribute to overall academic performance.

Acceptance test: A student has results of 80%, 70%, and 90% in three equally weighted subjects. The system calculates 80% and identifies the equal-subject method used.

ASM-033 — Missing Subject Results

The system must disclose missing subject results in overall academic calculations.

Acceptance test: A student has six subjects but only four finalised results. The system does not present an average of the four results as a final average across all six subjects.

ASM-034 — Class Average

The system must calculate class averages using the configured method and eligible student results.

Acceptance test: The system calculates a class average for Mathematics using the appropriate student-level Mathematics results, not by adding every raw mark across unrelated subjects.

ASM-035 — Visible Denominator

Academic averages should show the relevant number of eligible results or students wherever practical.

Acceptance test: A class average is based on 24 eligible student results. The report displays the average and identifies the 24 results used.

ASM-036 — Compatible Comparisons

The system must preserve the academic context needed to interpret comparisons.

Acceptance test: Two results use different grading scales. The system does not automatically treat the grades as equivalent solely because their labels match.

ASM-037 — Historical Progress

The system must preserve academic results across reporting periods and academic years.

Acceptance test: A student's result increases from 72% in one academic year to 84% in the next. Both results remain available and the system can calculate a 12-percentage-point increase.

ASM-038 — No Unsupported Causal Claims

The system must not claim that an assessment result proves the cause of a student's performance.

Acceptance test: A student's mark declines after an attendance decline. The system may display both records but does not claim that absence caused the decline.

18.6 Corrections, Security, and Auditability

ASM-039 — Audited Result Corrections

The system must preserve the history of authorised result corrections.

Acceptance test: A teacher changes a mark from 32 to 42. The system records the previous value, new value, user, timestamp, and required reason.

ASM-040 — Recalculation After Corrections

The system must update affected calculated results after an authorised correction.

Acceptance test: A finalised assessment result is corrected. The relevant subject result, overall average, and permitted dashboards reflect the corrected data under the applicable rules.

ASM-041 — Server-Side Permissions

The system must enforce assessment and result permissions on the server.

Acceptance test: A teacher cannot modify another teacher's restricted assessment by changing an assessment identifier in a direct request.

ASM-042 — Parent and Student Privacy

Students and parents must only access academic results they are authorised to view.

Acceptance test: A parent linked to Student A cannot retrieve Student B's assessment results by changing a student identifier.

ASM-043 — School and Organisation Scope

School and organisation administrators must be restricted to their authorised data scopes.

Acceptance test: An administrator from one organisation cannot retrieve another organisation's assessment results through the user interface or direct server request.

ASM-044 — Publication Controls

The system must apply configured permissions and workflow rules before publishing assessment results.

Acceptance test: A draft result remains unavailable to users who are not authorised to view drafts. After the authorised publication process, the result becomes visible to the appropriate users.

ASM-045 — Audited Rule Changes

Changes to assessment weights, grading scales, and calculation rules must be recorded.

Acceptance test: An administrator changes a grading boundary. The audit history identifies the changed rule, user, and time, and historical reports remain interpretable.

ASM-046 — Import Validation

If assessment imports are supported, the system must validate imported records before committing them.

Acceptance test: An imported file contains an unknown student, an invalid mark, and a duplicate result. The system identifies the affected rows and does not silently create inaccurate records.

ASM-047 — Export Permissions

Assessment exports must respect the user's permissions.

Acceptance test: A teacher exports marks for an assigned class. The file excludes restricted results from other classes and schools.

ASM-048 — Reproducible Calculations

The same eligible data and the same calculation rules must produce the same result.

Acceptance test: The system recalculates a subject result with unchanged source records and rule versions. The calculated result is identical.

19. Recommended Development Sequence

The assessment system should be implemented in a deliberate order.

First, establish the academic structure. The system must identify academic years, reporting periods, subjects, classes, and student enrolments.

Second, implement assessment definitions and student results. Teachers should be able to create assessments and record valid outcomes for their assigned students.

Third, implement result statuses and validation. Missing, absent, exempt, incomplete, and recorded results must remain distinguishable.

Fourth, implement percentages and simple aggregation. Verify raw marks, maximum marks, and percentage calculations with automated tests.

Fifth, implement configurable weights and category calculations. Ensure that the system never invents weights or silently changes the calculation method.

Sixth, implement grading scales and rounding. Test boundaries, decimal values, and the order in which rounding and grade assignment occur.

Seventh, implement subject and overall academic indicators. Confirm that missing results and incompatible grading systems are handled transparently.

Eighth, implement historical records and audit trails. Corrections must preserve history and trigger consistent recalculation.

Ninth, enforce permissions and publication rules. Test every portal's access separately, including direct requests that attempt to bypass the interface.

Tenth, build dashboards and comparisons. Only after the underlying calculations are reliable should the platform present more complex class averages, progress indicators, and school-level comparisons.

20. Final Product Principles

The assessment system must preserve the original result, identify the assessment context, and apply the correct calculation rules.

A mark is not meaningful without its maximum marks. A percentage is not automatically comparable across different assessments. A missing result is not zero. A grade is not universal. A subject average is not the same as a raw total. A provisional result is not a final result.

Optimus Orien should support numerical marks, configurable grading scales, and the assessment aggregation methods required by its initial schools. More advanced systems, such as GPA, CGPA, and rubric-based assessment, should be implemented when their requirements are clearly defined.

All calculations must be explainable, reproducible, and auditable.

The platform should preserve historical results and the rules used to calculate them, even when a school later changes its assessment policy.

Finally, academic indicators should help educators understand student performance without pretending to explain causes that the data does not establish.

A trustworthy academic-performance platform does not simply calculate grades. It preserves the evidence, explains the method, identifies missing information, and makes the result understandable.

What Part 10 Establishes

Part 10 establishes the conceptual and technical foundation for the assessment system in Optimus Orien.

It defines how the platform should represent assessments and individual results, validate marks, calculate percentages, apply assessment weights, handle incomplete results, use grading scales, and aggregate results into subject and overall academic indicators.

It also establishes the requirements for historical records, correction workflows, permissions, and reproducible calculations.

Part 11 focuses on three important areas of Optimus Orien: homework and assignment completion, classroom participation, and subject-linked skills. These indicators can help schools understand aspects of student performance that examination marks alone cannot show.

The most important principle is to **measure each indicator accurately without confusing activity, engagement, skill, and academic achievement**.

PART 11

HOMEWORK, CLASS PARTICIPATION, AND SUBJECT-LINKED SKILLS

How Optimus Orien Should Record Student Work, Classroom Participation, and Skills Beyond Examination Marks

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner and software developer
Primary objective: Build a transparent system for recording homework completion, classroom participation, and subject-linked skills without confusing these indicators with academic achievement.

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1. Why This Document Exists

Part 10 established how Optimus Orien should record assessment results, calculate marks and percentages, apply grading rules, and produce academic-performance indicators.

However, examination marks do not tell the entire story of a student's educational experience.

Consider three students.

Student A receives high examination marks but regularly fails to complete assigned homework.

Student B completes homework consistently but struggles in examinations.

Student C receives average examination marks, completes most assignments, communicates confidently in class, and demonstrates strong analytical skills.

If a school looks only at examination marks, it may miss important differences between these students.

Homework completion, classroom participation, and subject-linked skills can provide additional information about how students approach their learning and what they can demonstrate.

However, these indicators must be designed carefully.

A student who completes every homework task has demonstrated consistency in completing assigned work. That does not automatically prove that the student has mastered the subject.

A student who speaks frequently in class may be highly engaged, but speaking frequently does not automatically mean that the student's contributions are accurate or thoughtful.

A student who rarely speaks may still listen carefully, think deeply, and perform exceptionally well.

The software must therefore record each indicator according to what it actually measures.

The objective is not to create one simplistic score that claims to represent the whole student. The objective is to give schools a clearer, more complete, and more transparent picture of student performance.

2. Three Different Things: Homework, Participation, and Skills

Before implementation, the developer must understand that these are three separate concepts.

2.1 Homework and Assignment Completion

Homework completion records whether a student has completed assigned work according to the relevant rules.

It answers a question such as:

Did the student complete the assigned task?

It does not automatically answer whether the student understood the material or performed well.

2.2 Classroom Participation

Classroom participation records a student's observed contribution to learning activities.

Depending on the school's chosen framework, this may include asking relevant questions, contributing to discussions, collaborating with classmates, or participating in practical activities.

It answers a question such as:

What evidence is available about the student's participation in the relevant learning activities?

It must not be reduced to the number of times a student speaks.

2.3 Subject-Linked Skills

Subject-linked skills describe specific abilities relevant to a subject or educational programme.

For example, English may involve reading comprehension, writing, speaking, and listening.

Mathematics may involve reasoning, problem-solving, and mathematical communication.

Science may involve observation, experimentation, interpretation of evidence, and scientific reasoning.

It answers a question such as:

What can the student demonstrate in this particular skill, and what evidence supports that judgement?

These three indicators may complement academic marks, but they must remain distinguishable in the data model and user interface.

3. The Complete Homework and Assignment System

Homework is work assigned to students to complete outside a particular classroom activity. The exact meaning varies between schools.

Some schools may assign work to be completed at home. Others may use independent study, take-home projects, online tasks, or other forms of assigned work.

Optimus Orien should use a broader assignment model that can accommodate the relevant school practices.

The platform must not assume that every assignment is completed at home, requires a file upload, or receives a numerical mark.

3.1 Assignment Definition

An assignment is the task created by an authorised teacher or other permitted user.

For example, an English teacher may assign a reading exercise to Grade 7, Section B.

The assignment itself should contain information such as its title, instructions, subject, class, assigned students, assigned date, due date, and completion requirements.

Where relevant, the assignment may also include an assessment rubric, a maximum mark, or a link to a formal assessment record.

The assignment definition is separate from each student's completion record.

One assignment may have 30 students, each with a different completion status.

3.2 Student Assignment Record

The student assignment record represents the status of one student's relationship with a particular assignment.

It should identify the student, assignment, relevant enrolment, and current completion status.

It may also contain submission time, verification time, permitted feedback, and the identity of the users who recorded or confirmed the status.

The system should not create a separate assignment definition every time a teacher records a student's completion.

3.3 Assignment Versus Assessment

This distinction is particularly important.

An assignment is a task given to a student.

An assessment is an activity used to gather evidence about learning.

An assignment can also be an assessment, but the two terms are not interchangeable.

For example, a teacher may assign a worksheet purely for practice. The teacher may check whether the worksheet was completed without awarding marks.

Another assignment may be graded and contribute to the student's coursework result.

A third may be a project assessed using a rubric.

Optimus Orien should allow an assignment to be linked to an assessment when appropriate, without requiring every assignment to have marks.

If a graded assignment contributes to a subject result, the system must connect it to the assessment rules defined in Part 10 rather than independently inventing a second grading calculation.

4. Assignment Creation and Assignment Scope

The teacher must be able to define which students are expected to complete an assignment.

The assignment may apply to an entire class, a selected group, or individual students where the school's workflow permits it.

The platform should preserve the original assignment scope.

4.1 Assigning Work to a Class

Suppose a teacher assigns a task to Grade 7, Section B, which contains 30 students.

The system should identify the students expected to complete the assignment according to the relevant class and enrolment records.

If a student later transfers to another class, the system must preserve the historical assignment record rather than silently replacing the student's original class context.

4.2 Individual Assignments

Some students may receive different work because of their programme, schedule, or an authorised educational arrangement.

The system should support individual assignment scope where required.

It must not assume that every student in the same class has exactly the same assignment requirements.

4.3 Assignment Dates

An assignment may have:

Assigned date: When the assignment is given.

Due date: When the work is expected to be completed or submitted.

Submission date: When the student submits the work, if submission tracking is supported.

Verification date: When the teacher confirms the completion status, where confirmation is required.

These dates represent different events and must not be stored as though they were the same timestamp.

4.4 Due-Date Changes

Teachers may change an assignment's due date.

The system should record authorised changes and preserve enough history to explain which due date applied at the time.

If the due date changes, the system must not automatically mark students late based on the previous date unless the applicable policy explicitly requires that outcome.

5. Homework Completion Statuses

A homework system should use explicit statuses rather than only a checkbox that can mean several different things.

A practical starting set is described below. Schools should be able to configure additional statuses where needed.

5.1 Not Started or Not Submitted

This status indicates that no completion or submission has been recorded.

It does not automatically establish that the student has refused to do the work.

It may also mean that the student has not yet reached the due date.

The interface should therefore distinguish between work that is still in progress and work that is overdue.

5.2 Submitted

Submitted means the student has handed in the work or completed the required submission process.

It does not necessarily mean that the teacher has checked it.

A student can submit an incomplete answer, an incorrect answer, or a blank file.

Submission is evidence that a submission event occurred, not proof that the work was completed to the required standard.

5.3 Completed

Completed means that the assignment has met the school's configured completion requirements.

For Optimus Orien, the recommended default is that a teacher must confirm completion before the platform treats the assignment as officially completed.

A student may indicate that they have finished the work, but the teacher's confirmation should be stored separately.

This prevents an unverified student declaration from automatically becoming a verified completion record.

5.4 Incomplete

Incomplete means that work has been submitted or reviewed, but the required completion conditions have not been met.

For example, a student may submit a worksheet with several required questions unanswered.

The teacher should be able to mark the work Incomplete according to the school's rules.

5.5 Late

Late means that the relevant submission or completion occurred after the applicable deadline under the school's policy.

The system must use the relevant due date and timestamp.

It should not mark a submission late merely because a teacher verified it after the deadline when the student actually submitted it on time.

Submission time and verification time must remain separate.

5.6 Excused or Not Required

A school may decide that a particular assignment does not apply to a student or that a student is excused from completing it.

The system should represent this explicitly when supported.

An excused assignment must not automatically be treated as an incomplete assignment or a zero.

5.7 Overdue

Overdue describes an assignment that has passed its applicable due date without satisfying the required completion condition.

Overdue is often better treated as a derived state than as an independent permanent record.

For example, an assignment can be Not Submitted and become Overdue after the due date passes.

If the student later submits it, the system should preserve the submission history and update the current status according to the configured policy.

6. The Two-Stage Homework Confirmation Model

For Optimus Orien, the recommended default is a two-stage completion process.

Stage 1: Student declaration or submission.

The student indicates that the work has been completed or submits it through the school's chosen process.

Stage 2: Teacher verification.

The teacher reviews the submission or completion record and confirms whether it meets the assignment's completion requirements.

These stages must be represented separately.

For example, the student may submit a Mathematics worksheet on Monday. The teacher may verify it on Tuesday.

The system should preserve Monday as the submission date and Tuesday as the verification date.

It should not rewrite the submission date to Tuesday merely because the teacher confirmed it then.

6.1 Why Verification Matters

Verification helps prevent inaccurate completion records.

However, verification should not be confused with grading.

A teacher may verify that a worksheet has been completed without deciding that every answer is correct.

If the assignment is graded, the teacher can record the academic result separately through the assessment system.

This preserves the distinction between:

-

Whether work was submitted.

-
-

Whether the completion requirements were met.

-
-

Whether the work was academically correct.

-
-

What mark the student received.

-

6.2 Exceptions

Some schools may not require teacher confirmation for every task.

For example, a school might use automated completion records for a particular type of activity.

Optimus Orien may support different verification policies in the future, but the default workflow should follow the product's intended rule: **student completion claims do not become verified completion records without the required teacher confirmation.**

6.3 Corrections

If a teacher marks an assignment Completed by mistake, an authorised correction must be possible.

The platform should preserve the previous status, new status, user, timestamp, and reason where required.

A correction must update affected completion rates and reports.

7. Calculating Homework Completion Rates

Homework completion rates can be useful, but only if the system defines which assignments are eligible.

The platform must not simply divide the number of completed assignments by every assignment that has ever existed.

7.1 Basic Formula

A basic completion-rate formula is:

Homework completion rate = Eligible assignments completed ÷ Eligible assignments due × 100

However, both the numerator and denominator require explicit definitions.

The school must decide which statuses count as completed, how late submissions are treated, how excused assignments are handled, and whether the report concerns completion by the due date or eventual completion.

7.2 Example

Suppose a student has 20 eligible assignments that have reached their due dates.

The records show:

Completed on time: 14.

Completed late: 2.

Not completed: 3.

Excused: 1.

Assume the school excludes excused assignments from the denominator and calculates completion by the due date.

The eligible denominator is:

20 − 1 = 19 assignments.

The number completed on time is 14.

The on-time completion rate is:

$14 \div 19 \times 100 \approx 73.7\%$.

If the school calculates eventual completion and accepts the two late assignments as completed, the numerator is 16.

The eventual completion rate is:

$16 \div 19 \times 100 \approx 84.2\%$.

Both calculations may be useful, but they answer different questions.

The interface must clearly label them.

7.3 On-Time Completion Versus Eventual Completion

The system should distinguish:

On-time completion rate: The proportion of eligible assignments completed by the applicable deadline.

Eventual completion rate: The proportion of eligible assignments eventually completed under the school's policy.

A student may have a lower on-time rate but a higher eventual completion rate.

Combining the two into one unexplained percentage could hide this difference.

7.4 Missing Confirmation

Suppose a student declares that an assignment is complete, but the teacher has not yet verified it.

The system should not automatically count the assignment as verified Completed.

It should show that the student has declared completion and that teacher verification is pending.

The treatment of pending verification in provisional reports must be explicit.

The system should not silently treat pending verification as either confirmed completion or confirmed failure.

7.5 Missing Assignments and Denominators

An assignment should enter a completion-rate denominator only when it is eligible under the configured rules.

Depending on the policy, eligibility may require that:

-

The assignment applies to the student.

-
-

The due date has passed, if the report concerns due assignments.

-
-

The student was enrolled in the relevant class or programme.

-
-

The assignment has not been formally excused or excluded.

-
-

The assignment is not a draft or cancelled task.

-

The exact rules must be defined by the school.

A future assignment should not count as an incomplete assignment merely because it has not yet been completed.

7.6 Completion Is Not Mastery

A student may complete every assignment but perform poorly on assessments.

Another student may complete fewer assignments but demonstrate strong subject knowledge.

The system must not treat completion rate as an automatic substitute for academic performance.

Homework completion should remain a separate indicator in the student profile.

8. Homework Corrections, Extensions, and Special Circumstances

The system must support realistic changes to assignment requirements.

8.1 Extensions

A teacher may grant an extension to a student or group.

The platform should preserve the original due date and the authorised revised due date.

The completion calculation must use the due date that applies under the school's extension policy.

If the system overwrites the original due date without preserving history, it may become impossible to explain why a submission was or was not considered late.

8.2 Cancelled Assignments

A teacher may cancel an assignment or decide that it no longer applies.

The system should retain the record of the assignment and its cancellation rather than deleting it without history.

Cancelled assignments should be handled according to the configured completion-rate policy.

8.3 Student Transfers

If a student transfers to another class or school, historical assignment records should remain associated with the original class and enrolment.

The system must not automatically add every historical assignment from the new class to the student's old completion-rate denominator.

8.4 Group Work

Some assignments are completed by groups.

If group assignments are supported, the system must distinguish the group submission from the individual student's participation or completion record.

A single group submission does not automatically establish that every group member contributed equally.

The platform should not infer individual participation from the existence of a group submission alone.

8.5 Offline Work

Some schools may assign work completed on paper or outside the platform.

The system should allow an authorised teacher to record completion without requiring a digital file upload.

The presence or absence of an uploaded file must not determine completion unless the assignment explicitly requires one.

9. Classroom Participation: What Should Be Measured?

Classroom participation is the student's observable contribution to relevant learning activities.

It is a broad concept, and schools may define it differently.

Optimus Orien should help schools record participation systematically without assuming that one particular behaviour represents engagement for every student.

9.1 Participation Is Not Simply Speaking Frequency

A student who frequently speaks may contribute useful ideas, ask thoughtful questions, or help classmates understand a difficult concept.

Another student may speak frequently without contributing relevant information.

A student who speaks less may still demonstrate careful listening, thoughtful written contributions, or strong performance in practical work.

Therefore, a participation rating should not be based only on how many times a student speaks.

The school should define observable criteria for the behaviours it wants to record.

9.2 Examples of Observable Participation

Depending on the school's teaching methods, participation may include:

- Contributing relevant ideas to a discussion.

-
-

Asking questions that clarify or extend understanding.

-
-

Responding thoughtfully to a task.

-
-

Collaborating constructively with classmates.

-
-

Contributing to group problem-solving.

-
-

Participating in practical activities.

-
-

Listening and responding appropriately to others.

-
-

Completing assigned classroom activities.

-

These examples should not all be mandatory for every subject or lesson.

A science practical may require different participation evidence from a written language exercise.

The criteria should be relevant to the activity being observed.

9.3 Participation Is Not Personality

The system must not label a student as lazy, intelligent, disinterested, shy, or unmotivated based on a participation rating.

Those labels are interpretations about the student, not direct records of an observed contribution.

Optimus Orien should record educationally relevant observations using neutral language.

For example, a teacher might record that a student contributed to group problem-solving in three observed activities.

That is more precise than claiming that the student has a good attitude.

9.4 Participation and Cultural Differences

Students may express participation differently depending on the classroom environment, language proficiency, teaching method, accessibility needs, and cultural expectations.

A system that rewards only frequent verbal participation could unfairly disadvantage students whose learning contributions take other forms.

Optimus Orien should allow schools to define appropriate participation criteria and should avoid presenting one participation style as universally superior.

10. The Participation Rating Scale

Optimus Orien has considered teacher-entered participation ratings on a scale of 1–10.

This can be supported, but the number must have a clear meaning.

A rating of 8 should not mean one thing to one teacher and something entirely different to another without any guidance.

10.1 Define the Scale

The school should define what the scale represents.

For example, a school may use:

1: Very limited evidence of the defined participation behaviours during the observation period.

5: Some evidence of the defined participation behaviours, with inconsistent contributions.

10: Consistent and strong evidence of the defined participation behaviours during the observation period.

These descriptions are illustrative. The school should establish the complete rubric and the meaning of intermediate values.

The system should not present this example as a universal educational standard.

10.2 Define the Observation Period

A rating must refer to a defined period or set of observations.

For example, a teacher may rate participation across a term rather than during one isolated lesson.

The record should identify the relevant subject, reporting period, evaluator, and date.

A rating without an observation period is difficult to interpret.

10.3 Define the Evidence

The teacher should understand what evidence is relevant to the rating.

A structured rubric can help teachers apply similar criteria.

The system may provide an optional short observation note, subject to privacy and school policy.

The note should describe relevant evidence rather than make unsupported claims about the student's personality or motivation.

10.4 Do Not Treat a Rating as an Objective Measurement

A teacher rating is a professional judgement based on observations.

It may contain useful information, but it is not equivalent to a directly counted event such as the number of assignments submitted.

Optimus Orien should label the measure as a teacher rating and identify who entered it.

The platform should not present a rating as if it were an independently verified scientific measurement.

10.5 Multiple Teachers

If a student has different teachers for different subjects, each teacher may observe different behaviours.

The platform should preserve the subject and evaluator associated with each rating.

If a school chooses to calculate an overall participation indicator from several ratings, it must define how those ratings are combined and explain the limitations.

The system must not simply average every participation rating across subjects without considering the intended meaning of that average.

11. Subject-Linked Skills: What They Mean

A skill is an ability to perform a task or apply knowledge in a particular context.

A subject-linked skill is an ability relevant to a specific subject or learning area.

Examples include reading comprehension in English, mathematical reasoning in Mathematics, and interpreting evidence in Science.

Skills are related to academic knowledge, but they are not identical to examination marks.

A student may memorise facts successfully yet struggle to apply them to a new problem. Another may understand a concept but make mistakes under examination pressure.

A broader student record can help educators understand such differences, provided that the measures are designed appropriately.

11.1 Knowledge, Skill, and Performance

These concepts should be distinguished.

Knowledge concerns what a student knows or understands.

Skill concerns what the student can do with relevant knowledge or abilities.

Performance is the result observed under particular conditions.

For example, a student may know the rules of grammar but struggle to use them in spoken conversation.

An English examination may assess grammatical knowledge. A speaking activity may provide evidence about oral communication.

Neither result automatically captures every aspect of the student's English ability.

The platform should preserve the context of the evidence.

11.2 Skills Depend on Context

A skill label is not meaningful unless the school defines what it represents.

Consider the term Critical Thinking.

One school might use it to describe evaluating evidence and identifying logical weaknesses.

Another might use it more broadly to describe problem-solving.

Optimus Orien should allow administrators to define skill names, descriptions, and assessment criteria rather than assuming that every label has an identical meaning across institutions.

11.3 Skills Should Be Connected to Subjects

The system should connect a skill to the subject or learning area in which it is assessed.

For example:

English:
Reading comprehension, writing, speaking, listening.

Mathematics:
Logical reasoning, problem-solving, mathematical communication.

Science:
Observation, experimentation, evidence interpretation, scientific reasoning.

These are illustrative examples, not a mandatory universal list.

Schools should be able to configure their own skill frameworks.

A skill may be used across multiple subjects when appropriate, but the platform must preserve the subject context of each rating or observation.

12. The Skill Rating Scale

Optimus Orien has considered a 1–10 scale for subject-linked skills.

A numerical scale can help summarise teacher observations, but it needs clear criteria.

12.1 A Rating Is Not Self-Explanatory

Suppose a teacher gives a student a score of 6 out of 10 for analytical thinking.

What does 6 mean?

Does it indicate that the student completed 60% of a task correctly? Does it mean the student demonstrates the skill at an intermediate level? Does it represent the teacher's general impression?

Without a defined rubric, the score is ambiguous.

The platform must therefore allow the school to define the scale's meaning.

12.2 A Possible Skill-Rating Framework

A school might define the scale as follows:

1–2: The student demonstrates limited evidence of the defined skill in the observed tasks.

3–4: The student demonstrates emerging ability but requires substantial support.

5–6: The student demonstrates developing ability and can complete some relevant tasks with varying levels of support.

7–8: The student demonstrates consistent ability in the defined tasks.

9–10: The student demonstrates strong and consistent ability across the relevant tasks and contexts.

This is only an illustrative framework. The exact descriptions must be defined and approved by the school.

The developer must not hardcode these descriptions as an official universal scale.

12.3 Rating Versus Assessment Score

A skill rating of 8 out of 10 is not automatically the same as an assessment score of 8 out of 10.

An assessment score may represent marks awarded against a specific task.

A skill rating may summarise a teacher's judgement across several observations.

Although both use a numerical range, their meanings differ.

The database should distinguish these result types.

12.4 Who Assigns the Rating?

The system should identify the authorised person who entered the skill rating.

For the initial Optimus Orien model, subject teachers are a reasonable default because they are responsible for teaching and observing the relevant subject.

However, the product should allow authorised schools to define the roles permitted to assess skills.

The system must not allow an unrelated user to change a student's skill ratings simply because they can view the student profile.

12.5 When Should Skills Be Rated?

Skill ratings need a defined observation period.

Optimus Orien has previously considered biannual skill ratings. That can be used as a starting configuration, but it should not be hardcoded as a universal requirement.

Some schools may review skills once per term. Others may use a semester or another reporting cycle.

The system should allow schools to configure the frequency and relevant reporting periods.

13. Evidence, Rubrics, and Reliability of Ratings

A skill rating becomes more useful when it is supported by observable evidence.

For example, a teacher might assess analytical thinking through several tasks involving evidence interpretation, reasoning, and problem-solving.

The teacher can then assign a rating based on the defined criteria.

This is more interpretable than assigning a number without explaining what was observed.

13.1 What Is a Rubric?

A rubric is a structured set of criteria and performance descriptions used to judge the quality of work or demonstrated ability.

A rubric may define different performance levels for each criterion.

For example, a reading-comprehension rubric might assess whether a student can identify the main idea, interpret supporting details, and draw reasonable conclusions.

The school can define what performance at each level looks like.

13.2 Why Rubrics Help

Rubrics can make expectations clearer.

They can help teachers apply more consistent criteria and explain why a student received a particular rating.

However, rubrics do not automatically eliminate subjective judgement.

Teachers may still interpret evidence differently, and the quality of the rubric matters.

Optimus Orien should present rubric-based ratings as structured educational judgements, not as perfectly objective measurements.

13.3 Multiple Observations

A rating based on several relevant observations may provide a broader picture than a rating based on one isolated event.

The system should allow the evaluator to identify the observation period and, where appropriate, the evidence considered.

The platform should not require excessive documentation for every rating if the school does not need it.

The goal is sufficient context to make the rating interpretable.

13.4 Rating Changes

A skill rating may change as the student develops or as the teacher gathers new evidence.

The system should preserve previous ratings rather than replacing them.

A change from 5 to 7 should be visible as a change in the recorded rating.

However, the platform must not automatically claim that the student's underlying skill improved by exactly 40%. The scale may be ordinal or rubric-based rather than a precise ratio measurement.

A change in a rating is evidence of a changed recorded judgement, not necessarily a precise numerical measurement of learning growth.

14. Measuring Progress in Skills

Longitudinal skill information can be valuable when the platform preserves comparable observations.

Suppose a student receives these analytical-thinking ratings:

Term 1: 4 out of 10.

Term 2: 6 out of 10.

Term 3: 7 out of 10.

The platform can show the recorded rating trend.

It may describe the ratings as increasing from 4 to 7.

However, it should not automatically conclude that the student's analytical ability increased by 75%.

That calculation assumes that the scale has meaningful ratio properties and equal numerical intervals, which may not be true for a teacher-rating scale.

The platform should therefore distinguish between:

- A change in the recorded rating.
- A change in observed performance against a defined rubric.
- A formally validated measurement of skill growth.

The first two may be supported by the available data. The third requires stronger measurement evidence.

14.1 Comparable Ratings

Historical ratings are most interpretable when they use the same skill definition, rubric, scale, and relevant observation conditions.

If the school changes the skill definition or rating rubric, the system should preserve the version used for each historical record.

It should not automatically compare the old and new ratings as if their meanings were identical.

14.2 Missing Skill Ratings

If a teacher has not yet assessed a skill, the system should show the rating as unavailable or not yet assessed.

It must not automatically assign zero.

Zero might be a valid rating only if the school's configured scale explicitly includes zero and the evaluator has recorded that result.

An absent rating is not the same as a recorded rating of zero.

15. Avoiding Bias and Unfair Interpretation

Homework, participation, and skill ratings can provide useful context, but they also introduce risks if interpreted carelessly.

15.1 Homework Completion and Access

Students may have different access to time, devices, internet connectivity, quiet study spaces, or other resources.

A low completion rate does not automatically prove that a student is lazy or uninterested.

The platform should record the completion data without claiming to know why the work was not completed.

Where schools record relevant circumstances, access should be limited to authorised users and sensitive information should be minimised.

15.2 Participation and Communication Style

A participation framework that rewards only frequent speaking may disadvantage students who contribute in other ways.

Schools should define participation criteria appropriate to the activity and allow different forms of relevant contribution where practical.

The software should not automatically treat quietness as lack of engagement.

15.3 Skill Ratings and Teacher Judgement

Different teachers may apply rating scales differently.

Clear criteria, rubrics, and observation periods can improve consistency, but the system must not assume that all ratings are perfectly objective.

The platform should preserve who made the rating and the framework used.

It should not hide uncertainty behind a polished numerical dashboard.

15.4 Do Not Combine These Indicators Into an Unexplained Score

Suppose a student has:

Homework completion: 90%.

Participation rating: 7 out of 10.

Analytical-thinking rating: 8 out of 10.

Academic performance: 75%.

The platform must not automatically average these four values into a single score.

They represent different constructs and use different measurement methods.

A combined score would require a clearly defined and justified framework. For the current product scope, these indicators should remain separate.

16. How These Indicators Should Appear in Optimus Orien

The interface should make the indicators understandable without overwhelming the user.

16.1 Student Profile

The student profile may show separate sections for:

Academic performance: Marks, percentages, grades, subject averages, and historical trends.

Homework and assignment completion: Eligible assignment counts, on-time completion, eventual completion, pending teacher verification, and overdue work.

Classroom participation: Subject-linked participation ratings, observation periods, evaluator information, and rubric descriptions where configured.

Subject-linked skills: Skill definitions, current ratings, historical ratings, and the relevant subject and reporting period.

These sections should not be collapsed into a single score.

16.2 Teacher Portal

Teachers should be able to manage assignments and record completion for students within their authorised classes.

They should also be able to enter participation and skill ratings for the students and subjects they are permitted to assess.

The interface should show the applicable rubric or rating descriptions so that teachers understand what a value represents.

16.3 Principal Portal

Principals should be able to review relevant school-level patterns, such as assignment completion by class, missing verification records, or distributions of skill ratings where the data is sufficiently complete.

The system should not use these summaries to label entire classes or teachers as poor performers without an explicitly defined and justified evaluation framework.

16.4 Organisation Administrator Portal

An organisation administrator may review aggregated indicators across authorised schools.

Comparisons must account for differences in assignment practices, skill definitions, reporting periods, and rating rubrics.

A school that assigns homework daily should not automatically be judged against a school that assigns homework weekly using an unexplained completion-rate comparison.

Likewise, skill ratings from different rubrics should not be treated as directly equivalent merely because both use a 1–10 scale.

16.5 Parent and Student Portals

Students and authorised parents or guardians should be able to understand the relevant records.

The interface should explain the meaning of statuses and ratings in simple language.

For example, it should distinguish between Submitted, Teacher Verification Pending, and Completed.

For skill ratings, it should show the relevant skill description and rating framework where appropriate.

17. Recommended Data Model

The following is a conceptual starting point for the database. The implementation may use a different technical structure, but it must preserve these relationships and meanings.

17.1 Assignment

Represents the task created by a teacher or authorised user.

It should include the assignment's identity, instructions, subject, relevant class or group, due date, and applicable completion requirements.

17.2 Assignment Eligibility

Represents which students are expected to complete an assignment.

This may be derived from class enrolments or stored explicitly when the assignment scope needs to be preserved historically.

The system must know which students belong in the completion-rate denominator.

17.3 Student Assignment Record

Represents the student's status for an assignment.

It should preserve submission status, completion status, relevant timestamps, and verification information.

17.4 Assignment Verification Record

Represents the teacher's confirmation or rejection of a declared completion status.

It should identify the verifier, verification time, decision, and correction history where required.

17.5 Participation Rating

Represents a teacher's participation rating for a student within a defined subject, period, and observation framework.

It should identify the evaluator and preserve the relevant rating scale or rubric version.

17.6 Skill Definition

Represents a skill that the school wants to track.

It should include the skill's name, description, subject or learning-area connection, and applicable rating framework.

17.7 Skill Rating

Represents a student's recorded result for a particular skill.

It should preserve the student, subject context, evaluator, observation period, rating value or rubric result, and applicable framework version.

17.8 Rubric Definition and Rubric Result

If structured rubrics are supported, the system should preserve the criteria, performance levels, descriptions, and student results.

17.9 Audit Record

Records relevant changes to assignments, completion statuses, participation ratings, skill ratings, and configuration rules.

It should identify who made the change, when it happened, and what changed.

18. Software Implementation Requirements and Acceptance Tests

The following requirements define the minimum expected behaviour for the assignment, participation, and subject-linked skills modules.

18.1 Assignment Management

HW-001 — Assignment Creation

Authorised teachers must be able to create assignments for their permitted classes or students.

Acceptance test: A teacher assigned to Grade 7, Section B creates an English assignment. The assignment is associated with the correct subject, class, and academic context.

HW-002 — Assignment Scope

The system must identify the students expected to complete each assignment.

Acceptance test: An assignment is assigned to a class of 30 eligible students. The system identifies the expected students and does not automatically include unrelated students.

HW-003 — Individual Assignment Scope

The system must support individual assignment scope where enabled.

Acceptance test: A teacher assigns a task to one authorised student. The task does not automatically become an assignment for every student in the class.

HW-004 — Assignment and Assessment Separation

The system must distinguish an assignment from a formal assessment result.

Acceptance test: A teacher creates a practice worksheet without marks. The system records completion without inventing a numerical assessment result.

HW-005 — Due-Date Validation

The system must validate and preserve applicable assignment dates.

Acceptance test: A teacher changes an assignment's due date. The system records the change and applies the configured deadline rules to completion reports.

HW-006 — Assignment Cancellation

The system must support authorised assignment cancellation or exclusion where required.

Acceptance test: A teacher cancels an assignment. The system preserves the assignment history and applies the configured denominator rules rather than silently deleting the record.

HW-007 — Student Transfer History

The system must preserve historical assignment context when a student changes class or school.

Acceptance test: A student transfers to another class. Previous assignments remain associated with the original class and are not replaced by the new class's assignment list.

18.2 Homework Completion and Verification

HW-008 — Separate Submission and Verification

The system must distinguish student submission from teacher verification.

Acceptance test: A student submits an assignment. The system records the submission but does not mark the assignment as verified Completed until the required teacher confirmation occurs.

HW-009 — Verification Identity

The system must record who verified the completion status and when.

Acceptance test: A teacher verifies an assignment on Tuesday after the student submitted it on Monday. Both timestamps remain available.

HW-010 — Configurable Completion Statuses

The system must support configured assignment statuses, including the required distinction between unsubmitted, submitted, completed, incomplete, excused, and late where applicable.

Acceptance test: A teacher marks an assignment Incomplete. The system does not treat it as verified Completed.

HW-011 — Overdue Status

The system must calculate overdue status using the applicable due date and completion rules.

Acceptance test: An eligible assignment passes its due date without meeting the completion requirement. The system identifies it as overdue under the configured policy.

HW-012 — Late Submission

The system must preserve the actual submission time and determine lateness using the applicable deadline rules.

Acceptance test: A student submits work before the due date, but the teacher verifies it afterward. The system does not automatically label the student's submission Late merely because verification occurred later.

HW-013 — Pending Verification

The system must distinguish pending teacher verification from confirmed completion.

Acceptance test: A student declares an assignment complete, but the teacher has not verified it. The system displays the pending state and does not silently count it as verified Completed.

HW-014 — Assignment Extensions

The system must support authorised extensions and preserve the relevant date history.

Acceptance test: A teacher grants a student an extension. The system uses the applicable revised deadline according to policy and retains the original due date in the history.

HW-015 — Group Assignments

If group assignments are supported, the system must distinguish group submission from individual participation.

Acceptance test: One group member submits a project. The system does not automatically claim that every group member contributed equally.

HW-016 — Completion Corrections

The system must preserve the history of authorised completion-status corrections.

Acceptance test: A teacher changes a status from Completed to Incomplete. The system records the previous status, new status, user, timestamp, and required reason.

18.3 Completion Calculations

HW-017 — Eligible Assignment Denominator

The system must calculate completion rates using eligible assignments under the configured policy.

Acceptance test: A student has an upcoming assignment and a cancelled assignment. Neither is automatically counted as a missed due assignment if the configured eligibility rules exclude them.

HW-018 — On-Time Completion Rate

The system must calculate on-time completion separately where supported.

Acceptance test: A student completes 14 of 19 eligible assignments by the due date. The system calculates approximately 73.7% and displays the relevant counts.

HW-019 — Eventual Completion Rate

The system must distinguish eventual completion from on-time completion where both are reported.

Acceptance test: Two additional assignments are completed late. The system can report eventual completion separately from the on-time completion rate.

HW-020 — Missing Data

The system must distinguish missing assignment records or pending verification from confirmed non-completion.

Acceptance test: A student has a pending teacher verification. The system does not silently treat the assignment as a confirmed failure.

HW-021 — Excused Assignments

The system must apply the configured rule for excused or non-applicable assignments.

Acceptance test: An assignment is formally excused. The system excludes it from the denominator when the configured policy requires exclusion.

HW-022 — No Mastery Inference

The system must not treat completion status as proof of academic mastery.

Acceptance test: A student completes an assignment but receives a low mark. The system preserves both records without automatically changing completion to Incomplete solely because the academic mark is low, unless the assignment's explicit completion criteria require that outcome.

HW-023 — Recalculation After Corrections

The system must update affected completion rates after an authorised status or eligibility change.

Acceptance test: A correction changes an assignment from Incomplete to Completed. The relevant completion reports update consistently.

18.4 Participation Ratings

PAR-001 — Configurable Participation Framework

The system must support a school-defined participation framework.

Acceptance test: An administrator defines participation criteria and a rating scale. Authorised teachers can use the framework without a code change.

PAR-002 — Defined Rating Scale

The system must preserve the meaning and boundaries of the configured participation scale.

Acceptance test: A school uses a 1–10 scale with defined descriptions. The interface displays the relevant descriptions to authorised teachers when entering ratings.

PAR-003 — Subject and Period Context

Every participation rating must identify its subject and observation or reporting period.

Acceptance test: A student receives a participation rating in Science for Term 1. The system does not display it as a Mathematics rating or a Term 2 result.

PAR-004 — Evaluator Identity

The system must record the user who entered the rating.

Acceptance test: A principal reviews a participation rating and can identify the authorised evaluator.

PAR-005 — No Automatic Speaking-Frequency Score

The system must not automatically equate speaking frequency with participation quality unless a specific, approved framework explicitly defines that measure.

Acceptance test: A student has few recorded verbal contributions but strong participation evidence through another configured activity. The system does not automatically assign a low rating based solely on the number of times the student spoke.

PAR-006 — Rating Corrections

The system must preserve authorised changes to participation ratings.

Acceptance test: A teacher corrects a participation rating from 5 to 7. The audit history preserves the previous and new values.

PAR-007 — Historical Participation Ratings

The system must preserve previous participation ratings and their applicable framework versions.

Acceptance test: A school changes its participation rubric. Earlier ratings remain connected to the previous rubric version.

PAR-008 — Optional Evidence Notes

If participation notes are supported, the system must apply the configured privacy and permission rules.

Acceptance test: A user without permission to view restricted notes cannot retrieve them through the interface or a direct server request.

18.5 Subject-Linked Skills

SKL-001 — Configurable Skill Definitions

The system must allow authorised users to define skills and descriptions.

Acceptance test: An administrator creates an Analytical Thinking skill with a description and associates it with Mathematics. The skill becomes available within the authorised academic context.

SKL-002 — Subject Context

The system must preserve the subject or learning-area context of a skill rating.

Acceptance test: A student receives a reading-comprehension rating in English. The system does not automatically reuse that value as a Science rating.

SKL-003 — Configurable Rating Scale

The system must support the configured skill-rating framework.

Acceptance test: A school uses a 1–10 scale with defined criteria. The system validates values against the configured scale and displays the relevant descriptions.

SKL-004 — Rubric-Based Ratings

If rubric-based skill assessment is supported, the system must preserve the rubric and criterion-level results.

Acceptance test: A teacher records results for three rubric criteria. The system retains each criterion's result instead of storing only an unexplained overall number.

SKL-005 — Evaluator Identity

The system must identify the authorised user who entered the skill rating.

Acceptance test: An authorised reviewer can identify who recorded a student's skill rating.

SKL-006 — Observation Period

The system must associate skill ratings with the relevant observation or reporting period.

Acceptance test: A skill rating recorded for Term 1 remains distinguishable from a rating recorded for Term 2.

SKL-007 — Missing Skill Rating

The system must distinguish a missing skill rating from a recorded zero where zero is permitted.

Acceptance test: No rating has been entered for a student's skill. The system displays Not Yet Assessed or an equivalent state rather than automatically assigning zero.

SKL-008 — Historical Rating Version

The system must preserve the skill definition and rating framework used for historical ratings.

Acceptance test: A school revises its Analytical Thinking rubric. Historical ratings remain interpretable under the previous rubric.

SKL-009 — Rating Corrections

The system must preserve authorised changes to skill ratings.

Acceptance test: A teacher changes a skill rating from 6 to 7. The audit history records the previous and new values, user, and timestamp.

SKL-010 — No Unsupported Growth Claims

The system must not automatically convert changes in an ordinal or rubric-based rating into a precise percentage of skill growth.

Acceptance test: A rating increases from 4 to 7. The system can show the rating change but does not automatically claim that the underlying skill improved by 75%.

18.6 Dashboards, Permissions, and Privacy

IND-001 — Separate Indicators

The system must keep homework completion, participation, subject-linked skills, and academic marks distinguishable.

Acceptance test: A student has high homework completion and low examination marks. The dashboard displays both indicators without combining them into an unexplained overall score.

IND-002 — Student and Parent Access

Students and authorised parents or guardians must only access records permitted by their relationships and roles.

Acceptance test: A parent linked to Student A cannot retrieve Student B's assignment, participation, or skill records by changing an identifier.

IND-003 — Teacher Scope

Teachers must be restricted to assignments, ratings, and students within their authorised scope.

Acceptance test: A teacher cannot alter another teacher's restricted skill rating through a direct request.

IND-004 — Principal Scope

Principals must be restricted to the schools they are authorised to manage.

Acceptance test: A principal from School A cannot retrieve restricted assignment records from School B.

IND-005 — Organisation Scope

Organisation administrators must be restricted to their authorised organisational records.

Acceptance test: An administrator cannot retrieve another organisation's homework, participation, or skill data.

IND-006 — Comparable Aggregation

The system must not automatically compare completion rates or skill ratings whose definitions are incompatible.

Acceptance test: Two schools use different skill rubrics. The organisation dashboard identifies the difference and does not present their ratings as directly equivalent without an approved comparison method.

IND-007 — Reproducible Calculations

The same eligible records and calculation rules must produce the same completion rates and configured aggregates.

Acceptance test: Recalculating a completion report without changing the source records or rules produces the same result.

IND-008 — Public Privacy

Public pages must not expose individual homework records, participation ratings, skill ratings, or restricted notes.

Acceptance test: An unauthenticated visitor cannot retrieve individual student indicators through a public endpoint.

IND-009 — Server-Side Permissions

The server must enforce all permissions for creating, reading, updating, exporting, and deleting supported records.

Acceptance test: A user without permission attempts to alter a restricted record through a direct request. The server rejects the operation.

IND-010 — Audited Configuration Changes

Changes to completion rules, participation scales, skill definitions, and rating rubrics must be traceable.

Acceptance test: An administrator changes a skill-rating rubric. The audit history identifies the relevant change and preserves the context of historical ratings.

19. Recommended Development Order

The developer should implement these features in a deliberate sequence.

Stage 1: Build assignment definitions and eligibility. The system must know what was assigned, to whom, and when it was due.

Stage 2: Build completion statuses. Implement submission, teacher verification, completion, incompletion, lateness, excusal, and relevant correction workflows.

Stage 3: Implement completion calculations. Test the denominator, on-time completion, eventual completion, and missing-verification cases.

Stage 4: Build participation frameworks. Allow the school to define the rating scale, observation period, and criteria before teachers begin recording ratings.

Stage 5: Build subject-linked skill definitions. Implement skills, subject connections, rating frameworks, and historical versions.

Stage 6: Implement rating history. Preserve who entered each rating, when it was entered, and which framework applied.

Stage 7: Enforce permissions. Test teacher, principal, organisation administrator, student, and parent access separately.

Stage 8: Build the student profile. Present the indicators separately and explain what each one measures.

Stage 9: Build aggregated dashboards. Add class-level and organisation-level summaries only after the underlying definitions and calculations are reliable.

Stage 10: Test changes and edge cases. Test student transfers, due-date extensions, assignment cancellation, incomplete verification, rubric changes, and corrections to historical records.

20. Final Principles

Homework completion, classroom participation, and subject-linked skills can enrich a student's educational record.

However, each indicator has a different meaning.

Homework completion shows whether assigned work met the applicable completion requirements. It does not automatically prove mastery.

Classroom participation records relevant contributions or observations. It should not be reduced to speaking frequency or personality judgements.

Subject-linked skills describe abilities demonstrated in a particular context. Their ratings need clear definitions, relevant evidence, and an identifiable assessment framework.

Optimus Orien must preserve these distinctions in its database, calculations, and user interface.

It must not invent missing data, turn unverified declarations into confirmed records, or treat teacher ratings as perfectly objective measurements.

It should support configurable school policies, preserve historical records, apply permissions correctly, and explain what each indicator means.

Most importantly, the platform should help schools understand students more fully without reducing them to a single number.

The goal is not to measure everything about a student. The goal is to record useful educational evidence accurately and present it in a way that supports better-informed human decisions.

What Part 11 Establishes

Part 11 establishes the conceptual and technical foundation for recording homework and assignment completion, classroom participation, and subject-linked skills in Optimus Orien.

It defines how assignments and student completion records should be separated, why teacher verification matters, how completion rates should be calculated, how participation and skill-rating frameworks should work, and how these indicators should be preserved over time.

It also defines software requirements for configuration, calculations, historical records, privacy, permissions, and acceptance testing.

Part 12 brings together the concepts established in the previous documents. It explains how Optimus Orien should combine a student's academic results, attendance, homework completion, participation, skills, and historical progress into one clear profile—without confusing different measurements or overwhelming the people using it.

PART 12

THE COMPLETE STUDENT PERFORMANCE DASHBOARD

How to Combine Academic Results, Attendance, Homework, Participation, Skills, and Progress Into One Clear Student Profile

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner, software developer, and future product team
Primary objective: Design a reliable, understandable, and secure student performance dashboard that helps schools understand a student's educational progress through transparent data and visualisations.

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1. Why This Document Exists

In the previous parts, we established the foundations of an educational information system.

We explained how education systems differ between countries, how schools and academic structures are organised, how users and permissions should work, how student performance records should be maintained, how school comparisons should be calculated, and how homework, participation, and skills should be recorded.

We also discussed attendance, assessment, grading, and the importance of using fair comparison rules.

However, these features are not very useful if they remain isolated from one another.

Imagine that a principal wants to understand the performance of a particular student.

The principal might need to open one page to see examination marks, another to check attendance, another to review homework, and another to examine skills.

If the information is scattered across the platform, understanding the student's situation becomes unnecessarily difficult.

Optimus Orien needs a central student performance dashboard that brings relevant information together.

But there is an important challenge.

Combining information on one screen does not mean combining every measurement into one score.

Examination marks, attendance, homework completion, classroom participation, and skill ratings represent different aspects of a student's educational experience.

They should be presented together while retaining their individual meanings.

The dashboard should help users answer five questions:

1.

How is the student performing academically?

2.
3.

How has the student's performance changed over time?

4.
5.

What do attendance and homework records show?

6.
7.

What evidence is available about participation and subject-linked skills?

8.
9.

Does the available information suggest that a teacher or school leader should review the student's situation?

10.

The dashboard should answer these questions using transparent calculations, clear labels, appropriate comparisons, and access-controlled information.

It must not pretend to know more than the recorded evidence supports.

2. What Is a Student Performance Dashboard?

A dashboard is a visual interface that brings important information together so that users can understand a situation without examining every individual record.

In Optimus Orien, the student performance dashboard is a structured view of a student's educational records.

It should present current performance, historical progress, relevant indicators, and the underlying information needed to interpret them.

For example, a dashboard might show that a student achieved 68% in Mathematics during the current reporting period, compared with 74% in the previous period.

It might also show that the student's attendance rate has declined and that several assignments remain unverified.

These observations may help a teacher decide to investigate further.

However, the dashboard should not automatically conclude that declining attendance caused the lower Mathematics result.

That conclusion would require additional evidence.

The dashboard is therefore a tool for **understanding recorded evidence and supporting human decisions**, not a system that automatically explains every educational outcome.

3. The Difference Between a Student Record and a Student Dashboard

These two concepts are related but different.

3.1 The Student Record

The student record contains the underlying information about a student.

Depending on the platform's configured features, it may include:

- Student identity and profile information.
- Enrolment and school history.
- Academic-year and class assignments.
- Subject enrolments.
- Assessment results.
- Attendance records.
- Homework and assignment records.
- Participation ratings.
- Subject-linked skill ratings.
- Historical results.

-
-

Relevant corrections and audit records.

-

The student record is the underlying collection of information.

3.2 The Student Dashboard

The dashboard selects and presents relevant information from that record.

It may show:

-

Current academic performance.

-
-

Subject-by-subject results.

-
-

Attendance trends.

-
-

Homework completion rates.

-
-

Participation ratings.

-
-

Skill ratings.

-
-

Comparisons with appropriate reference groups.

-
-

Changes between reporting periods.

-
-

Data completeness.

-
-

Indicators requiring human review.

-

The dashboard should not create a separate, inconsistent version of the student's data.

It should use the same underlying records and calculation rules as the rest of Optimus Orien.

If an authorised teacher corrects an assessment result, the dashboard should reflect the corrected result according to the applicable calculation rules.

The dashboard must not continue showing an old result simply because the interface stored a separate copy.

4. The First Important Design Principle: Separate Data From Interpretation

One of the most important concepts in educational information systems is the difference between raw data, calculated indicators, and interpretations.

These three layers should remain distinguishable.

4.1 Layer One: Recorded Data

Recorded data is information entered into or imported into the platform.

Examples include:

A Mathematics assessment result of 68 out of 100.

A recorded absence on a particular school day.

A homework assignment marked Completed by an authorised teacher.

A teacher-entered analytical-thinking rating of 7 out of 10.

These are individual records.

They may be corrected if the original entry was wrong, but the platform should preserve an appropriate history of authorised changes.

4.2 Layer Two: Calculated Indicators

A calculated indicator is a result produced by applying a defined calculation to recorded data.

Examples include:

A subject percentage of 68%.

An attendance rate of 91%.

An on-time homework completion rate of 78%.

A change in Mathematics performance from 74% to 68%.

A count of five recorded absences during a reporting period.

These values are derived from the underlying records.

Their accuracy depends on the correctness of both the data and the calculation rules.

4.3 Layer Three: Interpretation or Review Indicator

An interpretation uses recorded data and calculated indicators to suggest what the information may mean or what action may be appropriate.

For example:

"Mathematics performance has declined by 6 percentage points compared with the previous reporting period."

Or:

"Attendance is below the school's configured review threshold. Staff review is recommended."

The first statement describes a calculated change.

The second applies a school-defined rule to the available data.

Neither statement automatically explains why the situation occurred.

The platform should distinguish a factual result from an interpretation and distinguish both from a final decision made by an authorised educator.

4.4 Why This Separation Matters

Suppose a student has low attendance and declining academic marks.

If the dashboard simply labels the student "Unmotivated," it has moved beyond the evidence.

The records show attendance and academic results. They do not directly establish motivation.

A better dashboard would display the attendance trend, academic trend, relevant dates, and a transparent review indicator where appropriate.

A teacher could then examine the circumstances and decide what support is needed.

Optimus Orien should display evidence clearly, calculate indicators transparently, and leave unsupported conclusions out of the system.

5. The Student Dashboard Must Have a Defined Context

Before showing results, the system must establish which student records the user is viewing.

It must also establish the educational context of those records.

A student may have attended several schools, studied different subjects, or participated in different educational programmes.

Results from these different contexts should not be mixed carelessly.

5.1 Student Identity

Every student should have a stable internal identifier.

The internal identifier should remain stable when the student changes classes or academic years.

A student's name alone is not a reliable database identifier because two students may share the same name, and a student's name may change.

The dashboard should retrieve records using the correct student identifier and authorised access scope.

5.2 School and Enrolment Context

The dashboard should identify the relevant school, campus or branch where applicable, academic year, and enrolment period.

If a student transfers between schools, the platform must preserve the historical records associated with each enrolment.

A principal should not automatically see a complete historical record from every school the student has attended unless the principal's permissions and applicable policies allow that access.

5.3 Academic Year and Reporting Period

The dashboard should clearly identify the academic year and reporting period currently being viewed.

For example:

Academic year: 2026–2027.

Reporting period: Term 1.

Class: Grade 8, Section A.

Programme: The programme configured for the student's enrolment.

These are illustrative values, not universal school structures.

Different schools may use semesters, terms, quarters, trimesters, or other reporting arrangements.

The platform should use the configured academic structure rather than hardcoding one calendar for every school.

5.4 Subject Context

A student may study different subjects under different programmes or curricula.

The dashboard should retrieve the subjects associated with the student's relevant enrolment and academic context.

It should not assume that every student studies the same subjects.

6. The Recommended Student Dashboard Structure

The student dashboard should be organised into clear sections.

The exact visual arrangement can evolve, but the information architecture should remain understandable.

A recommended structure is:

Section 1: Student overview.

Shows the student's identity, school, class, academic year, and reporting period.

Section 2: Academic performance.

Shows current results, subject-level performance, relevant averages, and assessment history.

Section 3: Attendance.

Shows attendance rates, absence counts, late arrivals where supported, and trends.

Section 4: Homework and assignments.

Shows completion rates, overdue work, pending verification, and eligible assignment counts.

Section 5: Participation.

Shows the relevant participation ratings, observation periods, and applicable rating framework.

Section 6: Subject-linked skills.

Shows configured skills, current ratings, historical ratings, and their subject context.

Section 7: Progress over time.

Shows how comparable academic results and other indicators have changed across reporting periods.

Section 8: Review indicators.

Shows transparent school-defined conditions that may require attention, together with the evidence that triggered them.

Section 9: Data quality and completeness.

Shows whether the information is complete enough to support reliable interpretation.

These sections may appear on one long page, separate tabs, or a combination of both.

The design should be selected according to usability and the amount of information each role needs.

It should not place every available metric on the screen at once.

7. Student Overview: Establishing the Context

The top of the dashboard should answer a simple question:

Whose record am I viewing, and which period does this information cover?

It should display only the profile information necessary for the user's role.

For example, the overview may contain the student's name, student identifier, school, class, academic year, and current reporting period.

Additional information may be displayed where it serves a legitimate educational purpose and is permitted by the school's policies.

Sensitive personal information should not be displayed merely because it exists in the database.

The overview should also communicate the status of the displayed data.

For example, the current reporting period might be marked:

-

In progress.

-
-

Submitted.

-
-

Under review.

-
-

Finalised.

-

These labels should correspond to actual system states.

The dashboard must not imply that results are final if teachers are still entering or correcting them.

8. Academic Performance Section

Academic performance is one of the central components of the student dashboard.

It should present the student's assessment results clearly and preserve the grading rules applicable to the educational programme.

8.1 Subject-Level Results

For each relevant subject, the dashboard should show the information needed to understand the result.

Depending on the school's configuration, this may include:

-

Marks obtained.

-
-

Maximum marks.

-
-

Percentage.

-
-

Grade or achievement level, where configured.

-
-

Relevant class average.

-
-

Previous-period result.

-
-

Change from the previous comparable period.

-
-

Assessment or reporting-period status.

-

For example, a Mathematics result might show:

Marks obtained: 68.

Maximum marks: 100.

Percentage: 68%.

Previous comparable result: 74%.

Change: −6 percentage points.

This is more informative than showing only "68" because the user can understand the scale and comparison.

8.2 Do Not Confuse Marks and Percentages

A student who receives 45 out of 50 has earned 90%.

A student who receives 45 out of 100 has earned 45%.

The raw mark of 45 does not have a complete meaning without the maximum possible mark.

The dashboard must display the relevant maximum or a clearly defined percentage.

It must also preserve the applicable assessment and grading rules.

8.3 Overall Academic Performance

Schools may want an overall academic indicator.

However, an overall result must be calculated using a clearly defined method.

Suppose a student has these subject percentages:

English: 80%.

Mathematics: 60%.

Science: 70%.

A simple average is:

(80 + 60 + 70) ÷ 3 = 70%.

This gives each subject equal weight.

But suppose the school's approved programme gives different weights to the subjects.

The overall result may then need a weighted calculation.

The dashboard must use the method configured for the applicable programme and reporting period.

It must not silently switch between calculation methods.

If no valid overall calculation rule exists, the platform should show the individual subject results rather than inventing an overall score.

8.4 Subject Results Should Remain Visible

An overall average can hide important differences.

Suppose a student has an overall average of 75%.

The student may have scored 90% in English, 85% in Science, and 50% in Mathematics.

The overall average alone does not show where the student is struggling.

The dashboard should therefore present the overall result, where valid, alongside the subject-level breakdown.

8.5 Assessment History

The dashboard should allow authorised users to understand how a subject result was produced.

Where supported, users should be able to inspect the underlying assessment records and the relevant calculation.

For example, the Mathematics result might include tests, quizzes, assignments, and examinations.

The dashboard should not require the user to navigate through unrelated pages merely to understand the components of a displayed result.

However, access to individual assessment records must still follow role permissions.

9. Attendance Section

Attendance helps the school understand whether the student was present for the relevant scheduled learning opportunities.

The dashboard should show the attendance measure and explain how it was calculated.

A percentage without a denominator may be misleading.

9.1 Attendance Rate

A configured attendance rate can be expressed as:

Attendance rate = Eligible present-equivalent attendance opportunities ÷ Eligible scheduled attendance opportunities × 100

The exact definition of a present-equivalent attendance opportunity must follow the school's policy.

For example, the school may define how late arrivals, approved school activities, authorised absences, and other attendance statuses affect the calculation.

The dashboard should not assume that every jurisdiction uses the same rules.

9.2 Show Counts Alongside Rates

Consider two students.

Student A has attended 18 of 20 eligible school days.

Student B has attended 180 of 200 eligible school days.

Both have a 90% attendance rate.

However, the number of recorded attendance opportunities differs.

Showing the numerator and denominator helps users understand the percentage.

The dashboard should display counts such as:

Eligible scheduled days: 20.

Present-equivalent days: 18.

Attendance rate: 90%.

The exact counts and labels should match the configured attendance model.

9.3 Absence and Lateness

Where the data supports it, the dashboard may show absence counts, consecutive absences, and late arrivals.

These indicators should remain distinguishable.

An absence is not the same as a late arrival, and the school may apply different rules to each.

The system should not automatically treat a late arrival as a full absence unless the applicable attendance policy explicitly requires that treatment.

9.4 Attendance Trends

The dashboard should show whether attendance has remained stable, improved, or declined across comparable periods.

For example:

Term 1: 96%.

Term 2: 92%.

Term 3: 87%.

The system can display a declining recorded attendance rate.

It should not automatically conclude that the student is disengaged or that attendance decline caused an academic decline.

The purpose is to make the pattern visible so authorised staff can investigate where appropriate.

9.5 Missing Attendance Data

A missing attendance register must not automatically be interpreted as an absence.

If a teacher has not submitted a register, the system should distinguish missing attendance data from recorded absences.

This is essential because an attendance rate calculated from incomplete records may not represent the student's actual attendance.

10. Homework and Assignment Section

The homework section should show whether the student has completed assigned work under the school's configured rules.

It should not present homework completion as a substitute for academic achievement.

10.1 Recommended Indicators

Depending on the school's workflow, the dashboard may show:

-

Eligible assignments due.

-
-

Assignments completed on time.

-
-

Assignments completed late.

-
-

Assignments awaiting teacher verification.

-
-

Assignments marked incomplete.

-
-

Assignments currently overdue.

-
-

On-time completion rate.

-
-

Eventual completion rate.

-

Not every school needs every indicator immediately.

The initial interface should prioritise the measures that are meaningful and supported by reliable records.

10.2 Show the Denominator

Suppose a dashboard shows a homework completion rate of 80%.

That figure is more interpretable when the user can see that the student completed 16 of 20 eligible assignments.

The platform should make the underlying counts available.

It should also indicate the relevant reporting period and completion-rate definition.

10.3 Pending Verification

Suppose a student declares that 18 assignments are complete, but the teacher has verified only 15.

The dashboard should distinguish the student-declared count from the teacher-verified count.

It should not silently treat all 18 as verified completions.

10.4 Academic Results Remain Separate

A student may complete every assignment but perform poorly on examinations.

Another student may perform well academically while missing several assignments.

The dashboard should display both indicators so that educators can understand the difference.

It should not automatically lower the student's examination result because homework completion is low, unless the formal assessment rules explicitly make homework part of that result.

11. Participation Section

Participation information can help educators understand how students contribute to relevant classroom activities.

However, participation ratings are generally based on observations and professional judgement.

They should be presented with appropriate context.

11.1 Show the Rating Framework

If the school uses a 1–10 scale, the dashboard should identify the scale and provide its configured meaning.

A rating of 7 is not self-explanatory.

The user should be able to understand what the school intended the rating to represent.

11.2 Show the Subject and Reporting Period

A participation rating should identify the subject and relevant observation period.

For example:

Subject: Science.

Reporting period: Term 1.

Rating: 7 out of 10.

Evaluator: Authorised Science teacher.

This provides more useful context than a single number displayed without explanation.

11.3 Do Not Treat Ratings as Perfectly Objective

The dashboard should identify participation ratings as teacher-entered or rubric-based measures where appropriate.

It should not imply that a rating of 8 from one teacher is automatically equivalent to a rating of 8 from another teacher using a different framework.

Where ratings are not comparable, the platform should display them separately.

12. Subject-Linked Skills Section

The skills section should show the student's recorded performance against configured subject-linked skills.

Examples might include reading comprehension, mathematical reasoning, scientific reasoning, or analytical thinking.

The exact skills should come from the school's configured framework.

12.1 Recommended Display

For each relevant skill, the dashboard should show:

Skill name.

Skill description.

Associated subject or learning area.

Current rating or result.

Rating scale or rubric.

Relevant observation period.

Previous rating, where a valid comparison exists.

Evaluator information, where appropriate.

This allows users to understand the meaning of the rating rather than treating it as an unexplained number.

12.2 Show Historical Change Carefully

Suppose a student's analytical-thinking rating changes from 5 to 7.

The dashboard may show that the recorded rating increased by two points.

It must not automatically claim that the underlying skill improved by 40%.

The rating scale may be ordinal or rubric-based, meaning that the difference between two numerical values does not necessarily represent an equal amount of skill growth.

12.3 Do Not Mix Different Skills Into One Score

A student might have:

Reading comprehension: 8.

Writing: 6.

Speaking: 9.

Listening: 7.

The platform should not automatically average these values into an overall English skill score unless the school has defined an appropriate method.

The skills measure different abilities, and their scales may not support a meaningful combined result.

A subject-level skill summary may be possible, but it requires an explicit and appropriate calculation rule.

13. Progress Over Time: Understanding Change

One of the most useful features of Optimus Orien is the ability to preserve and visualise student progress across reporting periods and academic years.

However, progress must be measured using comparable information.

13.1 What Is Progress?

Progress refers to a change in a student's recorded performance or another defined indicator over time.

For example, a Mathematics percentage may change from 62% in Term 1 to 70% in Term 2.

The difference is:

70% − 62% = 8 percentage points.

The platform can display this as an increase of eight percentage points.

13.2 Percentage Points Versus Relative Percentage Change

These two concepts must not be confused.

A change from 60% to 70% is an increase of 10 percentage points.

The relative percentage increase is:

(70 − 60) ÷ 60 × 100 ≈ 16.7%.

Both calculations are mathematically valid, but they describe different things.

For the student dashboard, percentage-point change is often easier to interpret when comparing percentages.

If the platform displays relative percentage change, it must label it clearly.

13.3 Comparability Matters

Suppose a student receives 80% on a relatively easy assessment in one term and 70% on a much harder assessment in the next term.

The numerical result has declined, but the two assessments may not be directly comparable.

The dashboard should avoid claiming a precise change in underlying knowledge unless the assessment design supports that interpretation.

It can still show the recorded marks and their difference, with appropriate context.

13.4 Progress Across Different Academic Years

The system should preserve historical records across years.

However, the student's subjects, curriculum, grading rules, and school context may change.

A historical Mathematics percentage from one programme should not automatically be treated as fully equivalent to a result from a different programme.

The platform should identify changes in context and compare records only where the comparison method is defensible.

13.5 No Historical Record

If the platform lacks a previous result, it should display the current result without inventing a trend.

For example:

"Current result: 72%. Previous comparable result: Not available."

It must not assume that a missing historical result represents zero.

14. Review Indicators: Helping Staff Notice Patterns

A student dashboard may include indicators that suggest a record deserves further attention.

These are useful because teachers and school leaders may manage many students and may not notice every important change immediately.

However, a review indicator is not a diagnosis, a prediction, or proof that a student has a problem.

For the current Optimus Orien scope, these indicators should use transparent, configurable rules rather than Artificial Intelligence (AI).

14.1 What Is a Review Indicator?

A review indicator is a status generated when defined conditions are met.

For example:

A student's attendance rate falls below a school-defined threshold.

A student's Mathematics performance declines by a configured number of percentage points.

Several assignments remain overdue.

The system may flag these records for authorised staff review.

The indicator should explain the condition that triggered it.

14.2 Example: Attendance Review

Suppose a school configures an attendance review threshold of 85%.

A student's attendance rate is 78%.

The system may display:

"Attendance review recommended: current attendance is 78%, below the configured 85% threshold."

The indicator is transparent because it identifies both the observed value and the rule.

The school should be able to configure the threshold according to its policy.

The platform must not present 85% as a universal scientific boundary that automatically defines student risk.

14.3 Example: Academic Decline

Suppose the school configures a rule that identifies a decline of at least 10 percentage points between comparable reporting periods.

A student's result changes from 78% to 66%.

The decline is 12 percentage points.

The system may display:

"Academic review recommended: the recorded result declined by 12 percentage points compared with the previous comparable period."

It should also provide access to the underlying results.

It should not automatically claim that the student has lost motivation or that a particular factor caused the decline.

14.4 Multiple Indicators

A student may have low attendance, overdue assignments, and declining academic results.

The dashboard can display each condition separately.

The platform should not automatically turn these conditions into an unexplained combined risk score.

A combined score would require a separate, justified model and validation process.

That is not required for the current implementation.

14.5 Human Review Is Required

A review indicator should support a human decision.

An authorised educator should be able to inspect the evidence, assess its relevance, and decide whether action is necessary.

The platform should not automatically punish a student, change their academic results, or make a high-stakes educational decision simply because a threshold was crossed.

14.6 Review Statuses

If Optimus Orien implements a review workflow, useful statuses may include:

New: The condition has been identified and has not yet been reviewed.

Under review: An authorised staff member is examining the information.

Action planned: A relevant follow-up action has been recorded.

Monitoring: The situation is being observed over time.

Resolved: The authorised reviewer has closed the review according to the school's policy.

These statuses describe the review process, not the student's personal worth or ability.

The school should be able to configure the workflow where necessary.

15. Data Completeness: Can the Dashboard Be Trusted?

A dashboard may look professional while containing incomplete information.

For example, a student may have recorded examination marks but missing attendance registers for several weeks.

The academic section may be reliable while the attendance section is incomplete.

The dashboard should communicate this difference.

15.1 What Is Data Completeness?

Data completeness describes whether the expected records are available for a particular purpose.

It does not necessarily mean that the available records are accurate.

For example, a school may have an attendance entry for every scheduled day, but some entries may still contain errors.

Completeness and accuracy are separate data-quality dimensions.

15.2 Examples of Incomplete Data

A dashboard may encounter:

-

Missing examination results.

-
-

Attendance registers that have not been submitted.

-
-

Assignments awaiting teacher verification.

-
-

Skills that have not yet been assessed.

-
-

Missing historical results.

-

-

  Results awaiting approval or finalisation.

-

The system should identify these conditions when they affect interpretation.

15.3 Missing Data Must Not Become Zero

If a student has not been assessed in a skill, the platform must not automatically assign zero.

If a teacher has not submitted an attendance register, the platform must not automatically mark all students absent.

If a result has not been entered, the platform must not automatically treat it as a failed assessment.

Missing data is a separate state.

15.4 Completeness Indicators

The dashboard may display labels such as:

"Attendance data incomplete for this reporting period."

"Two assignments are awaiting teacher verification."

"Skill ratings have not yet been entered for this period."

"Academic results are provisional."

These messages should be generated from actual system states.

They should not appear merely because a generic field is empty when that field is not required for the relevant context.

16. The Difference Between Current Performance and Final Results

A reporting period may still be in progress when a user opens the dashboard.

Teachers may still be entering results, verifying assignments, or correcting records.

The dashboard should distinguish between provisional information and finalised information.

16.1 Provisional Results

A provisional result is a result that has not yet reached the relevant finalisation stage.

The exact workflow depends on the school's policy.

A provisional result may be recalculated when additional assessments are entered or corrections are made.

16.2 Finalised Results

A finalised result has reached the configured finalisation state.

This does not mean that it can never be corrected.

It means that the result has been approved or locked according to the relevant workflow.

Any authorised post-finalisation correction should be traceable and should update affected calculations according to policy.

16.3 Consistency Across the Platform

If the same academic result appears in the student dashboard, teacher portal, and principal portal, those views should use the same underlying record and applicable calculation rules.

They may display different levels of detail because their permissions differ.

They must not silently calculate conflicting results.

17. Designing the Dashboard Without Overwhelming Users

The student dashboard will contain several categories of information.

A common interface-design mistake is to display every available number, chart, table, warning, and historical result on one screen.

This can make the dashboard difficult to understand.

Optimus Orien should prioritise the information according to the user's role and purpose.

17.1 Visual Hierarchy

Visual hierarchy means arranging information so that users can distinguish the most important elements from supporting details.

For example, the current reporting period and academic results may appear near the top of the principal's student dashboard.

Detailed assessment histories can be placed in a separate section that users open when they need more information.

The aim is to make the main information easy to understand without removing access to the underlying evidence.

17.2 Use Clear Labels

Every important metric should have a meaningful label.

For example:

"Attendance rate" is clearer than "Attendance Index."

"On-time homework completion" is clearer than "Homework Score."

"Recorded skill rating" is clearer than "Skill Intelligence."

Labels should explain what the platform actually measures.

17.3 Explain Unfamiliar Terms

The dashboard should provide concise explanations for terms that users may not understand.

For example, a tooltip or help panel could explain:

"Percentage-point change: the difference between two percentages. A change from 60% to 70% is an increase of 10 percentage points."

A similar explanation could clarify how the attendance denominator is calculated.

Tooltips should be concise, and more detailed explanations can be available through a help section.

17.4 Use Colours Carefully

Colours can help users distinguish categories or status levels.

For example, a school might use its configured colours to indicate a review status.

However, colours should not carry meaning without text.

A red indicator should have a clear label explaining the condition.

The system should not assume that every school wants the same colours or uses the same thresholds.

The interface should also remain usable for people with colour-vision deficiencies.

17.5 Charts Must Preserve Meaning

A chart should have a clear title, labelled axes where applicable, a defined time period, and an understandable scale.

A line chart may be suitable for showing attendance over several reporting periods.

A bar chart may be suitable for comparing subject percentages.

A table may be better for displaying individual assessments and marks.

The visual format should serve the information rather than being selected merely because it looks attractive.

18. Role-Specific Student Dashboards

Different users need different levels of detail.

The dashboard should adapt to the user's role and authorised scope without changing the meaning of the underlying measurements.

18.1 Teacher View

The teacher should be able to view and manage records for assigned students, classes, and subjects.

The teacher view should prioritise information relevant to teaching and assessment.

Depending on permissions, it may include assessment results, attendance information, homework status, participation ratings, skills, and review indicators.

A teacher should not automatically have permission to alter every record visible in the student profile.

Viewing a record and editing a record are different permissions.

18.2 Principal View

The principal should be able to review relevant student records within the authorised school.

The principal may need to understand overall performance, historical trends, and review indicators across different subjects.

The principal should also be able to inspect the underlying evidence where permitted.

The system must not assume that a principal should automatically access every sensitive note or restricted record merely because they manage the school.

18.3 Organisation Administrator View

An organisation administrator may need to inspect students across multiple authorised branches.

The administrator may compare student performance patterns across schools and identify records requiring review.

However, organisation-wide access must follow the permissions and policies defined for that role.

The dashboard should not expose records from unrelated organisations.

18.4 Student View

A student should be able to understand their own academic results and relevant educational indicators.

The interface should explain results in clear language and distinguish current performance from historical change.

A student should not be exposed to another student's private records merely because both appear in the same class.

18.5 Parent or Guardian View

An authorised parent or guardian should be able to view the records of the student or students linked to their account, within the permissions granted by the school.

The dashboard should present useful information without exposing unrelated students' data.

The system must verify the parent-student relationship on the server rather than relying only on the interface.

18.6 Public View

A public homepage may show approved school-level information, such as published school rankings or school health indicators.

It must not expose individual students' marks, attendance records, homework records, participation ratings, or skill ratings.

The public school leaderboard and the private student dashboard serve different purposes and require separate access controls.

19. Student Review and Intervention Workflow

The dashboard can identify conditions that may deserve attention, but the school needs a process for reviewing those conditions.

This process should be transparent and controlled.

19.1 From Indicator to Review

Suppose the system identifies a student whose attendance has declined below a configured threshold.

The system creates or displays a review indicator according to the school's policy.

An authorised staff member examines the relevant attendance records and considers whether the data is complete and accurate.

The staff member may decide that no action is necessary, that additional information is needed, or that a supportive follow-up is appropriate.

The platform should record the review status if the school uses a formal workflow.

19.2 Record Evidence Separately From Actions

The attendance records are evidence.

The review decision is a separate record.

A follow-up action is another record.

These should not be collapsed into one field.

For example, correcting an attendance entry should not automatically close a separate review unless the workflow explicitly requires that outcome.

Likewise, creating an intervention record should not rewrite the underlying academic results.

19.3 Follow-Up Records

If intervention tracking is enabled, the platform may record:

The reason for review.

The relevant indicator or evidence.

The responsible staff member.

The date of review.

The planned action.

The follow-up date.

The review outcome.

Any authorised status changes.

The school should define what information is necessary and who may access it.

Sensitive family or medical details should not be collected unnecessarily.

19.4 Human Decisions Must Remain Human Decisions

The platform should not automatically label a student as having a behavioural problem, a learning disorder, or a lack of motivation.

It should not automatically impose penalties because an indicator is triggered.

Its role is to make recorded information visible, explain the applicable rule, and support authorised staff in deciding what to do next.

20. Data Security and Privacy

The student performance dashboard brings together information that may be sensitive.

Even when each individual indicator appears harmless, the combined profile can reveal substantial information about a student.

Security must therefore be built into the underlying data and server logic.

20.1 Server-Side Access Control

The server must verify the user's permissions for every protected request.

Hiding a button is not sufficient.

For example, a parent should not be able to change a student identifier in a request and retrieve another student's dashboard.

The server must validate the user's role, scope, and relationship to the requested record.

20.2 Least Privilege

Least privilege means giving a user only the access needed to perform their authorised tasks.

A teacher may need to edit assessment results for an assigned subject but may not need permission to change school-wide settings.

A parent may need to view their child's performance but should not be able to edit official marks.

The platform should separate view, create, edit, approve, export, and administrative permissions where appropriate.

20.3 Audit History

Important changes to student records should be traceable.

The audit history should record who made the change, when it happened, what changed, and the reason where required.

This is particularly important for assessment corrections, attendance changes, skill ratings, and finalised results.

20.4 Data Minimisation

The dashboard should show information needed for the user's task.

It should not display sensitive information simply because the information exists in the database.

Access to restricted notes and intervention details should be separately controlled.

20.5 Data Retention

The platform should preserve educational records according to the school's policies and applicable legal requirements.

It should not assume that one retention period is suitable for every country or institution.

Deletion, archiving, and historical-record rules should be defined before implementation.

21. Recommended Database and Service Architecture

The developer should avoid implementing the student dashboard as a separate collection of manually maintained values.

Instead, the dashboard should draw on the underlying modules.

21.1 Source Modules

The dashboard may retrieve information from:

The student and enrolment module.

The academic structure module.

The assessment and grading module.

The attendance module.

The assignment and homework module.

The participation-rating module.

The subject-linked skills module.

The comparison and review-indicator module.

These modules should maintain their own records and rules.

21.2 Shared Calculation Logic

Where the same calculation is required in multiple places, the platform should use a shared and tested calculation service or equivalent implementation.

For example, if the Mathematics percentage is calculated in the student dashboard, teacher portal, and principal portal, all three should use the same applicable grading rules.

The developer should not independently reimplement the formula in each interface.

This reduces the chance that different pages show different results for the same student.

21.3 Dashboard Aggregation

A dashboard aggregation service combines relevant values from the source modules for display.

It should:

Retrieve records within the user's authorised scope.

Apply the requested student and reporting-period filters.

Use the applicable calculation rules.

Preserve the status of provisional and finalised records.

Identify missing or incomplete information.

Return clear values and supporting context.

Avoid calculating unsupported combined scores.

21.4 Performance and Caching

A dashboard may require information from several modules.

The developer should avoid loading every historical record and recalculating everything unnecessarily each time a user opens a page.

Where appropriate, the system may use optimised queries or cached derived values.

However, any cached value must be refreshed or invalidated when relevant source records or calculation rules change.

Caching must never override permission checks.

A user must not receive another user's dashboard because the system reused a cached response without considering access scope.

22. Example: Reading a Complete Student Dashboard

Consider a fictional student named Sara.

Sara is enrolled in Grade 8 at a school using a term-based academic calendar.

The following values are illustrative and are not intended to represent a universal educational standard.

22.1 Academic Performance

English: 82%.

Mathematics: 64%.

Science: 71%.

Overall academic average: 72.3%, using a simple average across the three subjects for this example.

The dashboard also shows that Mathematics declined from 74% in the previous comparable reporting period to 64% in the current period.

This represents a decline of 10 percentage points.

22.2 Attendance

Eligible scheduled days: 100.

Present-equivalent days: 88.

Attendance rate: 88%, under the school's configured attendance rules.

The previous comparable period's attendance rate was 95%.

The dashboard shows a decline of seven percentage points.

22.3 Homework

Eligible assignments due: 20.

Completed on time: 13.

Completed late: 3.

Not completed: 4.

Excused assignments are excluded from this illustrative denominator.

On-time completion rate:

13 ÷ 20 × 100 = 65%.

Eventual completion rate:

16 ÷ 20 × 100 = 80%.

The dashboard shows both values because they describe different aspects of completion.

22.4 Participation

English participation rating: 8 out of 10.

Mathematics participation rating: 6 out of 10.

Science participation rating: 7 out of 10.

The ratings are linked to the school's configured participation framework and the relevant reporting period.

22.5 Subject-Linked Skills

Reading comprehension: 8 out of 10.

Mathematical reasoning: 6 out of 10.

Scientific reasoning: 7 out of 10.

These ratings are shown using the applicable skill framework.

22.6 What the Dashboard Can Reasonably Say

The dashboard can report that:

Mathematics performance declined by 10 percentage points compared with the previous comparable period.

Attendance declined by seven percentage points.

The on-time homework completion rate is 65%.

The recorded mathematical-reasoning rating is 6 out of 10 under the school's configured framework.

If the school has configured appropriate review rules, one or more of these indicators may require staff review.

22.7 What the Dashboard Must Not Conclude

The dashboard must not automatically conclude that Sara is lazy, unmotivated, incapable of Mathematics, or experiencing a psychological disorder.

It must not claim that low attendance caused the decline in Mathematics.

It must not assume that incomplete homework explains the skill rating.

Those conclusions go beyond the displayed data.

The dashboard should make the evidence understandable so that an authorised educator can investigate the situation.

23. Software Implementation Requirements and Acceptance Tests

The following requirements define the expected behaviour of the complete student performance dashboard.

23.1 Student Context

STD-001 — Stable Student Identity

Every student dashboard must use a stable internal student identifier.

Acceptance test: Two students share the same name. Each dashboard retrieves the correct student's records without confusing their identities.

STD-002 — Enrolment Context

The dashboard must respect the student's school, enrolment, and academic context.

Acceptance test: A student transfers to another school. Historical records remain associated with the appropriate enrolment and school.

STD-003 — Reporting-Period Selection

The dashboard must identify the selected academic year and reporting period.

Acceptance test: A user selects Term 2. The dashboard shows the correct period's records and does not silently mix Term 1 and Term 2 values.

STD-004 — Configurable Academic Structures

The dashboard must work with the academic periods and programme structures configured for the school.

Acceptance test: A school uses semesters rather than terms. The dashboard displays the configured reporting periods without requiring a code change.

STD-005 — Permission-Scoped History

The dashboard must only retrieve historical records the current user is authorised to view.

Acceptance test: A principal cannot retrieve restricted records from another school by requesting a previous academic year.

23.2 Academic Performance

STD-006 — Correct Assessment Results

The dashboard must display assessment results from the authoritative assessment records and applicable grading rules.

Acceptance test: An authorised teacher corrects an assessment result. The dashboard reflects the corrected result according to the configured recalculation workflow.

STD-007 — Marks and Maximum Marks

Where raw marks are displayed, the dashboard must identify the maximum possible marks.

Acceptance test: A student receives 45 out of 50. The interface displays both values and, where supported, the correct percentage of 90%.

STD-008 — Defined Overall Average

The dashboard must only display an overall academic average when a valid calculation rule exists.

Acceptance test: A school has no configured overall aggregation rule. The dashboard displays the subject results without inventing an overall average.

STD-009 — Subject Breakdown

The dashboard must preserve subject-level results even when an overall average is available.

Acceptance test: A student has high results in two subjects and a low result in one. The subject-level difference remains visible.

STD-010 — Historical Comparability

The dashboard must identify the context used for historical academic comparisons.

Acceptance test: A student changes programmes. The dashboard does not silently treat results from incompatible grading systems as directly comparable.

STD-011 — Percentage-Point Change

When comparing percentages, the dashboard must calculate and label percentage-point change correctly.

Acceptance test: A result changes from 60% to 70%. The dashboard reports an increase of 10 percentage points.

STD-012 — Provisional and Final Results

The dashboard must distinguish provisional results from finalised results where the workflow supports those states.

Acceptance test: A reporting period is still in progress. The dashboard does not label its provisional results as finalised.

23.3 Attendance

STD-013 — Attendance Calculation

The dashboard must use the configured attendance calculation rules.

Acceptance test: A school configures how late arrivals and approved activities affect attendance. The displayed rate follows those rules.

STD-014 — Attendance Denominator

The dashboard must show or make available the relevant attendance denominator and numerator.

Acceptance test: A student has attended 18 of 20 eligible school days. The dashboard displays the counts and the corresponding 90% rate.

STD-015 — Missing Attendance Records

The dashboard must distinguish missing registers from recorded absences.

Acceptance test: A teacher has not submitted an attendance register. The system does not automatically count every student as absent for that register.

STD-016 — Attendance Trend

The dashboard must calculate trends using the selected and comparable reporting periods.

Acceptance test: A student's attendance changes from 95% to 88%. The dashboard displays a decline of seven percentage points.

23.4 Homework and Assignments

STD-017 — Completion-Rate Context

The dashboard must display the applicable completion-rate definition and relevant counts.

Acceptance test: A student completes 13 of 20 eligible assignments on time. The dashboard displays the 65% on-time completion rate and its underlying counts.

STD-018 — On-Time and Eventual Completion

The dashboard must distinguish on-time completion from eventual completion when both are supported.

Acceptance test: A student completes three additional assignments late. The two rates remain separate and correctly calculated.

STD-019 — Pending Verification

The dashboard must distinguish pending teacher verification from confirmed completion.

Acceptance test: A student declares an assignment complete, but the teacher has not verified it. The dashboard shows the pending state rather than treating it as verified completion.

STD-020 — Missing Assignment Records

The dashboard must not automatically treat missing assignment records as confirmed failures.

Acceptance test: An assignment has not been created or assigned to the student. The system does not count it as an incomplete assignment.

23.5 Participation and Skills

STD-021 — Participation Context

The dashboard must display participation ratings with their relevant subject, period, and configured framework.

Acceptance test: A student receives a Science participation rating. The dashboard identifies it as a Science rating and displays the applicable framework.

STD-022 — Skill Context

The dashboard must display skill ratings with the relevant skill definition, subject context, period, and scale.

Acceptance test: A student receives a mathematical-reasoning rating of 6 out of 10. The dashboard identifies the skill and the framework used.

STD-023 — Missing Skill Ratings

The dashboard must distinguish an unrecorded rating from a recorded zero.

Acceptance test: No skill rating has been entered. The dashboard displays the appropriate missing or not-yet-assessed state.

STD-024 — Historical Frameworks

The dashboard must preserve the framework used for historical participation and skill ratings.

Acceptance test: A school changes its skill-rating rubric. Previous ratings remain associated with the earlier rubric version.

STD-025 — No Unsupported Growth Calculation

The dashboard must not automatically convert changes in ordinal or rubric-based ratings into precise percentages of underlying skill growth.

Acceptance test: A skill rating changes from 4 to 7. The dashboard shows the recorded change without automatically claiming a 75% increase in the student's underlying skill.

23.6 Review Indicators and Data Quality

STD-026 — Transparent Review Rules

Every rule-based review indicator must identify the condition that triggered it.

Acceptance test: Attendance falls below a configured threshold. The dashboard shows the observed attendance rate, the configured threshold, and the reason for the review indicator.

STD-027 — No Automatic Diagnosis

The dashboard must not label a student with a psychological disorder or unsupported behavioural trait based solely on performance indicators.

Acceptance test: A student's attendance and marks decline. The system may display configured review indicators but does not automatically label the student as lazy, unmotivated, or disordered.

STD-028 — Human Review

Review indicators must not automatically trigger unsupported high-stakes decisions.

Acceptance test: A student crosses a review threshold. The system displays the indicator but does not automatically change academic marks or impose a penalty.

STD-029 — Data Completeness

The dashboard must identify missing or provisional information where it affects interpretation.

Acceptance test: Attendance registers are missing for part of the reporting period. The dashboard identifies incomplete attendance data.

STD-030 — No Missing-to-Zero Conversion

The system must not automatically convert missing assessment results, attendance records, assignment statuses, or skill ratings into zero values.

Acceptance test: A student has no recorded skill rating. The dashboard does not display zero unless zero was explicitly recorded under a valid configured scale.

STD-031 — Source Record Access

Where authorised, users must be able to inspect the source records underlying a displayed indicator.

Acceptance test: A principal opens an attendance review indicator and can inspect the relevant attendance records within their authorised scope.

STD-032 — Consistent Calculations

The dashboard and other authorised portals must use consistent calculation rules for the same underlying records.

Acceptance test: The student dashboard and principal dashboard show the same subject result for the same student, period, and calculation version.

23.7 Security and Performance

STD-033 — Server-Side Permissions

The server must enforce access control for every protected dashboard request.

Acceptance test: A user attempts to retrieve another student's dashboard by changing the student identifier. The server rejects the unauthorised request.

STD-034 — Parent-Student Relationship

A parent or guardian must only access student records permitted by their verified relationship and role.

Acceptance test: A parent linked to Student A cannot retrieve Student B's dashboard through a direct request.

STD-035 — Organisation Isolation

The system must prevent unauthorised cross-organisation access.

Acceptance test: An organisation administrator attempts to retrieve a student from an unrelated organisation. The server rejects the request.

STD-036 — Cache Isolation

Any cached dashboard response must respect the current user's access scope and the selected reporting context.

Acceptance test: Two users with different permissions open student dashboards. The second user does not receive the first user's restricted data from a shared cache.

STD-037 — Refresh After Corrections

The dashboard must reflect relevant authorised changes to source records and calculation rules.

Acceptance test: An attendance record is corrected. The dashboard updates the affected attendance indicator according to the recalculation workflow.

STD-038 — Audit History

The system must preserve appropriate audit records for changes to underlying data and relevant configurations.

Acceptance test: An authorised user changes a skill rating. The audit history identifies the previous value, new value, user, and timestamp.

STD-039 — No Public Student Data

Public pages must not expose individual student dashboards or private indicators.

Acceptance test: An unauthenticated visitor attempts to retrieve a student's performance record through a public endpoint. The server rejects the request.

STD-040 — Usable Error States

The dashboard must handle unavailable, incomplete, and failed data requests clearly.

Acceptance test: The attendance service is temporarily unavailable. The dashboard does not display an invented attendance rate of zero; it identifies the information as temporarily unavailable.

24. Recommended Development Sequence

The developer should not attempt to build every chart and review indicator before the underlying data is reliable.

The recommended sequence is:

Stage 1: Establish the student context.

Confirm that student identity, enrolment, school, academic year, subjects, and reporting periods are correctly connected.

Stage 2: Connect academic results.

Retrieve authoritative assessment records and apply the configured grading rules.

Stage 3: Connect attendance.

Use the attendance module's established calculations and display the relevant counts and completeness information.

Stage 4: Connect homework.

Display assignment counts, verified completion, pending verification, and configured completion rates.

Stage 5: Connect participation and skills.

Display ratings with their subject, observation period, evaluator, and applicable framework.

Stage 6: Implement progress comparisons.

Compare compatible periods and distinguish percentage-point changes from changes in rating scales.

Stage 7: Implement data-quality indicators.

Show incomplete, provisional, or unavailable records without inventing values.

Stage 8: Implement review indicators.

Use transparent, configurable rules and provide access to the supporting evidence.

Stage 9: Enforce permissions.

Test every role and access scope at the server level.

Stage 10: Build the visual interface.

Organise the information into clear sections, with concise labels and charts that support understanding.

Stage 11: Test consistency and corrections.

Confirm that changes to source records are reflected correctly across all authorised views.

Stage 12: Test realistic scenarios.

Test students with missing records, transfers, incomplete verification, different programmes, changing grading rules, and provisional results.

25. Final Product Principles

The student performance dashboard is one of the central features of Optimus Orien because it brings together information that would otherwise remain scattered across the platform.

However, its value depends on more than its appearance.

The dashboard must use reliable source records, correct calculations, appropriate historical comparisons, and clearly defined indicators.

It must preserve the differences between academic performance, attendance, homework completion, participation, and skills.

It must show missing information honestly.

It must identify the rules behind review indicators.

It must enforce permissions on the server and protect private student records.

It must support educators in understanding the evidence without pretending to explain every cause behind a student's performance.

A well-designed dashboard does not need to display every possible metric at once. It needs to show the right information, in the right context, with enough explanation for the user to understand it.

Optimus Orien should turn educational records into understandable information—not turn students into unexplained numbers.

What Part 12 Establishes

Part 12 defines the student performance dashboard as a presentation layer built on reliable underlying records.

It establishes how the dashboard should combine academic results, attendance, homework, participation, subject-linked skills, historical progress, review indicators, and data-completeness information.

It also establishes the distinction between recorded data, calculated indicators, and interpretations; defines role-specific dashboard requirements; and provides 40 implementation requirements with acceptance tests..

Part 13 moves from the individual student to the educators responsible for teaching and the school leaders responsible for understanding school performance. It explains how Optimus Orien should present teacher-related information and school-level patterns without confusing student outcomes with teacher quality or encouraging unfair comparisons.

PART 13

TEACHER PERFORMANCE AND SCHOOL LEADERSHIP DASHBOARDS

Understanding Teacher Responsibilities, Classroom Performance, Subject Outcomes, and School-Level Patterns Without Unfairly Evaluating Educators

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner, software developer, and future product team
Primary objective: Define how Optimus Orien should help teachers, principals, and organisation administrators understand educational performance at classroom, subject, teacher, school, and organisation levels.

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1. Why This Document Exists

In Part 12, we designed the complete student performance dashboard.

We established how Optimus Orien should bring together academic results, attendance, homework completion, participation, subject-linked skills, historical progress, and review indicators.

However, students do not learn in isolation.

Their educational experience takes place within classrooms, subjects, teaching assignments, academic programmes, and schools.

Teachers record assessments, monitor attendance where authorised, assign homework, assess skills, and observe students' learning. Principals oversee school operations and academic performance. Organisation administrators may need to understand patterns across multiple schools or branches.

Optimus Orien must therefore provide dashboards for educators and school leaders as well.

But there is a significant design challenge.

A student's academic result is not automatically a measurement of the teacher's quality.

A class average can tell us something about the recorded academic performance of a group of students. It cannot, by itself, tell us how effectively a teacher taught them.

Student results can be influenced by many factors, including prior knowledge, attendance, assessment difficulty, curriculum alignment, available resources, language background, and learning opportunities.

The platform should help school leaders investigate educational patterns without pretending that one number explains everything.

This part defines how to achieve that.

2. Three Different Things: Teacher Activity, Student Outcomes, and Teaching Quality

Before designing the dashboards, the developer must understand three distinct concepts.

2.1 Teacher Activity

Teacher activity refers to actions and responsibilities recorded in the system.

Examples include:

A teacher entered Mathematics assessment results.

A teacher submitted an attendance register.

A teacher verified homework completion.

A teacher recorded a subject-linked skill rating.

A teacher completed a required reporting task.

These records can help a school understand whether administrative and academic processes are being completed.

However, activity is not the same as effectiveness.

A teacher may enter all marks on time without necessarily providing effective instruction. Another teacher may provide excellent instruction but experience technical problems that delay data entry.

Activity indicators must therefore be labelled accurately.

2.2 Student Outcomes

Student outcomes are the results or indicators recorded for students.

Examples include examination marks, attendance rates, homework completion, and skill ratings.

These outcomes can be summarised for a class or subject.

For example, the average Mathematics result for a class may be 72%.

That tells us about the class's recorded academic performance under the relevant assessment and grading rules.

It does not directly establish why the class achieved that result.

2.3 Teaching Quality

Teaching quality concerns the quality of the educational experience and teaching practice.

It may involve instructional clarity, subject knowledge, lesson planning, classroom management, feedback, inclusion, assessment design, and the ability to support different learners.

Some aspects can be examined through structured observations, professional reviews, or validated evaluation frameworks.

They cannot be reliably measured by class averages alone.

For example, a teacher who works with students who begin the year with substantial learning gaps may have a lower class average than a teacher whose students already have strong foundations.

That difference does not automatically mean the second teacher is more effective.

Product rule: Optimus Orien should distinguish recorded teacher activity, student outcomes, and formal teaching-quality evaluation. These concepts must not be combined into one unexplained teacher score.

3. The Main Teacher and Leadership Dashboards

Optimus Orien should support different views for different educational responsibilities.

The recommended structure includes five related dashboards.

3.1 Teacher Dashboard

The teacher dashboard helps an educator understand their assigned teaching responsibilities and the relevant performance of their students.

It may show:

Assigned classes and subjects.

Students within the teacher's authorised scope.

Assessment-entry status.

Subject-level academic results.

Class averages.

Student progress patterns.

Attendance information where the teacher has permission to access it.

Homework completion and verification status.

Subject-linked skill ratings.

Records requiring review.

Incomplete or overdue administrative tasks.

The dashboard should prioritise information that helps the teacher perform their actual responsibilities.

It should not automatically expose other teachers' private performance records.

3.2 Principal Dashboard

The principal dashboard provides a school-level view.

It helps the principal understand academic performance across classes, subjects, and reporting periods.

It may show:

Overall school academic indicators.

Class-level and subject-level averages.

Attendance patterns.

Homework completion patterns.

Assessment-entry completeness.

Student review indicators.

Participation and skill-rating coverage.

Teacher reporting status.

Historical performance trends.

School health indicators.

The principal should be able to move from a school-wide summary to a relevant class, subject, or student record when authorised.

3.3 Academic Leadership Dashboard

Some schools assign academic oversight to a head of academics, director of studies, head of assessment, or similar role.

These roles may require more detailed academic information than a general administrative dashboard.

For example, an academic leader may need to examine assessment completion, subject outcomes, reporting-period consistency, and differences between classes studying the same subject.

Optimus Orien should support these responsibilities through configurable roles and permissions rather than assuming that every school uses the same job titles.

3.4 Organisation Administrator Dashboard

An organisation administrator may oversee several schools or branches.

This dashboard should support comparisons across the administrator's authorised organisation.

It may show:

Performance indicators by school.

Attendance patterns by branch.

Academic reporting completeness.

School health indicators.

Changes over time.

Differences between programmes or educational structures.

Areas requiring further review.

The organisation dashboard must distinguish genuine differences in recorded performance from differences in curriculum, assessment rules, school calendars, and data completeness.

3.5 Future Government or Oversight Dashboard

A government or oversight dashboard may eventually support education-system monitoring across districts, regions, or schools.

This is a separate product scope requiring its own governance, permissions, data-sharing rules, and reporting definitions.

The current teacher and school leadership dashboards should not automatically grant government users access to identifiable student records.

A future oversight portal must be designed and authorised separately.

4. The Teacher's Work Must Be Represented Correctly

To build a useful teacher dashboard, Optimus Orien must model teaching assignments accurately.

4.1 What Is a Teaching Assignment?

A teaching assignment connects a teacher to the educational work they are responsible for.

It may identify:

The teacher.

The school or campus.

The academic year.

The class or student group.

The subject.

The relevant reporting period.

The assignment's start and end dates.

The status of the assignment.

For example, a teacher may teach Mathematics to Grade 8, Section A, during the 2026–2027 academic year.

Another teacher may teach Science to the same class.

The system must preserve these distinctions.

4.2 One Teacher Can Have Several Assignments

A teacher may teach multiple classes or subjects.

For example:

Mathematics in Grade 7, Section A.

Mathematics in Grade 8, Section A.

Science in Grade 7, Section B.

The dashboard should show the correct students and subjects for each assignment.

It must not assume that a teacher has only one class or one subject.

4.3 Several Teachers Can Share Responsibility

A class or subject may involve more than one teacher.

One teacher might lead the class, while another supports practical sessions or specialist instruction.

If the school permits shared assignments, the system must represent them explicitly.

It should not arbitrarily assign all student outcomes to whichever teacher's name happens to appear first in the database.

4.4 Teaching Assignments Change Over Time

Teachers may join a school, leave a school, change subjects, or take over a class partway through an academic year.

The system must preserve historical assignments.

If a teacher takes over a class in January, the platform should not automatically attribute every result from the entire academic year to that teacher.

Where the data supports it, reporting should distinguish results recorded before and after the relevant assignment dates.

If the available information does not support fair attribution, the dashboard should present the student or class results without claiming that they belong exclusively to one teacher.

5. Teacher Dashboard: The Recommended Layout

The teacher dashboard should provide a clear overview of the teacher's current responsibilities.

A practical structure is described below.

5.1 Section One: Teaching Overview

This section identifies the teacher's current assignments.

It may show the number of assigned classes, subjects, and students.

It should also show the academic year and reporting period.

These counts must be based on active, relevant assignments.

A teacher should not continue seeing a former class as a current responsibility simply because the historical assignment remains in the database.

5.2 Section Two: Academic Results

This section shows academic outcomes for the teacher's assigned subjects and classes.

For each relevant class or subject, the teacher may see:

The number of enrolled students.

The number of students with valid recorded results.

The applicable average.

The distribution of results, where meaningful.

The previous comparable reporting period.

The change from the previous period.

The status of assessment entry and finalisation.

The interface should make it clear whether a result is a class average, subject average, assessment result, or overall academic measure.

5.3 Section Three: Student Progress

The teacher should be able to identify changes in individual student performance.

For example, the teacher may see that a student's Mathematics result declined from 75% to 63%.

The teacher should be able to open the student's authorised dashboard to inspect the underlying results and relevant context.

The system should not assume that every decline is caused by poor teaching, lack of effort, or a particular personal circumstance.

The purpose is to help the teacher understand what has changed.

5.4 Section Four: Attendance

Where authorised, the teacher may see attendance information for assigned classes or students.

This may include attendance rates, missing registers, recorded absences, late arrivals, and trends.

The dashboard should use the school's configured attendance rules.

It should not mix whole-day attendance with lesson-level attendance without a clear definition.

5.5 Section Five: Homework and Skills

The teacher may need to review assignment completion, pending verification, participation ratings, and subject-linked skill ratings.

These measures should remain separate.

The teacher should be able to identify which assignments await verification and which skill ratings remain unrecorded.

A missing rating should not be treated as evidence that a student lacks the skill.

5.6 Section Six: Reporting Tasks

The teacher dashboard may show required actions, such as:

Attendance registers awaiting submission.

Assessment results awaiting entry.

Assignments awaiting verification.

Skill ratings awaiting completion.

Records requiring correction.

These indicators should be based on actual assigned responsibilities and school deadlines.

The system should not show a task as overdue if no applicable deadline has been configured.

6. Class and Subject Performance: What the Numbers Mean

Class and subject summaries are important because they help educators understand group-level patterns.

However, their interpretation depends on the calculation method.

6.1 Class Average

A class average summarises the results of students in a defined class under a specified calculation rule.

For example, suppose five students have the following Mathematics percentages:

60%, 70%, 75%, 80%, and 85%.

The simple average is:

(60 + 70 + 75 + 80 + 85) ÷ 5 = 74%.

The dashboard may display a Mathematics class average of 74%.

It should also identify the relevant class, subject, reporting period, and eligible student population.

6.2 The Denominator Matters

Suppose a class has 30 enrolled students, but only 20 have valid Mathematics results.

An average calculated from those 20 students does not describe the results of all 30 students.

The dashboard should display both the average and the coverage.

For example:

Enrolled students: 30.

Students with valid results: 20.

Result coverage: 66.7%.

Average among students with valid results: 74%.

This prevents users from assuming that every enrolled student is represented in the calculation.

6.3 Missing Results Must Not Become Zero

If a student has not yet taken an assessment or the result has not been entered, the platform must not automatically treat the result as zero.

The system should distinguish among missing, pending, absent, exempt, and recorded zero results according to the configured assessment rules.

A recorded zero may be a valid result.

A missing result is a different state.

6.4 Class Average Versus Average Subject Performance

Optimus Orien may display several types of averages.

A subject average summarises results for one subject.

An assessment average summarises results for a particular assessment.

An overall academic average combines subject results using a defined rule.

A school-wide average summarises an appropriate set of student or class results.

These averages are not automatically interchangeable.

The dashboard should label the calculation it uses.

If the platform calculates a class average from average subject performance, it must define which subjects are included, how missing subjects are handled, and whether subjects have equal or different weights.

It should not silently change the method from one report to another.

7. Understanding Result Distributions

An average tells us about the centre of a group of results, but it does not describe the entire group.

Consider two classes.

Class A has results of:

70%, 71%, 72%, 73%, and 74%.

Class B has results of:

40%, 55%, 72%, 89%, and 104%.

If the last value is an invalid percentage for the configured assessment, the system must first identify the data-quality problem rather than use it unquestioningly. Even when all results are valid, different distributions can produce similar averages.

The important principle is that a group average can conceal meaningful differences between students.

Where appropriate and supported by the data, Optimus Orien should allow teachers and school leaders to examine distributions.

Possible visualisations include:

A histogram showing the distribution of percentages.

A bar chart showing the number of students within configured result bands.

A table showing the count of students above, within, or below defined thresholds.

A trend chart showing class averages across comparable reporting periods.

The result bands must be defined and labelled.

For example, the school may configure achievement bands for its own reporting purposes.

The platform must not treat those bands as universal educational standards.

7.1 Why Distributions Matter

Suppose two classes both have an average of 70%.

One class may have most students clustered around 70%.

The other may contain several students with very high results and several with very low results.

The average alone hides that difference.

A distribution chart can help the educator identify whether the group contains a wide spread of results.

It still does not explain why the distribution exists.

8. Comparing Classes Fairly

School leaders may want to compare classes studying the same subject.

Such comparisons can be useful, but only when the compared results are sufficiently compatible.

8.1 Compare Like With Like

Before comparing two classes, the platform should consider:

Whether they study the same subject.

Whether the assessment or grading rules are compatible.

Whether the reporting periods align.

Whether the student populations are defined appropriately.

Whether the data is sufficiently complete.

Whether the same calculation method was used.

A comparison between two Mathematics classes using the same assessment framework may be more meaningful than a comparison between Mathematics and Art.

Even when the subject is the same, different assessments or teaching contexts may limit direct comparison.

8.2 Example of a Fairer Comparison

Suppose two Grade 8 Mathematics classes complete the same assessment under the same marking rules.

Class A has an average of 76%.

Class B has an average of 69%.

The platform can report that Class A's recorded average is seven percentage points higher.

It should identify the assessment, reporting period, number of students with valid results, and relevant calculation method.

It should not automatically conclude that the teacher of Class A is seven percentage points more effective.

The result establishes a difference in recorded assessment performance, not its cause.

8.3 Avoid Unfair Teacher Rankings

A list of teachers sorted by class average may look simple, but it can create misleading conclusions.

Teachers may work with students who have different prior knowledge, attendance patterns, language needs, or educational resources.

Their classes may also differ in size, subject allocation, and assessment context.

Therefore, Optimus Orien should not introduce a public or automatic teacher leaderboard based solely on student marks.

If a school later implements a formal teacher evaluation system, it should be a separately defined feature with appropriate evidence, validation, transparency, and governance.

9. Understanding Student Progress Within a Teacher's Classes

A useful teacher dashboard should help educators see changes over time.

However, progress comparisons require context.

9.1 Current Result Versus Previous Result

Suppose a student's Science result changes from 62% to 71%.

The recorded change is nine percentage points.

The dashboard may show the current result, previous result, and difference.

It should also identify the relevant reporting periods and whether the comparison is valid.

9.2 Progress Does Not Automatically Prove Teaching Effectiveness

A student's result may improve after a teacher begins teaching the class.

That is a useful observation, but it does not automatically prove that the teacher caused the improvement.

Other factors may have changed, including assessment difficulty, attendance, additional tutoring, curriculum coverage, or the student's prior learning.

Likewise, a student may experience a decline even when the teacher provides effective support.

The platform should report observed changes without claiming causation.

9.3 Baseline Information

A baseline is an initial measurement used as a reference point for later measurements.

For example, a school's Mathematics baseline might be the result recorded at the beginning of a programme.

A later result can then be compared with that baseline when the measures are sufficiently compatible.

The dashboard should clearly identify which result is being used as the baseline.

It should not silently select an arbitrary historical value.

9.4 Changing Class Membership

Suppose a class has 25 students at the beginning of the year and 30 students later.

If the class average changes, the difference may partly reflect the change in membership.

The dashboard should use the appropriate enrolment and reporting context.

Where relevant, it may show the number of students represented in each period and distinguish group-level change from progress among a consistent group of students.

The system should not assume that the students in the two periods are identical.

10. Teacher Activity and Reporting Completeness

Optimus Orien should also help schools monitor whether required records are being completed.

This is a separate function from measuring teaching quality.

10.1 Examples of Activity Indicators

The platform may track:

The proportion of required assessment results entered.

The proportion of required attendance registers submitted.

The number of assignments awaiting verification.

The proportion of required skill ratings recorded.

The number of outstanding correction requests.

The number of required reporting tasks completed by their deadlines.

These indicators should only be displayed where the underlying workflow and expectations have been defined.

10.2 Use the Correct Denominator

Suppose a teacher has submitted 18 of 20 required attendance registers.

The completion rate is:

18 ÷ 20 × 100 = 90%.

The dashboard should make clear that this is the register-submission completion rate.

It is not a measure of classroom teaching quality.

10.3 Distinguish Activity From Outcomes

Suppose Teacher A has entered every required result, while Teacher B has several outstanding entries.

The dashboard may report that Teacher A's reporting tasks are more complete.

It should not automatically conclude that Teacher A is a better teacher.

Likewise, a teacher with a high class average may still have incomplete attendance records.

The platform should show both dimensions separately.

10.4 Fair Interpretation of Delays

Before labelling a task overdue, the system should verify that:

The task was assigned to the teacher.

A valid deadline exists.

The deadline has passed.

The task has not been completed or excused under the applicable workflow.

The system should also preserve authorised adjustments to deadlines and responsibilities.

11. Principal Dashboard: Understanding the Whole School

The principal dashboard should provide an overview of school performance while allowing authorised users to investigate details.

It should not simply reproduce the teacher dashboard with larger numbers.

The principal needs to understand patterns across classes, subjects, and reporting periods.

11.1 School Overview

The overview may show:

The number of currently enrolled students.

The number of active teachers.

The number of active classes.

Overall assessment-data completeness.

Attendance indicators.

Academic performance indicators.

Homework completion indicators.

Outstanding reporting tasks.

Configured school health indicators.

These figures must use clearly defined counting rules.

For example, the number of students should reflect the selected date and relevant enrolment rules rather than including every student who has ever attended the school.

11.2 Academic Overview

The principal should be able to inspect academic results across subjects and classes.

The dashboard may show:

Subject averages.

Class averages.

Assessment completion.

Historical changes.

Distribution of results.

Subjects or classes with incomplete records.

Student groups requiring further review under the school's configured rules.

The principal should be able to open a relevant class or subject to understand how the summary was produced.

11.3 Attendance Overview

The principal may need to understand attendance across the school.

The dashboard may show:

School attendance rate.

Attendance by class or year group.

Absence and lateness patterns.

Attendance trends across reporting periods.

Missing attendance registers.

Areas where data completeness limits interpretation.

The calculation must use the school's configured attendance rules.

A school-wide rate should not be calculated by blindly averaging class percentages if a different aggregation method is required.

11.4 Homework and Skill Coverage

The principal may want to understand whether the school is consistently recording assignment completion and skill ratings.

For example, a school may have high assignment completion data in some subjects but very little recorded information in others.

The dashboard can reveal that difference.

It should distinguish between a student receiving a low skill rating and a student not yet being assessed.

It should also avoid treating the absence of a rating as evidence of poor student performance.

11.5 Reporting Completeness

The principal should be able to identify missing or delayed records.

For example, the dashboard might show that three classes have not submitted the required attendance registers.

This allows the principal to address the reporting process.

It does not require the platform to make unsupported judgements about the teachers involved.

12. Organisation Administrator Dashboard: Understanding Multiple Schools

An organisation administrator may need to oversee several branches or schools.

The organisation dashboard should support a broader view while preserving the differences between educational contexts.

12.1 Organisation-Wide Overview

The dashboard may display:

The number of active schools or branches.

Student and teacher counts.

School-level attendance indicators.

Academic reporting completeness.

Subject or class performance summaries where comparisons are valid.

School health indicators.

Changes over time.

Outstanding administrative tasks.

The administrator should be able to select a school and open its authorised dashboard.

12.2 School Comparison

Suppose an organisation operates three branches.

Branch A reports an attendance rate of 95%.

Branch B reports 91%.

Branch C reports 87%.

These figures may help the organisation identify a pattern requiring further investigation.

Before drawing conclusions, the platform should verify that the schools use compatible attendance definitions and reporting periods.

If one branch records lesson-level attendance while another records only whole-day attendance, the figures may not be directly comparable.

12.3 School Health Indicators

A school health indicator is a defined measure or status intended to summarise an aspect of school performance or operations.

It may include academic, attendance, reporting-completeness, or other configured indicators.

The platform should identify what each indicator measures and how it is calculated.

If Optimus Orien later implements a composite School Health Score, its formula, weights, minimum data requirements, and limitations must be explicitly defined.

The current dashboard should not invent a single school-quality score by combining unrelated metrics without justification.

12.4 Organisation-Wide Trends

The organisation administrator should be able to compare changes over time.

For example, the platform may show whether attendance has improved across several branches over three reporting periods.

The system should distinguish an actual change in recorded indicators from a change caused by different reporting definitions, data coverage, or calculation rules.

12.5 Data Isolation

An organisation administrator should only see the schools and branches included in their authorised scope.

The system must not expose unrelated organisations' student, teacher, or school records.

13. Teacher Attendance Is Not Student Attendance

Optimus Orien may record attendance for both students and staff.

However, the two systems should remain conceptually and technically distinct.

Student attendance measures whether a student attended relevant scheduled learning opportunities.

Staff attendance measures whether an employee attended according to the applicable work schedule and attendance policy.

The systems may share general technical components, but they have different rules and permissions.

For example, a teacher's staff attendance record should not automatically be used to calculate the attendance rate of the students they teach.

Likewise, a student absence should not be recorded as a teacher attendance event.

The dashboard should keep these data categories separate.

If staff attendance is displayed in the teacher or principal portal, it should follow the applicable employment policies and access permissions.

14. Comparing Teachers: What Optimus Orien Should and Should Not Do

Because Optimus Orien is a school performance intelligence platform, users may eventually request teacher comparisons.

The product should distinguish operational reporting from formal evaluation.

14.1 Comparisons That Can Be Useful

The platform can support comparisons of clearly defined administrative measures.

For example:

The percentage of required attendance registers submitted.

The number of assessments awaiting entry.

The proportion of assigned skill ratings completed.

The time taken to complete an assigned reporting task, where the data and policy support that measurement.

Even these measures require context and should not automatically be interpreted as teaching quality.

14.2 Comparisons That Need Greater Caution

Comparing teachers by student examination averages requires much greater caution.

The comparison may be affected by student selection, prior attainment, assessment difficulty, subject allocation, class size, attendance, and other factors.

The platform should not automatically rank teachers using raw student averages.

14.3 Formal Teacher Evaluation Is a Separate Feature

A formal teacher evaluation system would need a defined purpose, an appropriate evaluation framework, valid evidence, transparent procedures, and governance.

It might use structured classroom observations, professional standards, assessment evidence, and other approved sources.

Those components require separate educational and organisational decisions.

They are not automatically part of the initial Optimus Orien dashboard.

14.4 Recommended Product Decision

For the current implementation:

Show teacher responsibilities and reporting completeness.

Show student outcomes for the teacher's authorised classes and subjects.

Show changes over time when comparisons are appropriate.

Provide context and access to underlying records.

Do not introduce a public teacher leaderboard or an unexplained teacher-performance score.

15. Dashboard Visualisation: Selecting the Right Chart

Different questions require different visualisations.

The developer should not use the same chart for every metric.

15.1 Bar Charts

Bar charts are useful for comparing categories.

For example, a principal may compare the average Mathematics results of several classes, provided the assessment context is compatible.

Each bar should have a clear label, and the chart should identify the reporting period.

15.2 Line Charts

Line charts are useful for showing change over time.

For example, a school may examine its attendance rate across terms.

The chart should identify the time periods and should not imply continuous measurements between discrete reporting periods.

15.3 Tables

Tables are useful when users need exact values.

For example, a teacher may need to inspect each assessment result, the number of students with recorded marks, and the status of result entry.

Tables should have clear headings and should support practical filtering or sorting where needed.

15.4 Distribution Charts

Histograms or configured achievement-band charts can help users understand the spread of student results.

The chart should make the bands and the number of students in each group clear.

15.5 Indicator Cards

Indicator cards can present concise values, such as the current attendance rate or the number of incomplete registers.

Each card should include a meaningful label and, where necessary, the relevant period and comparison basis.

A card should not display a percentage without making its meaning understandable.

15.6 Avoid Visual Overload

The principal should not be forced to examine dozens of charts before understanding the school's main indicators.

The teacher should not have to scroll through organisation-wide information to find their assigned classes.

The organisation administrator should be able to begin with a high-level summary and drill down into authorised detail.

The interface should support the user's task rather than simply display all available data.

16. Data Quality and Calculation Consistency

Dashboards are only as reliable as the data and calculation rules behind them.

Optimus Orien should implement validation and consistency checks before relying on dashboard indicators.

16.1 Invalid Values

The system should validate values against the relevant rules.

For example, if an assessment has a maximum mark of 50, a recorded mark of 75 should trigger validation unless the configured assessment model explicitly permits that value.

The platform should not silently accept invalid values and display them as reliable performance data.

16.2 Duplicate Records

The system should identify duplicate or conflicting records where possible.

For example, two final results for the same student, assessment, and result context may require review.

The exact duplicate-detection rules depend on the data model.

16.3 Historical Corrections

If an authorised user corrects a mark, attendance record, or assignment status, affected dashboard calculations should be refreshed according to the configured workflow.

The platform should preserve an audit trail.

16.4 Versioned Calculation Rules

If a school changes its grading or attendance rules, the system should preserve enough information to interpret historical results correctly.

The developer should not recalculate every historical record using a new rule without an explicit migration or recalculation decision.

Historical reporting must remain understandable.

17. Recommended Technical Design

The developer should build teacher and leadership dashboards on top of the platform's existing modules.

The dashboards should not become separate databases containing manually maintained copies of student results.

17.1 Shared Source Modules

The relevant source modules include:

Student and enrolment records.

Academic structure and teaching assignments.

Assessment and grading.

Attendance.

Homework and assignments.

Participation and skills.

Comparison and review indicators.

School and organisation configuration.

User roles and permissions.

17.2 Shared Calculation Services

The same calculation rules should be used across dashboards.

For example, a subject average shown to a teacher should match the corresponding value shown to a principal when both views use the same student population, reporting period, and calculation method.

If the two dashboards use different scopes, the system should make those scopes clear.

17.3 Aggregation by Context

Dashboard queries should specify the relevant organisation, school, academic year, reporting period, subject, class, and teaching assignment as applicable.

The system must not retrieve every student record and rely on the interface to hide the ones the user should not see.

Authorisation and filtering should be enforced on the server.

17.4 Avoid Unnecessary Recalculation

The platform should use efficient queries and, where appropriate, cached or precomputed summaries.

However, cached values must be refreshed when relevant source data or calculation rules change.

The cache must respect the user's permissions and the reporting context.

17.5 Drill-Down Navigation

A drill-down allows users to move from a summary to the information behind it.

For example:

Organisation overview → School overview → Class overview → Subject results → Student dashboard.

Every step must preserve the authorised scope.

The user should not gain additional permissions simply by opening a more detailed page.

18. Software Implementation Requirements and Acceptance Tests

The following requirements define the expected behaviour of the teacher and school leadership dashboards.

18.1 Teaching Assignments

TDB-001 — Assignment-Based Access

The teacher dashboard must retrieve classes, subjects, and students according to active teaching assignments and authorised permissions.

Acceptance test: A teacher is assigned to Mathematics in Grade 8 but not Science in Grade 8. The teacher can access the assigned Mathematics records but cannot access restricted Science records.

TDB-002 — Multiple Assignments

A teacher must be able to have multiple classes and subjects.

Acceptance test: A teacher has three active assignments. All three appear in the dashboard with their correct class and subject contexts.

TDB-003 — Historical Assignments

The system must preserve previous teaching assignments.

Acceptance test: A teacher leaves a class midway through the year. Historical records remain associated with the correct assignment dates.

TDB-004 — Shared Teaching Responsibilities

Where the school supports shared teaching assignments, the system must represent each relevant teacher's assignment explicitly.

Acceptance test: Two teachers share a class. The platform does not arbitrarily attribute all records to only one teacher.

TDB-005 — Assignment Dates

Where teaching responsibilities change over time, reporting must use the relevant assignment dates.

Acceptance test: A teacher takes over a class halfway through the reporting period. The system does not automatically attribute all earlier results exclusively to that teacher.

18.2 Teacher Dashboard

TDB-006 — Current Teaching Overview

The teacher dashboard must show current authorised assignments and relevant student counts.

Acceptance test: A former class is no longer assigned to the teacher. It is not presented as a current assignment, although authorised historical access may remain available.

TDB-007 — Subject Results

The dashboard must show subject results using the configured assessment and grading rules.

Acceptance test: A teacher opens a Mathematics class. The displayed subject average matches the authoritative calculation for the selected context.

TDB-008 — Student-Level Drill-Down

The teacher must be able to open student details only within their authorised scope.

Acceptance test: A teacher opens an assigned class and selects a student. The correct student dashboard appears. A direct request for an unrelated student's dashboard is rejected.

TDB-009 — Reporting Tasks

The dashboard must show outstanding tasks that are actually assigned to the teacher.

Acceptance test: A teacher has two required attendance registers awaiting submission. The dashboard displays those tasks without inventing additional deadlines.

TDB-010 — Task Deadlines

A task must not be labelled overdue unless an applicable deadline exists and has passed.

Acceptance test: A skill-rating task has no configured deadline. The dashboard does not label it overdue solely because it remains incomplete.

TDB-011 — Activity Versus Quality

Administrative activity indicators must be labelled according to what they measure.

Acceptance test: A teacher has submitted every required attendance register. The dashboard identifies the completion rate without automatically assigning a teaching-quality score.

18.3 Class and Subject Performance

TDB-012 — Defined Class Average

The platform must calculate class averages using a documented method.

Acceptance test: Five valid Mathematics results are 60%, 70%, 75%, 80%, and 85%. Under the configured simple-average rule, the displayed average is 74%.

TDB-013 — Valid-Result Coverage

Class and subject summaries must identify the number of students represented in the calculation.

Acceptance test: A class has 30 enrolled students, but only 20 have valid results. The dashboard displays the average alongside the 20 valid results and the relevant coverage information.

TDB-014 — Missing Results

Missing results must not automatically become zero.

Acceptance test: Five students have not yet received recorded assessment results. The platform does not automatically add five zeros to the class average.

TDB-015 — Result Distributions

Where distribution charts are supported, the chart must represent the valid results in the selected context.

Acceptance test: A teacher selects a subject and reporting period. The distribution chart includes only the eligible results defined by the calculation rules.

TDB-016 — Compatible Class Comparisons

Class comparisons must identify the relevant subject, period, and calculation basis.

Acceptance test: Two classes use incompatible assessment rules. The platform does not silently present their averages as directly equivalent.

TDB-017 — Class Membership Changes

The system must preserve the enrolment context used in historical summaries.

Acceptance test: Five students join a class during the year. The dashboard identifies the relevant student population for each reporting period.

18.4 Progress and Teacher Comparisons

TDB-018 — Historical Result Comparison

The dashboard must identify the periods used in progress comparisons.

Acceptance test: A student's Mathematics result changes from 62% to 71%. The dashboard shows a nine-percentage-point increase and identifies both periods.

TDB-019 — No Unsupported Causal Claims

The platform must not automatically attribute student progress to teacher effectiveness.

Acceptance test: A class average improves. The dashboard reports the recorded change but does not state that the teacher caused the improvement.

TDB-020 — No Raw-Average Teacher Leaderboard

The initial product must not automatically rank teachers by student averages as if those averages measured teacher quality.

Acceptance test: Two teachers have different class averages. The system may display the relevant class results but does not generate a teacher-quality ranking solely from those averages.

TDB-021 — Separate Teacher Activity Indicators

Administrative completion rates must remain separate from student academic outcomes.

Acceptance test: A teacher's register-submission rate is 90%. The system displays that activity indicator separately from academic performance.

18.5 Principal and Academic Leadership

TDB-022 — School Overview

The principal dashboard must summarise relevant records within the authorised school.

Acceptance test: A principal opens the dashboard. Student, teacher, and class counts reflect the selected school and reporting context.

TDB-023 — School-Level Drill-Down

The principal must be able to inspect authorised class and subject summaries.

Acceptance test: The principal opens the school academic overview and selects a Mathematics class. The relevant class results appear without exposing unrelated schools' records.

TDB-024 — School Attendance Summary

School-level attendance must use the configured attendance rules.

Acceptance test: A school has different attendance statuses configured. The school-wide summary applies the defined policy and does not assume every status has identical treatment.

TDB-025 — Reporting Completeness

The principal dashboard must identify relevant missing or incomplete reporting records.

Acceptance test: Three required attendance registers remain unsubmitted. The dashboard identifies the missing registers without automatically marking students absent.

TDB-026 — Role Configuration

Academic leadership permissions must be configurable according to the school's organisational structure.

Acceptance test: A school grants a head of academics access to academic summaries but not restricted employment records. The dashboard enforces that distinction.

18.6 Organisation Dashboard

TDB-027 — Organisation Scope

The organisation administrator dashboard must include only authorised schools and branches.

Acceptance test: An administrator manages three branches. The dashboard shows those branches and does not retrieve records from an unrelated organisation.

TDB-028 — School Comparison Context

School comparisons must identify the reporting period and relevant calculation definitions.

Acceptance test: Two branches use incompatible attendance definitions. The system warns that their rates are not directly comparable or excludes the comparison according to the configured policy.

TDB-029 — Historical School Trends

The dashboard must preserve the context of historical school indicators.

Acceptance test: A school changes its attendance rules. Historical values remain interpretable under the earlier rule version.

TDB-030 — Composite School Health Score

The platform must not generate an unexplained composite School Health Score.

Acceptance test: No approved composite-score formula is configured. The dashboard displays the separate available indicators instead.

18.7 Security, Data Quality, and Performance

TDB-031 — Server-Side Authorisation

All protected dashboard requests must be authorised by the server.

Acceptance test: A teacher changes a school identifier in a request to access another school's data. The server rejects the request.

TDB-032 — Organisation Isolation

The system must prevent cross-organisation data leakage.

Acceptance test: An organisation administrator requests records from an unrelated organisation. The request is rejected.

TDB-033 — Calculation Consistency

Shared indicators must use the same calculation logic across authorised dashboards.

Acceptance test: The teacher and principal dashboards show the same class average when their student populations, periods, and calculation methods are identical.

TDB-034 — Data Validation

Invalid records must be detected according to the relevant module's validation rules.

Acceptance test: An assessment has a maximum mark of 50 and a mark of 75 is entered. The platform rejects or flags the record according to the configured assessment workflow.

TDB-035 — Audit History

Relevant changes to assessment, attendance, assignment, and configuration records must be traceable.

Acceptance test: An authorised user corrects a mark. The audit history records the change according to the platform's audit requirements.

TDB-036 — Cache Permission Isolation

Cached dashboard responses must respect user permissions and reporting context.

Acceptance test: A teacher and a principal open dashboards with different scopes. Neither receives data that is restricted to the other role.

TDB-037 — Responsive and Accessible Layout

The dashboard must remain usable across supported screen sizes and should provide accessible labels and non-colour indicators.

Acceptance test: A user opens the dashboard on a supported mobile screen and can identify the relevant values and statuses without relying solely on colour.

TDB-038 — Unavailable Data

The dashboard must distinguish unavailable information from a valid zero.

Acceptance test: The attendance service is temporarily unavailable. The dashboard shows an appropriate unavailable state rather than an attendance rate of zero.

19. Recommended Development Order

The developer should implement the dashboards in a sequence that reduces duplication and calculation errors.

First, verify that teaching assignments accurately connect teachers, classes, subjects, students, and reporting periods.

Second, connect the teacher dashboard to the authoritative assessment, attendance, homework, participation, and skill records.

Third, implement shared calculation services for class averages, result coverage, and historical changes.

Fourth, add the teacher's outstanding reporting tasks.

Fifth, build the principal dashboard using the same calculation services, with school-level aggregation and appropriate drill-down.

Sixth, implement the organisation dashboard using authorised branch-level summaries.

Seventh, add visualisations only after the underlying data and calculations are verified.

Eighth, test incomplete data, class membership changes, teacher transfers, changing reporting periods, and different educational configurations.

Ninth, test all permissions using direct requests, not merely the visible interface.

Tenth, test that all dashboards present consistent values when they use the same underlying data and calculation context.

20. Final Product Principles

The teacher and school leadership dashboards should help educators and administrators understand educational performance without reducing teaching and learning to a single number.

The teacher dashboard should prioritise assigned responsibilities and relevant student information.

The principal dashboard should reveal patterns across classes and subjects.

The organisation dashboard should support comparisons across authorised schools while preserving the context of each educational setting.

All three dashboards should use consistent calculations, explain their indicators, and preserve access controls.

Most importantly, Optimus Orien should distinguish three different things:

Teacher activity: what the system records about the completion of teaching and administrative responsibilities.

Student outcomes: what the recorded student results and indicators show.

Teaching quality: a broader professional judgement that cannot be established from student averages alone.

This distinction is essential to making Optimus Orien a credible school performance intelligence platform.

A useful dashboard does not simply show which number is highest. It helps the user understand what the number measures, how it was calculated, which records support it, and what conclusions can reasonably be drawn from it.

What Part 13 Establishes

Part 13 defines the teacher, principal, academic leadership, and organisation administrator dashboards.

It explains teaching assignments, class averages, result distributions, historical progress, reporting completeness, and fair comparisons. It also establishes why

student outcomes must not automatically be treated as measurements of teacher quality.

The document provides 38 software implementation requirements with acceptance tests.

PART 14

SCHOOL HEALTH AND EARLY INTERVENTION

How Optimus Orien Identifies Concerning Patterns, Supports Human Review, Tracks Follow-Up Actions, and Measures Outcomes Without AI-Based Predictions

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner, software developer, and future product team
Primary objective: Define a transparent, fair, and auditable system for identifying educational patterns that may require attention, supporting human review, recording follow-up actions, and evaluating changes over time.

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1. Why This Document Exists

In Part 12, we designed the student performance dashboard.

In Part 13, we designed the teacher and school leadership dashboards.

Those dashboards help users understand recorded educational information. They show academic results, attendance, homework completion, skills, class averages, historical trends, and reporting completeness.

But understanding information is only one part of improving school performance.

Sometimes the information reveals a pattern that deserves attention.

A student's attendance may have declined over several reporting periods. A class may have a large number of missing assessment results. A school may experience a sustained decline in attendance. A subject may have unusually incomplete skill assessments. A student may show declining academic results alongside several overdue assignments.

Optimus Orien should help authorised users identify these patterns and organise an appropriate response.

This is where two important concepts become relevant:

School health monitoring means monitoring defined indicators that help a school understand its current condition and identify areas requiring investigation.

Early intervention means responding to an emerging concern before it continues or becomes more difficult to address.

In this context, intervention does not necessarily mean a formal psychological or clinical intervention. It may be a simple educational action, such as a teacher reviewing missing assessment results, a principal arranging academic support, or a school contacting a parent about an attendance concern.

However, there is an important limitation.

A pattern in the data does not automatically explain why the pattern exists.

A student with declining marks may need additional support, but the marks alone do not tell us why they declined.

A school with low recorded attendance may have an attendance problem, incomplete registers, or a calculation issue.

A class with many low results may need additional learning support, but the assessment itself may also need to be reviewed.

Optimus Orien must therefore distinguish identifying a pattern from confirming a problem and confirming a problem from deciding what action to take.

This document defines that process.

2. What Is Early Intervention in an Educational Information System?

Consider a simple example.

A student has an attendance rate of 96% during the first reporting period.

In the second reporting period, the rate falls to 89%.

In the third reporting period, it falls to 78%.

A system that only displays the current attendance rate may show 78%.

A system that supports early review can also compare the value with previous periods, identify the decline, and indicate that the school's configured review rule has been triggered.

An authorised staff member can then inspect the underlying attendance records.

The staff member may discover that the records are correct and that the student has been absent frequently.

Alternatively, the staff member may discover that some attendance registers were never submitted or that the student transferred schools during the reporting period.

The same numerical pattern can therefore have different explanations.

The platform's responsibility is to make the pattern visible, explain why the indicator appeared, and support a suitable review.

It should not automatically decide what the pattern means.

2.1 The Four Stages

Optimus Orien should distinguish four stages.

Stage 1: Detection. The system identifies a defined condition in the recorded data.

Stage 2: Review. An authorised person examines the information and determines whether the indicator reflects a genuine concern.

Stage 3: Action. Where appropriate, the school records a follow-up action.

Stage 4: Follow-up and evaluation. The school checks whether the action was completed and examines subsequent information to understand what changed.

These stages should be represented separately in the data model.

A detected condition is not automatically a confirmed problem.

A confirmed problem does not automatically require the same action in every case.

An action being completed does not automatically mean that the underlying problem has been resolved.

3. The Difference Between an Indicator, a Concern, an Intervention, and an Outcome

The developer must understand these four concepts because they represent different states in the workflow.

3.1 Indicator

An indicator is a value or status that makes a particular pattern visible.

Examples include:

Attendance rate: 78%.

Mathematics result declined by 12 percentage points.

Five assignments are overdue.

Forty per cent of required assessment results are missing.

An indicator can be calculated directly from recorded data.

3.2 Concern

A concern is a matter that an authorised person has reviewed and decided requires attention.

For example, a principal may review a low attendance rate, confirm that the register data is complete, and determine that follow-up is appropriate.

The concern is now a reviewed case, not merely an automatically generated indicator.

The school should record the decision and its basis according to the configured workflow.

3.3 Intervention or Follow-Up Action

An intervention is an action intended to address a concern.

Examples include:

A teacher reviews a student's recent Mathematics assessments.

A school checks whether a student has access to the required learning materials.

A principal arranges an academic-support session.

An administrator asks a teacher to complete missing attendance registers.

A school contacts a parent through an approved communication process.

These actions have different purposes and should not all be treated as equivalent.

The system should record what action was planned, who was responsible, and whether it was completed.

3.4 Outcome

An outcome is a later observation used to understand what happened after an action.

For example, the student's attendance rate may increase from 78% to 86% during the next comparable reporting period.

That is a recorded change.

It does not automatically prove that the intervention caused the improvement. Other circumstances may have changed.

The system should therefore distinguish an observed outcome from a demonstrated causal effect.

Implementation principle: Indicators identify patterns, reviews establish whether attention is warranted, actions document what the school does, and follow-up records describe what happens afterward.

4. What Is School Health?

In Optimus Orien, school health should be understood as a structured view of selected educational and operational indicators.

It should not be treated as a medical diagnosis or a perfect measurement of a school's overall quality.

A school may perform well on some indicators and poorly on others.

For example, a school might have strong academic results but incomplete attendance records.

Another school might have high attendance but substantial differences in academic performance between subjects.

A third school might have improving results but a high number of unverified assignment records.

These situations cannot be represented accurately by one unexplained number.

4.1 Recommended School Health Categories

Optimus Orien may organise school health information into separate categories.

Academic performance: Results, averages, distributions, and progress over time.

Attendance: Attendance rates, absence patterns, and register completeness.

Assignment completion: Completion rates, overdue work, and verification status.

Participation and skills: Coverage and recorded results of configured participation and skill-assessment frameworks.

Data quality: Missing, invalid, duplicated, or incomplete records.

Reporting and administration: Completion of required academic and operational reporting tasks.

These categories should remain distinguishable.

A school may have an excellent data-completeness rate while still needing to improve academic outcomes. Conversely, a school may have promising academic results but incomplete records that make broader conclusions unreliable.

4.2 School Health Status

If Optimus Orien uses labels such as Green, Yellow, and Red, those labels must have explicit meanings.

For example, a school might define:

Green: The relevant indicator is within the school's configured expected range.

Yellow: The indicator requires review or is approaching a configured threshold.

Red: The indicator has crossed a configured threshold requiring priority review.

These are illustrative definitions.

The actual thresholds must be configured and documented by the school or organisation.

A red indicator should not automatically mean that a school is poor, unsafe, or failing. It means that a defined condition has been met and requires the interpretation specified by the relevant policy.

The dashboard must show which indicator produced the status.

4.3 Avoid One Universal School Health Formula

The initial implementation should display separate indicators and transparent statuses.

A composite School Health Score should only be introduced after its purpose, formula, weights, data requirements, and validation have been explicitly defined.

For example, combining attendance, academic performance, homework completion, and reporting completeness into one score requires decisions about how each measure is scaled and weighted.

Those decisions can materially change the result.

The platform should not make them silently.

5. Rule-Based Detection: How the System Identifies Patterns Without AI

The current Optimus Orien scope excludes AI features.

That does not prevent the system from identifying defined conditions.

A rule-based system applies explicit conditions to recorded data.

For example:

If a student's attendance rate falls below a configured threshold, create an attendance-review indicator.

If a student's result declines by at least a configured number of percentage points between comparable periods, create an academic-review indicator.

If a required register remains unsubmitted after its deadline, create a reporting-completeness indicator.

These rules are understandable because the platform can show the exact condition that was met.

5.1 Example of a Rule

A school configures an attendance review threshold of 85%.

The current attendance rate for a student is 79%.

The system evaluates the rule:

79% is below 85%.

The rule is triggered.

The platform creates or displays an indicator explaining:

"Attendance review recommended. The recorded attendance rate is 79%, below the configured threshold of 85%."

This is not an AI prediction.

It is a direct application of a configured condition.

5.2 A Threshold Is a Policy Choice

The system should not present every configured threshold as a scientifically established boundary.

A school may select 85% because it aligns with an internal attendance policy. Another school may use a different threshold.

The dashboard should explain that the rule is configured for the relevant school or organisation.

The developer should not hardcode one threshold for every institution.

5.3 Rules Must Consider Data Completeness

A rule should not generate a confident interpretation when the necessary data is incomplete.

Suppose a student appears to have an attendance rate of 70%, but several registers are missing.

The platform may identify the incomplete attendance data.

It should not treat the apparent attendance rate as reliable without applying the relevant completeness requirements.

The rule should specify the minimum information needed for evaluation.

5.4 Rules Must Be Versioned

Suppose a school changes its attendance review threshold from 85% to 90%.

Historical indicators should remain interpretable.

The system should preserve which rule version produced each indicator and when that version was active.

Changing a rule should not silently rewrite the historical explanation for a past decision.

6. Categories of Review Indicators

The platform should support clearly defined categories of indicators.

The first release should prioritise indicators that are supported by reliable source data and can be explained in plain language.

6.1 Academic Performance Indicators

Examples include:

A decline in a student's subject result across comparable reporting periods.

A subject average below a configured school threshold.

A substantial number of students with results below a configured achievement band.

A large difference between class results where the comparison is valid.

An unusual number of missing assessment results.

These indicators should identify the exact measurement and comparison used.

The system must not automatically assume that a low result means the student has not learned anything or that the teacher has performed poorly.

6.2 Attendance Indicators

Examples include:

Attendance below a configured threshold.

A sustained decline in attendance.

A configured number of consecutive recorded absences.

Missing or unsubmitted attendance registers.

A high rate of incomplete attendance data.

The system must distinguish recorded absence from missing attendance information.

It should also respect the school's attendance policy and the student's actual enrolment period.

6.3 Homework Indicators

Examples include:

On-time completion below a configured threshold.

Several assignments overdue.

A large number of assignments awaiting verification.

A class with incomplete assignment records.

These indicators should use the assignment eligibility and completion rules defined in the relevant module.

The platform must not count assignments that were never assigned to the student.

6.4 Participation and Skill Indicators

Examples include:

Required skill ratings have not been recorded.

A skill rating has changed across comparable observation periods.

A subject has incomplete participation records.

The platform should be particularly careful when interpreting subjective ratings.

A lower rating may reflect an actual change, a different evaluator, a revised rubric, or an incomplete observation period.

A missing rating must not automatically become a low rating.

6.5 Data-Quality Indicators

Examples include:

Missing assessment results.

Conflicting attendance entries.

Invalid marks.

Unsubmitted registers.

Duplicate records.

Unexpected gaps in reporting.

These indicators may be particularly important because incorrect source data can produce misleading academic or attendance summaries.

Sometimes the correct intervention is to fix the data rather than intervene with a student.

6.6 School-Level Indicators

Examples include:

A sustained decline in attendance across a school.

Incomplete assessment reporting in several classes.

A subject with a high proportion of missing results.

A substantial change in the distribution of recorded academic outcomes.

These indicators should be calculated at the relevant school or organisation level and should respect the selected reporting period.

The platform should not assume that a school-level pattern applies equally to every student or teacher.

7. How an Indicator Becomes a Review Case

The platform should not automatically create a full intervention case for every threshold crossing without considering the configured workflow.

For example, a small, temporary change may require monitoring rather than a formal intervention.

A missing attendance register may require an administrative correction rather than a student-support case.

The workflow should allow authorised staff to review an indicator and decide what should happen next.

7.1 Recommended Workflow

A suitable workflow is:

Step 1: Indicator generated. The system identifies a configured condition.

Step 2: Indicator displayed. An authorised user sees the indicator and its supporting evidence.

Step 3: Review started. An authorised person begins examining the situation.

Step 4: Review decision recorded. The person determines whether the indicator reflects a genuine concern, incomplete data, or a condition that does not require action.

Step 5: Action planned, if needed. The school records an appropriate follow-up action.

Step 6: Action completed or updated. The responsible person records the outcome of the action.

Step 7: Follow-up conducted. The school reviews subsequent information where appropriate.

Step 8: Case closed or kept open. An authorised person records the decision according to the school's policy.

Not every indicator needs to progress through every stage.

For example, a missing mark may be corrected and the indicator closed without a student-support intervention.

7.2 Do Not Confuse the Indicator With the Case

An indicator describes a condition.

A review case records the process of examining that condition.

One indicator may lead to no action.

One indicator may lead to a follow-up action.

Several related indicators may contribute evidence to one review case.

The database should represent these relationships explicitly.

7.3 Repeated Indicators

Suppose a student's attendance remains below the configured threshold for three consecutive reporting periods.

The system should not necessarily create three unrelated intervention cases.

Depending on the school's workflow, it may update an existing review case or create a new case linked to the earlier history.

The system should preserve the occurrence dates and evidence for each indicator.

This allows the school to distinguish a repeated pattern from a single event.

8. Prioritisation: Which Indicators Need Attention First?

Schools may have many indicators and limited staff time.

The platform should help users organise their review workload.

However, prioritisation should be transparent and should not pretend to measure a student's personal worth or future potential.

8.1 Recommended Priority Levels

A school may configure priority levels such as:

Routine: The indicator should be reviewed through normal procedures.

Priority: The indicator should be reviewed sooner under the school's policy.

Urgent: The indicator requires prompt attention according to a defined operational rule.

These labels must have documented meanings.

The developer should not assume that every indicator type requires the same urgency.

For example, a missing assessment result may require administrative follow-up, while a different category of concern may require a different response.

8.2 Priority Rules

A priority level may depend on:

The type of indicator.

The severity of the recorded condition under the school's policy.

The duration of the condition.

Whether the indicator has appeared repeatedly.

The completeness and reliability of the underlying data.

The existence of an open review case.

The relevant school policy.

The rules should be configured and visible to authorised users.

8.3 Avoid Unexplained Risk Scores

The platform should not automatically combine several indicators into a single student-risk score.

For example, a formula that adds attendance, homework, and academic values together would require a justified scale, weighting method, and validation process.

Without these, the resulting number could be misleading.

For the current release, separate indicators with transparent explanations are preferable.

8.4 Data Quality Comes Before Priority

An indicator based on unreliable data may need data correction before any student-support decision is made.

For example, if attendance registers are missing, the system should identify the reporting problem.

It should not confidently assign a high-priority student attendance concern using a rate that may be inaccurate.

The platform should communicate the uncertainty or incompleteness of the evidence.

9. Designing a Review Case

A review case is a structured record of an issue being examined by an authorised person.

It should contain enough information to support an accountable workflow without collecting unnecessary sensitive information.

9.1 Recommended Case Information

A review case may include:

A unique case identifier.

The relevant student, class, school, or organisation, depending on the case type.

The indicator or indicators that initiated the review.

The date the case was created.

The reporting period.

The underlying evidence or references to source records.

The responsible reviewer.

The current status.

The review decision.

The action plan, if one is required.

The follow-up date, if one is required.

The closure reason.

The relevant audit history.

Not every case needs every field.

For example, a school-level reporting issue may not be associated with an individual student.

9.2 Keep Evidence and Notes Separate

The platform should distinguish the source evidence from the reviewer's notes.

Source evidence might include a calculated attendance rate and the relevant attendance records.

A review note might explain that the reviewer has requested correction of an incomplete register.

These are different kinds of information.

The system should preserve the evidence and its source, while protecting any restricted notes according to the applicable permissions.

9.3 Avoid Unnecessary Sensitive Details

The platform should not encourage staff to record extensive medical, family, or psychological information when a simple operational note is sufficient.

If sensitive information is genuinely required for a legitimate educational process, the school must define the relevant access and retention rules.

Optimus Orien should collect only what is necessary for the supported workflow.

9.4 Assign Responsibility

A case should identify who is responsible for the next action.

Without a responsible person, the case may remain open indefinitely.

The responsible person should have the necessary permissions and should be able to update the case according to the workflow.

If responsibility changes, the system should preserve the assignment history.

10. Planning and Recording Interventions

An intervention record describes a planned or completed action intended to address a reviewed concern.

The system should support simple, practical actions rather than forcing schools into a complex intervention model.

10.1 Examples of Educational Actions

A teacher may review a student's recent assessment errors.

A student may be offered additional practice or academic support.

A school may review whether an assignment deadline was appropriate.

A principal may arrange a discussion about attendance procedures.

An administrator may request that missing attendance registers be completed.

A teacher may conduct additional observations before assigning a skill rating.

These examples illustrate different types of actions. They should not all be treated as equivalent interventions.

10.2 An Action Must Have a Clear Purpose

The action record should identify what the school intends to do and which concern it addresses.

For example:

Concern: Mathematics results have declined across two comparable reporting periods.

Action: The teacher reviews recent assessment records and identifies the topics requiring additional support.

Responsible person: The assigned Mathematics teacher.

Review date: The date configured by the school.

Expected evidence: The review is recorded and any appropriate follow-up is documented.

The platform should not prescribe the same action for every student with a declining result.

10.3 Planned, Started, and Completed Are Different States

The system should distinguish between an action that has been planned and one that has actually been completed.

For example, arranging an academic-support session does not mean the student attended it.

The workflow should record completion based on the evidence required by the school's policy.

10.4 Do Not Automatically Close a Case When an Action Is Completed

Completing an action does not necessarily resolve the concern.

A teacher may complete a review of assessment records but still need to arrange additional support.

The case should remain open until an authorised person determines that the required review and follow-up have been completed.

11. Follow-Up: How Does the School Know What Happened Next?

The platform should help schools examine what happened after an intervention or follow-up action.

This is important because recording an action does not establish whether it was useful.

11.1 Define the Measure Before Comparing It

Suppose the original concern was a decline in attendance.

The relevant follow-up measure may be attendance during the next comparable reporting period.

Suppose the original concern was declining Mathematics performance.

The relevant follow-up measure may be a subsequent Mathematics assessment or reporting-period result, provided it is sufficiently comparable.

The follow-up measure should relate to the concern being reviewed.

A change in an unrelated indicator should not automatically be treated as evidence that the intervention worked.

11.2 Before-and-After Comparison

Suppose a student's attendance rate was 78% before a follow-up action.

During the next comparable period, the rate is 86%.

The recorded change is:

86% − 78% = 8 percentage points.

The platform may report an eight-percentage-point increase.

It may also record that the increase occurred after the action.

However, the platform must not automatically claim that the action caused the increase.

Other circumstances may have changed.

11.3 Outcomes May Differ

After a review, several outcomes are possible.

The indicator may no longer be present.

The indicator may remain present.

The data may reveal that the original indicator was incorrect.

The available evidence may be insufficient to determine whether the situation changed.

The case may require further action.

The platform should allow authorised staff to record these distinctions.

11.4 No Change Does Not Automatically Mean Failure

A student may continue to experience difficulties despite receiving appropriate support.

An intervention may still have been completed correctly.

The result may indicate that the school needs to reassess the situation, change the approach, or gather additional information.

The platform should record the observed outcome without automatically labelling the student or the responsible educator as a failure.

12. Measuring Intervention Outcomes Responsibly

Optimus Orien should distinguish between monitoring change and proving effectiveness.

These are not the same task.

12.1 Monitoring Change

Monitoring change means comparing recorded information over time.

For example:

Attendance before follow-up: 78%.

Attendance after follow-up: 86%.

The platform can report that the attendance rate increased by eight percentage points.

12.2 Evaluating an Intervention

Evaluating an intervention requires a more structured question.

For example:

Did the intervention achieve its intended outcome?

Was the outcome measured correctly?

Was the follow-up period appropriate?

Were there other changes that could explain the result?

Was the intervention actually delivered as planned?

The school may be able to answer some of these questions through its records.

Other questions may require a more formal evaluation design.

12.3 Correlation and Causation

Correlation means that two measured things vary together or are associated.

Causation means that one factor produces or contributes to a change in another factor.

Suppose attendance improves after a parent meeting.

The two events occurred in sequence, but that alone does not prove that the meeting caused the improvement.

The student may also have recovered from an illness, changed schedules, or experienced other circumstances.

The platform should report the measured change and the intervention history without making unsupported causal claims.

12.4 Intervention Completion Rate

A school may want to monitor whether planned actions are completed.

For example:

Actions due during the reporting period: 40.

Actions completed by their configured deadlines: 34.

On-time completion rate:

34 ÷ 40 × 100 = 85%.

This is an operational indicator of action completion.

It is not a measure of intervention effectiveness.

12.5 Follow-Up Coverage

A school may also monitor whether closed or completed interventions have the required follow-up records.

For example, the school may require a follow-up observation for certain types of cases.

The dashboard can report the proportion of eligible cases with completed follow-up records.

The denominator must include only cases for which follow-up was actually required and due.

The system should not count a case as overdue if the relevant follow-up date has not arrived.

13. Fairness, Bias, and Responsible Use

Educational data can be incomplete, subjective, or influenced by circumstances that the platform does not record.

The design must account for these limitations.

13.1 A Low Score Is Not a Personal Label

The platform should not turn a low mark or attendance rate into a permanent label attached to a student.

A review indicator describes a condition under defined rules.

It should not define the student's identity or potential.

13.2 Context May Be Missing

The platform may not know why a student was absent, why an assignment was not completed, or why academic performance declined.

Staff should have a way to review relevant context without assuming that the system has already established the explanation.

13.3 Subjective Ratings Need Particular Care

Participation and skill ratings may reflect teacher judgement.

Different teachers may apply a rubric differently.

A school may also change the rubric over time.

The platform should preserve the rating framework and evaluator information where appropriate.

It should not automatically treat all ratings as perfectly comparable.

13.4 Small Groups Require Privacy Protection

A dashboard may display school-level percentages or counts.

However, a very small group may make an individual student identifiable even when the student's name is not displayed.

For example, a report stating that the only student in a particular category has a specific result may reveal private information.

Public and broadly shared reports should apply suitable disclosure controls.

The exact thresholds should follow the organisation's privacy policy and applicable legal requirements.

13.5 Avoid Discriminatory Rules

Review rules should be based on relevant educational or operational indicators.

The system should not automatically create negative labels based on protected or sensitive personal characteristics.

If a school has legitimate accessibility or accommodation requirements, those must be handled through appropriate, authorised processes rather than used as unexplained penalties.

13.6 Provide a Correction Process

Users with appropriate permissions should be able to report or correct inaccurate underlying records.

If an indicator was generated from incorrect data, the system should update the relevant calculations and preserve an audit trail.

Where a review case is already open, the correction should be visible to authorised reviewers.

14. Review Dashboards for Teachers and School Leaders

The platform should provide a central place for authorised staff to review indicators and manage follow-up work.

The interface should show enough information to support action without overwhelming users.

14.1 Teacher Review View

Teachers may see relevant indicators for assigned students and subjects.

For example, a teacher may review declining Mathematics results, incomplete assignment records, or missing skill ratings.

The teacher should only access students and records within the authorised scope.

14.2 Principal Review View

The principal may review school-level indicators and authorised student-level cases.

The dashboard may organise cases by:

Priority.

Indicator category.

Current status.

Responsible person.

Due date.

School or class.

Reporting period.

The principal should be able to identify which cases require attention and which have already been reviewed.

14.3 Organisation Administrator Review View

The organisation administrator may review indicators across authorised schools.

For example, the administrator may identify branches with incomplete attendance reporting or repeated school-level review indicators.

The organisation dashboard should show the relevant evidence and definitions rather than merely presenting a list of red and green labels.

14.4 Separate Private and Public Information

Review cases and intervention notes must not appear on public leaderboards.

They should only be accessible to authorised users with the relevant permissions.

The public school leaderboard may display approved school-level indicators, but it must not expose private case details or identifiable student information.

15. Notifications and Deadlines

A review workflow may benefit from reminders, but notifications should follow clear rules.

The current Optimus Orien scope excludes in-app chat. A notification or reminder system, if implemented, should be a separate, explicitly scoped feature.

15.1 Reminder Examples

A responsible teacher may receive a reminder that a required register is incomplete.

A principal may receive a reminder that a priority review case has reached its follow-up date.

An authorised reviewer may receive a reminder that an assigned action remains incomplete.

These reminders should only be generated where a valid responsibility and deadline exist.

15.2 Avoid Excessive Notifications

If every minor indicator generates an urgent notification, users may begin ignoring them.

The system should allow authorised configuration of which events generate reminders and how frequently reminders are sent.

Repeated notifications should not create duplicate review cases or duplicate action records.

15.3 Notification Security

A notification should not expose sensitive student information to an unauthorised recipient.

The system should verify the recipient's permissions and avoid including restricted details in channels where they may be exposed.

The content and delivery method must follow the school's communication and privacy policies.

16. Review Case Lifecycle and Historical Records

The review workflow must preserve its history.

A case may change status several times before it is closed.

For example:

A new attendance indicator appears.

A teacher reviews the evidence.

The teacher discovers that a register was incomplete.

The register is corrected.

The attendance rate is recalculated.

The original indicator is reassessed.

The case is closed because the original concern was based on incomplete information.

The system should preserve this sequence.

It should not simply delete the indicator or overwrite the case with the final status.

16.1 Recommended Case States

The platform may support the following states:

New: A review has been created or is awaiting assignment.

Assigned: A responsible reviewer has been identified.

Under review: The reviewer is examining the evidence.

Action planned: A follow-up action has been recorded.

Monitoring: The school is monitoring the relevant indicator or outcome.

Closed: The authorised person has closed the case according to the applicable policy.

Reopened: A previously closed case has been reopened under an authorised workflow.

Not every case needs every state.

The school should configure the workflow appropriate to its needs.

16.2 Closure Reasons

A case should have a clear closure reason.

Examples include:

The indicator was caused by incorrect data.

The review found that no further action was required.

The planned action and required follow-up were completed.

The case was merged into an existing review.

The case was closed under an approved administrative policy.

The platform should not require staff to disclose unnecessary sensitive information merely to close a case.

16.3 Reopening a Case

If the relevant concern returns, the school may decide to reopen the existing case or create a new one linked to the earlier case.

The choice should follow the configured workflow.

The platform should preserve the previous closure date, reason, and audit history.

17. Recommended Database Structure

The developer should model indicators, cases, actions, and outcomes as separate but connected records.

This is important because a single indicator may lead to a case, a case may include several indicators, and a case may involve several actions.

17.1 Indicator Record

An indicator record should contain information such as:

A unique identifier.

Indicator type.

Relevant student, class, school, or organisation, as applicable.

Reporting period.

Rule identifier and version.

Observed value.

Configured threshold or condition.

Date generated.

Data-completeness status.

Reference to supporting source records.

Current indicator status.

17.2 Review Case Record

A review case should contain:

A unique case identifier.

Case type.

Relevant scope.

Creation date.

Current status.

Assigned reviewer.

Linked indicators.

Review decision.

Action plan references.

Follow-up dates where applicable.

Closure information.

Audit history.

17.3 Action Record

An action record should contain:

A unique action identifier.

Linked review case.

Action description or configured action type.

Responsible person.

Creation date.

Due date where applicable.

Status.

Completion date.

Completion evidence or reference where appropriate.

Audit history.

17.4 Follow-Up Record

A follow-up record should identify:

The linked case or action.

The follow-up date.

The indicator being reassessed.

The relevant observation period.

The recorded outcome.

Whether the data was complete enough to support interpretation.

The reviewer or authorised person recording the result.

Any required next step.

17.5 Rule Configuration

A rule configuration record should preserve:

The rule identifier.

The rule name.

The indicator category.

The condition or formula.

The configured threshold.

The minimum data requirements.

The applicable scope.

The effective start and end dates.

The rule version.

The authorised person or organisation responsible for the configuration.

The change history.

The exact technical representation may differ, but the information needed to explain historical indicators must be preserved.

18. Security and Governance

Review cases may contain more sensitive information than ordinary dashboard summaries.

The platform must therefore enforce access control carefully.

18.1 Role-Based Permissions

A teacher may be able to review cases involving assigned students.

A principal may have access to authorised cases within the school.

An organisation administrator may have access to cases across authorised branches.

These are examples, not universal permission rules.

The actual access model should be configurable and enforced on the server.

18.2 Case-Level Access

Access to a student's ordinary performance dashboard should not automatically grant access to every restricted intervention note.

The system should support separate permissions for viewing indicators, opening cases, editing notes, assigning actions, closing cases, and exporting records where necessary.

18.3 Audit Logging

The platform should record important case changes.

For example:

A case was assigned to a reviewer.

The reviewer changed the status.

An action was added.

An action was completed.

The case was closed.

The case was reopened.

The audit trail should identify the relevant user, timestamp, and change.

18.4 Retention and Deletion

The school or organisation should define retention rules for review cases and associated notes according to applicable requirements.

The system should not assume that all case records should be retained forever or deleted after the same period.

The relevant policy must be documented and implemented.

18.5 No Public Case Disclosure

Review indicators, intervention actions, and private notes must never be exposed through public school leaderboards.

Any public school-level reporting must use approved information and appropriate privacy controls.

19. Worked Example: From an Attendance Indicator to Follow-Up

Consider a fictional student named Ali.

A school has configured an attendance review threshold of 85%.

The relevant reporting period has 100 eligible scheduled attendance days.

The attendance module records 78 present-equivalent days.

The calculated attendance rate is:

78 ÷ 100 × 100 = 78%.

19.1 Detection

The system compares 78% with the configured 85% threshold.

The condition is met.

The platform creates an attendance-review indicator explaining the observed rate and threshold.

19.2 Review

An authorised staff member examines the attendance records.

The reviewer checks whether the attendance register is complete and whether the records correspond to Ali's actual enrolment period.

The reviewer confirms that the available attendance data is complete enough for the configured calculation.

The reviewer records that follow-up is appropriate.

This decision belongs to the authorised person, not the automatic rule.

19.3 Action

The school records a follow-up action according to its policy.

For example, the responsible staff member may contact the parent through an approved communication process and arrange a review of the attendance situation.

The action record identifies the responsible person and any relevant deadline.

19.4 Follow-Up

During the next comparable reporting period, Ali's attendance rate is 86%.

The recorded change is:

86% − 78% = 8 percentage points.

The system records the new rate and the follow-up observation.

19.5 Interpretation

The platform may report:

"Attendance increased by eight percentage points in the reporting period following the recorded follow-up action."

It should not automatically claim:

"The intervention caused attendance to improve by eight percentage points."

The observed sequence is useful, but it does not establish causation.

The authorised reviewer may decide that the case can be closed, should remain under monitoring, or requires further action according to the school's policy.

20. Software Implementation Requirements and Acceptance Tests

The following requirements define the expected behaviour of the School Health and Early Intervention features.

20.1 Indicator Generation

INT-001 — Configurable Rules

The platform must support explicit, configurable rules for supported indicators.

Acceptance test: A school configures an attendance threshold of 85%. A valid attendance rate of 79% triggers the corresponding indicator.

INT-002 — Explainable Indicators

Every generated indicator must identify the condition that caused it.

Acceptance test: An attendance indicator appears. An authorised user can see the recorded rate, configured threshold, relevant period, and rule version.

INT-003 — Minimum Data Requirements

Rules must define the minimum data required for reliable evaluation.

Acceptance test: Attendance data is incomplete for the reporting period. The system identifies the data-quality limitation rather than treating the calculated rate as fully reliable.

INT-004 — Missing Data Separation

Missing data must not automatically be interpreted as a negative outcome.

Acceptance test: An assessment result has not been entered. The system does not create a low-performance indicator by treating the missing result as zero.

INT-005 — Multiple Indicator Categories

The platform must support separate indicator categories for relevant academic, attendance, assignment, skill, and data-quality conditions.

Acceptance test: A student has a declining Mathematics result and an attendance concern. The system identifies the two conditions separately.

INT-006 — Rule Versioning

The platform must preserve the rule version used to generate each historical indicator.

Acceptance test: A school changes its attendance threshold from 85% to 90%. Previously generated indicators retain their historical rule context.

INT-007 — Indicator Recalculation

Relevant source-data corrections must trigger indicator reassessment according to the configured workflow.

Acceptance test: An attendance register is corrected. The platform recalculates the affected attendance rate and updates the relevant indicator status without erasing its history.

INT-008 — No AI Dependency

The initial implementation must use the documented rule-based system rather than AI predictions.

Acceptance test: The indicator engine evaluates a configured threshold using the relevant data and produces the expected result without an AI prediction service.

20.2 Review Case Management

INT-009 — Separate Case Records

Review cases must be separate from the indicators that initiate them.

Acceptance test: An attendance indicator is generated. The system can display the indicator without automatically treating it as a confirmed problem.

INT-010 — Human Review Decision

The system must allow an authorised reviewer to record whether further action is required.

Acceptance test: A reviewer examines an indicator and determines that no further action is necessary. The platform records that decision without deleting the original indicator history.

INT-011 — Case Assignment

A case must support assignment to an authorised responsible person.

Acceptance test: A principal assigns a case to an authorised teacher. The case displays the responsible person and current status.

INT-012 — Multiple Indicators per Case

Where the workflow permits, one case must be able to reference several related indicators.

Acceptance test: A school creates one review case concerning a student's academic decline and attendance pattern. Both indicators remain separately identifiable and linked to the case.

INT-013 — Repeated Indicators

The platform must support repeated indicators without automatically creating unnecessary duplicate cases.

Acceptance test: An attendance indicator is generated in consecutive periods. The configured workflow can link it to an existing case or create a new linked case while preserving each occurrence.

INT-014 — Case Status History

The platform must preserve important case status changes.

Acceptance test: A case moves from New to Under Review to Action Planned to Closed. The history retains the sequence and relevant timestamps.

INT-015 — Closure Reasons

The system must support a closure reason where required by the configured workflow.

Acceptance test: A reviewer closes a case because the original indicator was caused by incomplete data. The reason is recorded.

INT-016 — Reopening Cases

The platform must support authorised reopening where the workflow permits it.

Acceptance test: A previously closed case is reopened. The system preserves the earlier closure history and records the reopening event.

20.3 Actions and Follow-Up

INT-017 — Separate Action Records

Intervention actions must be stored separately from review cases.

Acceptance test: A case includes two follow-up actions. Each action has its own status, responsible person, and relevant dates.

INT-018 — Action Responsibility

Every assigned action must identify an authorised responsible person.

Acceptance test: A principal creates a follow-up action. The system does not treat the action as assigned until an authorised responsible person is selected.

INT-019 — Action Deadlines

An action must not be marked overdue without an applicable deadline.

Acceptance test: An action has no due date. The system does not label it overdue solely because it remains incomplete.

INT-020 — Action Completion

The system must distinguish planned actions from completed actions.

Acceptance test: A support session is planned but has not occurred. The platform does not display it as completed.

INT-021 — Case Closure Control

Completing an action must not automatically close a case unless the configured workflow explicitly requires that behaviour.

Acceptance test: One of two required actions is completed. The case remains open when the workflow requires both actions or an authorised closure decision.

INT-022 — Follow-Up Records

The system must support follow-up records linked to relevant cases or actions.

Acceptance test: A reviewer records a later attendance observation. The follow-up record identifies the relevant case, observation period, and recorded outcome.

INT-023 — Outcome Interpretation

The platform must distinguish observed changes from proven causal effects.

Acceptance test: Attendance increases after a follow-up action. The dashboard reports the recorded change but does not automatically state that the action caused it.

INT-024 — Intervention Completion Metrics

Where intervention completion rates are displayed, the platform must use a defined denominator and completion rule.

Acceptance test: Of 40 actions due in the period, 34 were completed on time. The platform displays 85% and identifies the relevant period and calculation basis.

20.4 School Health

INT-025 — Separate Health Indicators

The initial school health dashboard must present its component indicators separately.

Acceptance test: A school has strong academic results but incomplete attendance records. The dashboard shows both conditions rather than concealing the missing attendance data within one score.

INT-026 — Configurable Status Thresholds

Green, Yellow, Red, or equivalent status labels must be based on documented configuration.

Acceptance test: A school changes a status threshold. The platform applies the new rule according to its effective date and preserves historical context.

INT-027 — Composite Score Protection

The platform must not generate an unexplained composite School Health Score.

Acceptance test: No approved composite formula is configured. The dashboard displays separate indicators instead.

INT-028 — School-Level Scope

School health indicators must be calculated using the correct school, reporting period, and data scope.

Acceptance test: A principal views the school's attendance indicator. Records from unrelated schools are not included.

INT-029 — Organisation-Level Scope

Organisation-wide indicators must include only authorised branches.

Acceptance test: An organisation administrator oversees three branches. The dashboard excludes unrelated organisations' records.

20.5 Fairness, Privacy, and Security

INT-030 — No Automatic Diagnosis

The platform must not diagnose psychological, behavioural, or medical conditions from educational indicators.

Acceptance test: A student has declining academic results and attendance. The platform may display review indicators but does not diagnose a disorder.

INT-031 — No Automatic Punishment

An indicator must not automatically impose a penalty or change a student's academic result.

Acceptance test: A student crosses an attendance threshold. The system creates the configured indicator but does not automatically reduce examination marks.

INT-032 — Restricted Case Notes

The platform must enforce permissions for private review notes and intervention records.

Acceptance test: A teacher without permission to view a restricted note cannot retrieve it through a direct server request.

INT-033 — Server-Side Authorisation

Every protected case, indicator, and action request must be authorised by the server.

Acceptance test: A user changes a case identifier to request an unrelated case. The server rejects the request if the user lacks permission.

INT-034 — Organisation Isolation

The system must prevent unauthorised cross-organisation access.

Acceptance test: An administrator attempts to retrieve a review case from an unrelated organisation. The server rejects the request.

INT-035 — Audit History

Important changes to indicators, cases, assignments, actions, and closure decisions must be traceable.

Acceptance test: A reviewer changes a case from Under Review to Closed. The audit history records the user, timestamp, and relevant change.

INT-036 — Public Privacy

Review cases, intervention notes, and identifiable student-level indicators must not appear on public leaderboards.

Acceptance test: An unauthenticated visitor requests a private review case. The server rejects the request.

INT-037 — Data Minimisation

The interface must not require unnecessary sensitive information for routine review cases.

Acceptance test: A reviewer records an attendance follow-up. The workflow does not require unnecessary medical or family details.

INT-038 — Accessible Review Interface

The review interface must present statuses in understandable text and not rely solely on colour.

Acceptance test: A user can distinguish New, Under Review, Action Planned, and Closed cases without relying on colour alone.

INT-039 — Unavailable Data

The system must distinguish unavailable source data from a valid zero or a confirmed negative result.

Acceptance test: The attendance module is temporarily unavailable. The review engine does not create a zero-attendance indicator solely because the service failed.

INT-040 — Notification Permissions

If reminders are implemented, they must be delivered only to authorised recipients and must not expose restricted case information.

Acceptance test: A reminder is generated for a restricted case. An unrelated teacher does not receive the private case details.

21. Recommended Development Order

The developer should implement the feature in a controlled sequence.

Stage 1: Confirm source-data reliability.

Verify that academic results, attendance, homework, participation, skills, and data-quality states are correctly represented.

Stage 2: Define the initial indicators.

Start with a small set of transparent indicators, such as attendance below a configured threshold, declining academic results, and missing required records.

Stage 3: Implement configurable rules.

Define thresholds, minimum data requirements, applicable scopes, and rule versions.

Stage 4: Build the indicator engine.

Evaluate the configured rules against the relevant source records and generate traceable indicators.

Stage 5: Build the review-case model.

Separate indicators from cases and implement case assignment, review decisions, status changes, and closure reasons.

Stage 6: Implement actions.

Allow authorised staff to create, assign, update, and complete follow-up actions.

Stage 7: Implement follow-up records.

Record later observations and distinguish observed changes from causal conclusions.

Stage 8: Build review dashboards.

Provide role-specific views for teachers, principals, and organisation administrators.

Stage 9: Add school health indicators.

Display separate academic, attendance, reporting, and data-quality measures.

Stage 10: Enforce security and audit logging.

Test permissions for every indicator, case, action, and note.

Stage 11: Test realistic scenarios.

Test incomplete data, repeated indicators, incorrect source records, changing thresholds, transfers, overdue actions, and reopened cases.

Stage 12: Validate usability.

Confirm that staff can understand why an indicator appeared, what action is required, and how to record the outcome.

22. Final Product Principles

Optimus Orien should help schools notice educational patterns, investigate them responsibly, and organise follow-up actions.

It should not treat every unusual value as a confirmed problem.

It should not assume that a low mark, declining attendance rate, or incomplete assignment record reveals the reason behind the situation.

It should not automatically diagnose students, assign blame, or impose penalties.

Instead, the platform should:

Identify clearly defined conditions.

Explain the evidence and rules behind each indicator.

Distinguish indicators from reviewed concerns.

Allow authorised staff to decide whether action is necessary.

Record actions and responsibility.

Support follow-up and outcome monitoring.

Preserve historical context and audit records.

Protect student privacy.

Report observed changes without making unsupported causal claims.

These principles make the system more useful because they allow schools to respond to information without mistaking a calculation for a complete explanation of a student's circumstances.

The purpose of early intervention is not to label students earlier. It is to help schools notice relevant evidence earlier and respond more thoughtfully.

What Part 14 Establishes

Part 14 defines the complete rule-based early intervention workflow, from generating an indicator to reviewing it, planning an action, recording follow-up, and interpreting the outcome.

It also defines school health categories, review cases, priority levels, intervention records, follow-up measures, privacy controls, and 40 implementation requirements with acceptance tests.

Part 15 focuses on how Optimus Orien turns educational records into reliable reports, comparisons, and downloadable information. The central goal is to ensure that a principal, teacher, school owner, and organisation administrator can examine the same underlying data without receiving conflicting numbers or accidentally exposing private student information.

PART 15

THE COMPLETE REPORTING, ANALYTICS, AND DATA EXPORT SYSTEM

How Optimus Orien Generates Accurate, Consistent, Understandable, and Secure Educational Reports

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner, software developer, and future product team
Primary objective: Define how the platform should calculate, present, filter, generate, export, and secure reports across its different portals.

1. Why This Document Exists

Optimus Orien collects information about student performance, school operations, and educational progress.

The platform may contain assessment marks, attendance records, assignment completion, participation ratings, subject-linked skills, class averages, school health indicators, historical trends, and intervention records.

However, collecting information is not enough.

Schools need to turn this information into reports that help them answer practical questions.

A teacher may want to know which students have not received all their assessment results.

A principal may want to compare the performance of different classes.

A school owner may want to understand how several branches are performing.

An organisation administrator may want to examine changes in attendance across the organisation.

A parent may want to understand their child's academic progress.

A school may also need to download a report for a meeting, retain a record, or transfer information to another system.

These activities require a reporting and analytics system.

The important distinction is that **a report is not simply a table of database records**.

A reliable report must define what information it contains, how its numbers are calculated, which records are included, which records are excluded, what period it covers, and who is permitted to access it.

If these rules are unclear, two dashboards may display different numbers for the same measure. A school may appear to improve simply because its reporting population changed. A downloaded report may omit records that appear on the screen. A principal may accidentally receive information belonging to another branch.

This document defines how Optimus Orien should prevent these problems.

2. What Is the Difference Between Data, Information, Analytics, and Reporting?

Before implementation, the developer must understand four related concepts.

2.1 Data

Data consists of recorded facts or values.

Examples include:

A student obtained 78 marks out of 100.

A student was present for a scheduled attendance session.

A teacher confirmed that an assignment was completed.

A student received a skill rating of 7 out of 10.

An assessment was scheduled for a particular reporting period.

These records are the foundation of the reporting system.

2.2 Information

Information is data organised so that it becomes easier to understand.

For example, a single attendance record is data.

A table showing a student's attendance across an academic term is information.

A report showing attendance rates for every class in a school is also information.

2.3 Analytics

Analytics means examining data to understand patterns, differences, distributions, or changes.

Examples include:

Comparing attendance between reporting periods.

Examining the distribution of assessment results.

Identifying subjects with a high proportion of missing marks.

Comparing class averages.

Measuring the change in school-level indicators over time.

These activities can be performed using ordinary calculations and defined rules. They do not require artificial intelligence.

The initial Optimus Orien implementation should use transparent calculations and clearly defined comparisons.

2.4 Reporting

Reporting means presenting selected information in a structured form for a particular purpose.

A report may appear as an interactive table, a dashboard, a printable document, or a downloadable spreadsheet.

For example, a principal may generate a term performance report containing student results, subject averages, attendance rates, and data-completeness indicators.

The report should clearly state its reporting period, calculation rules, and included population.

2.5 How They Connect

Consider the following sequence:

Raw assessment records are entered into Optimus Orien.

The platform calculates percentages and subject averages.

The platform organises the results into a class performance table.

The principal filters the table to examine one reporting period.

The principal exports the report as a spreadsheet.

Each stage depends on the correctness of the stage before it.

A beautifully designed report cannot correct an inaccurate source record or an incorrectly defined formula.

Implementation principle: Reporting quality depends on the quality of the source data, the correctness of calculations, the clarity of definitions, and the security of access.

3. The Main Questions Reports Should Answer

The reporting system should be designed around questions that schools need to answer.

This is more useful than creating dozens of reports simply because they are technically possible.

3.1 Student-Level Questions

A student or authorised parent may need to understand:

What were the student's results during the selected period?

How did the student perform in each subject?

What was the student's attendance rate?

Which assignments were completed, overdue, or awaiting verification?

What skill ratings were recorded?

How have comparable results changed over time?

Are any required records missing?

Which reporting period does each value represent?

3.2 Teacher-Level Questions

A teacher may need to understand:

How did assigned students perform in a particular subject?

Which students have missing assessment results?

What was the class average for a selected assessment or reporting period?

Which attendance or assignment records require attention?

How have subject results changed across comparable periods?

How complete are the teacher's required records?

3.3 Principal-Level Questions

A principal may need to understand:

How are the school's classes performing?

What are the subject averages?

How do academic results differ across classes?

What are the attendance rates for different year groups?

Which required reports remain incomplete?

How have school-level indicators changed over time?

Which school health indicators require review?

How much of the available data is complete and reliable enough to support comparison?

3.4 Organisation-Level Questions

An organisation administrator may need to understand:

How do the organisation's branches compare?

Which branches have complete reporting records?

How do academic and attendance indicators change across branches?

Are there differences between schools that require further investigation?

Which branches have not completed required data entry?

How have the organisation's indicators changed over several academic years?

The system should only include branches within the administrator's authorised scope.

3.5 Government or Oversight Questions

A future government or oversight portal may require broader reporting, such as comparisons between districts, regions, or school groups.

However, this must be treated as a separately defined product capability.

It should not be assumed that an organisation administrator's permissions automatically allow government-level access.

Government reporting requires its own access rules, data definitions, privacy controls, and legal assessment.

4. What Is a Report Definition?

A report definition specifies what a report means and how it should be produced.

It is similar to a recipe.

If two people follow the same recipe using the same ingredients and measurements, they should produce the same result.

Likewise, if two authorised users generate the same report using the same source data, period, filters, and calculation rules, their results should agree.

A report definition should specify:

The report's purpose.

The records or entities included.

The reporting period.

The filters available.

The measures displayed.

The calculation rules.

The treatment of missing data.

The rules for grouping records.

The permissions required to access the report.

The export formats supported.

The relevant calculation or configuration version.

For example, an attendance report must define whether its attendance rate is based on scheduled school days, scheduled lessons, or another configured attendance unit.

Without that definition, a figure such as 90% is incomplete information.

5. Raw Records, Calculated Measures, and Interpretations

One of the most important reporting principles is to distinguish three different kinds of information.

5.1 Raw Records

Raw records are the underlying entries.

Examples include an examination mark, an attendance status, an assignment submission time, or a teacher's skill rating.

These records should remain traceable to their source.

5.2 Calculated Measures

A calculated measure is produced from source records using a defined formula.

Examples include:

An assessment percentage.

A class average.

An attendance rate.

An assignment completion rate.

A percentage-point change between reporting periods.

The platform must document the formula used for each measure.

5.3 Interpretations

An interpretation describes what a value may mean within a defined policy or context.

Examples include:

Attendance below the school's configured review threshold.

A subject result that has declined across comparable periods.

A reporting completeness status requiring attention.

These interpretations must not be presented as raw facts.

For example, an attendance rate of 78% is a calculated measure.

The statement that it falls below the school's configured threshold of 85% is a rule-based interpretation.

The statement that the student is absent because of a particular family circumstance would require evidence that the platform may not possess.

The system must not confuse these three levels.

6. Reporting Periods and Time Context

A report must clearly identify the period it covers.

Educational records may be organised by day, week, month, term, semester, examination session, academic year, or another configured period.

Different schools may use different academic calendars.

The platform must not assume that all schools begin their academic year in the same month or use the same number of terms.

6.1 Date Range Versus Academic Period

A date range is a period defined by calendar dates.

An academic period is a period defined by the school's educational structure.

For example, a user may request records between 1 January and 31 March.

Another user may request the school's second term.

These periods may overlap, but they are not necessarily identical.

The system should support the relevant reporting methods without silently treating them as equivalent.

6.2 Academic-Year Boundaries

Some academic years cross calendar years.

A school may have an academic year running from September to June.

Another school may use a different calendar.

Reports should display the school's configured academic-year label and the relevant dates where useful.

6.3 Time Zones

Dates and times should be stored and interpreted consistently.

For example, an assignment submitted shortly before midnight may fall on different calendar dates if the system uses the wrong time zone.

The platform should apply the relevant school's configured time zone when displaying local dates and times.

Where exact timestamps matter, the system should preserve them rather than storing only a formatted date.

6.4 Historical Configuration

A report covering a past academic year should use the relevant historical records and calculation definitions.

If the school changes its grading policy, attendance rules, or skill-rating rubric, the change must not silently rewrite the meaning of historical reports.

The system should preserve sufficient configuration history to reproduce or explain past results.

7. The Difference Between a Filter, a Grouping, and a Comparison

These concepts are related but different.

7.1 Filter

A filter determines which records are included.

Examples include:

Only Grade 8.

Only the Mathematics subject.

Only one school branch.

Only the selected academic year.

Only assignments with a particular status.

A filter changes the population being examined.

7.2 Grouping

A grouping organises included records into categories.

For example, a report may group student results by subject.

Another report may group attendance by class.

An organisation report may group indicators by branch.

Grouping does not necessarily exclude records. It organises them.

7.3 Comparison

A comparison examines differences between two or more values or groups.

Examples include:

Comparing two reporting periods.

Comparing the average result of two classes.

Comparing attendance between branches.

Comparing a student's results across comparable assessments.

A comparison requires the measures to be sufficiently compatible.

A filter selects records, a grouping organises them, and a comparison examines differences.

The developer should implement these as distinct reporting operations.

8. Report Categories for Optimus Orien

The first reporting release should prioritise reports that support the platform's main purpose.

8.1 Student Performance Report

This report presents the selected student's academic and educational records for a specified period.

It may include:

Student identity and current school enrolment.

Academic year and programme.

Subject-level marks and percentages.

Assessment details.

Class averages where comparison is appropriate.

Attendance rate and relevant attendance counts.

Assignment completion information.

Participation ratings.

Subject-linked skill ratings.

Historical trends.

Missing-data indicators.

The report should separate the different performance dimensions rather than combining them into an unexplained overall score.

It should show which reporting period each measure represents.

8.2 Class Performance Report

This report summarises the results of students in a selected class.

It may include:

Student-level results for authorised staff.

Subject averages.

Assessment averages.

Counts of students with complete results.

Counts of students with missing results.

Performance distributions.

Attendance summaries.

Assignment completion summaries.

Relevant progress measures.

The report must define whether a class average is a simple average of student percentages, a weighted calculation based on marks, or another explicitly configured method.

8.3 Subject Performance Report

This report focuses on a particular subject.

It may include:

Student results.

Assessment-level results.

Class average.

Distribution of marks.

Changes across comparable periods.

Missing assessment records.

Relevant skill ratings.

The report should not assume that the same subject name automatically represents identical curriculum content or assessment difficulty across every school or programme.

8.4 Attendance Report

This report may show:

Attendance rate.

Present-equivalent count.

Eligible scheduled attendance opportunities.

Recorded absences.

Late arrivals where supported.

Incomplete attendance registers.

Attendance trends.

The report must follow the attendance rules documented in Part 9.

It must distinguish missing registers from recorded absence and identify the denominator used in the attendance calculation.

8.5 Homework and Assignment Report

This report may show:

Assignments assigned during the selected period.

Eligible students.

Verified completions.

Pending teacher verification.

Overdue assignments.

Late submissions.

Excused or not-required assignments.

On-time completion rate.

Eventual completion rate, if supported.

The report must follow the rules documented in Part 11.

It must not count an assignment as completed merely because a student has declared completion if teacher verification is required by the configured policy.

8.6 Participation and Skill Report

This report may show:

Participation ratings.

Skill ratings by subject.

The relevant rubric.

Rating dates.

Evaluator information where appropriate.

Changes across comparable periods.

Missing ratings.

The report must distinguish missing data from a low rating and must preserve the context of the rubric used.

8.7 School Health Report

This report presents the school's defined health indicators.

It may include:

Academic performance measures.

Attendance indicators.

Assignment completion measures.

Data-quality indicators.

Reporting completeness.

Configured review statuses.

The report should display the component indicators separately.

It should not introduce an unexplained composite score.

8.8 Intervention and Follow-Up Report

This report may show:

Number of review cases opened.

Number of cases currently open.

Number of cases closed during the period.

Actions due.

Actions completed.

Actions overdue under the configured rules.

Follow-up coverage.

Recorded changes in the relevant measures.

The report must distinguish operational activity from educational outcomes.

For example, a high number of completed actions does not automatically prove that those actions improved student performance.

8.9 Organisation Performance Report

This report summarises authorised branches.

It may show:

Branch-level indicators.

Reporting completeness.

Academic performance measures.

Attendance measures.

Historical changes.

Configured school health statuses.

The report should make the reporting period and branch inclusion rules clear.

It should not rank branches using incompatible or incomplete measures without clearly identifying the limitation.

9. The Most Important Reporting Rule: Define the Denominator

Many educational reporting errors occur because the numerator and denominator are unclear.

The **numerator** is the number or quantity being counted.

The **denominator** is the total eligible number or quantity against which that count is measured.

Consider a homework completion report.

A class has 100 eligible assignment obligations.

Students have verified completion for 80 of them.

The verified completion rate is:

80 ÷ 100 × 100 = 80%.

But suppose 20 of the 100 assignments were formally excused and the school's policy excludes excused assignments from the denominator.

The denominator becomes 80.

If the same 80 assignments were completed, the resulting rate would be:

80 ÷ 80 × 100 = 100%.

Both calculations are mathematically correct, but they answer different questions.

The report must therefore state how eligibility and exclusions are defined.

The same principle applies to attendance, assessment coverage, intervention completion, and skill-assessment coverage.

The developer must never calculate a percentage without defining which records are eligible for inclusion.

10. Averages: Why One Number Can Be Misleading

Averages are useful, but the method matters.

10.1 Simple Average

Suppose three students have Mathematics percentages of 60%, 70%, and 80%.

The simple average is:

(60 + 70 + 80) ÷ 3 = 70%.

Each student contributes equally to the calculation.

10.2 Weighted Average

Suppose two assessments have different maximum marks.

Assessment A: 80 marks obtained out of 100.

Assessment B: 18 marks obtained out of 20.

The combined percentage based on total marks is:

(80 + 18) ÷ (100 + 20) × 100 = 81.67%.

This differs from the simple average of the two assessment percentages:

(80% + 90%) ÷ 2 = 85%.

Neither formula should be used automatically for every educational measure.

The correct method depends on what the report is intended to represent and on the school's assessment rules.

10.3 Student Percentages Versus Pooled Marks

A class average can mean different things.

It might be the simple average of the students' percentages.

It might be the total marks obtained by all students divided by the total possible marks.

It might be a weighted average defined by the assessment framework.

These calculations can produce different values.

The report definition must specify which calculation is used.

The interface should explain the method when the distinction could affect interpretation.

10.4 Do Not Average Incompatible Values

The platform should not average marks with different maximum values without applying an appropriate conversion or explicitly defined weighting method.

It should not average skill ratings from different rubrics as though the scales were identical.

It should not combine attendance rates and academic percentages into one number without a defined and justified formula.

11. Missing Data and Reporting Completeness

Missing information is one of the most important problems in school analytics.

A missing result is not the same as a low result.

A missing attendance register is not the same as a day when every student was absent.

A missing skill rating is not the same as a rating of zero.

The reporting system must preserve these distinctions.

11.1 Common Types of Missing Data

Examples include:

A teacher has not entered an assessment result.

A school has not submitted an attendance register.

A required skill assessment has not been conducted.

A student joined after the reporting period began.

A student transferred to another school.

An assignment has not yet reached its deadline.

A data import failed.

These situations have different meanings and should not be treated as one universal missing-value state.

11.2 Reporting Completeness

A report may show how much required information has been recorded.

For example:

Required assessment results: 200.

Valid recorded results: 184.

Assessment-result completeness:

184 ÷ 200 × 100 = 92%.

This is a measure of data completeness.

It is not a measure of academic achievement.

The platform should label it accordingly.

11.3 Separate Result Coverage From Result Quality

A school might have recorded every assessment result but contain data-entry errors.

Another school might have accurate recorded results but many missing records.

These are different problems.

The system should distinguish:

Completeness: How much required information is present?

Validity: Does the information meet the defined rules for acceptable data?

Accuracy: Does the recorded information correctly represent the real-world event or result?

The system can automatically check some forms of validity, such as whether a mark exceeds the assessment maximum.

It cannot always establish accuracy without additional evidence.

11.4 Do Not Silently Exclude Missing Records

If a report excludes missing records from an average, that exclusion must be clear.

For example, a class average based on 18 recorded results out of 25 eligible students should not be presented as though all 25 students had results.

The report should show the number of included results and the number still missing.

12. Comparing Results Fairly

Reports often become misleading when users compare values that are not sufficiently comparable.

12.1 Comparable Reporting Periods

A student's result from one term should not automatically be compared with a result from a different academic year without considering the context.

The subject, assessment method, grading rules, and reporting period may have changed.

The system should preserve enough context to determine whether the comparison is appropriate.

12.2 Comparable Subjects and Assessments

The same subject label does not guarantee that two assessments measure the same content or difficulty.

A school may change its curriculum or assessment structure.

The platform should not automatically describe every numerical difference as improvement or decline.

12.3 Percentage Points Versus Relative Percentage Change

Suppose a student's attendance rate increases from 70% to 80%.

The change is 10 percentage points.

The relative increase is:

(80 − 70) ÷ 70 × 100 = approximately 14.29%.

These figures answer different questions.

Educational dashboards should generally use percentage-point changes when comparing two rates, while making the measure clear.

If relative percentage change is shown, the report should label it explicitly.

12.4 Group Composition

Suppose a school has a class average of 80% in one term and 75% in the next.

The decline may reflect changes in student performance.

However, it may also reflect a change in which students were enrolled, which assessments were included, or which records were available.

The report should identify important differences in the population or data coverage where the information is available.

12.5 Rankings Are Not Complete Explanations

A ranking indicates relative position within a defined comparison group.

It does not automatically explain why one school or student ranks above another.

Reports should identify the measure, period, eligible population, and missing-data rules used in the ranking.

They should avoid treating a rank as a complete judgement of educational quality.

13. Dashboard Versus Report

These terms are often used interchangeably, but they serve different purposes.

13.1 Dashboard

A dashboard is an interactive view designed to help users monitor information and explore relevant details.

For example, a principal's dashboard may show attendance indicators, subject averages, reporting completeness, and school health statuses.

Users may select a class, subject, or reporting period to examine the data more closely.

13.2 Report

A report is a structured presentation of information for a defined purpose.

It may be interactive, printable, or downloadable.

A term performance report may contain a table of results, subject averages, attendance measures, and reporting notes.

13.3 They Must Use Consistent Calculations

The dashboard and report should not independently implement different versions of the same formula.

For example, if the dashboard displays a class attendance rate of 91%, the exported attendance report should produce the same figure when it uses the same scope, period, and calculation rules.

If the report uses a different filter or a different attendance definition, that difference must be visible.

Recommended architecture: Define each measure once in a shared calculation or reporting layer and reuse it across dashboards, tables, and exports.

14. Report Filters and User Controls

Users should be able to select relevant report parameters without changing the underlying calculation rules.

Possible filters include:

Organisation.

School or branch.

Academic year.

Reporting period.

Programme.

Grade or year group.

Class.

Subject.

Assessment.

Student, where permitted.

Teacher, where permitted.

Assignment status.

Attendance status.

Date range.

Data-completeness status.

Not every report needs every filter.

The available filters should depend on the report's purpose and the user's permissions.

14.1 Filters Must Respect Permissions

A user should not be able to access unauthorised information by changing a filter or modifying a request.

For example, a teacher should not gain access to another teacher's students merely by changing a class identifier in the report URL.

The server must enforce the user's access scope.

14.2 Filter Combinations

The system should define what happens when users combine filters.

For example, a principal selects Grade 8, Mathematics, and a particular term.

The report should include records satisfying all selected conditions.

If a filter combination produces no records, the platform should display a clear empty-state message rather than an unexplained zero.

14.3 Preserve Report Context

When a report is displayed or exported, the selected filters should be identifiable.

A downloaded report should not simply say "Student Performance Report."

It should include enough context to identify the selected school, academic year, reporting period, and relevant filters.

15. Visualising Report Data

Reports should use charts when a visual representation helps users understand a pattern.

Charts should not be added merely to make the interface look sophisticated.

15.1 Tables

Tables are useful when users need exact values for multiple records.

For example, a teacher may need to examine individual assessment marks.

Tables should include clear column headings, appropriate sorting, and understandable missing-data labels.

15.2 Bar Charts

Bar charts are useful for comparing categories.

For example, a principal may compare average Mathematics results across classes.

The report should make the measure, period, and population clear.

15.3 Line Charts

Line charts are useful for showing changes over time.

For example, a school may examine attendance rates across several reporting periods.

The chart should identify the time intervals and should not connect incompatible values in a way that suggests a continuous trend when the underlying periods are not comparable.

15.4 Distribution Charts

A distribution chart shows how values are spread across a group.

For example, a report may show how many students fall within defined assessment-percentage bands.

This can be more informative than an average alone.

The platform should explain the bands and make clear that grouping students into bands does not define their overall ability.

15.5 Avoid Misleading Visuals

The system should not:

Use an unexplained composite score.

Use colours as the only way to communicate status.

Hide missing data without explanation.

Use an axis that exaggerates a small difference without making the scale clear.

Display precise-looking values that the underlying data cannot support.

Compare groups with incompatible definitions.

Every chart should help users answer a specific question.

16. Export Formats: What Should Optimus Orien Support?

A reporting system becomes more useful when users can take information outside the application.

However, exporting data creates additional privacy and security responsibilities.

The initial implementation should prioritise a small number of useful formats.

16.1 PDF

PDF stands for **Portable Document Format**.

PDF is useful for printable reports, formal summaries, and documents intended to preserve a consistent page layout.

Examples include a student performance summary, a term report, or a school health report.

PDF is not ideal for large-scale data manipulation.

16.2 XLSX

XLSX is the standard spreadsheet file format used by modern versions of Microsoft Excel and supported by other spreadsheet applications.

It is useful when school staff need to sort, filter, inspect, or analyse tabular information.

Examples include a class results report or a branch comparison table.

The exported workbook should have clear column names, appropriate data types, and understandable missing-value representations.

16.3 CSV

CSV stands for **Comma-Separated Values**.

It is a simple text format used to represent rows and columns.

CSV is useful for data exchange between software systems.

However, it does not preserve rich formatting, multiple worksheets, or the full structure of a spreadsheet workbook.

The system should define the character encoding and how values containing commas, quotation marks, and line breaks are handled.

16.4 Recommended Initial Export Strategy

A practical first release may support:

PDF for formal, printable reports.

XLSX for structured reports that school staff need to inspect and manipulate.

CSV for approved data-export workflows where a simple interoperable format is useful.

The platform does not need to support every possible format immediately.

The exact formats should be prioritised according to user needs, privacy requirements, and implementation cost.

17. Exporting Is a Security-Sensitive Action

A user who can view a dashboard does not automatically need permission to export all its information.

A downloadable file can be copied, forwarded, uploaded elsewhere, or stored outside the platform.

The system must therefore treat export permissions seriously.

17.1 Export Permission

The platform should define which roles can export which categories of data.

For example, a teacher might be allowed to export results for assigned students but not an entire school's student records.

A parent might be allowed to download an approved report for their linked child but not a report containing other students' information.

The actual permissions should follow the platform's access-control model.

17.2 Scope Enforcement

Export queries must use the same server-side authorisation rules as ordinary report requests.

The system must not rely on the interface to hide unauthorised rows.

17.3 Sensitive Columns

Some exports may contain information that is unnecessary for the report's purpose.

For example, a class performance report may not need to contain personal contact details.

The export should include only the fields required for its defined purpose.

17.4 Export Audit Trail

For sensitive exports, the system should record relevant information such as:

Who requested the export.

Which report was generated.

The authorised scope.

The reporting period.

The export time.

The export format.

Whether generation succeeded.

The exact audit fields should follow the organisation's security and retention requirements.

17.5 File Expiry and Access

If exports are generated asynchronously and stored temporarily, the system should use protected file access and an appropriate expiry policy.

A downloadable report must not become publicly accessible simply because its URL is difficult to guess.

The server must verify access before providing the file.

18. Report Generation and Performance

Large schools may have thousands of students, multiple academic years, many subjects, and extensive historical records.

A reporting system that works for a small demonstration may become slow when used by a real organisation.

The developer should plan for both correctness and performance.

18.1 Avoid Loading Everything at Once

A class table may contain many students and several measures.

The application should not automatically load every historical record for every student when a user opens a report.

Instead, it should retrieve the records needed for the selected report and provide pagination or other suitable navigation for large result sets.

18.2 Pagination

Pagination divides a large result set into manageable pages.

For example, a report may display 50 rows at a time while allowing the user to move to the next page.

The chosen page size should be configurable or selected according to usability and performance requirements.

18.3 Large Exports

Some exports may take longer than a normal page request.

For these, the platform may use background processing.

The system should clearly show whether an export is pending, processing, completed, or failed.

If background processing is not necessary for a small report, a simpler synchronous approach may be sufficient.

18.4 Avoid Conflicting Numbers

If the source data changes while a large report is being generated, the exported file should not contain an uncontrolled mixture of different data states.

Where necessary, the system should use a consistent database snapshot or another documented strategy for generating a coherent report.

The generated report should record when it was created and which reporting period it represents.

18.5 Caching

Caching means temporarily storing a previously calculated result so it can be reused.

It may improve performance for frequently viewed reports.

However, cached results must be invalidated or refreshed when relevant source data or configuration changes.

The system must also prevent cached information from being served to users who lack permission to view it.

Correctness and security take priority over the convenience of caching.

19. Report Accuracy, Corrections, and Historical Reproducibility

Reports may change when source records are corrected.

For example, a teacher may enter the wrong mark and later correct it.

The current report should reflect the corrected value according to the applicable rules.

However, the system may also need to preserve evidence of what was previously recorded.

19.1 Current Report

A current report should use the currently valid source records and applicable calculation rules.

It should not include superseded values as if they were still current.

19.2 Historical Report

A historical report may refer to a completed reporting period.

The platform should preserve the records and configuration needed to interpret that period correctly.

If a correction is made later, the system should follow a documented policy for historical reports.

Depending on the report type, the platform may regenerate a report using corrected source data or preserve a previously issued report as an archived snapshot.

These are different purposes and should not be confused.

19.3 Report Snapshots

A report snapshot is a preserved copy of a report as it existed at a particular point in time.

Snapshots can be useful when a school needs to retain an issued report exactly as it was delivered.

The system should not create permanent snapshots of every dashboard view without a clear purpose.

Snapshot creation, retention, and access should be governed by policy.

19.4 Reproducibility

A report is reproducible when the system can explain or recreate the result using the relevant source records, filters, calculation rules, and configuration versions.

The platform should preserve sufficient metadata for important reports to be explained later.

This is particularly important for school comparisons, published rankings, and formal academic records.

20. Report Templates and Customisation

Different schools may require different report layouts.

However, unrestricted customisation can create inconsistent definitions and misleading comparisons.

The system should separate presentation customisation from calculation logic.

20.1 Presentation Customisation

A school may be allowed to configure:

Report title.

Logo and approved branding.

Visible columns.

Column order.

Page orientation.

Selected charts.

Relevant explanatory notes.

These changes should not silently alter the meaning of the underlying measures.

20.2 Calculation Configuration

Calculation rules should be configured through the relevant academic, attendance, assignment, and analytics settings.

A report template should not independently redefine attendance or academic formulas.

Otherwise, two templates may display different values for the same measure.

20.3 Standard Templates

The initial release should provide standard report templates for the most common needs.

For example:

Student performance report.

Class performance report.

Subject performance report.

Attendance report.

Assignment completion report.

School health report.

Organisation comparison report.

These templates should use shared calculation rules and role-based access controls.

20.4 Custom Reports

A future custom-report builder may allow authorised users to select measures, filters, and groupings.

This should be treated as a separate capability with its own validation, permissions, and testing requirements.

It is not necessary to implement a fully flexible report builder before the core standard reports are reliable.

21. Accessibility and Understandability

Reports are useful only when users can interpret them.

The platform should use plain language and explain specialised terms.

For example, instead of displaying only "Data Coverage: 92%", the interface may explain:

"Valid assessment results have been recorded for 92% of the required student-assessment records in this report."

Where appropriate, the report should identify the numerator and denominator.

Charts should have descriptive titles, readable labels, and text alternatives where necessary.

Tables should use meaningful column headings and support keyboard navigation.

Status indicators should not rely on colour alone.

The report should distinguish zero, missing, not applicable, and unavailable values.

A parent should not need advanced statistical knowledge to understand the basic meaning of a student performance report.

A school leader should be able to inspect the definitions and limitations of more advanced comparisons.

22. Recommended Reporting Architecture

The developer should implement reporting as a shared service rather than building every report independently.

A useful conceptual architecture has six layers.

Layer 1: Source Records

This layer contains the underlying academic, attendance, assignment, skill, participation, enrolment, and configuration records.

Layer 2: Data Validation

This layer checks relevant data constraints and identifies invalid, incomplete, or conflicting records.

Layer 3: Shared Calculations

This layer calculates defined measures using documented formulas.

Examples include assessment percentages, class averages, attendance rates, and assignment completion rates.

Layer 4: Report Definitions

This layer specifies which measures, filters, groupings, periods, and fields belong in each report.

Layer 5: Presentation

This layer presents the results through dashboards, tables, charts, printable reports, and downloadable files.

Layer 6: Permissions and Audit

Security must apply throughout the architecture.

The server must authorise data access before returning report information or generating a file.

The audit layer records the important actions that need to be traceable.

These layers are conceptual responsibilities. The developer may implement them using the application's existing architecture rather than creating six separate services.

23. Worked Example: A Class Performance Report

Consider a fictional Grade 8 Mathematics class.

The class has 25 enrolled students during the selected reporting period.

The school has recorded valid results for 20 students.

The average of those 20 recorded percentages is 74%.

Five required results are missing.

23.1 What the Report Should Display

The report should show:

Eligible students: 25.

Valid recorded results: 20.

Missing required results: 5.

Recorded-result coverage: 80%.

Average of recorded student percentages: 74%.

23.2 What the Report Should Not Imply

It should not imply that all 25 students have an average result of 74%.

The average was calculated using 20 recorded results.

The five missing results may change the eventual class average.

The platform should therefore label the figure as the average of recorded results and show the coverage.

23.3 If a Result Is Added

Suppose the teacher later records the remaining five results.

The report should recalculate the average using the defined formula.

The coverage should become 100% if all 25 results are valid and all required records are now present.

The report should not require a separate manual recalculation by the teacher.

23.4 If the Report Is Exported

If the principal exports the same report using the same filters and data state, the file should contain the same included population, valid-result count, missing-result count, and calculated average.

If the source data changes before export generation completes, the system should use a consistent data state or clearly record the state used.

24. Software Implementation Requirements and Acceptance Tests

The following requirements define the expected behaviour of Optimus Orien's reporting and export system.

24.1 Report Definitions and Calculations

REP-001 — Standard Report Definitions

The system must provide defined report templates for the principal supported reporting needs.

Acceptance test: An authorised principal can generate a standard class performance report for a selected academic period.

REP-002 — Shared Calculation Logic

Dashboards, tables, and exports must use shared definitions for equivalent measures.

Acceptance test: The dashboard and exported report use the same source data, filters, and attendance formula. They display the same attendance rate.

REP-003 — Calculation Transparency

Every calculated measure must have a documented formula or calculation rule.

Acceptance test: An authorised user can determine how the displayed class average was calculated.

REP-004 — Defined Denominators

Every rate or percentage must use an explicitly defined eligible population.

Acceptance test: An assignment completion report excludes excused assignments when the configured policy excludes them from the denominator.

REP-005 — Average Method

Each average must use its defined calculation method.

Acceptance test: A report that uses the simple average of student percentages does not silently substitute a pooled-marks calculation.

REP-006 — Missing-Data Handling

The system must distinguish missing data from valid zero values.

Acceptance test: A student with no recorded assessment result is not assigned zero merely to complete a report.

REP-007 — Completeness Reporting

Reports must show relevant counts of included, missing, or invalid records where these affect interpretation.

Acceptance test: A class has 20 valid results and five missing results among 25 eligible students. The report displays both counts and 80% coverage.

REP-008 — Calculation Versioning

The system must preserve the relevant calculation or configuration version for reports where historical interpretation requires it.

Acceptance test: A school changes an applicable grading rule. A historical report retains sufficient information to explain the original calculation context.

REP-009 — Source Traceability

Important calculated measures must be traceable to the relevant source records.

Acceptance test: An authorised user investigates a class average and can identify the underlying results included in the calculation.

REP-010 — Consistent Rounding

The system must apply documented rounding rules consistently.

Acceptance test: The same percentage displayed on the dashboard and in the exported report uses the same defined rounding method.

24.2 Periods, Filters, and Comparisons

REP-011 — Academic Period Selection

The system must support the reporting periods configured for the relevant school or programme.

Acceptance test: A school with configured terms can generate a report for a selected term.

REP-012 — Date-Range Reporting

Where supported, reports must allow authorised users to select a valid date range.

Acceptance test: A user selects a date range and the report includes only records meeting the report's date rules.

REP-013 — Time-Zone Consistency

Date and time values must be interpreted using the relevant configured time zone.

Acceptance test: An assignment submitted shortly before midnight is displayed with the correct local date for the school.

REP-014 — Filter Composition

The system must apply supported filters consistently.

Acceptance test: A principal selects Grade 8, Mathematics, and one term. The report includes only records meeting all three conditions.

REP-015 — Empty Results

The interface must clearly explain when a valid filter combination produces no matching records.

Acceptance test: A user selects a subject with no recorded assessments during the period. The system displays a clear empty state rather than presenting an unexplained zero average.

REP-016 — Comparable Measures

The system must not silently compare incompatible reporting periods, assessment scales, or rubrics.

Acceptance test: Two skill ratings use incompatible rubric versions. The system identifies the comparability limitation rather than presenting an unexplained progress percentage.

REP-017 — Percentage-Point Changes

Where rates are compared, the system must distinguish percentage-point changes from relative percentage changes.

Acceptance test: Attendance increases from 70% to 80%. The report identifies the increase as 10 percentage points.

REP-018 — Group Composition

Where available and relevant, reports must disclose material differences in the eligible population or data coverage that could affect interpretation.

Acceptance test: A class average is based on 20 valid results out of 25 eligible students. The report identifies the five missing results.

24.3 Report Categories

REP-019 — Student Performance Reports

The system must provide an authorised student performance report.

Acceptance test: A permitted user generates a report showing the selected student's relevant academic and configured performance dimensions for a specified period.

REP-020 — Class Performance Reports

The system must provide class-level performance summaries.

Acceptance test: A principal generates a class report containing the configured measures and valid-result counts.

REP-021 — Subject Reports

The system must support reports focused on a selected subject.

Acceptance test: A teacher generates a report for an assigned subject and sees only authorised student records.

REP-022 — Attendance Reports

Attendance reports must use the configured attendance definitions and denominators.

Acceptance test: An attendance report displays the calculated rate and relevant eligible attendance count.

REP-023 — Assignment Reports

Assignment reports must follow the configured eligibility, verification, and completion rules.

Acceptance test: An assignment awaiting teacher verification is not counted as verified completion when the policy requires teacher confirmation.

REP-024 — Skill and Participation Reports

Reports must preserve the relevant rating context.

Acceptance test: A skill report identifies the applicable rubric and does not treat a missing rating as zero.

REP-025 — School Health Reports

School health reports must show the defined component indicators and their reporting context.

Acceptance test: A school has strong academic results and incomplete attendance data. The report displays both conditions separately.

REP-026 — Intervention Reports

Intervention reports must distinguish cases, actions, and recorded outcomes.

Acceptance test: A report shows the number of actions completed without claiming that the same number of educational concerns were resolved.

REP-027 — Organisation Reports

Organisation reports must summarise only authorised branches and applicable measures.

Acceptance test: An organisation administrator generates a branch comparison report that excludes unrelated organisations.

24.4 Visualisation and Usability

REP-028 — Appropriate Chart Selection

Charts must be selected according to the purpose of the measure.

Acceptance test: A report comparing category values uses a suitable chart and identifies the compared measure and period.

REP-029 — Chart Labels

Charts must have clear titles, labels, and relevant units.

Acceptance test: An attendance chart identifies the reporting periods and expresses attendance as a percentage.

REP-030 — Colour-Independent Status

Status information must not rely exclusively on colour.

Acceptance test: A user can distinguish a priority review status from a routine status using text or another non-colour cue.

REP-031 — Accessible Tables

Report tables must use meaningful headings and support the applicable accessibility requirements.

Acceptance test: A user navigating by keyboard can move through the report table and identify its column headings.

REP-032 — Missing and Unavailable States

The system must distinguish zero, missing, not applicable, and unavailable values where relevant.

Acceptance test: An unavailable attendance service does not display a valid zero attendance rate.

REP-033 — Report Context

Reports must identify the relevant period, scope, and calculation context.

Acceptance test: A downloaded report includes the selected school and reporting period.

24.5 Exporting and File Security

REP-034 — PDF Export

Where enabled, authorised users must be able to export supported reports to PDF.

Acceptance test: A permitted principal exports a class performance report to a readable PDF.

REP-035 — XLSX Export

Where enabled, authorised users must be able to export supported tabular reports to XLSX.

Acceptance test: An authorised user downloads an XLSX report with the expected rows, column headings, and data values.

REP-036 — CSV Export

Where enabled, the system must export approved tabular data to CSV using documented encoding and escaping rules.

Acceptance test: A report containing commas and quotation marks in a text field exports and imports without corrupting the row structure.

REP-037 — Export Scope

Exports must use server-enforced permissions and scope restrictions.

Acceptance test: A teacher attempts to export an entire school's records without permission. The server rejects the request.

REP-038 — Data Minimisation

Exports must contain only fields needed for the report's purpose.

Acceptance test: A class academic report does not include unrelated private contact information.

REP-039 — Export Audit

Sensitive exports must generate an appropriate audit record.

Acceptance test: An authorised user exports a sensitive report. The audit record identifies the user, report type, scope, time, format, and result.

REP-040 — Protected Download Access

Stored export files must not be publicly accessible without authorisation.

Acceptance test: An unrelated user attempts to retrieve an export file. The server rejects the request.

REP-041 — Temporary File Expiry

Where temporary export storage is used, files must follow the configured expiry policy.

Acceptance test: A temporary export reaches its expiry time. An unauthorised or expired download request is rejected.

REP-042 — Export Integrity

An export must contain a coherent set of records.

Acceptance test: Source records change during a large export. The resulting report uses the documented consistency strategy rather than combining incompatible data states without explanation.

24.6 Performance, History, and Security

REP-043 — Pagination

Large report tables must support appropriate pagination or an equivalent scalable loading strategy.

Acceptance test: A report contains more rows than the configured page size. The user can access the remaining records without loading the entire dataset into one page.

REP-044 — Large Export Processing

Where exports require background processing, the system must communicate the generation status.

Acceptance test: A large export is queued. The user can distinguish Pending, Processing, Completed, and Failed states.

REP-045 — Cache Invalidation

Cached reports must be refreshed or invalidated when relevant source data or configuration changes.

Acceptance test: A mark is corrected. The affected current report does not continue displaying an obsolete average beyond the documented refresh policy.

REP-046 — Report Snapshots

Where snapshots are supported, the system must preserve their creation time and relevant report context.

Acceptance test: An authorised user opens an archived report and can identify when it was created and which period it covers.

REP-047 — Historical Reproducibility

The system must preserve sufficient metadata to explain important historical reports.

Acceptance test: A historical ranking is reviewed after a configuration change. The system identifies the calculation rules used for that ranking.

REP-048 — Server-Side Authorisation

All protected report requests must enforce permissions on the server.

Acceptance test: A user modifies a report request to include an unauthorised student identifier. The server rejects the request.

REP-049 — Organisation Isolation

Reports must not include records belonging to unrelated organisations.

Acceptance test: An organisation administrator attempts to generate a report using another organisation's identifier. The server rejects the unauthorised request.

REP-050 — Parent and Student Scope

Student and parent reports must be restricted to the permitted student records.

Acceptance test: A parent requests a report for a student who is not linked to their account. The server rejects the request.

REP-051 — Report Auditability

Important report-generation and export actions must be traceable according to the configured audit policy.

Acceptance test: A sensitive report is generated and exported. The system records the required audit events.

REP-052 — Error Handling

The reporting system must communicate report-generation failures without exposing internal technical details or private data.

Acceptance test: A report-generation request fails. The user receives an understandable error message, while sensitive server details remain protected.

25. Recommended Development Order

The developer should implement the reporting system in the following order.

Stage 1: Identify the source records.

Confirm which academic, attendance, assignment, skill, participation, enrolment, and configuration records each report needs.

Stage 2: Define the measures.

Document formulas, eligible populations, exclusions, rounding, missing-data treatment, and comparison rules.

Stage 3: Build the shared calculation layer.

Implement the calculations once and reuse them across relevant reports and dashboards.

Stage 4: Build standard report definitions.

Begin with student, class, subject, and attendance reports.

Stage 5: Add completeness and data-quality reporting.

Make missing and invalid records visible.

Stage 6: Build school health and intervention reports.

Use the definitions established in Part 14 and other relevant modules.

Stage 7: Add filters and groupings.

Ensure that report parameters behave consistently and remain within the user's authorised scope.

Stage 8: Build standard visualisations.

Use tables and charts that serve a clear informational purpose.

Stage 9: Implement PDF, XLSX, and CSV exports where required.

Validate file contents, formatting, encoding, and permissions.

Stage 10: Add performance improvements.

Implement pagination, appropriate queries, and background export processing where needed.

Stage 11: Add auditability and historical context.

Preserve the information required to explain important reports.

Stage 12: Test security and consistency.

Compare dashboard values with exported values and test unauthorised access attempts.

Stage 13: Test realistic school scenarios.

Include missing marks, transfers, corrected results, incomplete registers, different academic calendars, and changing configurations.

26. Final Principles

Optimus Orien's reporting system should not merely produce attractive charts.

It should produce information that schools can understand, verify, compare, and use responsibly.

Every report should have a defined purpose.

Every percentage should have a defined denominator.

Every average should have a documented calculation method.

Every comparison should have a clear basis.

Every missing value should retain its actual meaning.

Every export should respect permissions.

Every important historical report should retain enough context to explain how its results were produced.

The system should also distinguish between a dashboard and a formal report, between raw records and calculated measures, and between observed changes and conclusions about why those changes occurred.

These distinctions are essential for a platform intended to support educational decisions.

The goal is not to generate the largest number of reports. The goal is to generate the smallest set of reliable reports that answers the most important questions clearly.

What Part 15 Establishes

Part 15 defines the reporting architecture, report definitions, calculation consistency, filters, comparison rules, data completeness, visualisation, export formats, report security, historical reproducibility, and 52 implementation requirements with acceptance tests.

Part 16 focuses on one of the most important foundations of Optimus Orien: **data quality and integrity**. It explains how the platform should prevent incorrect records, detect inconsistencies, manage corrections, validate imported information, and protect school decisions from misleading data.

PART 16

DATA QUALITY, VALIDATION, CORRECTIONS, IMPORTS, AND DATA INTEGRITY

Project: Optimus Orien
Document type: Educational-domain guide and software implementation specification
Audience: Product owner, software developer, and future product team
Primary objective: Ensure that the information stored by Optimus Orien is valid, consistent, traceable, and reliable enough to support educational reporting and decision-making.

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1. Why This Document Exists

Optimus Orien is designed to help schools understand student performance, monitor school health, identify patterns, and support early intervention.

However, every calculation the platform performs depends on the information it receives.

Imagine that a student actually scores 78 out of 100, but a teacher accidentally enters 87.

The platform may calculate the percentage correctly as 87%, display it on the student dashboard, include it in the class average, use it in a school performance report, and export it to a spreadsheet.

Every calculation could be mathematically correct while the final information is still wrong.

Now imagine that a student transfers to another school but remains enrolled in the previous school's class. The previous school may continue including the student in its attendance and performance calculations.

Or imagine that a school imports the same student twice. The platform may count one real student as two separate students, affecting enrolment totals, averages, attendance rates, and school comparisons.

These are not merely technical problems. They can change how schools understand their students and operations.

Data quality is therefore not a separate administrative feature. It is a foundation of the entire Optimus Orien platform.

This document explains what good data means, how errors occur, how the system should detect them, how corrections should work, and how developers should implement validation and data-integrity protections.

It also provides concrete software requirements and acceptance tests.

2. What Is Data Quality?

Data quality describes how well recorded information meets the requirements of its intended use.

A record can be technically valid but still unsuitable for a particular calculation.

For example, a mark of 75 out of 100 is mathematically valid.

However, if that mark belongs to the wrong student, it is not reliable for the student's performance report.

Similarly, a recorded attendance status may be valid in isolation but incorrect if the student was not enrolled in that school on the relevant date.

Data quality has several dimensions.

2.1 Accuracy

Accuracy means that a record correctly represents the real-world fact or event it is intended to represent.

Example: A student's recorded mark matches the mark awarded by the teacher.

A system can detect some possible accuracy problems, but it cannot automatically know whether every entered value is factually correct.

For example, a mark of 78 out of 100 is within the permitted range, but the system cannot establish from that fact alone whether the teacher awarded 78 or 87.

2.2 Completeness

Completeness means that the required information is present.

Example: Every student who was required to receive a result has an appropriate assessment record, or the missing result is explicitly identified.

Completeness does not mean that every optional field must always be filled in.

A student's middle name may be optional, while the student identifier and enrolment relationship may be essential.

Required fields should depend on the type of record and the operation being performed.

2.3 Validity

Validity means that information satisfies defined rules.

Examples include:

A mark cannot exceed the applicable maximum unless the assessment model explicitly supports another arrangement.

An assessment must belong to an appropriate subject or programme.

A date must follow the expected format.

A student identifier must reference an existing student record.

A skill rating must fall within the configured scale.

Validation checks whether a record follows the system's rules. It does not guarantee that the record is accurate.

2.4 Consistency

Consistency means that related records agree with one another according to the system's rules.

For example, a student should not simultaneously be recorded as enrolled in two mutually exclusive full-time class placements if the school's configuration prohibits this.

An assessment result should not reference one student while displaying another student's name.

A historical report should not use a grading configuration that conflicts with the configuration it claims to represent.

Some differences are legitimate, so consistency rules must reflect actual educational arrangements rather than assuming that every school operates identically.

2.5 Uniqueness

Uniqueness means that a record that should occur only once is not duplicated.

For example, importing the same assessment result twice should not create two independent results for the same student and assessment attempt unless the system explicitly supports multiple attempts.

A student may have multiple enrolments across their educational history, but the system should not accidentally create duplicate identities for the same person.

Uniqueness rules must reflect the type of record being stored.

2.6 Timeliness

Timeliness describes whether information is available when it is needed and whether it reflects an appropriate point in time.

A school may have accurate assessment records but enter them several weeks after the reporting deadline.

The records may still be accurate, but the report was incomplete when school leadership needed it.

The platform should distinguish delayed data entry from incorrect data.

2.7 Traceability

Traceability means that the system can identify where information came from and how it changed.

For example, the platform should be able to show that a mark was originally entered as 87, later corrected to 78, and changed by an authorised teacher.

Traceability is particularly important when reports are used in academic reviews, school comparisons, or formal records.

3. Data Quality Is Not the Same as Data Entry

It is tempting to think that data quality means asking teachers to enter information carefully.

Careful data entry matters, but it is not enough.

Errors can enter the system through:

Manual entry.

Spreadsheet imports.

Incorrect student selection.

Duplicate accounts.

Incorrect class assignments.

Misconfigured assessment rules.

Changes to academic calendars.

Incorrect subject mappings.

Software bugs.

Failed integrations.

Outdated cached results.

Incorrect calculations.

Unauthorised changes.

A reliable system must prevent some errors, detect others, and preserve enough evidence to investigate the rest.

The developer should therefore treat data quality as a combination of:

Prevention.

Validation.

Detection.

Correction.

Auditability.

Monitoring.

4. A Practical Example: One Error Affecting Multiple Reports

Suppose a teacher enters an incorrect assessment mark for one student.

The correct mark is 68 out of 100, but the entered mark is 86.

The system calculates the student's percentage as 86%.

It then includes that result in the subject average.

The subject average contributes to the class report.

The class result contributes to a school performance indicator.

The school indicator appears in an organisation-level comparison.

The incorrect value may now influence several screens and reports.

This illustrates an important concept: **errors can propagate through connected calculations**.

A reliable platform must identify the original record, determine which calculations depend on it, and refresh the affected results after correction.

It should not require an administrator to manually edit every report where the incorrect number appeared.

Calculated results should be derived from authoritative source records whenever practical, or refreshed using a documented recalculation process.

5. The Data Lifecycle

A useful way to understand data quality is to follow a record through its lifecycle.

5.1 Creation

A record is created through manual entry, an import, or an approved integration.

Example: A teacher enters a student's assessment result.

5.2 Validation

The system checks whether the record satisfies the applicable rules.

Example: The mark must be numeric and must not exceed the assessment's permitted maximum.

5.3 Storage

The valid record is saved and linked to the correct entities.

Example: The result is linked to the correct student, assessment, subject, class context, and reporting period.

5.4 Use

The record is used by dashboards, calculations, and reports.

Example: The result contributes to the student's assessment history and a defined class average.

5.5 Correction

If the record is later found to be incorrect, an authorised user corrects it through a controlled process.

5.6 Recalculation

The platform refreshes affected calculations and reports according to its defined rules.

5.7 Historical Traceability

The system retains the required history of the change.

This allows authorised users to understand what changed, when it changed, who changed it, and why.

The system should not treat every saved record as permanently correct merely because it passed the first validation check.

6. Validation: Preventing Incorrect Records Before They Are Saved

Validation is the process of checking information against defined rules.

Optimus Orien should perform validation at the point where data enters the system and, where appropriate, again before the data is used.

6.1 Required-Field Validation

Some records cannot be meaningfully created without particular information.

For example, an assessment result generally needs a student, an assessment, and a value or an explicitly supported result status.

The exact requirements depend on the record type.

The interface should identify missing required fields before submission where practical.

The server must enforce the same essential rules.

6.2 Type Validation

A field intended to store a date should not accept arbitrary text.

A field intended to store a numerical rating should not accept an unrelated string.

A field intended to store a unique identifier should follow its defined format.

Type validation prevents values from being stored in incompatible forms.

6.3 Range Validation

A range rule defines acceptable minimum and maximum values.

Examples include:

A skill rating must fall within the configured rating scale.

A percentage must follow its defined range.

An assessment mark must comply with the applicable assessment rules.

However, ranges should not be hardcoded where schools may configure different scales.

If a school uses a 1–10 skill-rating scale, the system should not assume that every educational programme uses that same scale.

6.4 Relationship Validation

A record must reference the correct related entities.

For example:

An assessment result must reference an existing student and assessment.

The assessment must be available within the relevant school or programme context.

The student must satisfy the applicable eligibility rules for that result.

A teacher should not be able to attach a mark to a student outside their authorised scope.

6.5 Date Validation

Dates must be checked against the relevant rules.

For example, an attendance record should not ordinarily refer to an unscheduled attendance opportunity.

An assessment result should not silently appear in a reporting period that does not match the assessment's defined context.

Historical corrections may be legitimate, so the system should distinguish an invalid date from an unusual date that requires review.

6.6 Cross-Field Validation

Sometimes individual fields are valid, but their combination is not.

For example, a record may contain:

Assessment maximum: 100.

Marks obtained: 120.

Both values are valid numbers, but the combination violates the ordinary assessment rule that the obtained marks cannot exceed the maximum.

Another example is a completed assignment with no required submission or verification evidence under a policy that requires both.

Cross-field validation checks relationships between fields.

6.7 Validation Messages

Errors should be understandable to the person entering the information.

Instead of displaying:

```
HTTP 400: constraint violation
```

the application might display:

"The entered mark is greater than the maximum allowed for this assessment. Check the mark and assessment maximum before saving."

The system should not expose database structure, secret identifiers, or internal technical details in user-facing messages.

7. Warnings Versus Errors

Not every unusual record should be blocked.

A reliable system must distinguish between a record that cannot be accepted and a record that may be valid but deserves attention.

7.1 Error

An error prevents the requested operation because the record violates a mandatory rule.

Examples:

A mark exceeds the permitted maximum.

A required student identifier does not exist.

A user attempts to change a record outside their permissions.

An import row has no way to identify the intended student.

The system should explain the error and how it can be resolved.

7.2 Warning

A warning identifies a possible problem but does not necessarily prevent saving.

Examples:

A mark is substantially different from the student's previous result.

A school has an unusually high number of missing attendance registers.

A teacher enters several identical marks.

A student appears to have overlapping enrolment records that may be legitimate but need checking.

The system should avoid blocking legitimate records solely because they are unusual.

7.3 Review Status

Some issues should be recorded for investigation.

For example, a report may show that several assessment results require verification.

The platform should record the issue, its status, and its resolution where the workflow supports this.

A review status must not automatically mean that the record is wrong.

7.4 Avoid Arbitrary Anomaly Rules

A result that differs greatly from a student's previous result may be correct.

The student may have improved, struggled, changed programmes, or taken a different assessment.

A warning can help a user review the record, but it should not automatically change the mark or label the student negatively.

Initial Optimus Orien data-quality checks should be transparent and rule-based. They do not require AI.

8. Duplicate Records

Duplicate records are a major source of errors in school information systems.

However, duplication must be understood in context.

8.1 Duplicate Student Identity

A school might accidentally create two student profiles for the same person.

One profile may contain the student's current class, while another contains historical assessment records.

The platform may then split the student's history across two identities.

This can create misleading dashboards and reports.

8.2 Duplicate Enrolment

A student may have duplicate enrolment records for the same school, class, and period.

This can inflate student counts or create contradictory attendance eligibility.

The system should prevent or flag duplicate enrolments according to the applicable rules.

8.3 Duplicate Assessment Results

An import may accidentally create the same assessment result twice.

If the reporting system counts both results, the average or assessment coverage may become incorrect.

The platform should define what uniquely identifies an assessment result, taking into account whether multiple attempts are allowed.

8.4 Duplicate Attendance Records

The same student may accidentally receive two attendance records for the same attendance opportunity.

The system should define the uniqueness rule for the configured attendance model.

Whole-day attendance and lesson-level attendance have different natural record keys.

8.5 Duplicate Detection Versus Automatic Merging

The system may identify two records that appear to describe the same person.

It should not automatically merge them merely because their names match.

Two different students can have the same name.

A student may also have spelling variations, changed names, or incomplete identity information.

The platform should use appropriate identifiers and an authorised review process for identity resolution.

9. Student Identity, Enrolment, and Data Integrity

Optimus Orien must distinguish a student's identity from their enrolment in a particular educational setting.

This principle was established in Part 4 and is essential for reliable records.

9.1 Student Identity

The student identity represents the individual within the platform's identity model.

It should have a stable internal identifier that does not change simply because the student changes schools or classes.

9.2 Enrolment

An enrolment represents the student's participation in a school, programme, year group, or class during a defined period.

A student may have multiple enrolments over time.

9.3 Why the Difference Matters

Suppose a student transfers from School A to School B.

The student's historical assessment records should not disappear.

School A should retain authorised access to the historical records it is permitted to maintain.

School B should receive only the information it is authorised to access.

Current attendance calculations should follow the relevant enrolment and attendance eligibility rules.

Historical school-level reports should not silently assign all of the student's past results to the new school.

9.4 Historical Relationships

Changes to a student's class, programme, or school should not rewrite the historical context of earlier records.

For example, a student who moves from Grade 7 to Grade 8 should retain the correct year-group context for their earlier Grade 7 records.

The platform should use explicit relationships and effective dates rather than assuming that the student's current class applies to every historical record.

10. Data Validation for Marks and Assessments

Assessment data deserves specific attention because it influences academic reports and comparisons.

10.1 Assessment Configuration

Before a result can be validated, the system needs to know the assessment's applicable rules.

These may include:

Assessment identity.

Subject.

Programme or class context.

Reporting period.

Maximum marks.

Permitted result statuses.

Whether multiple attempts are allowed.

Applicable grading rules.

The relevant configuration must be preserved for historical interpretation.

10.2 Marks Obtained and Maximum Marks

A result should preserve the distinction between marks obtained and maximum possible marks.

For example:

Marks obtained: 42.

Maximum marks: 60.

Percentage:

42 ÷ 60 × 100 = 70%.

The system should not assume that every assessment is out of 100.

10.3 Missing, Absent, Exempt, and Zero

These states must remain distinct.

A student may receive zero marks.

A student may be absent from an examination.

A student may be exempt from an assessment.

A result may be pending.

A mark may not yet have been entered.

The exact statuses depend on the educational arrangement.

The platform should not convert all of them to zero.

10.4 Correcting an Assessment Maximum

Suppose an assessment was mistakenly configured as out of 50 when it should have been out of 60.

Changing the maximum may affect percentages, grades, class averages, and historical reports.

The system should not silently change a completed assessment's calculation context without following the applicable correction and recalculation rules.

The correction should be authorised, recorded, and traceable.

10.5 Grade Boundaries

If the school uses grades derived from percentages, the platform must apply the configured grade boundaries.

For example, one configuration might assign a particular grade to percentages within a defined interval.

Another programme may use different boundaries or a different grading system.

The system must not hardcode one country's grading structure as universal.

11. Attendance Data Quality

Attendance records require different validation rules from assessment results.

11.1 Scheduled Attendance Opportunities

The system should determine whether an attendance record corresponds to a legitimate scheduled attendance opportunity.

Depending on the configured attendance model, that opportunity may be a school day, session, or lesson.

The system should not count an attendance record for a holiday or an unscheduled lesson as an ordinary scheduled opportunity.

11.2 Valid Attendance Statuses

Attendance statuses should follow the school's configured attendance codes.

The system should reject an unknown code unless an authorised mapping or configuration change makes it valid.

11.3 Duplicate Attendance

The system should prevent duplicate attendance records for the same student and attendance opportunity unless the configured model explicitly supports multiple events.

For lesson-level attendance, two different lessons on the same day are not duplicates.

For whole-day attendance, multiple records for the same student and school day may indicate a data-integrity problem.

11.4 Missing Registers

A missing attendance register should not be treated as proof that students were absent.

The system should identify the missing register and keep its data-completeness status separate from the attendance rate.

11.5 Corrections

If an attendance status changes from absent to present, the system should preserve the required audit history and update affected reports according to the defined recalculation process.

12. Assignment, Participation, and Skill Data Quality

Part 11 established that homework completion, participation, and subject-linked skills are different measures.

Their validation rules must respect those differences.

12.1 Assignment Records

The system should verify that an assignment belongs to the appropriate class, subject, reporting period, or other configured scope.

It should validate the assignment deadline and the student's eligibility.

If teacher verification is required, the system should distinguish student-declared completion from verified completion.

12.2 Assignment Status Conflicts

The platform should prevent contradictory status combinations.

For example, a record should not simultaneously be marked as not submitted and verified as completed unless the workflow explicitly defines a legitimate transition or exception.

A corrected status should preserve the relevant history.

12.3 Participation Ratings

Participation ratings should fall within the configured rating scale.

The system should identify the evaluator and relevant period where required.

A missing participation rating must not be converted to zero.

A warning that a rating is unusual should not automatically change it.

12.4 Subject-Linked Skill Ratings

Skill ratings should reference the applicable skill definition and rubric.

If a skill scale changes from one version to another, the platform should preserve the context of historical ratings.

A rating of 7 under one rubric should not automatically be treated as equivalent to a rating of 7 under every other rubric.

13. Data Imports: Why Schools Need Them

Schools may already have information stored in spreadsheets or existing software.

Entering every student, assessment, and attendance record manually may be impractical.

An import function allows users to transfer supported data into Optimus Orien.

For example, a school may upload a spreadsheet containing student identifiers, assessment marks, and reporting-period information.

However, importing data is not simply a matter of reading a spreadsheet and saving every row.

The platform must understand what each column means, which records each row refers to, how values should be interpreted, and what should happen when a problem is detected.

A safe import process should include:

File selection.

File validation.

Column mapping.

Record matching.

Data validation.

Duplicate detection.

Error reporting.

Import preview.

Confirmation.

Controlled processing.

A summary of the results.

Audit history.

14. The Import Workflow

A reliable import should proceed through clearly defined stages.

14.1 Stage One: Upload

The user selects a supported file format.

The server checks the file type, size, structure, and other applicable security constraints.

A file should not be trusted merely because its filename ends in .xlsx or .csv.

14.2 Stage Two: Column Mapping

The system determines which file columns correspond to which platform fields.

For example:

Student ID maps to the platform's approved student identifier.

Assessment Code maps to the assessment reference.

Marks Obtained maps to the result value.

Maximum Marks may be supplied only if the import workflow permits it and the value is consistent with the assessment configuration.

The mapping should be visible to the user.

The system should not silently guess ambiguous columns and import them as if the mapping were certain.

14.3 Stage Three: Record Matching

The system identifies the student, assessment, class, or other entities referenced by each row.

Stable identifiers are preferable to names alone.

For example, two students may share the same name.

If a row cannot be matched reliably, it should be flagged for resolution rather than assigned to an arbitrary student.

14.4 Stage Four: Validation

Each row is checked against the relevant rules.

Possible problems include:

Missing required fields.

Invalid dates.

Unknown student identifiers.

Unknown assessment identifiers.

Marks outside the permitted range.

Duplicate records.

Records outside the user's authorised scope.

Conflicting assessment configurations.

The system should identify errors before committing the affected records.

14.5 Stage Five: Preview

Before final submission, the user should see a summary of the proposed changes.

For example:

Rows detected: 500.

Rows ready to import: 470.

Rows with errors: 20.

Rows requiring review: 10.

These figures are illustrative. The actual counts must come from the import process.

The preview should allow the user to inspect errors and understand which records will be created or updated.

14.6 Stage Six: Confirmation

An authorised user confirms the import.

For sensitive or large imports, the platform may require additional confirmation or approval according to its policies.

The system should clearly state whether the operation will create new records, update existing records, or perform both actions.

14.7 Stage Seven: Processing

The system processes the validated rows.

It should use a documented strategy to prevent partially completed imports from leaving the database in an inconsistent state.

For a small import, a single database transaction may be appropriate.

For a large import, the system may process controlled batches with explicit progress, failure handling, and recovery rules.

14.8 Stage Eight: Results

The user receives a summary of the completed operation.

It should identify:

Rows processed.

Records created.

Records updated.

Rows skipped.

Rows rejected.

Rows requiring review.

Any processing failures.

The system should provide a way to inspect rejected rows and correct them without requiring the user to repeat the entire import unnecessarily.

15. Import Modes: Create, Update, and Upsert

The platform must define what an import does when a matching record already exists.

15.1 Create

Create mode adds new records.

If a record already exists and the operation would create a duplicate, the system should reject or flag that row.

15.2 Update

Update mode changes existing records.

The system must identify the exact record being updated and verify that the user has permission to change it.

15.3 Upsert

Upsert means updating an existing record when a reliable match is found and creating a new record when no match exists.

This can be useful, but it is risky if the matching rules are weak.

The platform should not perform an upsert based only on a student's name where that could identify the wrong person.

15.4 Recommended Initial Policy

For the first release, use clearly separated create and update operations wherever possible.

Support upsert only for record types with reliable matching identifiers and well-defined behaviour.

For every import mode, the user must know whether existing records may be changed.

16. Partial Success, Failed Imports, and Recovery

Large imports can contain both valid and invalid rows.

The platform must define how these cases are handled.

16.1 All-or-Nothing Import

An all-or-nothing import succeeds only if the entire operation satisfies its required conditions.

If one blocking error occurs, no records from that operation are committed.

This can be appropriate for small, tightly connected data changes.

16.2 Partial Import

A partial import processes valid rows while rejecting invalid ones.

For example, 470 valid rows may be imported while 30 rows are rejected or flagged.

This can be practical for large spreadsheets, but the user must receive an accurate summary of what happened.

16.3 Do Not Leave Hidden Partial Changes

The platform should never report that an import failed while silently leaving an unknown number of records changed.

If a large import uses batches, the system must record which batches succeeded and which failed.

The user must be able to identify the resulting state.

16.4 Retrying Failed Imports

A retry must not create duplicate records from rows that were already successfully processed.

The import process should use reliable record matching and, where appropriate, an import identifier or other mechanism to make retries safe.

The exact approach depends on the database and application architecture.

17. Import Security

Uploaded files can contain incorrect, sensitive, or malicious content.

The platform must validate them before processing.

17.1 File Restrictions

The server should enforce permitted formats, file-size limits, and other necessary constraints.

It should reject unsupported file types.

17.2 Malicious Content

The system should not execute macros, scripts, or embedded content from uploaded spreadsheets.

Imported values must be handled as data.

17.3 Spreadsheet Formula Injection

Some spreadsheet applications interpret text beginning with particular characters as a formula.

If untrusted values are exported into a spreadsheet, those values may be interpreted as formulas when opened.

The export process should neutralise dangerous cell values according to the chosen format and security policy.

17.4 Data Privacy

Imports may contain student names, identifiers, marks, attendance, and other sensitive information.

Files should be available only to authorised users.

Temporary files should follow the configured retention and deletion policies.

The system should avoid logging full student records or sensitive spreadsheet contents unnecessarily.

18. Data Corrections: Fixing Errors Without Losing History

No system can prevent every error.

A reliable platform must provide a controlled way to correct records.

18.1 Why Simple Overwriting Is Risky

Suppose a teacher changes a mark from 86 to 78.

If the platform simply replaces 86 with 78 and deletes every trace of the previous value, it becomes difficult to understand why a previously issued report showed 86.

The correction may be legitimate, but the system has lost the history needed to explain the difference.

For important records, corrections should preserve appropriate audit information.

18.2 Recommended Correction Information

Depending on the record type and applicable policy, a correction history may include:

The record identifier.

The previous value.

The corrected value.

The person making the change.

The date and time of the change.

The reason for the correction, where required.

The relevant approval, where required.

The resulting record status.

Not every field needs to be copied into every audit event. The system should record enough information to support traceability without collecting unnecessary sensitive information.

18.3 Correction Versus New Record

Some changes should update an existing record.

Others should create a new event or attempt.

For example, correcting a typing mistake in an assessment result is different from recording a student's second examination attempt.

The system must define which operation applies to each record type.

It should not treat every new result as a correction or every correction as a new assessment attempt.

18.4 Correction Permissions

The ability to view a record does not automatically grant permission to correct it.

For example, a teacher may be allowed to enter marks for assigned classes but not alter a finalised assessment belonging to another teacher.

The system should define correction permissions according to the role, record status, and authorised scope.

18.5 Approval for Sensitive Corrections

Some corrections may require approval.

Examples may include:

Changes to finalised results.

Changes to important school-level configuration.

Corrections that affect an already issued formal report.

The platform should support a configurable approval workflow where justified.

It should not require an administrator to approve every harmless correction if that would create unnecessary operational burden.

19. Recalculating Reports After Corrections

A corrected source record may affect multiple calculated values.

For example, changing a mark may affect:

The student's assessment percentage.

The student's subject average.

The class average.

A reporting-period average.

A school-level performance measure.

A comparison or ranking.

A school health indicator that depends on the changed measure.

The platform must identify the relevant dependencies and update the affected current calculations.

19.1 Recalculate From Source Data

Where practical, the system should calculate current measures from the authoritative source records.

This reduces the risk of accumulated inconsistencies.

For large datasets, calculated summaries may be stored for performance reasons, but their update rules must be explicit.

19.2 Avoid Double Application

Suppose a class average has already been calculated.

A correction occurs.

The system must not add the corrected value to the existing average without properly removing the superseded contribution.

The correct method is to recalculate the result or update it through a mathematically sound and tested procedure.

19.3 Recalculation Failure

If a correction is saved but a dependent summary fails to update, the system should not silently present an outdated value as current.

The application should have a reliable strategy for identifying and retrying failed recalculations.

Where immediate recalculation is not practical, the affected measure should be marked stale or temporarily unavailable according to the product's design.

19.4 Historical Reports

A current dashboard may reflect the corrected value.

An archived report may need to preserve the values originally issued.

The system should distinguish these cases rather than overwriting every historical report automatically.

20. Data Quality Monitoring

Validation at entry time is important, but some problems appear only when records are examined collectively.

For example, a school may have valid attendance records for individual days but an unusual number of missing registers across the academic term.

A class may have valid marks but a high percentage of missing assessment results.

An organisation may have several branches that have not completed required reporting.

These situations require data-quality monitoring.

20.1 Types of Data-Quality Checks

The platform may monitor:

Missing required records.

Duplicate identities or enrolments.

Invalid marks.

Unknown attendance codes.

Missing attendance registers.

Assessments without expected results.

Records linked to inactive or unrelated entities.

Unusual numbers of corrections.

Failed imports.

Reports generated from incomplete data.

Stale calculated summaries.

20.2 Completeness Indicators

A report may show the percentage of required records that have been entered.

For example:

Required records: 200.

Valid recorded records: 184.

Completeness: 92%.

The report should explain what counts as a required record and what makes a record valid.

20.3 Data-Quality Status

A report may show a status such as:

Complete.

Incomplete.

Requires review.

Unavailable.

These labels should have documented definitions.

A school should not be labelled as having poor educational performance merely because it has incomplete reporting data.

Data quality and educational performance are related but distinct dimensions.

20.4 Review Workflow

Some data-quality problems need a human decision.

For example, two student profiles may refer to the same person, but the evidence may be insufficient for automatic merging.

The system should allow authorised staff to review the issue, record the decision, and preserve the relevant history.

20.5 Rule-Based Monitoring Only

The initial implementation should use clear, configurable rules.

It does not need AI to identify a missing mark, an invalid date, a duplicate attendance entry, or a result exceeding the assessment maximum.

Rules should be documented so users can understand why a record was flagged.

21. Data Quality Across Organisations and Branches

Optimus Orien supports organisations that may manage multiple schools or branches.

Data-quality rules must respect this structure.

21.1 Local Validation

Some rules belong to a particular school.

For example, an attendance code may be valid in one school but not another.

A school may also use a different assessment scale or academic calendar.

21.2 Organisation-Wide Rules

Other rules may be defined centrally.

For example, an organisation may require every branch to complete a particular report by a specified date.

The system should distinguish central requirements from school-specific configurations.

21.3 Configuration Inheritance

An organisation may provide default settings for its branches.

However, the system should make it clear whether a setting is inherited, overridden, or independently configured.

An organisation administrator should not accidentally change historical calculations for every branch by modifying a current default.

21.4 Cross-Branch Data Integrity

The system must prevent records from being accidentally linked to the wrong branch.

For example, an attendance record belonging to Branch A should not contribute to Branch B's attendance rate.

If a student transfers between branches, the system should preserve the correct historical relationship.

21.5 Comparability

Even when two branches belong to the same organisation, their records may not be directly comparable.

They may use different programmes, academic calendars, or configured grading rules.

Reports should identify these differences rather than assume that shared ownership guarantees identical educational arrangements.

22. Imports, Exports, and Data Lineage

Data lineage means understanding where data originated and how it was transformed.

For Optimus Orien, this includes knowing whether a record was entered manually, imported from a spreadsheet, or created through an approved integration.

22.1 Source Information

For relevant records, the platform should preserve appropriate source information.

This might include:

Creation method.

Import job identifier.

Source system reference.

Creation time.

Responsible user or process.

Correction history.

Not every source field is required for every record.

The system should record enough information to support troubleshooting and accountability.

22.2 Import Mapping History

If a school imports a spreadsheet using a particular column mapping, the system should preserve the mapping or sufficient import metadata to explain how the file was processed.

This is particularly useful when a school later discovers that two spreadsheet columns were mapped incorrectly.

22.3 Reconciliation

Reconciliation means comparing information from two sources to identify differences.

For example, a school may compare the number of students in its existing system with the number of active students in Optimus Orien.

A reconciliation report may identify:

Records present in both systems.

Records missing from Optimus Orien.

Records present only in Optimus Orien.

Potential duplicates.

Unmatched identifiers.

The system should not automatically delete or overwrite records simply because the counts differ.

A difference may reflect timing, transfers, or different inclusion rules.

22.4 No Unauthorised Destructive Import

An imported spreadsheet should not be able to delete all existing records simply because the file omits them.

Deletion or mass deactivation should require an explicitly defined operation, appropriate permissions, validation, and confirmation.

The system should make the effect of such an operation clear before it is executed.

23. Data Retention, Deletion, and Privacy

Data integrity does not mean retaining every piece of information forever.

The platform should preserve records according to applicable educational, contractual, legal, and organisational requirements.

23.1 Retention Rules

Different record types may have different retention needs.

For example, a school may need to retain formal academic results longer than temporary import files.

The platform should support an appropriate retention policy rather than applying one arbitrary duration to every type of data.

23.2 Deletion Versus Deactivation

Some records should be deleted when permitted and appropriate.

Others may need to be deactivated or retained as historical records.

For example, a student who leaves a school should not ordinarily have their entire historical identity deleted merely because their current enrolment has ended.

The system should distinguish a person's identity, historical enrolment, current access, and retention obligations.

23.3 Audit History

Audit history can contain sensitive information.

It must have appropriate access controls and retention rules.

The platform should not assume that every audit event must be kept permanently.

23.4 Data Minimisation

The platform should collect and retain only information needed for defined purposes.

Import files should not remain available indefinitely without a reason.

Logs should not unnecessarily contain complete marks spreadsheets, personal contact information, or other sensitive records.

23.5 International Deployment

Schools in different countries may be subject to different privacy, education-record, and data-protection requirements.

The platform should be designed to support appropriate retention, access, correction, export, and deletion policies.

Legal obligations must be assessed for the jurisdictions in which the product operates.

A software setting alone does not guarantee legal compliance.

24. Recommended Data Integrity Architecture

The developer should implement the following responsibilities.

24.1 Input Layer

Receives manual entries, imported files, and approved external data.

24.2 Validation Layer

Checks field types, required information, ranges, relationships, and other applicable rules.

24.3 Authorisation Layer

Verifies that the user or process has permission to create or change the relevant record.

Authorisation must be enforced on the server.

24.4 Persistence Layer

Saves accepted records while preserving database constraints and transactional consistency.

A database constraint is a rule enforced by the database itself, such as preventing duplicate values for a field or combination of fields that must be unique.

24.5 Audit Layer

Records relevant changes and administrative operations according to the configured audit policy.

24.6 Calculation and Recalculation Layer

Updates affected derived measures when source records or applicable configurations change.

24.7 Monitoring Layer

Identifies incomplete, invalid, duplicate, stale, or otherwise questionable records.

These responsibilities may be implemented within the existing application architecture. They do not require separate services for every layer.

The important requirement is that the responsibilities exist and are tested.

25. Worked Example: Importing and Correcting Assessment Results

Consider a fictional school importing 500 Mathematics assessment results.

The import process identifies:

470 rows that satisfy all required checks.

15 rows with invalid student identifiers.

10 rows with marks exceeding the assessment maximum.

5 rows that appear to duplicate existing results.

These numbers are illustrative.

The system should not import all 500 rows blindly.

It should show the user the detected problems and apply the configured import policy.

Suppose the user corrects the invalid identifiers and marks, then reviews the potential duplicates.

After validation, the import proceeds according to the chosen processing mode.

Later, a teacher discovers that one imported mark should have been 68 rather than 86.

The teacher submits a correction.

The platform records the correction according to the applicable audit policy.

The student's current assessment percentage is recalculated.

The relevant subject and class averages are updated.

Affected current reports are refreshed.

An archived report that was issued before the correction is handled according to the school's historical-report policy.

The result is a controlled process in which the error can be corrected without losing the history needed to explain the change.

26. Software Implementation Requirements and Acceptance Tests

The following requirements define the expected behaviour of Optimus Orien's data-quality and integrity system.

26.1 Validation

DQ-001 — Required Fields

The system must validate required fields for each supported record type.

Acceptance test: A user attempts to save an assessment result without the required student reference. The server rejects the request and returns an understandable validation error.

DQ-002 — Type Validation

The system must reject values that do not match the field's expected data type.

Acceptance test: A user submits non-numeric text for a field that requires a numerical mark. The record is not saved.

DQ-003 — Configurable Ranges

Where scales are configurable, validation must use the applicable configuration.

Acceptance test: A school uses a configured 1–10 skill scale. A rating of 11 is rejected, while a rating of 8 is accepted.

DQ-004 — Assessment Maximums

The system must validate assessment marks against the applicable assessment rules.

Acceptance test: A result of 105 out of 100 is rejected unless the configured assessment model explicitly permits it.

DQ-005 — Relationship Validation

Records must reference valid related entities within the permitted context.

Acceptance test: A result references an assessment that does not exist. The server rejects the record.

DQ-006 — Enrolment Eligibility

The system must validate enrolment eligibility where the record type requires it.

Acceptance test: A user attempts to create an ordinary attendance record for a student who was not eligible for that school and attendance opportunity. The system rejects or flags the record according to the configured rules.

DQ-007 — Cross-Field Validation

The system must validate combinations of related fields.

Acceptance test: An assessment result contains marks obtained greater than the permitted maximum. The system rejects the record.

DQ-008 — Server-Side Validation

Essential validation must not depend solely on browser-side checks.

Acceptance test: A user bypasses the interface and sends an invalid request directly to the server. The server rejects the invalid data.

DQ-009 — Understandable Errors

Validation failures must produce understandable user-facing messages without exposing sensitive technical details.

Acceptance test: An invalid mark produces a message explaining the permitted range rather than displaying raw database error text.

DQ-010 — Warnings Versus Errors

The system must distinguish blocking errors from non-blocking warnings.

Acceptance test: A valid but unusual assessment mark produces a configured warning without being automatically changed.

26.2 Duplicate Prevention and Record Relationships

DQ-011 — Stable Student Identifiers

Student identities must have stable internal identifiers.

Acceptance test: A student's class changes. The student's internal identity remains unchanged.

DQ-012 — Duplicate Identity Review

Potential duplicate student identities must be identified without automatically merging uncertain matches.

Acceptance test: Two student profiles have the same name but different reliable identifiers. The system does not merge them automatically.

DQ-013 — Duplicate Enrolment Prevention

The system must prevent or flag enrolments that violate the applicable uniqueness rules.

Acceptance test: A duplicate enrolment is submitted for a student, class, and period where only one enrolment is permitted. The system rejects or flags it according to the defined rule.

DQ-014 — Assessment Result Uniqueness

The system must prevent duplicate results where the assessment model permits only one result for the relevant student and attempt.

Acceptance test: The same result is imported twice under a configuration allowing one result per student and attempt. The second row does not create a duplicate result.

DQ-015 — Attendance Uniqueness

The system must apply the configured uniqueness rules for attendance records.

Acceptance test: The same student and whole-day attendance opportunity are imported twice. The system identifies the duplicate.

DQ-016 — Multiple Assessment Attempts

The system must support multiple attempts only when the assessment configuration permits them.

Acceptance test: An assessment allows two attempts. The system stores the attempts separately and does not treat the second as an accidental duplicate.

DQ-017 — Historical Enrolment Context

Historical records must remain linked to the relevant historical enrolment context.

Acceptance test: A student transfers schools. Earlier attendance and assessment records retain their original school context.

DQ-018 — Referential Integrity

The system must prevent important records from referencing deleted or invalid related entities in ways that break the data model.

Acceptance test: An administrator attempts to delete an assessment that still has protected historical results. The system follows the defined restriction or archival policy rather than leaving broken references.

26.3 Assessment, Attendance, and Other Educational Data

DQ-019 — Missing Versus Zero

The system must distinguish a missing assessment result from a recorded zero.

Acceptance test: A student has no result entered. The report shows a missing result rather than zero marks.

DQ-020 — Assessment Statuses

The system must preserve the configured distinction between pending, absent, exempt, zero, and other supported result statuses.

Acceptance test: An exempt student is not automatically assigned zero marks.

DQ-021 — Assessment Configuration History

The system must preserve the relevant assessment configuration for historical interpretation.

Acceptance test: An assessment's maximum marks are corrected. The system preserves the correction history and follows the defined recalculation policy.

DQ-022 — Attendance Code Validation

Attendance records must use valid configured attendance codes.

Acceptance test: An import contains an unrecognised attendance code. The row is rejected or flagged according to the import rules.

DQ-023 — Missing Register Distinction

The system must distinguish a missing attendance register from recorded absence.

Acceptance test: A register has not been submitted. The report identifies missing attendance data rather than counting all students as absent.

DQ-024 — Assignment Status Consistency

The system must prevent contradictory assignment statuses that violate the configured workflow.

Acceptance test: A record is marked both not submitted and verified completed without an allowed exception. The system rejects or flags the conflict.

DQ-025 — Skill Scale Validation

Skill ratings must comply with the applicable scale and rubric.

Acceptance test: A skill uses a configured 1–10 scale. A rating of 0 is rejected unless the scale explicitly permits it.

DQ-026 — Historical Rubric Context

Historical skill ratings must preserve the relevant rubric version.

Acceptance test: A school changes a skill rubric. Earlier ratings remain linked to the rubric used when they were recorded.

26.4 Import Processing

DQ-027 — Supported File Types

The system must accept only explicitly supported import formats.

Acceptance test: A user uploads an unsupported file type. The system rejects it with an understandable message.

DQ-028 — File Size Limits

The server must enforce configured file-size limits.

Acceptance test: A file exceeds the permitted size. The server rejects it before processing the data.

DQ-029 — Column Mapping

The system must identify and validate import column mappings.

Acceptance test: A user uploads a spreadsheet with an ambiguous student identifier column. The system requires a valid mapping before processing.

DQ-030 — Reliable Record Matching

Imports must use reliable matching rules for existing records.

Acceptance test: Two students share a name. The system does not assign a result to one of them solely because their names match.

DQ-031 — Import Preview

The system must preview proposed changes before confirmation where the import workflow requires it.

Acceptance test: A spreadsheet contains valid and invalid rows. The user can inspect the counts and relevant errors before confirming the import.

DQ-032 — Import Mode

The system must make clear whether an import creates records, updates records, or supports both operations.

Acceptance test: Before an update import, the user can identify that matching existing records may be changed.

DQ-033 — Duplicate Import Handling

Imports must detect duplicate records according to the relevant record rules.

Acceptance test: A spreadsheet contains the same assessment result twice. The system identifies the duplicate before it is silently counted twice.

DQ-034 — Import Summary

The system must report the outcome of each completed import.

Acceptance test: After processing, the user can see the number of records created, updated, skipped, rejected, and flagged where applicable.

DQ-035 — Partial Import Visibility

If partial imports are supported, the system must clearly identify which rows succeeded and which failed.

Acceptance test: Some rows succeed while others fail validation. The import summary accurately reflects the final state.

DQ-036 — Safe Retry

Retrying an import must not accidentally duplicate previously processed records.

Acceptance test: A user retries an import after a processing interruption. Successfully processed rows are not created a second time.

DQ-037 — Import Recovery

The system must have a documented recovery strategy for failed or interrupted imports.

Acceptance test: An import fails during batch processing. The system identifies the processed portion and does not report an unknown final state.

DQ-038 — Destructive Import Protection

An import must not delete or deactivate unrelated records unless the operation explicitly supports this and the user is authorised.

Acceptance test: A spreadsheet omits existing students. Importing it does not automatically delete those students.

DQ-039 — Import Auditability

Important import operations must be traceable.

Acceptance test: An authorised user reviews an import record and can identify the responsible user, time, operation, and processing outcome.

DQ-040 — Uploaded File Security

Uploaded files must be validated and processed safely.

Acceptance test: A file with a misleading extension or unsupported content is rejected, and embedded spreadsheet macros are not executed.

26.5 Corrections and Recalculation

DQ-041 — Controlled Corrections

Important records must be corrected only by authorised users.

Acceptance test: A teacher attempts to correct a finalised result outside their permitted scope. The server rejects the request.

DQ-042 — Correction History

The system must preserve the required history of important record changes.

Acceptance test: A mark changes from 86 to 78. An authorised reviewer can identify the previous value, corrected value, responsible user, and change time.

DQ-043 — Correction Reason

Where the configured policy requires a correction reason, the system must collect it.

Acceptance test: A finalised result requires a reason for correction. The system prevents submission until a valid reason is supplied.

DQ-044 — Approval Workflow

Where configured, sensitive corrections must receive the required approval.

Acceptance test: A finalised result requires principal approval. The corrected result does not become final until the required approval is recorded.

DQ-045 — Recalculation Dependencies

The system must update affected calculated measures when source data changes.

Acceptance test: A mark is corrected. The student's relevant percentage and affected class average are updated according to the defined recalculation process.

DQ-046 — No Double Counting During Recalculation

Corrections must not cause a source result to be counted twice.

Acceptance test: A mark changes from 86 to 78. The class average uses the corrected result once and does not count both values.

DQ-047 — Recalculation Failure Handling

The system must detect and handle failed recalculation jobs.

Acceptance test: A dependent summary cannot be refreshed. The system identifies the failure and follows the defined retry or stale-data procedure.

DQ-048 — Historical Report Policy

The system must follow the defined policy for archived reports after corrections.

Acceptance test: A mark is corrected after a report has been issued. The current report reflects the corrected data, while the archived report is handled according to its retention and snapshot policy.

26.6 Monitoring, Privacy, and Integrity

DQ-049 — Completeness Monitoring

The system must identify relevant missing required records.

Acceptance test: A required assessment result is missing. The relevant completeness report identifies the missing record.

DQ-050 — Data-Quality Status Definitions

Data-quality statuses must have documented meanings.

Acceptance test: A school is marked incomplete because attendance registers are missing. The report identifies the missing records and does not label the school academically unsuccessful for that reason alone.

DQ-051 — Reviewable Data Issues

The system must support an appropriate review process for data-quality issues that require human judgement.

Acceptance test: Two student profiles may be duplicates. An authorised user can review the issue and record the decision without automatic merging.

DQ-052 — Branch Isolation

Data-quality checks and imports must respect organisation and branch boundaries.

Acceptance test: An administrator imports records into Branch A. The records do not accidentally become associated with Branch B.

DQ-053 — Configuration History

Relevant configuration changes must be traceable where they affect historical data or calculations.

Acceptance test: A school changes an attendance rule. An authorised reviewer can identify the applicable rule version for the relevant report.

DQ-054 — Data Minimisation

Import files, logs, and audit records must not retain unnecessary sensitive information.

Acceptance test: A failed import is logged. The log records the necessary error details without unnecessarily reproducing the entire student spreadsheet.

DQ-055 — Retention Policy

Records and temporary files must follow their applicable retention rules.

Acceptance test: A temporary import file reaches its configured retention limit. It is no longer available through ordinary application access.

DQ-056 — Server-Side Scope Enforcement

Data-quality and import operations must enforce user permissions on the server.

Acceptance test: A user modifies an import request to reference another organisation. The server rejects the unauthorised operation.

DQ-057 — Audit Protection

Ordinary users must not be able to alter or erase protected audit history outside authorised administrative processes.

Acceptance test: A teacher attempts to delete the audit record of a corrected mark. The system denies the operation.

DQ-058 — Data Reconciliation

Where reconciliation is supported, the system must display differences without automatically treating every difference as an error requiring deletion.

Acceptance test: A school has 500 students in its existing system and 490 in Optimus Orien. The reconciliation report identifies the discrepancy and permits authorised review.

DQ-059 — Stale Calculation Detection

The system must identify important calculated results that have not been refreshed after relevant source changes.

Acceptance test: A mark is corrected but the dependent summary has not yet refreshed. The system follows the documented stale-data handling rule instead of silently claiming that the summary is current.

DQ-060 — Data Integrity Testing

The implementation must include automated tests for important data constraints and relationships.

Acceptance test: Automated tests verify that duplicate protected records, invalid relationships, and unauthorised cross-organisation operations are rejected.

27. Recommended Development Order

The developer should build these capabilities in a controlled sequence.

Stage 1: Define the source data model.

Confirm the identities, enrolments, assessment records, attendance records, assignments, skill ratings, and configuration relationships.

Stage 2: Define validation rules.

Document required fields, permitted values, relationships, uniqueness rules, and valid record states.

Stage 3: Enforce validation on the server.

Do not rely only on frontend checks.

Stage 4: Add database constraints.

Use appropriate database-level protections for uniqueness and referential integrity.

Stage 5: Implement controlled corrections.

Preserve relevant history and enforce correction permissions.

Stage 6: Implement recalculation handling.

Identify which measures depend on changed source records and ensure they refresh reliably.

Stage 7: Build the import preview and validation process.

Start with a small number of supported file formats and record types.

Stage 8: Add import processing and recovery.

Make partial success, failure, and retries understandable and safe.

Stage 9: Add data-quality reports.

Show missing records, invalid data, duplicate candidates, and other important issues.

Stage 10: Add reconciliation where required.

Help schools identify differences between existing records and Optimus Orien.

Stage 11: Test security and privacy.

Test role restrictions, organisation isolation, protected files, and audit controls.

Stage 12: Test realistic school scenarios.

Include transfers, corrected marks, incomplete registers, duplicate imports, changing assessment configurations, and historical reports.

28. Final Principles

Optimus Orien should never assume that data is correct simply because it exists in the database.

It should validate records before accepting them, identify questionable patterns, prevent duplicates where appropriate, and preserve the relationships needed to interpret historical information.

When an error is discovered, authorised users should be able to correct it without losing the history needed to explain the change.

When source data changes, dependent calculations should be refreshed reliably.

When a school imports a spreadsheet, the system should validate and preview the proposed changes instead of blindly accepting them.

When information is missing, the platform should say that it is missing rather than inventing a zero.

When two records appear to be duplicates, the platform should investigate according to defined rules rather than merging them recklessly.

When a report is incomplete, the system should disclose the limitation.

When a user attempts to change data outside their permissions, the server should reject the request.

The central principle is this: Optimus Orien must preserve not only the values stored in its database, but also the relationships, history, and context that make those values meaningful.

What Part 16 Establishes

Part 16 defines the foundations of data quality, validation, duplicate prevention, student identity and enrolment integrity, educational-record validation, imports,

corrections, recalculation, data-quality monitoring, data lineage, privacy, and retention.

It also establishes 60 implementation requirements with acceptance tests.

Part 17

Notifications, Alerts, Reminders, and Communication Workflows in Optimus Orien

1. Purpose of This Document

Optimus Orien collects and analyses school information. However, collecting information is not enough. The people responsible for acting on that information must also know when something requires their attention.

For example, a teacher may forget to submit an attendance register. A principal may need to review a student whose attendance has declined. A school administrator may need to correct incomplete records before the end-of-term report is generated.

If Optimus Orien only displays these situations on dashboards, users may overlook them. Notifications can bring important information to their attention.

However, notifications must be designed carefully. Sending too many alerts can make people ignore all of them. Sending sensitive information to the wrong person can violate student privacy. Sending an alert without explaining what the recipient should do can create confusion rather than solve a problem.

This document explains how notifications, alerts, reminders, and communication workflows should work in Optimus Orien. It also defines the requirements that the developer must implement.

The system's initial version should use understandable, rule-based logic. It does not need artificial intelligence to decide which notifications to send.

2. Understanding Notifications, Alerts, Reminders, and Messages

These four concepts are related, but they are not identical.

2.1 Notification

A notification informs a user that something relevant has happened or that information requiring their attention is available.

For example, a principal might receive a notification saying that three attendance registers have not been submitted.

The notification should explain what happened, when it happened, why it matters, and what the user can do next.

2.2 Alert

An alert communicates a condition that may require attention or action.

For example, a student's attendance rate may fall below a threshold configured by the school.

The alert should not automatically conclude that the student is irresponsible, that the teacher has failed, or that the family has neglected its responsibilities. It should communicate the observable condition and, where appropriate, recommend a human review.

An alert is therefore a signal, not a final judgement.

2.3 Reminder

A reminder tells someone that an expected action is approaching or has not yet been completed.

For example, a teacher might receive a reminder that attendance registers must be submitted by 3:00 PM.

Reminders are often time-based. Alerts are usually condition-based, although some alerts may also depend on deadlines.

2.4 Message

A message contains communication written by one person for another person or group.

For example, a principal might send a message to teachers explaining a change in the school's assessment schedule.

A notification can inform someone that a message exists, but the notification itself is not necessarily the message.

These distinctions matter because each feature requires different permissions, data structures, and rules.

Initial product decision: Optimus Orien should support system-generated notifications, condition-based alerts, scheduled reminders, and basic administrative announcements where approved. A full in-app messaging or chat system is outside the initial scope.

3. A Simple Example

Imagine that a school has 500 students and 25 teachers.

At the end of the school day, three teachers have not submitted their attendance registers.

Optimus Orien detects the missing submissions according to the school's configured attendance schedule.

The system creates a record of the problem. It then sends an appropriate reminder to each teacher responsible for an incomplete register.

If the deadline passes, the system may notify the principal or authorised school administrator, depending on the school's escalation policy.

Once a teacher submits the register, the reminder should no longer be treated as an outstanding task. If an alert has already been sent, the system should update its status or mark the underlying condition as resolved.

The principal should be able to see the number of outstanding registers without receiving a separate notification for every minor update.

This example illustrates an important principle: **a notification should be connected to a real condition or task, and its lifecycle should reflect what happens afterward.**

4. The Difference Between an Event, a Condition, and a Notification

The developer must distinguish three concepts.

An **event** is something that happened. For example, a teacher submitted an attendance register.

A **condition** is a situation that exists or becomes true. For example, an attendance register remains unsubmitted after its deadline.

A **notification** is the communication generated because an event occurred or a condition requires attention.

Consider the following sequence.

At 2:00 PM, a teacher creates an attendance register. At 3:00 PM, the submission deadline passes. The register is still incomplete. Optimus Orien identifies the outstanding condition and creates a reminder.

At 3:15 PM, the teacher submits the register. The condition is no longer outstanding.

The notification history may still show that the reminder was sent, because the historical event really happened. However, the associated task must now be marked as completed or resolved.

The system must not delete the history simply because the problem was fixed.

This distinction is important because a school may need to investigate repeated delays, understand whether reminders are useful, or determine whether deadlines are realistic.

5. The Main Types of Notifications Optimus Orien Needs

Notifications should be introduced according to their usefulness and the maturity of the relevant feature.

5.1 Attendance Notifications

Attendance notifications can relate to missing registers, late submissions, attendance corrections, or attendance patterns requiring review.

For example, a teacher might be reminded to complete a register for a scheduled class. A principal might receive a summary of outstanding registers across the school.

An authorised staff member might also be notified when a student's attendance falls below a configured review threshold.

The system must distinguish between missing attendance data and actual student absence. An unsubmitted register does not prove that students were absent.

Likewise, an absence recorded in a submitted register is different from a register that has not been submitted.

The notification must reflect the actual situation.

5.2 Academic Performance Notifications

These notifications relate to marks, assessment records, and academic performance.

Examples include an assessment deadline approaching, an assessment's marks remaining incomplete, or a student's recorded performance meeting a school's configured review criteria.

The system must not send a warning merely because a student's marks are low if the assessment data is incomplete or the relevant comparison is invalid.

For example, a student who transferred into a school halfway through a term may have no results for assessments that took place before enrolment. The absence of those results must not automatically be treated as academic failure.

Where an alert concerns an individual student, it must only reach authorised recipients.

5.3 Homework Notifications

Homework notifications may remind teachers to verify student submissions or notify authorised users about overdue assignments.

Suppose a student marks an assignment as submitted, but the teacher has not yet verified its completion.

Optimus Orien may remind the teacher to review the submission if the relevant deadline or reminder rule has been reached.

The system must not incorrectly report the assignment as uncompleted simply because teacher verification is pending.

It must also distinguish between an assignment that was never submitted and one that was submitted but is awaiting verification.

5.4 Participation and Skill-Assessment Notifications

These notifications may remind teachers to complete scheduled participation or subject-linked skill assessments.

For example, if a school's configuration requires teachers to review certain skill ratings at the end of each term, the system may remind the responsible teacher before the deadline.

The notification should identify the relevant class, subject, reporting period, and incomplete task.

It should not imply that a student lacks a skill simply because the teacher has not entered a rating.

5.5 School Health and Early-Intervention Notifications

Optimus Orien can notify authorised staff when a transparent, configured rule identifies a situation requiring review.

For example, a school might define a rule that a student with an attendance rate below 80% during a specified reporting period should be reviewed.

The value of 80% is only an illustrative example. It must not become a universal threshold hardcoded into the software.

The actual threshold, reporting period, eligible attendance opportunities, and calculation method must be configurable.

The notification should explain the condition that triggered it and provide access to the supporting information for authorised users.

It must not claim that the student will fail, diagnose a problem, or assert that a particular intervention is guaranteed to work.

5.6 Data-Quality Notifications

Data-quality notifications inform users about missing, inconsistent, or potentially incorrect records.

Examples include an assessment containing marks greater than the maximum possible marks, a register with unresolved errors, or an import that failed validation.

These notifications are particularly important because inaccurate records can affect dashboards, comparisons, and reports.

A data-quality notification should explain what needs correction, who is authorised to correct it, and whether the affected information can currently be used in reporting.

5.7 Administrative and System Notifications

Administrative notifications may include school configuration changes, user invitations, account changes, or the availability of a completed export.

The system should only introduce notifications for actions that are actually supported by the implemented product.

For example, it should not notify users that an artificial intelligence analysis is ready because AI analysis is not part of the current development scope.

6. Who Should Receive Each Notification?

A notification must reach the people who can appropriately respond to it.

Optimus Orien has several user roles, and their access must remain separate.

6.1 Organisation Administrator

An organisation administrator may oversee multiple schools or branches.

This user may receive organisation-level notifications about incomplete data, unresolved school-level issues, or other conditions within their authorised scope.

The administrator should be able to see which school requires attention without automatically receiving unrestricted access to every student's private records.

Access to detailed information must follow the organisation's approved permissions.

6.2 School Administrator

A school administrator may receive notifications about operational tasks such as incomplete registers, missing assessment records, unresolved data-quality issues, or approaching reporting deadlines.

The administrator should only receive information within the permissions granted to their role.

6.3 Principal

A principal may receive school-level summaries and alerts requiring leadership review.

For example, the principal may see that five attendance registers remain incomplete or that several student review cases require follow-up.

A summary notification should normally be preferred over many separate notifications for the same underlying problem.

6.4 Teacher

A teacher may receive notifications about assigned classes, subjects, students, attendance registers, assignments, or skill assessments.

A teacher must not receive notifications about another teacher's private records unless an explicit permission allows it.

The system should determine the recipient using current teaching assignments and permissions, not simply because the person is employed by the same school.

6.5 Student

Students may receive appropriate notifications about their own records or tasks, depending on the school's policy.

For example, an authorised student-facing notification might remind a student about an upcoming assignment deadline.

Student notifications should use clear, respectful language. They should not publicly label students as weak, problematic, or at risk.

6.6 Parent or Guardian

A parent or guardian may receive permitted notifications about a linked student's records.

The system must verify the relationship between the parent account and the student before disclosing any information.

A parent linked to one child must not automatically gain access to another student's information.

The school must also configure which categories of student information may be communicated to parents and under what circumstances.

6.7 Future Government or Oversight Users

A future government or oversight portal may require aggregated notifications about missing data or school-level patterns.

However, this role must be designed separately. Government-level access must not be assumed to include every student's identifiable records.

Government notification workflows should only be implemented after their responsibilities, permissions, legal basis, and reporting requirements have been explicitly defined.

7. The Notification Lifecycle

Every system-generated notification should have a clear lifecycle.

A recommended lifecycle is:

Condition detected → Notification created → Notification queued → Delivery attempted → Notification delivered or failed → User opens or acknowledges it → Underlying task resolved or condition cleared.

Not every notification needs every state.

For example, a notification about a completed report may simply be delivered and read. An alert about an incomplete register may also be associated with an outstanding task.

The developer should distinguish between the notification's status and the status of the underlying issue.

A notification may have been delivered successfully even though the issue remains unresolved.

Likewise, a user may read a notification without completing the required action.

For this reason, the system must not treat "read" as equivalent to "resolved."

Recommended notification statuses include:

-

Created

-
-

Queued

-
-

Sent

-
-

Delivered, where the delivery channel supports confirmation

-
-

Failed

-
-

Read

-
-

Acknowledged, where acknowledgement is required

-
-

Cancelled, where appropriate

-

These statuses are not all interchangeable. For example, a successful email handoff does not necessarily prove that the recipient read the email.

For issue tracking, separate statuses may include:

-

Open

-
-

In progress

-
-

Resolved

-
-

Dismissed with an authorised reason

-
-

Reopened

-

Not every notification needs a separate issue record. However, alerts connected to ongoing review cases or tasks should reference the relevant record rather than creating unrelated duplicate issues.

8. In-App Notifications

An in-app notification is displayed within Optimus Orien when a user is signed in.

A suitable implementation includes a notification icon in the application header. A number or badge may indicate unread notifications.

Selecting the icon should open a notification panel or page.

Each notification should display the information necessary for the user to understand it, including:

-

A short title.

-
-

A clear description.

-
-

The date and time.

-
-

The notification category.

-
-

The relevant school, class, subject, or reporting period, where applicable.

-
-

The priority, if the notification has one.

-
-

Its current status.

-
-

An action button or link, where an appropriate action exists.

-

For example, a teacher might see:

Title: Attendance register requires submission

Description: The attendance register for Class 7A on 19 October remains incomplete.

Action: Open attendance register

The example must only appear when the register actually exists and is associated with that teacher's authorised responsibilities.

The action button should open the relevant record directly, provided the user still has permission to access it.

If the user's access has been revoked, the system must deny access even if the user still has the notification.

8.1 Unread Counts

Unread counts should be calculated consistently.

If a user marks a notification as read on one device, the unread count should update on their other active sessions when synchronisation occurs.

Opening a notification should not automatically resolve its underlying issue.

8.2 Notification History

Users should be able to review recent notifications that remain relevant to their role.

The history should preserve important information about previously sent notifications according to the system's retention policy.

The developer must distinguish between archiving a notification from a user's normal view and deleting the underlying audit record.

Where retention rules require records to be deleted, that process must follow the approved privacy and retention policy.

9. Email, SMS, Push Notifications, and Other Channels

Notifications can be delivered through different channels. Each channel has different costs, privacy risks, technical requirements, and delivery limitations.

9.1 In-App Delivery

In-app delivery should be the first supported channel.

It keeps the user inside the existing platform and avoids requiring the school to configure an external messaging provider before using the notification system.

However, in-app notifications cannot reach a user who never signs in. Additional channels may therefore be useful later.

9.2 Email

Email may be useful for invitations, important administrative announcements, selected reminders, and notices that do not require the recipient to sign in immediately.

Email should not automatically include sensitive student information.

A safer approach is to send a minimal message explaining that an authorised user should sign in to review an item.

For example:

"You have a new student-performance review requiring your attention. Sign in to Optimus Orien to view the details."

The system must verify the recipient's address and the user's current permissions before including any additional information.

Email delivery should use a trusted provider, secure transport, and appropriate failure handling.

9.3 SMS

Short Message Service (SMS) can be useful where recipients have limited internet access or where a school needs to deliver a brief reminder.

However, SMS may expose information on shared devices or lock screens. Messages can also be delayed, and sending them may incur charges.

The initial version should not send sensitive student details through SMS. If SMS is introduced later, the school should explicitly enable it, define its permitted uses, and configure any relevant costs and consent requirements.

9.4 Push Notifications

Push notifications are messages delivered through a supported browser or mobile operating system, often while the application is not open.

They may be useful if Optimus Orien later offers a suitable progressive web application or native mobile application.

Push notifications introduce additional permission, subscription, device-management, and privacy considerations.

They should not be assumed to be available simply because the web application displays in-app notifications.

9.5 Recommended Initial Decision

Build the notification system so that it can support multiple channels in the future, but begin with in-app notifications.

Email may be added for carefully selected workflows once the required email provider, configuration, security, and delivery monitoring are implemented.

SMS, mobile push, and external messaging integrations should remain separately scoped.

10. Notification Rules

A notification rule defines the conditions under which the system should create a notification.

For example, a rule might state:

"When an assigned teacher has not submitted an attendance register by the configured deadline, create a reminder for that teacher."

A different rule might state:

"When a student's attendance rate falls below the school's configured review threshold for the selected reporting period, create a review alert for authorised staff."

Rules should be understandable by school administrators. The system must not require them to understand programming.

10.1 Components of a Rule

A configurable notification rule should define the event or condition, the scope in which it applies, the recipients, the timing, the priority, and any restrictions on repeated notifications.

It should also define what happens when the underlying condition is resolved.

Where a rule depends on a calculated indicator, it must reference the same approved calculation used by the relevant dashboard or report.

For example, the attendance threshold rule must not calculate attendance differently from the attendance dashboard.

10.2 Configurable Thresholds

Thresholds must be configurable within permitted limits.

A school might set a review threshold of 80% attendance for a particular period, but another school may adopt a different policy.

The platform should allow authorised users to configure relevant thresholds without changing the underlying software.

Changes must be validated, recorded, and versioned.

The system should also prevent a user from creating a rule that they are not authorised to manage.

10.3 Rule Scope

Rules may apply at different levels:

-

Organisation-wide.

-
-

School or branch.

-
-

Programme or educational pathway.

-
-

Year group or grade.

-
-

Class or subject.

-
-

Individual workflow, where permitted.

-

A rule inherited from an organisation should not silently override a school-specific configuration when the system is designed to permit local settings.

The system must clearly identify which rule is active and where it came from.

10.4 Rule Changes

When a rule changes, the platform must preserve enough history to explain why a notification was created.

For example, suppose the attendance review threshold changes from 80% to 75%.

The system should not silently rewrite the historical explanation for an alert that was created under the 80% rule.

Each generated alert should retain its relevant rule version, calculation period, trigger information, and creation time.

11. Preventing Notification Overload

Notification overload occurs when users receive so many notifications that they stop paying attention to them.

This is a serious usability problem for school platforms.

If a principal receives 150 separate notifications every morning, important issues may become difficult to identify.

Optimus Orien should therefore use controlled, relevant, and grouped notifications.

11.1 Group Similar Notifications

Suppose 12 teachers have incomplete attendance registers.

Instead of sending the principal 12 separate notifications, the system may create a summary:

"12 attendance registers remain incomplete across the school."

The principal can open the summary to view the individual registers they are authorised to access.

Each teacher can still receive their own relevant reminder.

Grouping should not expose records outside the recipient's permissions.

11.2 Avoid Repeated Notifications for the Same Condition

A persistent problem should not create a new notification every time a dashboard refreshes.

For example, if an attendance register remains incomplete for four hours, the system should not generate hundreds of alerts because the underlying condition is checked repeatedly.

The rule should define a reasonable reminder interval or maximum frequency.

A notification may be repeated after a meaningful event, such as a missed deadline or an approved escalation interval, but repetition must be controlled.

11.3 Deduplication

Deduplication means recognising that two attempts to generate a notification relate to the same underlying event or condition.

For example, a background process might retry after a temporary network failure. The retry should not create a second logical notification if the original notification record already exists.

The developer should use a stable identifier or idempotency mechanism to prevent duplicate notifications.

Idempotency means that repeating the same operation does not create unintended additional effects.

11.4 Digest Notifications

A digest combines several related notifications into one summary.

For example, a principal might receive a daily summary of outstanding registers and incomplete assessment records.

Daily or weekly digests may be useful for non-urgent information. Urgent matters should not be delayed merely to include them in a digest.

Digest frequency must be configurable where appropriate, and the system should avoid sending empty summaries.

11.5 User Preferences

Where appropriate, users should be able to control optional notification preferences.

For example, a teacher might choose to receive a daily summary of routine reminders rather than separate in-app notifications for every minor update.

However, users must not be allowed to disable essential security notices or mandatory administrative communications where policy requires delivery.

User preferences must never grant or remove access to data. They only affect notification delivery.

12. Priority and Escalation

Not every notification has the same importance.

Optimus Orien may use a small, consistent priority system.

Informational: Something happened, but no immediate action is required.

Routine: A task or reminder needs attention within the normal workflow.

Important: A condition requires timely review by an authorised person.

Urgent: A situation requires prompt attention under an explicitly defined school or organisational policy.

The system should not label ordinary low marks, missing optional information, or routine administrative delays as urgent without a justified rule.

Priority must describe the urgency of the required response, not the worth of the student or teacher.

12.1 Escalation

Escalation means directing an unresolved issue to another authorised person after a defined condition or deadline is met.

For example:

A teacher receives a reminder to submit a register. If the register remains incomplete after the configured deadline, the system may notify the principal.

Escalation should be configured according to the school's workflow.

The system must not automatically escalate every notification to the organisation administrator. That would create unnecessary noise and could expose information beyond what is needed.

12.2 Escalation Requirements

Every escalation rule should specify the original responsible user, the next authorised recipient, the waiting period, the unresolved condition, and the maximum escalation level.

The system should also stop escalation when the underlying issue is resolved or when the rule is deactivated by an authorised user.

Where a record is reassigned, the notification should reflect the current responsibility rather than continuing to alert a person who no longer owns the task.

12.3 No Automatic Punishment

An overdue task or repeated alert must not automatically trigger a penalty against a student, parent, or teacher.

Notifications communicate information and prompt review. Disciplinary or other consequential decisions require their own authorised process and should not be hidden inside notification logic.

13. Acknowledgement Is Not Resolution

Some important alerts may require the recipient to acknowledge that they have seen them.

Acknowledgement means the person confirms awareness of the alert.

Resolution means the underlying issue has been addressed or the relevant review has been completed.

These are different actions.

For example, a principal may acknowledge an alert about a missing attendance register but still need to contact the responsible teacher.

The system should allow acknowledgement where required without marking the register as complete.

For alerts associated with early-intervention cases, the existing case workflow should remain the source of truth for the case's status. The notification should link to that case instead of creating an unrelated copy of the same issue.

The developer must avoid creating multiple competing records of the same problem.

14. Actionable Notifications

A useful notification should help the recipient take the next appropriate action.

A notification that says "Something is wrong" is not sufficient.

Where appropriate, it should explain:

What happened?

Which record is affected?

When did it happen?

Why was the notification generated?

What should the recipient do next?

Who is responsible for the action?

Is there a deadline?

For example, a teacher receiving an attendance reminder should be able to open the relevant register directly.

A principal receiving a school-level summary should be able to open the authorised list of outstanding registers.

A user receiving a data-quality notification should be directed to the relevant record or correction workflow, if they have permission to access it.

The system should avoid links that open an unrelated dashboard or require users to search manually for the affected record.

If a record has been deleted, moved, resolved, or made inaccessible, the notification should display an appropriate message rather than exposing stale private information.

15. Notification Language and Tone

Optimus Orien should use clear, neutral, and professional language.

The purpose is to communicate an observable fact and encourage an appropriate response.

Prefer:

"Attendance is below the school's configured review threshold for this period."

Avoid:

"This student is irresponsible."

Prefer:

"Three assessment records are missing required marks."

Avoid:

"The teacher has failed."

Prefer:

"The assignment submission is awaiting teacher verification."

Avoid:

"The student has not completed the assignment."

The second statement in the final example may be incorrect if the student has submitted the assignment and only the teacher's verification is outstanding.

Notifications should not present a possible explanation as an established fact.

They should not shame students, make unsupported predictions, or imply that a single indicator tells the complete story of a person's academic performance.

16. Privacy and Access Control

Notifications can expose information even when the main platform is secure.

For example, an email may be forwarded, a phone may be shared, or a notification may appear on a lock screen.

The notification system must therefore apply privacy protections at every stage.

16.1 Minimum Necessary Information

Include only the information required for the recipient to understand that action is needed.

An email notification might say that a new review is available without naming the student or displaying their marks.

Detailed information can be shown after the user signs in and passes the normal permission checks.

16.2 Recheck Permissions

Permissions must be checked when a notification is generated and again when the user opens the linked record.

This is important because a user's role or assignment may change after a notification is created.

For example, a teacher who is reassigned to a different class must not retain access to the previous class's private records simply because an old notification still exists.

16.3 Parent and Guardian Protection

A parent notification must use the correct verified parent-child or guardian relationship.

If a guardian's access is revoked, the platform must stop sending new student-specific notifications to that account.

Previously generated notifications must not bypass the revoked relationship when opened.

16.4 Shared Devices and Accounts

The system should not assume that a student or parent is the only person who can view a device's notifications.

Sensitive information should not be placed in public announcements, shared notification feeds, or other recipients' accounts.

16.5 Public Leaderboards

Public leaderboards must not expose individual student alerts, attendance problems, assessment details, intervention cases, or other private records.

Public school-level information must follow the platform's publication rules and privacy protections.

16.6 Audit Records

Important notification actions should be recorded, including generation, delivery attempts, acknowledgement, resolution references, cancellation, and relevant administrative changes.

Audit access must itself be restricted.

The audit log should not become an unrestricted way to browse student information.

17. Scheduling and Time Zones

Notifications and reminders often depend on dates and deadlines.

Optimus Orien must not assume that every school operates in the same time zone or follows the same daily schedule.

For example, an organisation might operate schools in Pakistan and the United Arab Emirates. A reminder intended for 3:00 PM at a school in Pakistan must not accidentally be scheduled for 3:00 PM in the other country's time zone.

The system should store timestamps in a consistent technical format, commonly Coordinated Universal Time (UTC), while displaying the appropriate local time to users.

Each school or relevant operational unit should have an explicitly configured time zone.

Scheduled rules should also consider school calendars, weekends, holidays, closures, and configured deadlines.

A reminder should not be sent merely because a scheduled time has passed if the underlying task was cancelled or is no longer applicable.

17.1 Scheduled Jobs

A scheduled job is a process that runs at a particular time or according to a recurring schedule.

Examples include checking for incomplete registers after a deadline or preparing a daily summary.

Scheduled jobs must be reliable and safe to retry.

If a process stops unexpectedly and restarts, it should not create duplicate reminders for the same condition.

The system should record failed executions and provide a way for authorised technical staff to investigate them.

18. Technical Design for the Developer

The following design separates notification rules, generated notifications, delivery attempts, and underlying tasks.

18.1 Notification Rule

A notification rule defines the condition, scope, recipients, priority, timing, repetition policy, and applicable actions.

A rule should also have a stable identifier, a version, an active status, and a record of who created or changed it.

18.2 Notification Record

A notification record represents one logical notification for a particular recipient or intended audience.

It should contain a stable identifier, notification type, title, safe summary, recipient, priority, creation time, relevant rule version, delivery status, read status, and optional references to the affected record.

It should not copy unnecessary sensitive student information into the notification record.

18.3 Delivery Attempt

A delivery attempt records an effort to send a notification through a channel.

For example, the same logical notification might be displayed in-app and sent through email if both channels are enabled.

Each delivery attempt should record the channel, attempt time, result, and safe error information.

A delivery failure should not necessarily create a new logical notification.

18.4 Underlying Task or Issue

Where a notification relates to an actionable task or an ongoing issue, it should link to the relevant record.

Examples include an attendance register, an assessment record, a data-quality issue, or an early-intervention case.

The notification system should not create a second, competing source of truth for those records.

18.5 Notification Preference

A preference record may define a user's permitted choices for optional notification categories and channels.

Preferences must be validated against mandatory communication rules and organisational policy.

They must not override permissions or allow a user to receive records they are not authorised to see.

18.6 Audit Record

An audit record captures important events affecting the notification or its configuration.

The implementation should protect audit records against unauthorised changes and preserve enough information to investigate failures or explain what happened.

19. Recommended Processing Architecture

The notification system should follow a reliable sequence.

First, a relevant event occurs or a configured condition becomes true.

Second, the system validates the event or condition and checks the relevant rule.

Third, it determines the authorised recipients and applies notification preferences and privacy restrictions.

Fourth, it creates a logical notification if an equivalent notification has not already been generated for the same condition.

Fifth, the notification is placed into the appropriate delivery process.

Sixth, the system attempts delivery through each enabled channel.

Seventh, delivery results are recorded. Failed attempts may be retried according to a controlled policy.

Finally, the notification remains connected to the relevant task or issue so that the platform can show whether the underlying condition is still open.

This design prevents notification delivery from becoming tightly coupled to the process that updates a student's marks or attendance.

For example, a temporary email-provider failure should not prevent a teacher's attendance register from being saved successfully.

Notification processing should not silently roll back a valid school-data transaction simply because an external delivery service is unavailable.

20. Notification Reliability and Failure Handling

The developer must assume that failures will occur.

An email provider may be unavailable. A scheduled job may stop. A network connection may fail. A delivery request may time out even though the external service has accepted it.

The system should handle these situations without losing important records or generating excessive duplicates.

20.1 Retry Policy

Retries should be limited and controlled.

The system may retry temporary failures after a configured delay. It should not retry permanent failures indefinitely.

For example, an invalid recipient address should be recorded as a delivery failure and made available for appropriate correction.

20.2 Idempotency

The notification creation process should use a stable event or condition identifier, together with relevant rule and recipient information, to prevent repeated processing from creating duplicates.

A practical uniqueness strategy must account for repeated legitimate events. For example, two different attendance registers should not be treated as the same issue merely because they belong to the same teacher.

20.3 Monitoring

The platform should monitor important notification-processing failures.

Technical administrators should be able to determine whether notifications are being created, queued, delivered, delayed, or failing.

Monitoring should avoid exposing private student details to people who only need to diagnose the notification service.

20.4 Expired Notifications

Some notifications become irrelevant over time.

For example, a reminder for a cancelled assessment may no longer be useful.

The system should define how such notifications are cancelled or marked obsolete while retaining appropriate historical records.

The system must not treat every old notification as resolved automatically. Its expiry behaviour should depend on the notification type and the underlying task.

21. Reporting on Notifications

Optimus Orien may eventually provide administrative reports showing whether notification workflows are functioning effectively.

Potential measures include the number of notifications created, successful delivery attempts, failures, unresolved alerts, acknowledgement rates where required, and the time taken to resolve linked tasks.

These measures must be interpreted carefully.

A high acknowledgement rate does not prove that the recipient acted appropriately. A high number of alerts does not necessarily indicate that a school is performing poorly. It may indicate that a threshold is poorly configured or that the school has more complete reporting than another school.

Likewise, a notification being delivered does not prove that the recipient read or understood it.

Notification analytics should therefore describe what the system can actually measure.

The initial version does not need a complex notification-analytics dashboard. Reliable operational logs and basic administrative monitoring are sufficient foundations.

22. Worked Example: A Student Attendance Review

Consider a fictional student whose calculated attendance rate for a particular reporting period is 76%.

The school has configured an attendance review threshold of 80% for that period.

The attendance calculation has sufficient valid data, and the rule is applicable to the student's current enrolment.

Optimus Orien identifies that the configured condition is met.

The system creates a review alert for the authorised staff member responsible for the student's case. It records the rule version, reporting period, calculation result, and the reason for generating the alert.

The notification says that the student's attendance is below the school's configured review threshold.

The authorised staff member opens the linked record, reviews the attendance details, and acknowledges the alert.

The staff member then investigates the circumstances. The platform must not assume the reason for the attendance pattern.

If the school opens an early-intervention case, the notification should link to that case. The case-management workflow should record the review, any agreed action, and subsequent follow-up.

Later, the student's attendance rate rises to 85%.

The system updates the current indicator according to the configured calculation rules. The previous alert remains in the historical record, while its current status reflects the applicable workflow.

The system must not claim that the intervention caused the improvement merely because the improvement occurred afterward.

This example demonstrates the relationship between a configured rule, an alert, a human review, and an outcome without using artificial intelligence or making unsupported conclusions.

23. Developer Implementation Requirements

The following requirements are intended to be used as a practical development specification. Each requirement includes an acceptance test that can be performed before the feature is considered complete.

Notification Foundations

NOT-001 — Separate notification concepts. The system must distinguish notification rules, generated notifications, delivery attempts, and linked tasks or issues.

Acceptance test: Create a rule that generates a reminder for an incomplete register. Confirm that the rule, notification, delivery attempt, and register are represented as distinct but connected records.

NOT-002 — Stable notification identity. Every logical notification must have a unique identifier.

Acceptance test: Generate two notifications for different registers and confirm that they have different identifiers.

NOT-003 — Traceable origin. A system-generated notification must record the rule, event, or condition that caused it.

Acceptance test: Open an alert and confirm that an authorised administrator can identify the relevant rule version and triggering condition.

NOT-004 — Clear lifecycle. The system must track notification status separately from the underlying issue's status.

Acceptance test: Mark a notification as read while leaving its linked register incomplete. Confirm that the register remains outstanding.

NOT-005 — Safe notification content. Notification titles and summaries must use clear, neutral language and avoid unsupported conclusions.

Acceptance test: Generate an attendance alert and verify that it describes the configured threshold condition without labelling the student as irresponsible or predicting failure.

Roles and Recipients

NOT-006 — Role-based recipients. Recipients must be selected according to their current role, assignment, permissions, and authorised data scope.

Acceptance test: Generate a teacher-specific reminder and confirm that an unrelated teacher cannot access the linked register.

NOT-007 — Parent-child verification. Student-specific parent notifications must require a valid authorised relationship.

Acceptance test: Send a notification to a guardian linked to Student A and confirm that it contains no information about Student B.

NOT-008 — Organisation isolation. Notifications must not cross organisation boundaries without explicit, authorised configuration.

Acceptance test: Create two separate organisations and confirm that users in one organisation cannot access the other's notifications through the interface or API.

NOT-009 — Permission revalidation. Access to linked records must be checked when the notification is opened.

Acceptance test: Create a notification for a teacher, remove the teacher's assignment, and confirm that the old notification does not provide access to the former class's private records.

NOT-010 — Least-privilege summaries. Summary notifications must disclose only information permitted for the recipient.

Acceptance test: A school administrator receives a summary of outstanding tasks. Confirm that the summary does not include detailed records beyond that administrator's permissions.

Rules and Configuration

NOT-011 — Configurable rules. Authorised administrators must be able to configure supported notification rules without editing application code.

Acceptance test: Enable a supported incomplete-register reminder through the approved configuration interface and confirm that it operates according to the saved rule.

NOT-012 — Configurable thresholds. Supported threshold-based alerts must use configured values rather than hardcoded universal thresholds.

Acceptance test: Change a test school's attendance review threshold from 80% to 75% and verify that the rule uses the new threshold for applicable calculations.

NOT-013 — Rule scope. Rules must respect their organisation, school, programme, class, or other configured scope.

Acceptance test: Configure a rule for one school and confirm that it does not unintentionally affect another school.

NOT-014 — Rule precedence. Where the platform permits inherited and local rules, precedence must be explicit and consistently applied.

Acceptance test: Configure an organisation default and a permitted school-specific override. Confirm that the active rule follows the documented precedence policy.

NOT-015 — Rule versioning. Changes to notification rules must preserve historical versions.

Acceptance test: Change a threshold and verify that a previously generated notification still identifies the version that originally triggered it.

NOT-016 — Rule activation. Only authorised users may activate, deactivate, or modify rules.

Acceptance test: Attempt to change a notification rule using an account without the required permission. Confirm that the change is rejected by the server.

Notification Delivery

NOT-017 — In-app notifications. The initial notification system must support in-app notification delivery.

Acceptance test: Generate a notification for a signed-in user and verify that it appears in the correct account's notification panel.

NOT-018 — Unread counts. Unread notification counts must update consistently when notification status changes.

Acceptance test: Mark a notification as read and verify that the unread count updates correctly.

NOT-019 — Delivery status. The system must record delivery outcomes appropriate to the selected channel.

Acceptance test: Simulate a failed delivery and confirm that the attempt is recorded as failed rather than silently treated as successful.

NOT-020 — Controlled retries. Temporary delivery failures may be retried according to a defined policy.

Acceptance test: Simulate a temporary failure followed by success. Confirm that the system retries within the configured limits and records the attempts.

NOT-021 — Permanent failure handling. Permanent failures must not be retried indefinitely.

Acceptance test: Simulate an invalid email address and confirm that the system records the failure without continuously retrying.

NOT-022 — Idempotent generation. Reprocessing the same event must not create unintended duplicate logical notifications.

Acceptance test: Process the same event twice and confirm that the configured deduplication strategy prevents a duplicate notification.

NOT-023 — Independent delivery. A notification delivery failure must not invalidate an otherwise successful school-data operation.

Acceptance test: Simulate a delivery-provider outage while a teacher submits an attendance register. Confirm that the register is saved successfully and the delivery failure is recorded separately.

Overload and Escalation

NOT-024 — Duplicate suppression. Repeated checks of the same unresolved condition must not generate uncontrolled duplicate notifications.

Acceptance test: Run the condition checker repeatedly without changing the underlying register and confirm that it does not create a new logical notification on every run.

NOT-025 — Frequency limits. Repeated reminders must follow configured frequency limits.

Acceptance test: Trigger a reminder rule repeatedly within its suppression window and confirm that the user receives no more than the permitted number of reminders.

NOT-026 — Notification grouping. Supported summaries must group related notifications without mixing unauthorised records.

Acceptance test: Generate multiple outstanding-register reminders for one school and verify that an authorised principal can see a summary with the correct count and permitted details.

NOT-027 — Digests. Where digest notifications are enabled, the system must group eligible items according to the configured schedule.

Acceptance test: Generate several eligible routine notifications and verify that they appear in the scheduled digest rather than generating unintended separate digest messages.

NOT-028 — Priority rules. Priority must follow documented and configurable criteria.

Acceptance test: Generate notifications for routine reminders and an explicitly configured urgent condition. Confirm that each receives the intended priority.

NOT-029 — Controlled escalation. Escalation must occur only when the defined condition, waiting period, and recipient rules are satisfied.

Acceptance test: Leave a register unresolved past its configured escalation deadline and confirm that the appropriate authorised recipient is notified.

NOT-030 — Stop escalation when resolved. The system must stop future escalation when the underlying issue is resolved or the applicable rule is deactivated.

Acceptance test: Resolve a register before the escalation deadline and confirm that the scheduled escalation does not proceed.

NOT-031 — Reassignment. Notifications must reflect current responsibility where a linked task is reassigned.

Acceptance test: Reassign a task and verify that subsequent reminders go to the new authorised owner according to policy.

Reading, Acknowledgement, and Resolution

NOT-032 — Read status. Users must be able to mark permitted notifications as read.

Acceptance test: Mark a notification as read and verify that its read status is saved and reflected in the interface.

NOT-033 — Acknowledgement. Notifications requiring acknowledgement must support a separate acknowledgement action.

Acceptance test: Acknowledge a review alert and verify that the acknowledgement is recorded without automatically resolving the underlying case.

NOT-034 — Linked resolution. Notifications connected to an existing issue must reference that issue rather than create an unrelated duplicate case.

Acceptance test: Generate a notification for an existing intervention case and confirm that the notification links to the original case identifier.

NOT-035 — Resolved-condition handling. The system must update the current status of an alert when its underlying condition is resolved, where the alert type supports this behaviour.

Acceptance test: Submit a previously incomplete register and verify that its outstanding-condition indicator is updated while the historical notification remains traceable.

NOT-036 — Cancellation. When a task is cancelled, related future reminders must be cancelled or marked obsolete according to the relevant rule.

Acceptance test: Cancel an assessment before its reminder time and verify that the system does not send the scheduled reminder as though the assessment were still active.

Privacy and Security

NOT-037 — Minimum necessary content. Notifications must not copy unnecessary student information into their titles, summaries, or external messages.

Acceptance test: Generate an external-channel notification for a student review and confirm that the default message does not reveal the student's marks or sensitive case details.

NOT-038 — Secure linked access. Notification links must require normal authentication and authorisation.

Acceptance test: Open a private notification link in an unauthenticated session and confirm that the protected record is not displayed.

NOT-039 — No public student alerts. Individual student alerts must never appear on public leaderboards or unauthenticated public pages.

Acceptance test: Generate an individual student alert and verify that no student-identifiable information is added to the public homepage.

NOT-040 — Audit trail. Important notification events and administrative changes must be auditable.

Acceptance test: Generate, deliver, acknowledge, and resolve a test notification. Confirm that the appropriate history is available to an authorised auditor.

NOT-041 — Revoked access. Revoking a user's access must prevent that user from opening protected records through previously received notifications.

Acceptance test: Revoke a guardian's student access and verify that a previously received notification no longer grants access to the student's private information.

NOT-042 — Sensitive audit protection. Audit records must be protected against unauthorised reading or modification.

Acceptance test: Attempt to access or modify audit records using an account without the required permission and confirm that the server rejects the request.

Scheduling and Monitoring

NOT-043 — School time zone. Scheduled notifications must use the configured local time zone for the relevant school.

Acceptance test: Configure two test schools in different time zones and verify that a local-time reminder is scheduled correctly for each.

NOT-044 — Calendar awareness. Scheduled reminders must respect the applicable school calendar and the task's validity.

Acceptance test: Configure a reminder that falls on a school closure and verify that the documented scheduling policy is applied.

NOT-045 — Reliable scheduling. Scheduled jobs must record execution status and handle safe retries.

Acceptance test: Interrupt a scheduled run, restart the worker, and verify that the system processes eligible notifications without unintended duplication.

NOT-046 — Failure monitoring. Technical administrators must be able to identify significant notification-processing failures.

Acceptance test: Simulate a failed notification job and verify that the failure appears in the authorised operational monitoring system.

NOT-047 — Expiry and obsolescence. Notification types with expiry rules must be marked obsolete or cancelled appropriately.

Acceptance test: Cancel an eligible task and verify that its outstanding reminder is no longer presented as an actionable task.

NOT-048 — Delivery reporting. The system must distinguish created notifications, attempted deliveries, successful deliveries, and failures.

Acceptance test: Generate several test notifications with different delivery outcomes and verify that the recorded counts match the underlying records.

Usability and Integration

NOT-049 — Actionable links. Notifications should link to the relevant authorised record or workflow when an appropriate destination exists.

Acceptance test: Select a register reminder and verify that it opens the correct register for an authorised user.

NOT-050 — Safe handling of inaccessible records. If a linked record becomes inaccessible or unavailable, the notification must display an appropriate message without revealing protected information.

Acceptance test: Remove a user's access to a linked record and confirm that opening the notification does not expose the record's private contents.

NOT-051 — Consistent calculations. Notifications based on academic or attendance indicators must use the same approved calculations as the relevant dashboard and reporting system.

Acceptance test: Configure a test attendance threshold and verify that the dashboard, report, and alert use the same attendance rate and reporting period.

NOT-052 — User preferences. Supported optional notification preferences must be saved and applied without overriding permissions or mandatory communication rules.

Acceptance test: Disable an optional reminder category and verify that the relevant optional notification is suppressed while mandatory security notices remain unaffected.

NOT-053 — Neutral wording. System-generated notifications must use approved templates or equivalent controlled wording.

Acceptance test: Review sample academic, attendance, and homework notifications and confirm that they describe observable conditions without unsupported predictions or blame.

NOT-054 — Notification history. Users must be able to review the notification history permitted for their role.

Acceptance test: A teacher views their history and sees their own authorised notifications but cannot browse another teacher's private notification records.

NOT-055 — Large notification volumes. Notification lists and processing must remain usable when an organisation has many users and notifications.

Acceptance test: Populate a test environment with a large notification dataset and verify that lists load through pagination or an equivalent efficient method, without exposing unrelated organisations' records.

NOT-056 — Data-source consistency. A notification must not be generated from invalid or incomplete source data unless the notification explicitly concerns that data-quality problem.

Acceptance test: Leave an attendance register unsubmitted and verify that the system can notify the responsible user about the missing register without incorrectly generating a student-absence alert from the missing data.

24. Recommended Development Order

The developer should build the notification system in stages rather than attempting every delivery channel and rule at once.

Stage 1: Core notification records. Create the notification data model, recipient relationships, basic statuses, timestamps, and audit history.

Stage 2: In-app notification interface. Build the notification panel, unread count, history, read status, and secure links to authorised records.

Stage 3: Initial system-generated notifications. Support a small number of useful workflows, such as incomplete attendance registers, pending teacher verification, missing assessment records, and important data-quality issues.

Stage 4: Rule configuration. Introduce supported rule settings, scopes, thresholds, versioning, and authorised configuration controls.

Stage 5: Reliability and duplicate prevention. Implement idempotency, controlled retries, frequency limits, scheduling, and failure monitoring.

Stage 6: Grouping and escalation. Add summaries, digests, priorities, and escalation rules after the basic workflows have been tested.

Stage 7: External channels. Add email only when the provider, security controls, consent or policy requirements, and delivery monitoring are ready. Consider SMS and push notifications as separate later projects.

Stage 8: Operational reporting. Provide basic information about delivery failures, unresolved alerts, and notification-processing health.

Each stage should pass its acceptance tests before the next stage is treated as complete.

25. What Should Not Be Built Yet?

To keep Optimus Orien manageable, the developer should not add every possible communication feature to the first release.

The initial release should not include a full chat application, unrestricted direct messaging between every role, AI-generated notification decisions, automated disciplinary actions, or external messaging integrations that have not been approved.

It should also avoid allowing users to create arbitrary rules that bypass access control or expose protected information.

The priority is to create a dependable notification foundation that can support additional workflows later.

26. Final Principles

The notification system should be designed around the following principles.

A notification must have a clear reason for existing.

Its recipient must be authorised to receive the information.

The wording must describe the condition accurately and respectfully.

The user must be able to understand what action is required.

The notification's status must not be confused with the status of the underlying task or issue.

Repeated processing must not generate uncontrolled duplicates.

Delivery failures must be recorded and handled safely.

Rules, thresholds, and escalation policies must be configurable where appropriate.

Historical records must remain traceable without unnecessarily preserving sensitive information.

Finally, the system must not confuse a notification with a decision. An alert may indicate that something deserves attention, but an authorised person must review the evidence before making a consequential judgement.

Optimus Orien should make important information easier to act upon, not simply generate more messages.

27. What Part 17 Establishes

Part 17 establishes how Optimus Orien should identify notification-worthy events and conditions, select authorised recipients, deliver notifications, manage reminders, control repetition, escalate unresolved tasks, protect private information, and record notification history.

It also defines the foundations for reliable processing, configurable rules, and future delivery channels.

PART 18

SECURITY, PRIVACY, AND PROTECTION OF STUDENT DATA

Project: Optimus Orien
Document type: Developer Implementation Specification
Priority: Critical
Applies to: Frontend, backend, database, APIs, authentication, reporting, exports, notifications, administration, integrations, and deployment.

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1. Implementation Objective

Implement a security and privacy architecture that protects educational records, personal information, organisational data, user accounts, and administrative operations throughout Optimus Orien.

Security must be enforced by the backend and database architecture wherever applicable. Frontend restrictions alone are insufficient.

The implementation must support:

- Organisation-level and school-level data isolation.
- Role-based permissions and record-level access controls.
- Secure authentication and session management.

Protection of student, parent, teacher, and school records.

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-

Secure storage, transmission, processing, and export of data.

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-

Auditable administrative and data-changing operations.

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-

Safe imports, validation, error handling, and file processing.

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-

Account suspension, permission revocation, and access termination.

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-

Backup, restoration, incident response, and data-retention controls.

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-

Configurable privacy settings for deployment across different jurisdictions.

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Current scope constraints: Do not introduce artificial intelligence, AI-based profiling, automated consequential decisions, or in-app chat as part of this implementation.

2. Security Architecture Requirements

SEC-001 — Centralised Authorisation

Implement a central backend authorisation layer.

Every protected API operation must verify the authenticated user, required permission, applicable organisation or school scope, and relationship to the requested records.

Acceptance criteria:

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Requests without authentication receive an appropriate unauthorised response.

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-

Authenticated users without the required permission receive a forbidden response.

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-

Changing an identifier in an API request does not grant access to another user's records.

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-

Authorisation checks cannot be bypassed by calling an API directly instead of using the interface.

-

SEC-002 — Deny by Default

Access must be denied unless an explicit permission grants it.

Do not assume that a user can access a record merely because the record exists in the same database or belongs to the same organisation.

Acceptance criteria:

-

Newly introduced endpoints have no implicit access.

-
-

A missing or invalid permission configuration results in denied access.

-
-

Unknown roles cannot access protected resources.

-

SEC-003 — Least Privilege

Grant users only the permissions required for their assigned responsibilities.

Avoid broad permissions such as unrestricted access to all student records when a narrower permission is sufficient.

Acceptance criteria:

- Teachers can access only their assigned classes, subjects, and authorised student records.
- Principals can access only the schools within their authorised scope.
- Organisation administrators can access only organisations assigned to their account.
- Student and parent accounts cannot access administrative records.

SEC-004 — Shared Authorisation Middleware

Use reusable backend authorisation services rather than implementing inconsistent permission checks independently in each endpoint.

The authorisation layer must support role, permission, scope, and record relationship checks.

Acceptance criteria:

- Equivalent endpoints apply consistent permission rules.
- A common authorisation test suite can be executed against protected API routes.

Code review can identify the central mechanism responsible for access decisions.

-

SEC-005 — Secure Object-Level Access

Verify access to each requested record, not merely access to its general category.

For example, permission to view a student record must not automatically grant access to every student's record.

Acceptance criteria:

-

Requests for another student's identifier are denied when the relationship is not authorised.

-
-

Requests for another school's identifier are denied when the user lacks access to that school.

-
-

Nested resources, including assessments, attendance, homework, skills, and intervention cases, enforce the same restrictions.

-

SEC-006 — No Security Through Hidden Interface Elements

Hiding navigation links, buttons, or dashboard sections must never be the only security control.

Acceptance criteria:

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Direct API requests remain protected even when a frontend button is hidden.

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-

Users cannot modify request parameters to gain higher privileges.

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-

UI visibility and backend permission decisions remain consistent.

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3. Multi-Tenant Data Isolation

Optimus Orien must support multiple independent organisations and their schools.

A tenant is an organisation or another explicitly defined isolation boundary. The system's tenant model must be documented and applied consistently.

SEC-007 — Explicit Tenant Ownership

Every tenant-owned record must have an unambiguous ownership path to its organisation.

Depending on the data model, this may be implemented using an organisation identifier, a school identifier with a verified relationship to its organisation, or a controlled relationship through another tenant-owned record.

Acceptance criteria:

-

Every protected record can be resolved to its authorised tenant.

-
-

Orphaned or ambiguous ownership relationships are rejected or flagged.

-
-

Tenant ownership cannot be changed by an ordinary user through an API request.

-

SEC-008 — Tenant-Scoped Queries

All tenant-owned data access must apply the appropriate tenant restrictions.

Do not retrieve unrestricted records and rely on the frontend to filter them.

Acceptance criteria:

-

List, search, detail, update, delete, export, and aggregate queries are tenant-scoped.

-
-

Pagination totals and search suggestions do not reveal records from other tenants.

-
-

Background jobs apply the same isolation rules.

-

SEC-009 — Cross-Tenant Identifier Attacks

Treat all client-supplied identifiers as untrusted.

Acceptance criteria:

-

Replacing a school, student, teacher, report, file, or export identifier with one belonging to another tenant does not expose or modify that resource.

-
-

The result is handled consistently without disclosing protected record details.

-
-

Automated security tests cover identifier substitution attacks.

-

SEC-010 — Tenant-Safe Aggregation

Dashboard metrics, class averages, school health indicators, rankings, and reports must respect the user's authorised scope.

Acceptance criteria:

-

A school principal's metrics include only the principal's authorised school.

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-

An organisation administrator's metrics include only the assigned organisation and permitted branches.

-
-

Global statistics do not accidentally expose restricted tenant-level information.

-
-

Every aggregate uses the intended scope and denominator.

-

SEC-011 — Background Job Isolation

Scheduled jobs, imports, exports, notifications, recalculations, and reporting workers must preserve tenant boundaries.

Acceptance criteria:

-

Every queued job carries validated tenant and scope context where applicable.

-
-

Workers do not trust arbitrary tenant identifiers supplied by an untrusted client.

-
-

Tests confirm that a job for one tenant cannot process another tenant's records unintentionally.

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SEC-012 — Tenant Isolation Testing

Create automated tests that deliberately attempt cross-tenant access.

Acceptance criteria:

-

Tests cover reads, writes, deletes, exports, search, aggregates, file downloads, and background processing.

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-

A failed cross-tenant request never returns protected record contents.

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-

Tenant isolation tests run in the automated test pipeline.

-

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4. Authentication and Account Security

SEC-013 — Established Authentication Framework

Use a maintained authentication framework or a secure identity provider rather than implementing password handling and authentication protocols from scratch.

The selected implementation must support the application's account model, session revocation, password recovery, and required security controls.

Acceptance criteria:

-

Authentication uses a documented, maintained implementation.

-
-

Passwords are never stored in plaintext or reversibly encrypted.

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-

Session invalidation works after account suspension or access revocation.

-

SEC-014 — Password Storage

If Optimus Orien manages passwords directly, store them using a recognised, appropriately configured password-hashing algorithm, such as Argon2id or another algorithm approved for the selected framework.

Use framework-maintained implementations and appropriate work factors.

Acceptance criteria:

- Database inspection reveals no plaintext passwords.
- Password hashes are salted through the selected hashing implementation.
- Password verification uses the framework's secure verification method.

SEC-015 — Password Policy

Implement a documented password policy that balances security and usability.

Do not impose arbitrary complexity rules without considering modern password-security guidance.

Support long passwords, reject commonly compromised passwords where a suitable mechanism is available, and prevent unreasonable password lengths from causing resource exhaustion.

Acceptance criteria:

- Password length and validation rules are enforced on the server.
- Passwords are not silently truncated.
- Error messages do not reveal sensitive authentication information.

SEC-016 — Rate Limiting

Apply rate limits to authentication-sensitive endpoints.

At minimum, consider login attempts, password recovery, account invitation acceptance, and other operations vulnerable to automated abuse.

Acceptance criteria:

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  Repeated failed login attempts trigger the configured protection.

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  Rate limits cannot be bypassed simply by changing a client-side parameter.

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-

  Rate limiting does not allow one user's activity to unnecessarily deny service to every user.

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SEC-017 — Password Recovery

Password recovery must use a secure, time-limited, single-use verification mechanism.

Do not disclose whether an email address belongs to an existing account through recovery responses.

Acceptance criteria:

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  Recovery responses do not reveal account existence.

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-

  Recovery tokens expire and cannot be reused.

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-

  Completing recovery invalidates the used token.

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-

  Existing sessions are revoked or handled according to the documented security policy.

SEC-018 — Account Invitations

Invitation links must be unpredictable, time-limited, single-use, and associated with the intended account and permitted role.

Acceptance criteria:

Expired invitations cannot be accepted.

A token cannot be used twice.

Accepting an invitation does not allow the recipient to select an arbitrary organisation or grant themselves a higher role.

Invitation cancellation prevents future use.

SEC-019 — Multi-Factor Authentication

Design authentication so that multi-factor authentication (MFA) can be enabled for privileged accounts and supported organisational policies.

Prefer enforcing MFA for high-privilege accounts when the selected authentication infrastructure supports it.

Acceptance criteria:

The authentication design documents MFA support or explicitly documents a security-approved alternative.

Privileged accounts cannot bypass mandatory MFA through an alternate login path.

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-

Recovery procedures do not silently weaken privileged-account security.

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SEC-020 — Privileged Account Protection

Protect organisation administrators and other high-privilege accounts with appropriate authentication and session controls.

Acceptance criteria:

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Privileged actions require an authenticated account with the relevant permission.

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Sensitive actions can require reauthentication or additional verification where justified.

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Privilege changes are audited.

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5. Session and Token Security

SEC-021 — Secure Sessions

Use secure session management appropriate to the chosen authentication architecture.

For browser-based sessions, prefer secure cookies where compatible with the selected design.

Acceptance criteria:

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Session identifiers are unpredictable.

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-

Authentication cookies use appropriate Secure, HttpOnly, and SameSite settings.

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-

Sensitive tokens are not exposed in application URLs or routine logs.

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SEC-022 — Session Expiration

Define inactivity and absolute session expiration policies.

Privileged accounts may require stricter settings.

Acceptance criteria:

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Expired sessions cannot access protected endpoints.

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-

Session expiration is enforced server-side.

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-

The frontend handles expired sessions without exposing protected data.

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SEC-023 — Logout and Revocation

Logout must invalidate the applicable session or refresh-token mechanism.

Acceptance criteria:

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A logged-out session cannot continue accessing protected APIs using the revoked credential.

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-

Account suspension invalidates active access according to the documented revocation design.

-

-

  Password resets trigger the required session-revocation behaviour.

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SEC-024 — Token Storage

Do not store long-lived authentication tokens in browser storage without a documented security assessment.

Avoid exposing bearer tokens to client-side scripts unnecessarily.

Acceptance criteria:

-

  The authentication design documents where tokens or session identifiers are stored.

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  Tokens are excluded from logs and analytics.

-
-

  Token revocation and expiration are tested.

-

SEC-025 — Cross-Site Request Forgery

If cookie-based authentication is used, implement appropriate Cross-Site Request Forgery (CSRF) protections for state-changing operations.

Acceptance criteria:

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  Unauthorised cross-site requests cannot perform protected state changes.

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  CSRF protection is compatible with the application's session and cookie configuration.

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Tests cover relevant state-changing endpoints.

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6. API and Application Security

SEC-026 — Input Validation

Validate all client-supplied data on the server.

Validation must cover types, required fields, lengths, permitted values, relationships, ranges, and business rules.

Acceptance criteria:

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Invalid payloads are rejected before persistence.

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-

The API returns controlled validation errors.

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-

The application does not trust frontend-only validation.

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SEC-027 — Output Encoding

Apply appropriate output encoding when rendering untrusted data.

Acceptance criteria:

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User-entered text cannot execute as script in the browser.

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-

Imported names, comments, notes, and other text fields are treated as data.

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-

Security tests cover relevant Cross-Site Scripting (XSS) scenarios.

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SEC-028 — Database Query Safety

Use parameterised queries or the selected framework's safe query mechanisms.

Do not concatenate untrusted input into database commands.

Acceptance criteria:

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Search fields cannot alter the intended database query.

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-

Security tests cover SQL injection where applicable.

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Dynamic query components use allowlisted values.

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SEC-029 — API Request Limits

Enforce sensible request-size limits, pagination limits, and query-complexity controls.

Acceptance criteria:

-

Excessively large requests are rejected.

-
-

List endpoints cannot be forced to return unlimited datasets.

-
-

Search and filtering endpoints have documented limits.

-

SEC-030 — CORS Configuration

Configure Cross-Origin Resource Sharing (CORS) according to the actual frontend and API deployment architecture.

Do not allow arbitrary origins to access authenticated resources by default.

Acceptance criteria:

-

  Production origins are explicitly configured.

-
-

  Unapproved origins cannot make credentialed cross-origin requests.

-
-

  Development exceptions are not unintentionally enabled in production.

-

SEC-031 — HTTP Security Headers

Configure appropriate security headers for the application.

Evaluate Content Security Policy (CSP), frame restrictions, MIME-type protections, and transport-security headers according to the deployment architecture.

Acceptance criteria:

-

  Production responses include the approved headers.

-
-

  The policy is tested against the actual application rather than copied blindly.

-
-

  Headers do not introduce avoidable application breakage.

-

SEC-032 — Error Handling

Return safe, structured error responses.

Do not expose stack traces, internal filesystem paths, SQL statements, secret values, or detailed infrastructure information to ordinary users.

Acceptance criteria:

-

  Production API errors omit internal implementation details.

-
-

  Detailed diagnostics are available only through authorised server-side logs.

-
-

  Errors do not disclose protected records belonging to other users.

-

SEC-033 — Dependency Security

Maintain an inventory of application dependencies and use automated vulnerability scanning where available.

Acceptance criteria:

-

  Dependencies are recorded in the appropriate lockfiles or equivalent manifests.

-
-

  Known critical vulnerabilities are reviewed and addressed before production deployment.

-
-

  The team documents any accepted exceptions and compensating controls.

-

SEC-034 — Secret Management

Store database credentials, signing keys, API keys, email-provider credentials, and other secrets in an approved secret-management mechanism or protected deployment configuration.

Acceptance criteria:

-

Production secrets are not committed to source control.

-
-

Secrets are not embedded in frontend bundles.

-
-

Rotation procedures are documented.

-
-

A leaked development secret can be replaced without rebuilding the application's security model.

-

---

7. Student Data and Privacy Controls

SEC-035 — Data Classification

Define data categories and apply protection appropriate to each category.

At minimum, distinguish public information, internal operational information, personal information, and sensitive educational or case information.

The classification policy must cover data stored in the database, files, logs, reports, exports, notifications, and backups.

Acceptance criteria:

-

Major data categories have documented access and handling rules.

-
-

Publicly approved school information is distinguishable from private student records.

-

-

Sensitive categories receive stronger access and retention controls where justified.

-

SEC-036 — Data Minimisation

Collect and retain only information needed for an approved product function, legal obligation, or documented operational purpose.

Acceptance criteria:

-

Every major personal-data field has a documented purpose.

-
-

Unnecessary sensitive fields are not introduced without approval.

-
-

Exports and notifications avoid copying data that the recipient does not need.

-

SEC-037 — Parent and Guardian Relationships

Represent parent or guardian relationships explicitly and enforce them in every relevant endpoint.

Acceptance criteria:

-

A guardian can access only linked students for whom access is currently authorised.

-
-

Revoking a relationship blocks future access.

-
-

Parent relationship checks apply to reports, exports, notifications, and historical records.

-

SEC-038 — Student Privacy in Public Pages

Public pages and school leaderboards must never expose individual student records or private student performance information.

Acceptance criteria:

-

Public endpoints return only explicitly approved public fields.

-
-

Student names, marks, attendance records, skill ratings, and intervention cases are excluded unless a separately approved lawful publication feature is implemented.

-
-

Public API responses are tested independently from authenticated dashboard responses.

-

SEC-039 — Small-Group Disclosure

Evaluate whether school-level or aggregate statistics could reveal an identifiable student's information, especially when groups are very small.

Acceptance criteria:

-

Reports with small groups are reviewed for re-identification risk.

-
-

Where required by the approved policy, small-group values are suppressed, grouped, or otherwise protected.

-
-

Public aggregate endpoints do not provide unrestricted filters that allow protected records to be inferred through repeated queries.

-

SEC-040 — Sensitive Notes

Restrict sensitive notes associated with student support, intervention, attendance reasons, or similar workflows.

Acceptance criteria:

-

Only explicitly authorised roles can view sensitive notes.

-
-

Ordinary academic or attendance permissions do not automatically grant access to all case notes.

-
-

Sensitive notes are excluded from public leaderboards and unrelated exports.

-

SEC-041 — Privacy-Safe Logging

Logs must not routinely contain passwords, session tokens, authentication secrets, full sensitive records, or unnecessary personal information.

Acceptance criteria:

-

Authentication and application logs are inspected for accidental secret exposure.

-
-

Request-body logging is disabled or appropriately redacted for sensitive endpoints.

-
-

Error tracking and monitoring integrations follow the same policy.

-

SEC-042 — Purpose-Limited Access

Access to data must be tied to an authorised product function.

Do not provide unrestricted data browsing merely because a role is considered senior.

Acceptance criteria:

-

  High-level roles still encounter scope and permission restrictions.

-
-

  Sensitive case records require their own applicable permissions.

-
-

  Export and reporting access is subject to the same purpose and scope controls as normal viewing.

-

---

8. Encryption and Key Management

SEC-043 — Encryption in Transit

Use HTTPS with current, supported Transport Layer Security (TLS) configuration for production application traffic.

Acceptance criteria:

-

  Production traffic is encrypted in transit.

-
-

  HTTP traffic is redirected or rejected according to the deployment design.

-
-

  Certificates are valid and renewal is monitored.

-

SEC-044 — Encryption at Rest

Use appropriate encryption-at-rest controls for databases, storage volumes, file storage, backups, and other persistent data stores.

The chosen implementation must be documented according to the hosting provider and deployment architecture.

Acceptance criteria:

-

Production data stores use approved encryption-at-rest controls.

-
-

Backups receive appropriate encryption protection.

-
-

Access to encryption keys is restricted.

-

SEC-045 — Key Separation

Keep encryption keys and application secrets separate from the encrypted data they protect.

Acceptance criteria:

-

Database or file access alone does not automatically provide unrestricted access to separately managed encryption keys.

-
-

Key access is limited to authorised services and operators.

-
-

Key rotation and recovery procedures are documented.

-

SEC-046 — Sensitive Field Encryption

Assess whether selected sensitive fields require application-level encryption beyond infrastructure-level encryption.

If application-level encryption is implemented, document how searching, reporting, recovery, key rotation, and access control will work.

Acceptance criteria:

-

The design identifies which fields, if any, receive additional encryption.

-
-

Encryption does not silently break required calculations or record retrieval.

-
-

Key loss and rotation risks are addressed before production use.

-

---

9. File Uploads, Imports, and Exports

SEC-047 — File Upload Validation

Validate uploaded files by permitted file type, size, structure, and content.

Do not trust a filename extension or browser-supplied MIME type alone.

Acceptance criteria:

-

Disallowed file types are rejected.

-
-

File-size limits are enforced server-side.

-
-

Invalid files cannot bypass validation by changing their extension.

-

SEC-048 — Safe File Storage

Store uploaded files outside publicly accessible directories unless public access is explicitly required and approved.

Acceptance criteria:

-

Private files require authenticated, authorised access.

-
-

Guessing a file URL does not reveal protected content.

-
-

File download permissions are checked server-side.

-

SEC-049 — Import Isolation

Imported records must pass the same tenant, relationship, and business-rule validation as ordinary user-entered records.

Acceptance criteria:

-

An import cannot assign records to another tenant by changing a spreadsheet identifier.

-
-

Invalid cross-school relationships are rejected.

-
-

The import summary identifies rejected records without exposing unrelated tenant data.

-

SEC-050 — Spreadsheet Formula Injection

Protect generated CSV and spreadsheet exports against formula injection.

Untrusted values beginning with characters interpreted as formulas by spreadsheet applications must be handled using a documented safe-export strategy.

Acceptance criteria:

- Malicious test values cannot execute unintended spreadsheet formulas when the exported file is opened in supported spreadsheet software.
- The original stored value is preserved where appropriate; sanitisation for export does not silently corrupt the source record.

SEC-051 — Export Authorisation

Apply record-level permissions to export jobs and downloaded files.

Acceptance criteria:

- Users cannot export data they cannot access through the corresponding authorised report.
- Export scope is revalidated when the job runs and when the file is downloaded, according to the documented policy.
- Cross-tenant export attempts are rejected.

SEC-052 — Export Expiration

Apply an expiration policy to temporary export files and generated download links.

Acceptance criteria:

-

Expired download links cannot retrieve protected files.

-
-

Temporary files are deleted or rendered inaccessible according to the retention policy.

-
-

Export expiration does not delete the underlying educational records.

-

SEC-053 — Export Audit

Record who initiated an export, the relevant scope, the export type, the time, and the result.

Avoid storing unnecessary exported student data in the audit record itself.

Acceptance criteria:

-

An authorised auditor can determine who generated a particular export.

-
-

Failed and successful export attempts are distinguishable.

-
-

Audit records do not contain unnecessary copies of the exported data.

-

---

10. Data Integrity and Privileged Changes

SEC-054 — Server-Side Business Rules

Enforce important business rules on the server, including mark limits, enrolment relationships, attendance rules, and permitted status transitions.

Acceptance criteria:

- Invalid values submitted directly through an API are rejected.
- The backend does not rely on a frontend dropdown or form validation to enforce business rules.
- Validation rules are consistent across relevant endpoints.

SEC-055 — Protected Administrative Actions

Require explicit permissions for changes to roles, school configurations, reporting rules, grading settings, attendance policies, and similar high-impact settings.

Acceptance criteria:

- Ordinary teachers cannot modify organisation-wide settings without an explicit permission.
- Role escalation attempts are rejected.
- Administrative changes are recorded in the audit trail.

SEC-056 — Immutable Audit History

Maintain an audit trail for important changes to educational records and security-sensitive settings.

Audit history should not be editable through ordinary application interfaces.

Acceptance criteria:

-

  Mark corrections, attendance corrections, role changes, and relevant configuration changes record the actor, timestamp, and affected record.

-
-

  Ordinary users cannot delete or rewrite the audit history.

-
-

  Audit access is separately permission-controlled.

-

SEC-057 — Concurrent Updates

Prevent concurrent edits from silently overwriting important changes.

Use an appropriate strategy, such as optimistic concurrency control, transactions, or version checks.

Acceptance criteria:

-

  Two users editing the same protected record cannot silently overwrite each other's changes.

-
-

  Conflicts return a clear message and a safe recovery path.

-
-

  Failed conflicting updates do not create inconsistent related records.

-

SEC-058 — Transactional Integrity

Use appropriate database transactions for operations that must succeed or fail together.

Acceptance criteria:

-

A failed multi-record operation does not leave unintended partial changes.

-
-

Import, enrolment, and related-record updates follow documented transaction boundaries.

-
-

External notification failures do not roll back otherwise valid school-data transactions.

-

---

11. Data Retention, Correction, and Deletion

SEC-059 — Retention Policy

Define retention rules for personal information, educational records, audit logs, uploaded files, exports, and backups.

Do not assume one universal retention period applies to all record categories or jurisdictions.

Acceptance criteria:

-

Each major record category has a documented retention rule or an explicitly identified pending policy decision.

-
-

Temporary exports and delivery logs have their own retention rules.

-
-

Retention processing does not unintentionally delete required historical academic records.

-

SEC-060 — Deletion and Deactivation

Distinguish account deactivation, access revocation, logical deletion, and permanent deletion.

Educational records may need to remain available for legitimate historical reporting even when a user account is deactivated.

Acceptance criteria:

-

Deactivating a user prevents authentication and relevant access.

-
-

Deactivating an account does not silently delete unrelated academic records.

-
-

Permanent deletion follows the approved retention, legal, and recovery policy.

-

SEC-061 — Correction History

Corrections to educational data must preserve appropriate historical traceability.

Acceptance criteria:

-

Correcting a mark or attendance record records the change and authorised actor.

-
-

Reports use the current approved value while preserving required correction history.

-
-

The system can identify whether an older report was generated before a correction.

-

SEC-062 — Data Subject Requests

Design an operational process for handling applicable requests concerning personal information, including access, correction, deletion, or restriction where required by the governing legal framework.

Acceptance criteria:

-

The responsible role and approval process are documented.

-
-

Requests can be tracked through completion.

-
-

The implementation supports applicable obligations without promising rights or deadlines that have not been legally verified.

-

---

12. Backup, Restoration, and Disaster Recovery

SEC-063 — Protected Backups

Back up production data according to a documented schedule and recovery requirement.

Backups must be protected from unauthorised access and accidental deletion.

Acceptance criteria:

-

Backup execution is monitored.

-
-

Backup access is restricted.

-
-

Backup copies receive appropriate encryption and retention controls.

-

SEC-064 — Restoration Testing

A backup is not considered reliable merely because a backup job completed successfully.

Acceptance criteria:

-

The team periodically restores a backup into an isolated test environment.

-
-

Restoration results are documented.

-
-

Restored data is checked for consistency and usability.

-

SEC-065 — Recovery Objectives

Define a Recovery Point Objective (RPO) and Recovery Time Objective (RTO) before production launch.

RPO defines the acceptable amount of data loss measured in time. RTO defines the target time for restoring service.

Acceptance criteria:

-

The product owner and technical lead approve documented RPO and RTO targets.

-
-

The backup and recovery design is assessed against those targets.

-
-

Recovery tests record actual restoration performance.

-

SEC-066 — Backup Isolation

Protect backups against compromise of ordinary application credentials.

Acceptance criteria:

-

  Ordinary application accounts cannot freely delete or download production backups.

-
-

  Backup permissions are separate from routine application permissions.

-
-

  Recovery procedures are documented and access-controlled.

-

---

13. Security Monitoring and Incident Response

SEC-067 — Security Event Logging

Record relevant security events, including authentication failures, privilege changes, permission denials, suspicious export activity, and sensitive administrative operations.

Acceptance criteria:

-

  Relevant events are recorded with timestamps and sufficient context for investigation.

-
-

  Logs do not unnecessarily include sensitive educational data.

-
-

  Log access is restricted.

-

SEC-068 — Suspicious Activity Detection

Implement practical detection for repeated authentication failures, unusual privileged operations, and other agreed high-risk activity.

Do not represent this as AI-powered threat detection.

Acceptance criteria:

-

  Configured thresholds generate operational signals.

-
-

  Signals are available to authorised technical administrators.

-
-

  Detection rules can be adjusted without changing unrelated application logic.

-

SEC-069 — Incident Response Procedure

Document how the team will respond to suspected data exposure, compromised accounts, lost credentials, unauthorised exports, and infrastructure compromise.

Acceptance criteria:

-

  The procedure identifies responsible contacts and escalation paths.

-
-

  The team can revoke compromised credentials.

-
-

  The procedure includes investigation, containment, recovery, and appropriate notification steps.

-
-

  Applicable legal reporting requirements are verified separately for each deployment jurisdiction.

-

SEC-070 — Security Incident Records

Maintain restricted incident records for material security events.

Acceptance criteria:

-

  Incident records include a timeline, affected systems, response actions, and resolution status where known.

-
-

  Access is limited to authorised personnel.

-
-

  Incident documentation does not become a general-purpose store of unnecessary student information.

-

---

14. Deployment and Operational Security

SEC-071 — Environment Separation

Separate development, testing, staging, and production environments.

Acceptance criteria:

-

  Production secrets are not reused casually across environments.

-
-

  Test environments cannot modify production data through normal credentials.

-
-

  Environment-specific configuration is documented.

-

SEC-072 — Production Data in Testing

Do not copy identifiable production student data into development or testing environments unless a documented, authorised, legally appropriate process explicitly requires it.

Prefer synthetic or appropriately de-identified test data.

Acceptance criteria:

-

The default test fixtures contain no real student personal information.

-
-

Any approved exception has restricted access and documented safeguards.

-
-

Test datasets do not expose production credentials or secrets.

-

SEC-073 — Secure Deployment Configuration

Disable development-only tools, verbose debugging, test endpoints, and insecure default accounts in production.

Acceptance criteria:

-

Production configuration is checked automatically or through a documented release checklist.

-
-

Default credentials are not active.

-
-

Debug output does not expose sensitive internals.

-

SEC-074 — Database Access Restrictions

Restrict database access to approved application services, operators, and maintenance processes.

Acceptance criteria:

- The database is not publicly exposed without an explicitly approved architecture.
- Application database accounts have only required permissions.
- Administrative database access is controlled and audited where supported.

SEC-075 — Security Review Before Release

Require a security review for significant changes to authentication, authorisation, exports, file access, role permissions, and tenant isolation.

Acceptance criteria:

- Security-sensitive changes include appropriate automated tests.
- Known critical unresolved issues block release unless a documented, authorised exception is approved.
- Review results are recorded.

---

15. Compliance and International Deployment

SEC-076 — Jurisdiction-Specific Privacy Assessment

Before deploying Optimus Orien in a country or region, identify the applicable personal-data protection, education-record, child-protection, contractual, and data-transfer requirements.

Do not assume that a single generic privacy policy satisfies every jurisdiction.

Acceptance criteria:

-

  Each deployment jurisdiction has a documented compliance assessment.

-
-

  Unresolved legal requirements are identified before launch.

-
-

  Product configuration and contracts reflect approved jurisdiction-specific decisions.

-

SEC-077 — Children's Data Protection

Treat student records as potentially sensitive, particularly where students are minors.

Determine the applicable consent, parental-authority, notice, access, and safeguarding requirements for each deployment.

Acceptance criteria:

-

  The onboarding and privacy design reflects the approved requirements.

-
-

  The system does not assume that every parent or guardian automatically has unrestricted access.

-
-

  Required notices, permissions, and records are implemented according to the approved policy.

-

SEC-078 — Data Location and Transfers

Document where production data, backups, logs, and exported files are stored and processed.

Acceptance criteria:

- Hosting locations and relevant service providers are recorded.
- International transfers are assessed where required.
- Changes to hosting regions or subprocessors trigger the appropriate review.

SEC-079 — Privacy Notice and Data Use

Maintain an approved privacy notice explaining the data processed, purposes, relevant recipients, retention approach, and applicable user rights.

Acceptance criteria:

- The deployed privacy notice matches actual product behaviour.
- New data uses are reviewed before implementation.
- The system does not claim certifications or compliance status that has not been verified.

SEC-080 — Third-Party Services

Maintain an inventory of third-party services that may process personal information, including hosting, email delivery, analytics, error tracking, and file storage.

Acceptance criteria:

- Each service has a documented purpose and data exposure assessment.
- Unnecessary personal information is not transmitted to third-party services.
- Service access and credentials can be revoked when no longer required.

---

16. Mandatory Security Test Scenarios

Before the initial production release, execute and record the following test scenarios.

Scenario A: Teacher Attempts to Access Another Class

Create two teachers with separate class assignments. Authenticate as the first teacher and attempt to access the second teacher's students, marks, attendance, reports, and exports.

Expected result: Every unauthorised operation is denied by the backend.

Scenario B: Parent Attempts to Access an Unlinked Student

Create two student accounts and a guardian linked to only one student. Attempt to retrieve the other student's profile, attendance, academic records, and reports.

Expected result: The guardian can access only the linked student's authorised records.

Scenario C: Cross-Organisation Identifier Manipulation

Create two organisations and records in both. Replace identifiers in API requests to reference the other organisation's records.

Expected result: The requests do not disclose or modify the other organisation's data.

Scenario D: Revoked Access

Create a notification for an authorised teacher. Remove the teacher's assignment and attempt to open the notification's linked record.

Expected result: The old notification does not bypass the updated permissions.

Scenario E: Public Data Exposure

Inspect unauthenticated homepage, leaderboard, search, and reporting endpoints.

Expected result: Only approved public information is returned. No private student records or intervention details are exposed.

Scenario F: Malicious File Upload

Upload disallowed file types, files with misleading extensions, oversized files, and malformed spreadsheets.

Expected result: The server rejects files that violate the configured upload policy and records appropriate safe error information.

Scenario G: Injection and Script Execution

Submit SQL-injection and script-injection test strings through relevant forms, search parameters, imports, and text fields.

Expected result: Queries remain safe and untrusted content cannot execute as script.

Scenario H: Export Access Control

Request an export containing records outside the user's authorised scope.

Expected result: The export is rejected or restricted to the permitted records according to the documented API contract. The user cannot obtain the protected records by downloading the generated file.

Scenario I: Session Revocation

Authenticate a user, revoke the session, and attempt to continue using the old credentials.

Expected result: Protected requests are denied after revocation takes effect.

Scenario J: Audit Integrity

Attempt to modify security audit records through the normal application interface and API.

Expected result: The operation is denied unless performed through a separately authorised maintenance process with appropriate controls.

Scenario K: Backup Restoration

Restore a production backup into an isolated recovery environment.

Expected result: The restored system meets the documented integrity checks and supports the approved recovery objectives.

Scenario L: Security Failure Monitoring

Simulate repeated failed logins, an unauthorised export attempt, and a failed background job.

Expected result: The configured security and operational monitoring mechanisms record the relevant events without exposing sensitive student information.

---

17. Definition of Done

Security implementation is not complete merely because login and logout work.

Part 18 is complete only when:

1.

Centralised server-side authentication and authorisation controls are implemented.

2.
3.

Role, permission, record relationship, and tenant-scope checks are enforced.

4.
5.

Cross-tenant and cross-user access tests pass.

6.
7.

Authentication, session revocation, and privileged-account protections are verified.

8.
9.

Input validation, safe query handling, and production error handling are implemented.

10.
11.

Student privacy, parent relationships, sensitive notes, and public-data restrictions are enforced.

12.
13.

File upload, import, export, and download security controls are tested.

14.
15.

Security-sensitive operations have an appropriate audit trail.

16.
17.

Encryption, key management, backup, and recovery controls are documented and verified.

18.
19.

Retention, deletion, and jurisdiction-specific privacy decisions are documented.

20.
21.

Security monitoring and incident-response procedures exist.

22.
23.

Production configuration has been reviewed for insecure defaults.

24.
25.

The security test scenarios have been executed and their results recorded.

26.
27.

Critical unresolved security issues have been addressed or handled through a documented, authorised release decision.

28.

Developer instruction: Implement security as a platform-wide foundation. Do not treat it as a final visual feature or a collection of frontend restrictions. Every subsequent Optimus Orien module must use these controls.

Part 18 is complete. The next document, Part 19, will specify developer requirements for school configuration, internationalisation, localisation, educational-system variations, currencies, dates, time zones, and country-specific settings.

Part 19 continues the developer-only specification. It defines how Optimus Orien must support different countries, languages, currencies, calendars, grading systems, time zones, and educational structures without requiring code changes for every new school or market.

PART 19

SCHOOL CONFIGURATION, INTERNATIONALISATION, AND LOCALISATION

Project: Optimus Orien
Document type: Developer Implementation Specification
Priority: High
Applies to: Application configuration, school onboarding, academic structures, grading, attendance, dates, times, currencies, languages, reports, dashboards, notifications, imports, exports, and APIs.

---

1. Implementation Objective

Implement a configurable, international-ready foundation that allows Optimus Orien to serve schools operating under different educational systems and local requirements.

The platform must support configuration at the appropriate levels:

1.

Platform defaults.

2.
3.

Country or jurisdiction.

4.
5.

Organisation.

6.
7.

School or branch.

8.
9.

Educational programme.

10.
11.

Academic year or reporting period, where historical settings must be preserved.

12.

Not every setting should be configurable at every level. Each setting must have a documented scope, permitted override rules, validation requirements, and historical behaviour.

The implementation must support:

-

Multiple languages and locale-specific formatting.

-
-

Country-specific and school-specific configurations.

-
-

Different academic calendars and school years.

-
-

Different grading systems and assessment structures.

-
-

Configurable attendance policies.

-
-

Local currencies and financial formatting where financial features are supported.

-
-

Local time zones and date conventions.

-
-

Different educational programmes within the same organisation.

-
-

Versioned configurations and historical reproducibility.

-
-

International imports, exports, reports, and notifications.

-

Critical rule: Country selection must not automatically determine every educational or operational rule. Schools in the same country may follow different programmes, exam boards, calendars, and assessment policies.

The initial development priority should be a reliable configuration engine, with Pakistan as the first deployment context. International expansion must be possible without hardcoding Pakistan-specific assumptions throughout the application.

---

2. Configuration Architecture

CFG-001 — Centralised Configuration Model

Implement a central configuration service or equivalent structured architecture.

Modules must retrieve their applicable settings through this configuration mechanism rather than independently hardcoding country, school, or programme assumptions.

Acceptance criteria:

-

  Academic, attendance, reporting, and display modules use the approved configuration source.

-
-

  Configuration changes do not require editing unrelated modules.

-
-

  Invalid configurations are rejected before becoming active.

-

CFG-002 — Explicit Configuration Scope

Every configurable setting must declare its permitted scope.

Examples include language preference at the user level, grading policy at the programme or school level, and an academic calendar at the school or programme level.

Acceptance criteria:

-

  Every setting has a documented scope.

-
-

  The backend validates which roles may modify the setting.

-
-

  A user cannot modify a setting outside their authorised scope.

-

CFG-003 — Configuration Precedence

Implement a documented precedence mechanism for settings that support inheritance.

Recommended default order:

Platform default → Country/jurisdiction default → Organisation configuration → School configuration → Programme configuration → Explicit period-specific configuration.

This order is a starting proposal, not a universal rule for every setting. The applicable order must be defined per setting because a programme's grading rules may need to override a school's general defaults, while a mandatory legal restriction must not be overridden by a lower-level setting.

Acceptance criteria:

-

The effective value can be traced to its source.

-
-

Permitted overrides behave consistently.

-
-

Restricted settings cannot be overridden by unauthorised users.

-
-

Conflicting values produce a clear validation result rather than silent behaviour.

-

CFG-004 — Effective Configuration Resolver

Provide a backend mechanism that resolves the effective configuration for a given organisation, school, programme, user, and relevant academic period.

Acceptance criteria:

-

The same input context produces the same effective configuration version.

-
-

The resolver returns only settings applicable to the requested context.

-
-

Configuration results are testable independently of the frontend.

-

CFG-005 — Configuration Validation

Validate configurations before activation.

Validation must cover required fields, valid ranges, relationships, conflicting settings, unsupported combinations, and missing dependencies.

Acceptance criteria:

-

Invalid configurations cannot become active.

-
-

Validation errors identify the affected setting and explain the correction required.

-
-

Existing active configurations remain usable when an invalid draft is rejected.

-

CFG-006 — Configuration Change Audit

Record important configuration changes, including the actor, timestamp, previous value, new value, scope, and applicable reason where required.

Acceptance criteria:

-

Changes to grading rules, attendance rules, academic calendars, and reporting settings are auditable.

-
-

Ordinary users cannot modify the audit history.

-
-

Audit records do not expose settings or records outside the viewer's permission.

-

CFG-007 — Configuration Versioning

Version configurations that affect calculations, educational structures, reporting, or historical interpretation.

Acceptance criteria:

-

A configuration change creates a new version when required.

-
-

Historical records retain the applicable configuration reference.

-
-

Existing academic results are not silently recalculated using new rules.

-

CFG-008 — Draft and Active Configurations

Where a configuration has significant operational consequences, support a draft-and-activation workflow.

Acceptance criteria:

-

Authorised users can review a draft before activation.

-
-

Invalid drafts cannot be activated.

-
-

Activation records the effective date and applicable scope.

-
-

A draft cannot silently alter production calculations.

-

---

3. Country and Jurisdiction Configuration

LOC-001 — Country Records

Maintain a structured country and jurisdiction registry.

Each record should contain a stable identifier, official or commonly accepted name, ISO country code where applicable, and any required default locale or time-zone suggestions.

Do not use a country name as a database primary key.

Acceptance criteria:

-

Country records have stable identifiers.

-
-

Renaming a displayed country label does not break related records.

-
-

Country selection is validated against the supported registry.

-

LOC-002 — Country Does Not Equal Educational System

Do not automatically assign a single educational model, grading scheme, or academic calendar solely from a country's selection.

Acceptance criteria:

-

A school can select a programme different from the country's suggested default.

-
-

Multiple programmes can coexist within one organisation.

-
-

A country's default is clearly labelled as a suggestion unless it has been formally verified as a required configuration.

-

LOC-003 — Jurisdiction-Specific Rules

Allow approved jurisdiction-specific policies to be configured where legally or operationally necessary.

These may include privacy requirements, data-retention rules, consent requirements, and reporting obligations.

Acceptance criteria:

-

Jurisdiction-specific settings are documented and versioned.

-
-

Users cannot override mandatory restrictions without authorised governance.

-
-

The system does not claim legal compliance merely because a country has been selected.

-

LOC-004 — Supported-Country Registry

Maintain an explicit list of countries and jurisdictions supported by the product.

A country appearing in the registry must not imply that every local qualification, grading system, or legal requirement is already implemented.

Acceptance criteria:

-

Unsupported features are identified before a school completes onboarding.

-
-

New country support can be introduced through configuration and verified implementation rather than hidden assumptions.

-
-

The product can distinguish a supported locale from a fully supported educational system.

-

---

4. Organisation and School Configuration

SCH-001 — Organisation Defaults

Allow an authorised organisation administrator to define organisation-wide defaults for supported settings.

Acceptance criteria:

-

Defaults apply only to the intended organisation.

-
-

School-specific overrides work only where permitted.

-
-

Changes are auditable and versioned where required.

-

SCH-002 — School-Specific Configuration

Each school must have its own applicable configuration.

This includes its calendar, time zone, enabled programmes, attendance policy, reporting periods, and other supported settings.

Acceptance criteria:

-

Schools in the same organisation can use different supported settings.

-
-

A change to one school's configuration does not unintentionally modify another school's records.

-
-

Effective settings are visible to authorised administrators.

-

SCH-003 — Multiple Programmes per School

Support multiple educational programmes within a single school where needed.

For example, a school may offer a national curriculum programme alongside Cambridge International qualifications.

Acceptance criteria:

-

Students, classes, subjects, assessments, and grading rules can be associated with the appropriate programme.

-

-

Programme-specific rules are applied to relevant records.

-
-

A programme's grading settings do not silently affect unrelated programmes.

-

SCH-004 — Configuration Wizard

Implement a school setup workflow that collects and validates the minimum required configuration.

The wizard should support:

-

Country or jurisdiction.

-
-

School and organisation identity.

-
-

Local time zone.

-
-

Preferred language and locale.

-
-

Educational programme or programmes.

-
-

Academic calendar and reporting periods.

-
-

Grading configuration.

-
-

Attendance configuration.

-
-

Applicable school-specific settings.

-

Acceptance criteria:

-

Required configuration is completed before the school performs dependent operations.

-
-

The wizard identifies incomplete or conflicting settings.

-
-

An authorised user can review and correct configuration before activation.

-
-

Optional settings are not falsely presented as mandatory.

-

SCH-005 — Configuration Completeness

Maintain a configuration-completeness status.

Acceptance criteria:

-

The system identifies missing settings that prevent a feature from operating correctly.

-
-

Incomplete configuration produces a clear explanation.

-
-

The system does not silently substitute a potentially misleading value.

-

SCH-006 — Configuration Change Impact

Before activating a change that could affect calculations, reports, or future workflows, identify the affected areas.

Acceptance criteria:

-

The interface indicates the relevant scope and potential consequences.

-
-

Historical records remain preserved.

-
-

Any required recalculation or migration follows an explicit process.

-

---

5. Internationalisation and Localisation

5.1 Technical Requirements

Internationalisation and localisation must be supported at the architecture level, not added through scattered conditional statements.

Internationalisation is commonly abbreviated as **i18n** because there are 18 letters between the first and last letters of the word. Localisation is commonly abbreviated as **l10n**.

The implementation must separate interface text from application logic and avoid assumptions about language, number formatting, text direction, or date presentation.

I18N-001 — Centralised Translation Resources

Store interface text in translation resources or the selected internationalisation framework.

Do not scatter user-facing strings across application components without a translation mechanism.

Acceptance criteria:

-

Major interface text can be translated without rewriting business logic.

-
-

Missing translations can be detected.

-
-

Translation keys remain stable enough to support ongoing maintenance.

-

I18N-002 — No User-Facing Text Hardcoding

New user-facing interface strings must use the approved translation mechanism.

Acceptance criteria:

-

Buttons, navigation, validation errors, empty states, tooltips, and relevant system messages use translation resources.

-
-

Automated or manual checks identify untranslated strings.

-
-

Internal developer logs may use a documented technical language, but must not expose private data.

-

I18N-003 — Language Preference

Support language preferences at the appropriate level, such as user preference and organisation defaults.

Acceptance criteria:

- A user can select a supported interface language when permitted.
- User preference does not change the language preference of unrelated users.
- A fallback language is defined.

I18N-004 — Language Fallback

Define deterministic fallback behaviour when a translation is missing.

Recommended fallback:

1. Requested language and regional variant.
2.
3. Base language.
4.
5. Configured organisation default, where applicable.
6.
7. Platform fallback language.
8.

Acceptance criteria:

-

Missing translations do not produce blank buttons or broken pages.

-
-

Fallback behaviour is consistent across the application.

-
-

Missing keys can be identified for correction.

-

I18N-005 — Locale-Specific Number Formatting

Format numbers according to the selected locale where appropriate.

Acceptance criteria:

-

Thousands separators and decimal separators are displayed correctly for supported locales.

-
-

Formatting changes do not alter stored numeric values.

-
-

Calculations operate on numeric values rather than formatted strings.

-

I18N-006 — Locale-Specific Dates

Support locale-appropriate date display.

For example, a date may be displayed in day-month-year order or month-day-year order depending on the chosen locale.

Acceptance criteria:

-

Date formatting follows the configured locale.

-
-

The underlying date remains unambiguous.

-
-

Parsing is not based on unreliable assumptions about the displayed format.

-

I18N-007 — Text Expansion

Design layouts to tolerate translated text that is longer than the original.

Acceptance criteria:

-

Common pages do not break when labels become longer.

-
-

Buttons, tables, navigation, and form layouts remain usable.

-
-

Truncated text can be accessed through an appropriate mechanism where necessary.

-

I18N-008 — Translation Plurals and Variables

Use the selected internationalisation framework's pluralisation and interpolation features.

Do not construct sentences by concatenating translated fragments where word order or grammar may vary.

Acceptance criteria:

-

Counts are rendered correctly for supported languages.

-
-

Dynamic values are safely inserted into translated messages.

-
-

User-supplied values are escaped appropriately.

-

I18N-009 — Language-Aware Search

Design search to support the character sets and language behaviour required by enabled locales.

Acceptance criteria:

-

Search does not assume that every user enters English text.

-
-

Names and supported scripts can be stored and retrieved correctly.

-
-

Search behaviour is documented for supported languages.

-

---

6. Right-to-Left and Multilingual Interface Support

I18N-010 — Right-to-Left Layout

Support right-to-left (RTL) layouts for languages such as Arabic and Urdu when enabled.

Acceptance criteria:

-

Page direction is set appropriately for the selected language.

-
-

Navigation, tables, forms, and major dashboard layouts remain usable in RTL mode.

-
-

Text direction is not simulated merely by reversing text strings.

-

I18N-011 — Logical CSS Properties

Prefer CSS logical properties such as margin-inline-start and padding-inline-end over hardcoded left/right assumptions where suitable.

Acceptance criteria:

-

Core layouts adapt to both left-to-right (LTR) and RTL directions.

-
-

Directional spacing does not require duplicated page implementations unless genuinely necessary.

-
-

Regression tests cover both layout directions.

-

I18N-012 — Bidirectional Text

Handle mixed-direction content, such as Arabic text containing Latin names, numbers, identifiers, or email addresses.

Acceptance criteria:

-

Mixed-direction text remains readable.

-
-

Identifiers and numeric values are not unintentionally reordered.

-
-

Direction-sensitive text fields are tested with representative multilingual data.

-

I18N-013 — Directional Icons

Review icons whose meaning depends on direction, such as back, forward, or directional navigation arrows.

Acceptance criteria:

-

Direction-sensitive icons behave consistently with the selected layout.

-
-

Non-directional icons are not unnecessarily mirrored.

-
-

Navigation remains understandable in both directions.

-

I18N-014 — Fonts and Script Coverage

Use fonts with appropriate glyph coverage for enabled languages.

Acceptance criteria:

-

Supported scripts render without missing-character boxes.

-
-

Font fallbacks are defined.

-
-

Line height and alignment remain usable with supported scripts.

-

---

7. Dates, Times, Calendars, and Time Zones

TIME-001 — Explicit School Time Zone

Each school must have an explicitly configured time zone using a maintained time-zone database identifier, such as an IANA time-zone name.

Acceptance criteria:

-

The system does not infer a school's time zone solely from the browser.

-
-

Scheduled operations use the school's configured time zone.

-
-

Invalid time-zone identifiers are rejected.

-

TIME-002 — UTC Timestamp Storage

Store event timestamps in a consistent, unambiguous representation, normally UTC.

Keep date-only values, such as a student's date of birth or a calendar date, distinct from timestamps representing moments in time.

Acceptance criteria:

-

Timestamp values can be converted reliably to the appropriate local time.

-
-

Date-only values do not shift to another calendar date because of time-zone conversion.

-
-

APIs document the format of date and timestamp fields.

-

TIME-003 — Local-Time Display

Display timestamps in the appropriate configured local time zone and locale.

Acceptance criteria:

-

A school event created at a specific moment displays correctly for authorised users.

-
-

User display preferences do not alter the underlying event timestamp.

-
-

Ambiguous or invalid input times are handled explicitly.

-

TIME-004 — Daylight-Saving Time

Use a maintained time-zone database rather than hardcoding fixed UTC offsets.

Acceptance criteria:

-

Regions with daylight-saving transitions display and schedule events correctly.

-
-

Future schedules use the applicable time-zone rules.

-
-

Changes in regional time-zone rules can be incorporated through supported database updates.

-

TIME-005 — Configurable Academic Calendar

Support configurable academic calendars, including academic-year boundaries, terms, semesters, holidays, closures, and reporting periods as required by the school.

Acceptance criteria:

-

A school can configure its academic periods without changing application code.

-
-

The system validates dates and period relationships.

-
-

Reports use the correct configured period boundaries.

-

TIME-006 — Academic Year Labels

Do not assume that every academic year corresponds to January through December.

Acceptance criteria:

-

Schools can define their academic-year start and end dates.

-
-

Academic year labels can differ from calendar years.

-
-

Historical academic periods remain identifiable after later configuration changes.

-

TIME-007 — Multiple Calendars

Support different calendars within the same organisation where required and approved.

Acceptance criteria:

-

Schools or programmes can use their applicable calendar.

-
-

Attendance and reporting calculations use the relevant calendar.

-
-

An unrelated calendar change does not silently change existing attendance records.

-

TIME-008 — Holiday and Closure Handling

Attendance and deadline calculations must respect the relevant calendar and applicable scheduled sessions.

Acceptance criteria:

-

Holidays and closures are excluded from attendance denominators where the configured policy requires it.

-
-

A school closure does not automatically create absence records for unscheduled attendance opportunities.

-
-

Calendar changes follow an explicit correction or recalculation process.

-

TIME-009 — Week Start and Weekend Configuration

Support configurable week-start conventions and school operating days where relevant.

Acceptance criteria:

-

Weekly views use the configured week start.

-
-

Reminder schedules respect configured operating days.

-
-

The application does not assume every school follows the same weekend.

-

---

8. Educational System and Programme Configuration

EDUCFG-001 — Programme Registry

Maintain a structured registry of educational programmes or pathways.

A programme record should have a stable identifier, display name, scope, status, and applicable configuration references.

Acceptance criteria:

-

Programme names are not used as primary keys.

-
-

Schools can select supported programmes.

-
-

Programme changes do not silently rewrite historical student records.

-

EDUCFG-002 — Configurable Academic Structure

Support configurable educational structures such as stages, year groups, grades, classes, subjects, and reporting periods.

Do not assume every system uses the same hierarchy or terminology.

Acceptance criteria:

-

Schools can configure supported structures without source-code changes.

-
-

Display labels can vary while stable identifiers remain unchanged.

-
-

The system rejects invalid relationships between configured entities.

-

EDUCFG-003 — Configurable Grade and Year Labels

Allow local labels such as Grade 8, Year 9, Class VIII, or another approved label.

Acceptance criteria:

-

Labels can be configured independently of internal identifiers.

-
-

Changing a display label does not change historical enrolment relationships.

-
-

Sorting follows an explicit ordering field rather than alphabetical assumptions.

-

EDUCFG-004 — Programme-Specific Subjects

Support subjects defined or selected within the applicable programme and academic context.

Acceptance criteria:

-

Subjects can be associated with the relevant programme and school.

-
-

Subject identifiers remain stable when display names change.

-
-

Subject records from different programmes are not automatically treated as equivalent.

-

EDUCFG-005 — Programme-Specific Grading

Assessment calculations must use the applicable grading configuration.

This may include marks-based grading, percentages, grade boundaries, letter grades, points, pass/fail outcomes, or other explicitly supported systems.

Acceptance criteria:

-

Grading rules are selected from the correct programme and period configuration.

-
-

Invalid or incomplete grading configurations are rejected.

-
-

Reports identify the applicable grading context.

-

EDUCFG-006 — Grading Configuration Versioning

Version grading rules that affect results.

Acceptance criteria:

-

Historical results retain their original grading context.

-
-

New grading boundaries do not silently rewrite historical grades.

-
-

Any approved historical recalculation is explicit and auditable.

-

EDUCFG-007 — Assessment Structure Configuration

Support configurable assessment categories, maximum marks, weighting rules, and reporting periods where required by the programme.

Acceptance criteria:

-

Assessment rules are validated before use.

-
-

Unsupported combinations are rejected or clearly flagged.

-
-

A change to an assessment rule does not silently alter existing results.

-

EDUCFG-008 — No Assumed Qualification Equivalence

Do not treat similarly named grades, subjects, or qualifications as equivalent without an explicitly approved mapping.

Acceptance criteria:

-

Cross-programme comparisons require compatible definitions or a documented mapping.

-
-

Reports do not imply official qualification equivalence merely because labels match.

-
-

Mappings are versioned and carry an explicit source and approval status where required.

-

EDUCFG-009 — Custom Educational Configurations

Allow authorised users to configure a custom educational arrangement when no supported template fits.

Acceptance criteria:

-

Custom configurations can be created and validated.

-
-

Custom configurations are labelled as custom, not as officially recognised templates.

-
-

Historical records retain their custom configuration reference.

-

EDUCFG-010 — Official Template Verification

Any template labelled as official must be supported by verified documentation from the relevant awarding organisation or authoritative body.

Acceptance criteria:

-

Official templates include their source and applicable version where available.

-
-

Unverified templates cannot be labelled official.

-
-

Updates to official requirements are reviewed before activation.

-

---

9. Grading, Marks, and Calculation Localisation

GRADELOC-001 — Store Raw Values Separately from Display Formats

Store marks and other numeric values in appropriate numeric database types.

Do not store formatted values such as "75,5%" as the underlying numeric result.

Acceptance criteria:

-

Locale changes do not change stored marks.

-
-

Calculations operate on numeric values.

-
-

Display formatting is applied at the presentation layer.

-

GRADELOC-002 — Configurable Decimal Precision

Support appropriate precision for assessment marks and calculated values.

Acceptance criteria:

-

Stored precision is sufficient for approved calculations.

-
-

Rounding behaviour is documented.

-
-

Display rounding does not unintentionally alter the underlying result.

-

GRADELOC-003 — Configurable Rounding Policy

Define rounding policies for calculated grades and reported percentages where applicable.

Acceptance criteria:

-

Rounding occurs at the specified calculation stage.

-
-

The same approved calculation produces consistent results across dashboards and exports.

-
-

Changes to rounding rules are versioned when they affect historical interpretation.

-

GRADELOC-004 — Percentage and Grade Boundaries

Grade boundaries must be configurable according to the applicable educational system.

Acceptance criteria:

-

Boundaries cannot overlap or leave unintended gaps unless explicitly supported by the grading model.

-
-

Boundary conditions are tested.

-
-

Grade labels are not hardcoded as universal across programmes.

-

GRADELOC-005 — Missing Results

Distinguish missing results from zero marks, absence, exemption, pending assessment, and other supported states.

Acceptance criteria:

-

Missing results are not automatically converted to zero.

-
-

The applicable state is preserved in calculations and reports.

-
-

Incomplete records are handled according to the configured policy.

-

GRADELOC-006 — Calculation Reproducibility

All grade calculations must use an explicit, versioned formula and configuration.

Acceptance criteria:

- A historical result can be explained using its source records and calculation configuration.
- Equivalent inputs and configuration produce reproducible outputs.
- Dashboard, report, and export values remain consistent.

---

10. Currency and Financial Localisation

Financial localisation applies only to financial features that are actually implemented in Optimus Orien. It must not introduce a full accounting or invoicing system into the current scope without separate approval.

FINLOC-001 — Currency Registry

Use a structured currency registry based on appropriate currency identifiers, such as ISO 4217 codes where applicable.

Acceptance criteria:

- Currency codes are not inferred solely from symbols.
- Supported currencies have defined display rules and precision.
- Unsupported currencies are rejected or clearly identified.

-

FINLOC-002 — Currency Display

Display currency values using the applicable locale and currency code or symbol.

Acceptance criteria:

-

Values are formatted consistently.

-
-

Ambiguous symbols are supplemented by an appropriate currency code where necessary.

-
-

Display formatting does not alter the underlying value.

-

FINLOC-003 — Currency Precision

Use currency-appropriate precision and appropriate decimal storage for financial values.

Acceptance criteria:

-

Calculations avoid binary floating-point errors for monetary values.

-
-

Rounding rules are explicit.

-
-

Currency precision is not assumed to be identical for every currency.

-

FINLOC-004 — No Implicit Currency Conversion

Do not combine values expressed in different currencies without an explicit, documented conversion process.

Acceptance criteria:

-

  Reports identify the currency of displayed financial values.

-
-

  Cross-currency totals are blocked unless a conversion rule is defined.

-
-

  Any conversion uses a documented exchange-rate source, effective date, and recorded rate.

-

FINLOC-005 — Subscription Pricing Configuration

If subscription pricing is implemented, separate pricing configuration from currency display and educational configuration.

Acceptance criteria:

-

  A displayed price identifies its applicable currency and billing basis.

-
-

  Pricing changes are versioned according to the billing design.

-
-

  Changing a school's locale does not silently change its contractual price.

-

---

11. Language, Locale, and User Preferences

PREF-001 — User Locale Preference

Support a user-specific locale preference where appropriate.

Acceptance criteria:

- Changing one user's locale does not change other users' display preferences.
- Locale changes do not alter the school's grading or attendance policy.
- Unsupported locale selections are rejected or handled using the documented fallback.

PREF-002 — School Default Locale

Allow an authorised school administrator to set a default locale for the school.

Acceptance criteria:

- New users receive the appropriate default where the onboarding policy specifies it.
- Existing user preferences are preserved unless the policy explicitly requires otherwise.
- The school default does not override mandatory system messages incorrectly.

PREF-003 — Separate Locale from Language

Do not treat language and locale as identical concepts.

A language determines interface text, while a locale can also determine number formatting, date presentation, and other conventions.

Acceptance criteria:

-

  Language and regional formatting can be configured separately where supported.

-
-

  Changing the display locale does not change the underlying educational programme.

-
-

  Configuration behaviour is documented.

-

PREF-004 — Stable API Formats

APIs must use explicit, documented formats for dates, timestamps, numbers, identifiers, and enumerated values.

Do not make machine-readable API values dependent on the user's display locale.

Acceptance criteria:

-

  API values remain stable across language and locale settings.

-
-

  Clients can distinguish numeric values from formatted display strings.

-
-

  Date-only fields and timestamps are represented unambiguously.

-

---

12. Reporting, Export, and Localisation

REPLOC-001 — Localised Reports

Reports must use the selected or configured language and appropriate locale-specific formatting where supported.

Acceptance criteria:

-

Report labels use translation resources.

-
-

Dates and numbers follow the approved locale.

-
-

The report preserves the underlying calculation values.

-

REPLOC-002 — Report Context

Reports must identify the relevant school, programme, academic period, and grading context when those details are necessary to interpret the results.

Acceptance criteria:

-

Reports do not present values without required context.

-
-

Programme-specific labels are displayed correctly.

-
-

Users can distinguish records from different reporting periods.

-

REPLOC-003 — Export Consistency

CSV, spreadsheet, and PDF exports must use documented formatting rules.

Machine-readable exports must remain suitable for data processing, while human-readable exports may use localised presentation.

Acceptance criteria:

-

Export formats are documented.

-
-

Numeric values are not corrupted by locale-specific formatting.

-
-

The same calculation produces consistent numeric results across supported export formats.

-

REPLOC-004 — Translation of Statuses

Translate user-facing status labels without changing their underlying stable codes.

Acceptance criteria:

-

A status can be displayed in different languages while retaining the same internal identifier.

-
-

Imports and APIs use documented stable values.

-
-

Translated labels do not become database keys.

-

REPLOC-005 — Localised Notification Templates

Notification templates must support translation, local time display, and locale-appropriate formatting.

Acceptance criteria:

-

Notifications use the configured language and formatting rules.

-
-

Required privacy restrictions apply regardless of language.

-
-

Dynamic text is safely interpolated.

-

---

13. Import and Export Compatibility Across Locales

DATA-LOC-001 — Explicit Import Formats

Import workflows must allow or clearly determine the expected date, number, encoding, and delimiter formats.

Acceptance criteria:

-

Ambiguous dates are not silently interpreted using the wrong locale.

-
-

Invalid or ambiguous numeric values are flagged for correction.

-
-

Import previews show how values will be interpreted before confirmation.

-

DATA-LOC-002 — Character Encoding

Support appropriate character encoding for enabled languages, normally UTF-8 for modern data exchange.

Acceptance criteria:

- Names and supported scripts survive import and export without corruption.
- Unsupported or malformed encodings produce a clear error.
- Round-trip tests cover multilingual data.

DATA-LOC-003 — Stable Identifiers

Imports must use stable record identifiers or approved matching rules rather than translated names alone.

Acceptance criteria:

- Changing a displayed school, subject, or programme name does not break reliable imports.
- Ambiguous matching is flagged rather than silently resolved.
- Duplicate records are not created merely because a label has changed.

DATA-LOC-004 — Locale-Safe Numeric Parsing

Parse imported numeric values using an explicitly selected or reliably detected format.

Acceptance criteria:

- A value such as `1,234` is not silently interpreted incorrectly when the source format is ambiguous.
- Ambiguous values require explicit confirmation where necessary.
- Validation occurs before records are committed.

---

14. Configuration Migration and Historical Preservation

MIG-LOC-001 — Versioned Configuration Migration

Changes to configuration structures must use controlled migrations.

Acceptance criteria:

- Database schema and configuration migrations are versioned.
- Migration failures are detectable.
- Existing records are not silently assigned incorrect educational settings.

MIG-LOC-002 — Historical Configuration References

Historical results must retain references to the configuration that governed their interpretation.

Acceptance criteria:

-

Old reports can identify their applicable programme, grading rules, and reporting period.

-
-

New defaults do not rewrite historical context.

-
-

Historical references remain valid after display labels change.

-

MIG-LOC-003 — Explicit Recalculation

If a configuration change requires recalculation of existing results, execute it through an explicit, authorised process.

Acceptance criteria:

-

The system identifies affected records.

-
-

Recalculation is auditable.

-
-

The original values or calculation history are preserved according to the data-retention design.

-
-

Failed recalculations do not leave partially updated results.

-

MIG-LOC-004 — Configuration Deprecation

Support deprecating a configuration without immediately deleting it when historical records still depend on it.

Acceptance criteria:

-

Deprecated configurations cannot be selected for new records when prohibited.

-
-

Historical records remain readable.

-
-

Users receive a clear explanation when a deprecated configuration cannot be used.

-

---

15. Performance and Configuration Caching

CFG-009 — Effective Configuration Caching

Cache resolved configuration where appropriate to reduce repeated database work.

Acceptance criteria:

-

Cached values are scoped to the correct organisation, school, programme, and period.

-
-

Configuration changes invalidate or refresh affected cached values.

-
-

A cache key cannot cause one school's configuration to be served to another school.

-

CFG-010 — Cache Invalidation

Implement explicit invalidation or version-based refresh for configuration changes.

Acceptance criteria:

-

Changes to grading rules are reflected in subsequent authorised operations according to the activation policy.

-
-

Users do not continue receiving obsolete configuration indefinitely.

-
-

Historical calculations still use their required historical configuration.

-

CFG-011 — Configuration Read Performance

Avoid loading every configuration record for every request.

Acceptance criteria:

-

Common configuration lookups are measured in performance tests.

-
-

Related configuration is fetched efficiently.

-
-

Performance optimisation does not weaken scope checks.

•

---

16. Recommended Configuration Data Model

The following entities describe the required logical model. Names may be adapted to the existing codebase, but their responsibilities must remain explicit.

Country or Jurisdiction

Stores country and jurisdiction identifiers, supported status, and approved defaults.

Organisation Configuration

Stores organisation-level settings and references to applicable versions.

School Configuration

Stores school-specific settings and permitted overrides.

Programme Configuration

Stores programme-specific academic structures, assessment settings, and grading policies.

Academic Calendar

Stores academic-year boundaries, periods, closures, holidays, and relevant scheduled dates.

Grading Scheme

Stores grading method, boundaries, calculation rules, display labels, and version.

Attendance Policy

Stores attendance codes, counting rules, eligible attendance opportunities, and reporting policies.

Locale Configuration

Stores supported language, regional formatting preferences, text direction, and fallback settings.

Currency Configuration

Stores supported currency identifiers, precision, and display metadata where financial features require them.

Configuration Version

Records version identifiers, activation status, effective dates, and relationships to affected configuration objects.

Configuration Audit Record

Records the actor, change, timestamp, scope, and relevant reason.

Effective Configuration Resolver

Provides the validated configuration applicable to a particular operation.

The implementation must avoid duplicating the same effective setting independently across every module.

---

17. Developer Test Scenarios

Scenario A — Two Schools, Different Academic Calendars

Create two schools in the same organisation with different academic-year boundaries.

Expected result: Each school uses its own configured periods, and changing one calendar does not silently modify the other school's records.

Scenario B — Multiple Programmes in One School

Configure a school offering two educational programmes with different grading rules.

Expected result: Results are calculated and displayed according to the correct programme configuration.

Scenario C — Language Change

Change a user's interface language.

Expected result: Interface labels change to the selected language where translations exist, but the user's marks, grading rules, and stored values remain unchanged.

Scenario D — Locale-Specific Numbers

Display and export the same numeric result under two supported locales.

Expected result: Presentation follows the selected locale while the underlying numeric value remains identical.

Scenario E — Right-to-Left Layout

Open core application pages in a supported RTL language.

Expected result: Navigation, forms, tables, charts, and major layouts remain usable, and mixed-direction identifiers remain readable.

Scenario F — Time-Zone Differences

Create scheduled events for schools in different time zones.

Expected result: Each event is stored unambiguously and displayed at the correct local time.

Scenario G — Missing Configuration

Attempt to activate a school without required grading or calendar settings.

Expected result: The system identifies the missing configuration and prevents dependent operations that cannot be performed safely.

Scenario H — Historical Grading Rules

Create historical results, change the grading scheme, and inspect the old results.

Expected result: Historical context is preserved, and the system does not silently rewrite previous grades.

Scenario I — Ambiguous Import

Import a spreadsheet containing ambiguous date or numeric formats.

Expected result: The system requests clarification or flags ambiguous values rather than silently importing potentially incorrect data.

Scenario J — Configuration Scope

Attempt to change an organisation setting through an account that only has school-level permissions.

Expected result: The backend rejects the operation.

Scenario K — Configuration Cache

Change an active configuration while configuration caching is enabled.

Expected result: Applicable new operations use the correct active version according to the activation policy, while historical records remain interpretable.

Scenario L — Official Template Labelling

Create a custom educational configuration without verified official documentation.

Expected result: The configuration is labelled custom and cannot be represented as an officially verified programme template.

---

18. Definition of Done

Part 19 is complete when:

1.

A central configuration architecture exists.

2.
3.

Configuration scopes and precedence rules are documented.

4.
5.

Organisation, school, and programme settings are separated appropriately.

6.
7.

School onboarding validates required configuration.

8.
9.

Educational structures and grading rules are configurable.

10.
11.

Configuration changes are audited and versioned where required.

12.
13.

Language and locale formatting are separated from business logic.

14.
15.

Core interface components support translation.

16.
17.

RTL support is implemented and tested for enabled languages.

18.
19.

Dates, timestamps, calendars, and time zones use explicit rules.

20.
21.

Currency handling is safe wherever financial features are implemented.

22.
23.

Reports, exports, imports, and notifications follow approved locale rules.

24.
25.

Historical records preserve the configuration needed to interpret them.

26.
27.

Cache invalidation and configuration migration are tested.

28.
29.

The developer test scenarios pass.

30.
31.

No country-specific assumption silently overrides a school's explicitly approved configuration.

32.

Implementation instruction: Build the configuration foundation before adding country-specific exceptions. Every new educational programme, language, or jurisdiction should use the established configuration architecture rather than introduce scattered hardcoded rules.

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19. Next Document

Part 20 — Technical Architecture and System Integration

The next specification will define the application architecture, frontend/backend boundaries, database design principles, API contracts, background processing, integration patterns, deployment environments, and technical implementation structure for Optimus Orien.

Part 19 is complete. Ask for **Part 20** when you're ready to continue.

Here is **Part 20**, shortened to focus on the developer's implementation decisions, architecture requirements, and acceptance criteria.

PART 20

TECHNICAL ARCHITECTURE AND SYSTEM INTEGRATION

Project: Optimus Orien
Audience: Developer
Priority: High
Objective: Establish a maintainable, secure, scalable architecture before expanding development.

1. Architecture Requirements

ARCH-001 — Inspect the Existing Codebase

Before making architectural changes, inspect the current repository, framework, database, authentication, existing modules, and deployment configuration.

Acceptance criteria: Existing functionality is documented, and the developer identifies reusable components before rebuilding anything.

ARCH-002 — Modular Architecture

Organise the application into clear modules, including:

-

Authentication and permissions

-
-

Organisations, schools, and enrolments

-
-

Academic programmes and configuration

-
-

Assessments and grades

-
-

Attendance

-
-

Homework, participation, and skills

-
-

Student performance dashboards

-
-

School health and interventions

-
-

Reports, analytics, and exports

-
-

Notifications, if implemented

-

Acceptance criteria: Modules have clear responsibilities and reuse shared services rather than duplicating business logic.

ARCH-003 — Frontend and Backend Separation

The frontend handles presentation and user interaction. The backend handles business rules, calculations, permissions, validation, and persistence.

Acceptance criteria: Critical rules cannot be bypassed by modifying frontend code or API requests.

ARCH-004 — Database as the Source of Truth

Persist authoritative records in the database. Do not use frontend state, cached data, or manually duplicated values as the source of truth.

Acceptance criteria: Refreshing a page or signing in on another authorised device returns consistent saved records.

2. Database and Data Integrity

ARCH-005 — Relational Data Model

Use a relational database where appropriate for structured educational records and relationships.

Use stable primary keys, foreign keys, appropriate indexes, and database constraints.

Acceptance criteria: Invalid relationships and prohibited duplicate records are prevented wherever database constraints can enforce them.

ARCH-006 — Historical Records

Preserve historical enrolments, assessment results, attendance, grading configurations, and relevant assignments.

Acceptance criteria: Moving a student to another class or school does not silently erase their historical records.

ARCH-007 — Transactions

Use database transactions for operations that must succeed or fail together.

Acceptance criteria: Failed multi-record operations do not leave unintended partial changes.

ARCH-008 — Schema Migrations

Version database schema changes and test migrations before deployment.

Acceptance criteria: Schema changes can be applied consistently across environments, and migration failures are detectable.

3. API Design

ARCH-009 — Consistent API Contracts

Use consistent API conventions for request validation, responses, pagination, filtering, and errors.

Document endpoints and expected payloads.

Acceptance criteria: Frontend and backend agree on field names, data types, validation rules, and error formats.

ARCH-010 — Server-Side Authorisation

Every protected endpoint must verify permissions, tenant scope, and record relationships.

Acceptance criteria: Direct API requests cannot bypass portal restrictions or access another school's data.

ARCH-011 — Shared Calculation Services

Implement grades, attendance rates, averages, progress measures, and school health indicators through reusable backend services.

Acceptance criteria: Dashboards, reports, and exports use the same approved calculation logic.

ARCH-012 — API Versioning

Introduce API versioning when changes would break existing clients or integrations.

Acceptance criteria: Breaking API changes follow a documented migration strategy rather than unexpectedly breaking deployed interfaces.

4. Background Processing and Integrations

ARCH-013 — Background Jobs

Use background processing for tasks that may be slow or resource-intensive, such as large exports, bulk imports, and substantial recalculations.

Acceptance criteria: Long-running tasks do not unnecessarily block ordinary user requests, and their status and failures can be tracked.

ARCH-014 — Safe Job Execution

Background jobs must preserve permissions, tenant boundaries, and data integrity.

Acceptance criteria: Failed jobs can be retried safely without duplicating records or applying changes twice.

ARCH-015 — External Integrations

Integrations with email providers, storage services, or other external systems must use documented interfaces and securely managed credentials.

Acceptance criteria: External-service failures are handled gracefully and do not expose secrets or protected student information.

Scope restriction: Do not introduce AI services, AI predictions, or in-app chat.

5. Performance and Scalability

ARCH-016 — Database Query Efficiency

Use appropriate indexes, efficient queries, pagination, and aggregation strategies.

Acceptance criteria: Large student lists and reports do not require loading every record into the browser.

ARCH-017 — Caching

Cache only where beneficial. Scope cached data correctly and invalidate it when relevant data or permissions change.

Acceptance criteria: Cached responses cannot expose another tenant's data or indefinitely display obsolete permissions.

ARCH-018 — Large Exports

Generate large exports asynchronously where necessary, with access-controlled downloads and expiration.

Acceptance criteria: Export generation does not exhaust application resources, and users can retrieve only authorised files.

ARCH-019 — Performance Testing

Establish measurable performance targets based on expected users, school sizes, and deployment resources.

Acceptance criteria: Representative dashboard, search, and reporting workloads are tested before production launch.

6. Environments and Deployment

ARCH-020 — Environment Separation

Maintain separate development, testing, staging, and production environments.

Acceptance criteria: Production credentials and data are not casually shared with development environments.

ARCH-021 — Configuration and Secrets

Keep environment-specific configuration and secrets outside source code.

Acceptance criteria: Credentials and private keys are not committed to the repository or exposed in frontend bundles.

ARCH-022 — Automated Build and Deployment

Where practical, automate dependency installation, tests, build validation, and deployment through a controlled pipeline.

Acceptance criteria: A release can be reproduced from the approved source revision and deployment configuration.

ARCH-023 — Logging and Monitoring

Implement structured application logs and operational monitoring.

Acceptance criteria: Developers can investigate failures without logging passwords, tokens, or unnecessary student information.

ARCH-024 — Backup and Recovery

Implement and test the approved backup and recovery strategy.

Acceptance criteria: Restoration is demonstrated in an isolated environment and measured against documented recovery objectives.

7. Required Architecture Deliverables

Before considering the architecture complete, the developer must provide:

1.

A concise architecture diagram showing frontend, backend, database, storage, background workers, and external services.

2.
3.

A database relationship diagram.

4.
5.

An API endpoint inventory.

6.
7.

A module and repository structure.

8.
9.

An environment and deployment configuration guide.

10.
11.

A list of dependencies and external integrations.

12.
13.

A documented authentication and authorisation approach.

14.
15.

Automated tests covering permissions, tenant isolation, calculations, and critical data operations.

16.

8. Definition of Done

Part 20 is complete when:

-

The architecture and major module boundaries are documented.

-
-

Frontend and backend responsibilities are clearly separated.

-
-

Database relationships and migrations are controlled.

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-

APIs enforce permissions and use consistent contracts.

-
-

Shared calculations prevent conflicting results.

-
-

Background jobs handle failures and retries safely.

-
-

Performance, logging, deployment, and recovery requirements are addressed.

-
-

Existing functionality remains operational after architectural changes.

-

Final instruction: Inspect the current codebase before selecting or replacing technologies. Prefer the simplest architecture that satisfies Optimus Orien's security, maintainability, and expected scale. Do not introduce unnecessary infrastructure or rewrite working modules without a clear technical justification.

Next: Part 21 — Testing, Deployment, Monitoring, and Reliability.